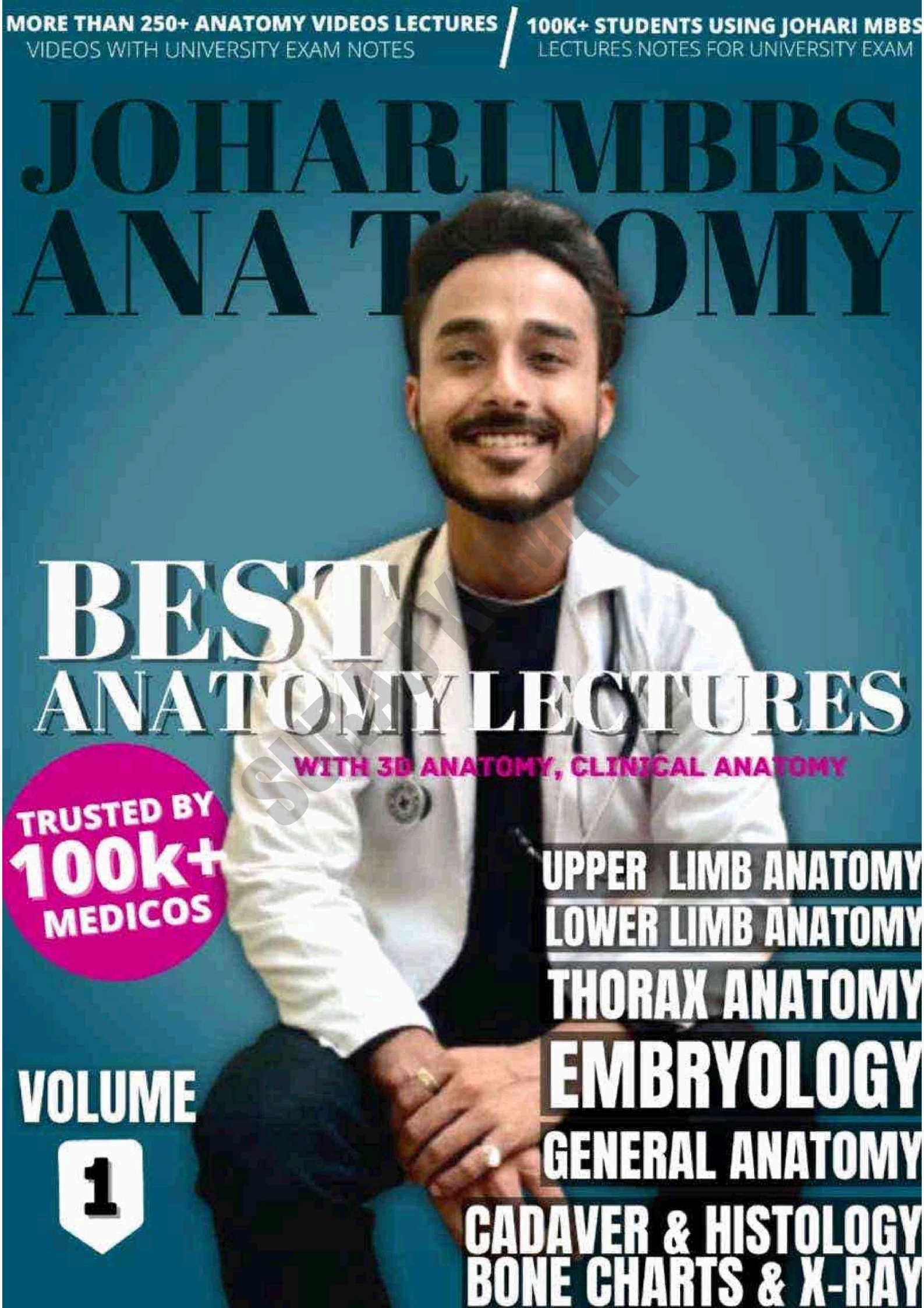
MORE THAN 250+ ANATOMY VIDEOS LECTURES
VIDEOS WITH UNIVERSITY EXAM NOTES

100K+ STUDENTS USING JOHARI MBBS
LECTURES NOTES FOR UNIVERSITY EXAM

JOHARI MBBS ANATOMY



BEST ANATOMY LECTURES

WITH 3D ANATOMY, CLINICAL ANATOMY

TRUSTED BY 100k+ MEDICOS

UPPER LIMB ANATOMY
LOWER LIMB ANATOMY
THORAX ANATOMY
EMBRYOLOGY
GENERAL ANATOMY
CADAVER & HISTOLOGY
BONE CHARTS & X-RAY

VOLUME 1

SUBJECT

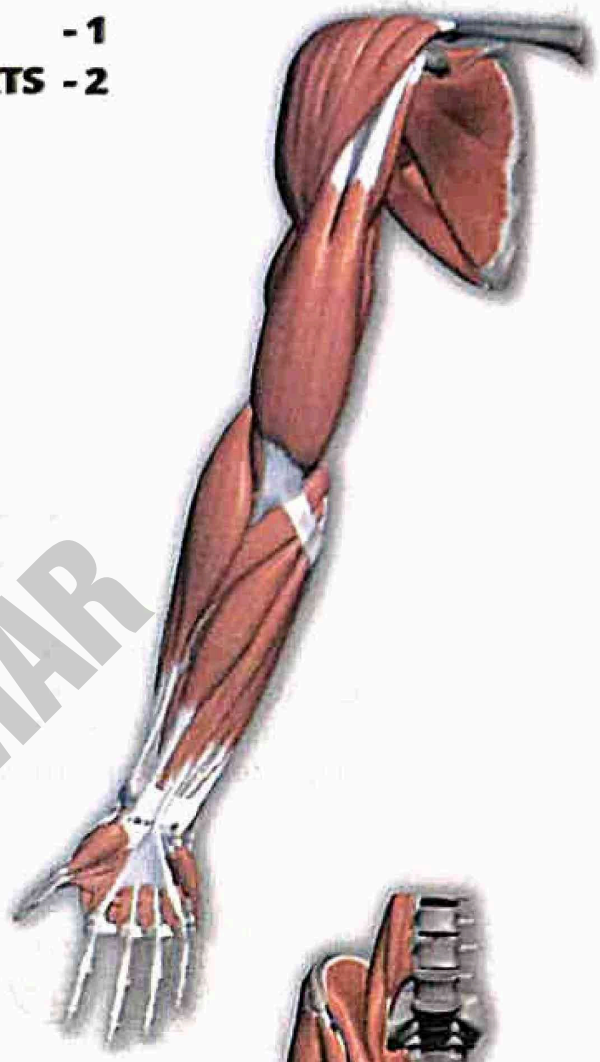
UPPER LIMB IMPS TOPICS

JOHARI MBBS | YOUTUBE

UPPER LIMB MUSCLE CHARTS - 1
BLOOD SUPPLY OF UPPER LIMB CHARTS - 2

1) BRACHIAL PLEXUS - 3
2) AXILLA - 4
3) AXILLARY ARTERY - 5
4) CUBITAL FOSSA - 6
5) MAMMARY GLAND - 8
6) RADIAL NERVE - 11
7) ULNAR NERVE - 13
8) MEDIAN NERVE - 15
9) ROTATOR CUFF - 17
10) FLEXOR RETINACULUM - 17
11) MID PALMAR SPACE - 19
12) SNUFF BOX - 19
13) LUMBRICAL - 20
14) THENAR - 20
15) SHOULDER JOINT - 21
14) ELBOW JOINT - 23
15) SUPINATION & PRONATION - 24

UNIVERSITY QUESTIONS - 26
MCQS OF UEPPR LIMB - 29



SURAJ KUMAR

LOWER LIMB IMPS

LOWER LIMB MUSCLE CHART - 30
NERVE SUPPLY OF LOWER LIMB CHARTS - 31

1) FEMORAL TRIANGLE - 32
2) FEMORAL SHEATH & CANAL - 34
3) FEMORAL ARTERY - 34
4) FEMORAL NERVE - 35
5) ADDUCTOR CANAL - 37
6) POPLITEAL FOSSA - 39
7) OBTURATOR NERVE - 39
8) SCIATIC NERVE - 41
9) ARCHE OF FOOT - 43
10) VENOUS DRAINAGE OF LOWER LIMB - 44
11) HAMSTRING - 47

Printer by : paramountbinding01@gmail.com

12) FLEXOR RETTINACULUM - 47
13) HIP JOINT - 49
14) LOCKING & UNLOCKING OF KNEE - 50
15) KNEE JOINT - 52

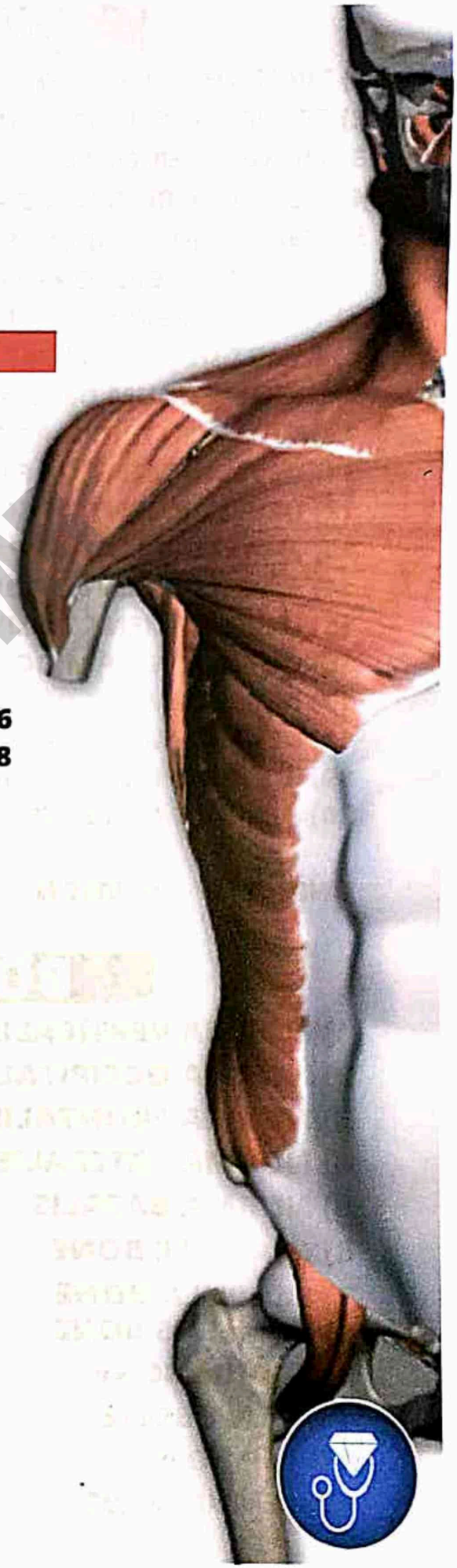
UNIVERSITY QUESTIONS - 55
MCQS OF LOWER LIMB - 56

THORAX IMPS

THORAX MUSCLE CHARTS - 57
ABDOMEN MUSCLE CHARTS - 58

1) PLEURA - 59
2) RECESSES OF PLEURA - 61
3) PULMONARY LIGAMENT - 61
4) FISSURE AND LOBE OF LUNG - 62
5) ROOT OF LUNG - 64
6) BRONCHOPULMONARY SEGMENTS - 66
7) SUPERIOR & INFERIOR MEDIASTINUM - 68
8) PERICARDIUM - 71
9) RIGHT ATRIUM - 73
10) ARTERY SUPPLY OF HEART - 74
11) VEINS OF HEART - 75
12) AZYGOS VEIN - 77
13) ARCH OF AORTA - 77
14) THORACIC DUCT - 79
15) INTERCOSTAL NERVE - 80
14) INTERCOSTAL SPACE - 82
15) INLET OF THORAX - 83
16) OUTLET OF THORAX - 84

UNIVERSITY QUESTION - 85
MCQ OF THORAX LIMB - 86

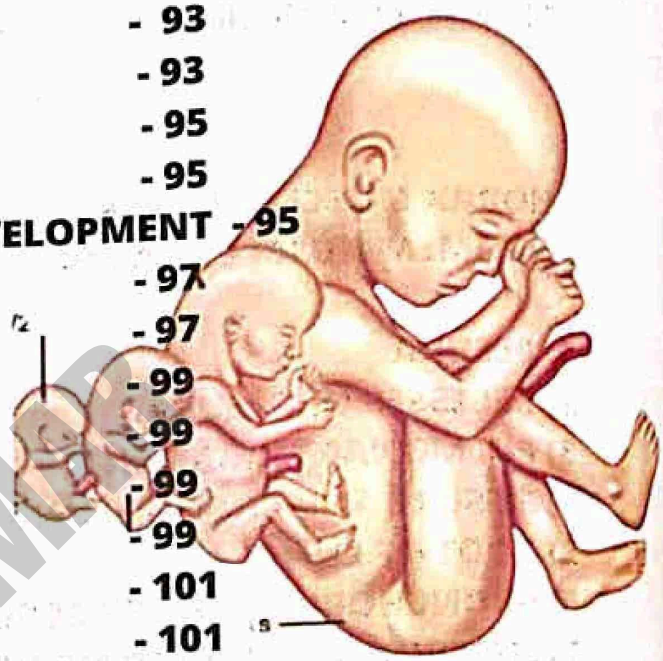


SURAJ KUMAR

EMBRYOLOGY

1) FIRST WEEK OF DEVELOPMET - 87
2) SECOND WEEK OF DEVELOPMENT - 89
3) THIRD WEEK OF DEVELOPMENT - 90
4) NOTOCHORD DEVELOPMENT - 90
5) PHARYNGEAL ARCHES - 92
6) THYROID DEVELOPMENT - 93
7) TONGUE DEVELOPMENT - 93
8) HEART DEVELOPMENT - 95
9) RIGHT ATRIUM DEVELOPMENT - 95
10) INTERVENTRICULAR SEPTUM DEVELOPMENT - 95
11) DEVELOPMENT OF FACE - 97
12) DEVELOPMENT OF SOFT PALATE - 97
13) DEVELOPMENT OF PANCREAS - 99
14) DEVELOPMENT OF DUODENUM - 99
15) DEVELOPMENT OF LIVER - 99
16) ROTATION OF MIDGUT - 99
17) DEVELOPMENT OF KIDNEY - 101
18) DEVELOPMENT OF HIDGUT - 101
19) DEVELOPMENT OF GONADS - 103
20) DESCENT OF TESTIS - 103

UNIVERITY QUESTION - 104



SURAJ KUMAR

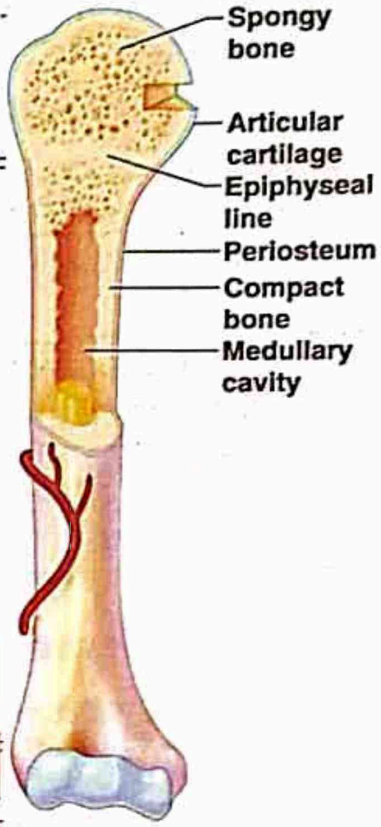
BONE CHARTS

1) NORMA VERTICALIS - 113
2) NORMA OCCIPITALIS - 113
3) NORMA FRONTALIS - 116
4) NORMA LATERALIS - 117
5) NORMA BASALIS - 118
6) CLAVICLE BONE - 119
7) SCAPULA BONE - 121
8) HUMERUS BONE - 125
9) RADIUS BONE - 128
10) ULNA BONE - 129
11) HIP BONE - 131
12) FEMUR BONE - 136

13) FIBULA BONE - 139
14) TIBIA BONE - 140

GENERAL ANATOMOY

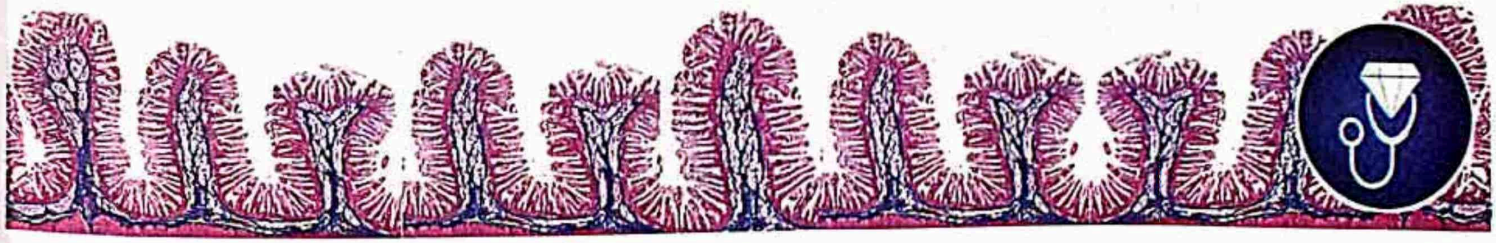
1) EPIPHYSIS - 105
2) BONE - 105
3) BLOOD SUPPLY OF BONE - 107
4) CARTILAGE - 108
5) JOINT - 109
6) SYNOVIAL - 110
7) KARYOTYPING - 111
8) DOWN SYNDROME - 112
9) KLINEFELTER SYNDROME - 112
10) TURNER SYNDROME - 112



X - RAY - 143 TO 145
CADAVER - 146 TO 148
HISTOLOGY - 149

1) HYALINE CARTILAGE , SEROUS GLAND - 149
2) SKELETON MUSCLE , CARDIAC MUSCLE - 149
3) CARDIAC MUSCLE , SPLEEN - 150
4) LIVER , PALATINE TONSIL - 150
5) PANCREAS , APPENDIX - 151
6) UROTHELIUM , RESPIRATORY EPITHELIUM - 151
7) DUODENUM , STOMACH - 152
8) KIDNEY , URINARY BALDDER - 152
11) PROSTRATE GLAND , UTERUS - 153
15) THYROID , CEREBELLUM - 153

SUNIL KUMAR



UPPER LIMB MUSCLES

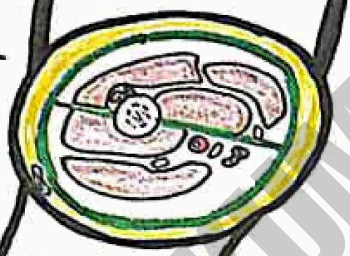
- Radial Nerve
- Radial Nerve
- Posterior interosseous Nerve

- Musculocutaneous Nerve
- Median Nerve
- Ulnar Nerve

ARM

POSTERIOR COMPARTMENT
- TRICEPS BRACHI

ANTERIOR COMPARTMENT
- BICEP BRACHII
- BRACHIALIS
- CORACOBRACHIALIS



FOREARM

POSTERIOR / EXTENSOR COMPARTMENT

SUPERFICIAL
- ANCONEUS
- BRACHIORADIALIS
- EXTENSOR CARPI RADIALIS LONGUS
- EXTENSOR CARPI RADIALIS BREVIS
- EXTENSOR DIGITORUM
- EXTENSOR DIGITI MINIMI
- EXTENSOR CARPI ULNARIS

DEEP
- SUPINATOR
- ABDUCTOR POLLICIS LONGUS
- EXTENSOR POLLICIS BREVIS
- EXTENSOR POLLICIS LONGUS
- EXTENSOR INDICIS

ANTERIOR / FLEXOR COMPARTMENT

SUPERFICIAL
- PRONATOR TERES
- FLEXOR CARPI RADIALIS
- PALMARIS LONGUS
- FLEXOR DIGITORUM SUPERFICIALIS
- FLEXOR CARPI ULNARIS

DEEP
- FLEXOR DIGITORUM PROFUNDA
  - MEDIAL 1/2
  - LATERAL 1/2
- FLEXOR POLLICIS LONGUS
- PRONATOR QUADRATUS

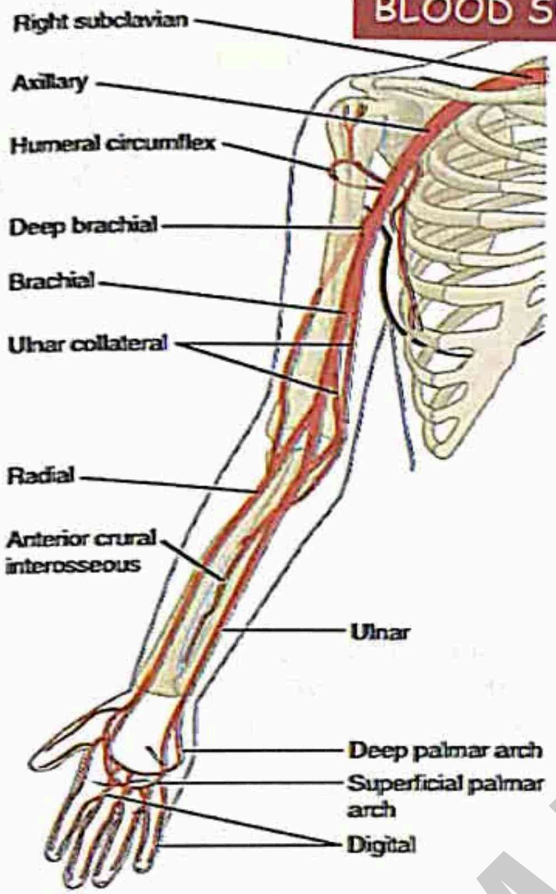
HAND

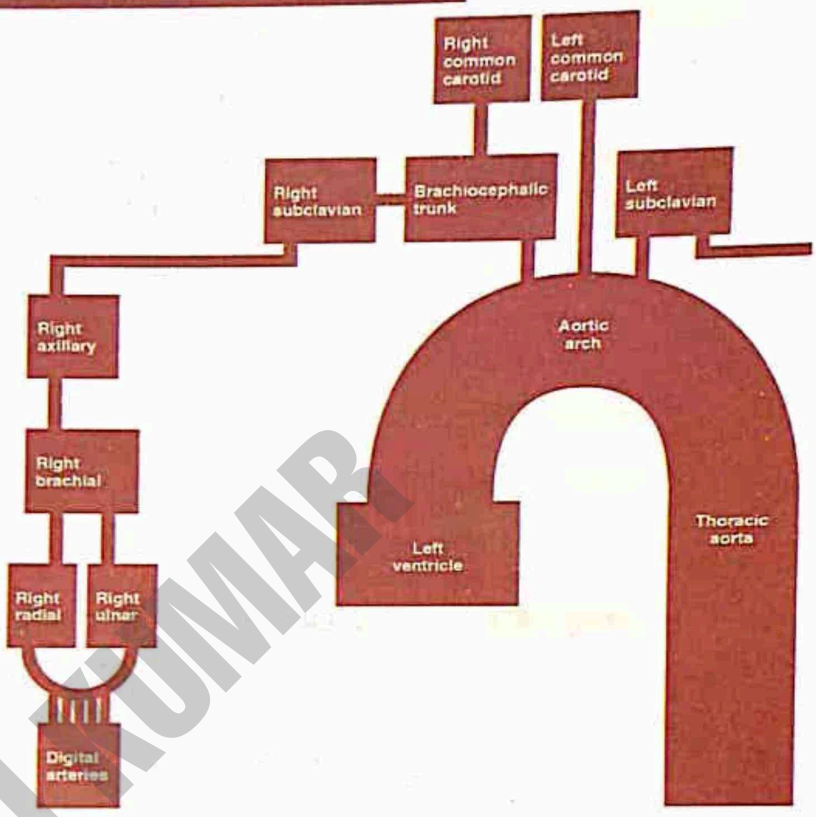
- THENAR
  - Abductor Pollicis brevis
  - Flexor Pollicis brevis
  - Opponens pollicis
  - Adductor pollicis
- HYPOTHENAR
  - Abductor digiti minimi
  - Flexor digiti minimi
  - Opponens digiti minimi
- LUMBRICALS
  - 1st & 2nd
  - 3rd & 4th
- INTEROSSEI
  - Palmar
  - Dorsal

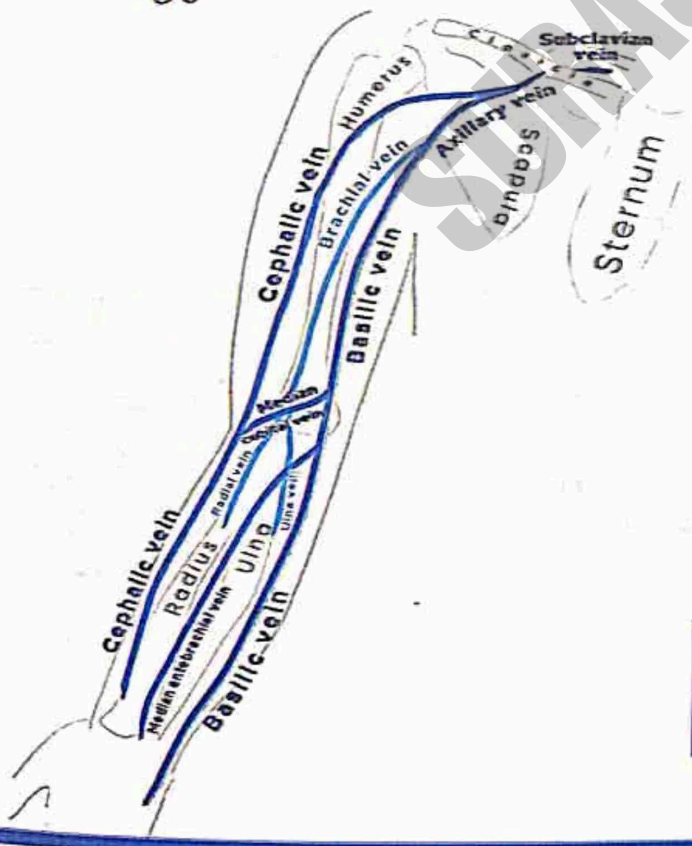
- Median Nerve
- Deep branch of Ulnar Nerve

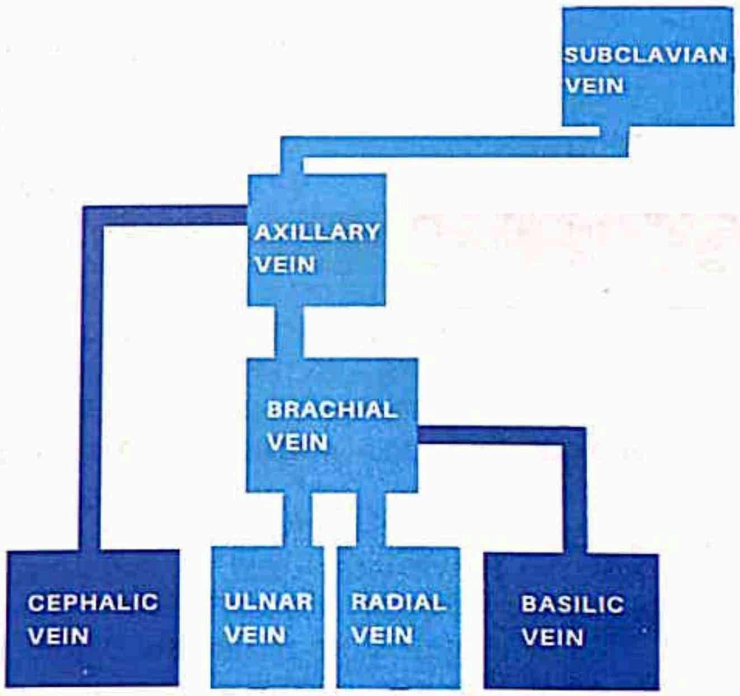
IRAJ KUMAR

BLOOD SUPPLY OF UPPER LIMB









SUBAJ KUMAR

BRACHIAL PLEXUS

- BRACHIAL PLEXUS IS A NETWORK OF NERVE WHICH SUPPLY UPPER LIMB
- CONSIST ⟶ ROOTS, TRUNKS, DIVISIONS, CORDS AND BRANCHES.
- ROOTS ⟶ CONSTITUTED BY ANTERIOR PRIMARY RAMI C5 - C8 AND T1.
  - ORIGIN OF PLEXUS MAY SHIFT UPWARD & DOWNWARD

PREFIXED PLEXUS ⟶ C4 IS LARGE AND T2 ABSENT
POSTFIXED PLEXUS ⟶ T1 LARGE, T2 PRESENT, C4 ADSENT C5 REDUCED IN SIZE

- TRUNK ⟶ C5 & C6 - UPPER TRUNK
  C7 - MIDDLE TRUNK
  C8 & T1 - LOWER TRUNK

- DIVISION ⟶ EACH TRUNK DIVIDE INTO VENTRAL & DORSAL DIVISION

- CORD ⟶ LATERAL CORD - UNION VENTRAL DIVISION OF UPPER & MIDDLE TRUNK
  MEDIAL CORD - VENTRAL DIVISON OF LOWER TRUNK
  POSTERIOR CORD - DORSAL DIVISION OF ALL 3 TRUNK

- BRANCHES ⟶ LATERAL CORD - LATERAL PECTORAL NERVE (C5-C7)
  - MUSCULOCUTANEOUS NERVE (C5-C7)
  - LATERAL ROOT OF MEDIAN (C5-C7)

⟶ MEDIAL CORD - MEDAIL PECTORAL (C8 , T1)
- MEDIAL CUTANEOUS NERVE OF ARM (C8,T1)
- MEDIAL CUTANEOUS NERVE OF FOREARM (C8,T1)
- ULNAR (C7,C8,T1)
- MEDIAL ROOT OF MEDIAN NERVE (C8-T1)

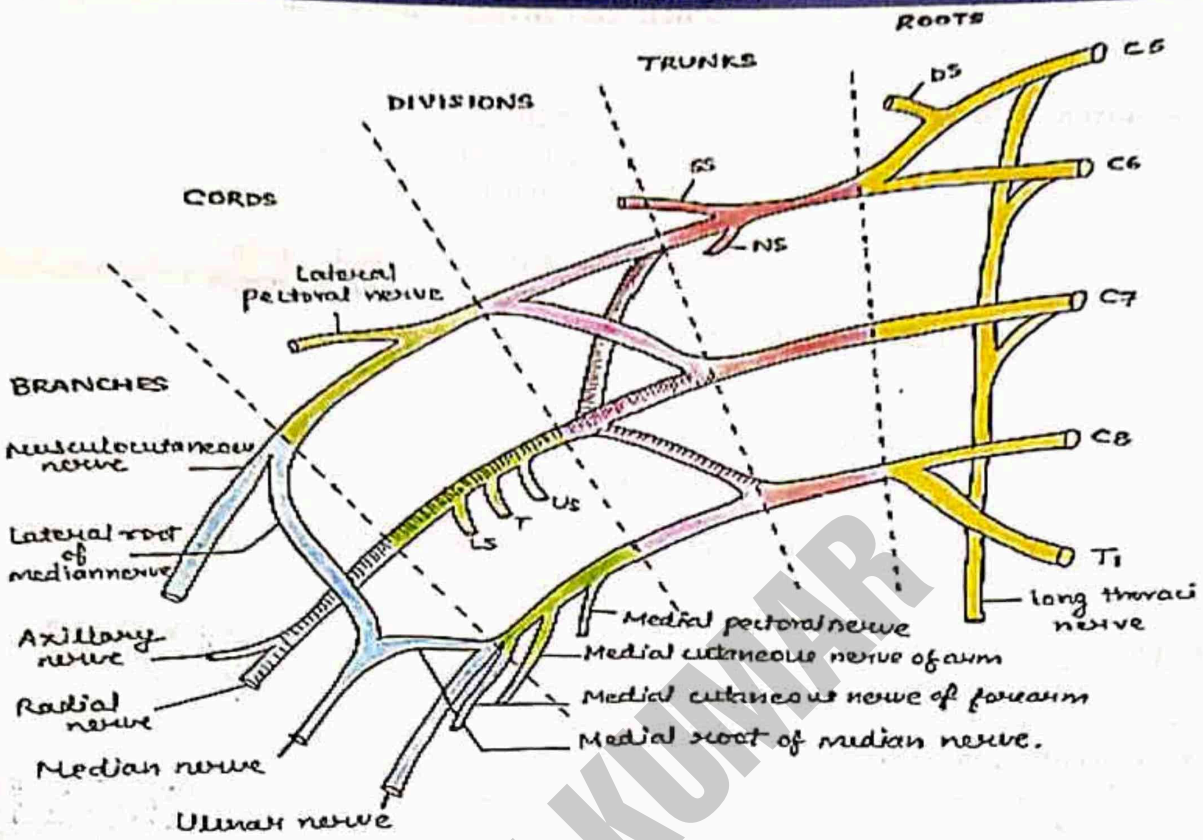
⟶ POSTERIOR CORD - UPPER SUBSCAPULAR (C5 , C6)
- NERVE TO LATISSIMUS DORSI (C6-C8)
- LOWER SUBSCAPULAR NERVE (C5-C6)
- AXILLARY NERVE (C5-C6)
- RADIAL NERVE (C5-C8 , T1)

SURAJ KUMAR

CLINICAL ANATOMY

1. ERB'S PARALYSIS - SITE OF INJURY - ERBS POINT IN UPPER TRUNK
   DEFORMITY - ARM - ADDUCTED AND MEDIALLY ROTATED
   FOREARM - EXTENDED & PRONATED
   THE DEFORMITY ALSO KNOWN AS POLICEMAN'S TIP HAND
   NERVE ROOT INVOLVE - MAINLY C5 & PARTIALLY C6

2. KLUMPKE'S PARALYSIS - SITE OF INJURY - LOWER TRUNK
   DEFORMITY - CLAW HAND - DUE TO UNOPPOSED ACTION OF LONG FLEXORS AND EXTENSORS
   CLAW HAND - HYPEREXTENSION AT METACARPOPHALANGEAL JOINT
   -FLEXION AT INTERPHALANGEAL JOINT
   NERVE ROOT INVOLVE - MAINLY T1 & PARTIALLY C8

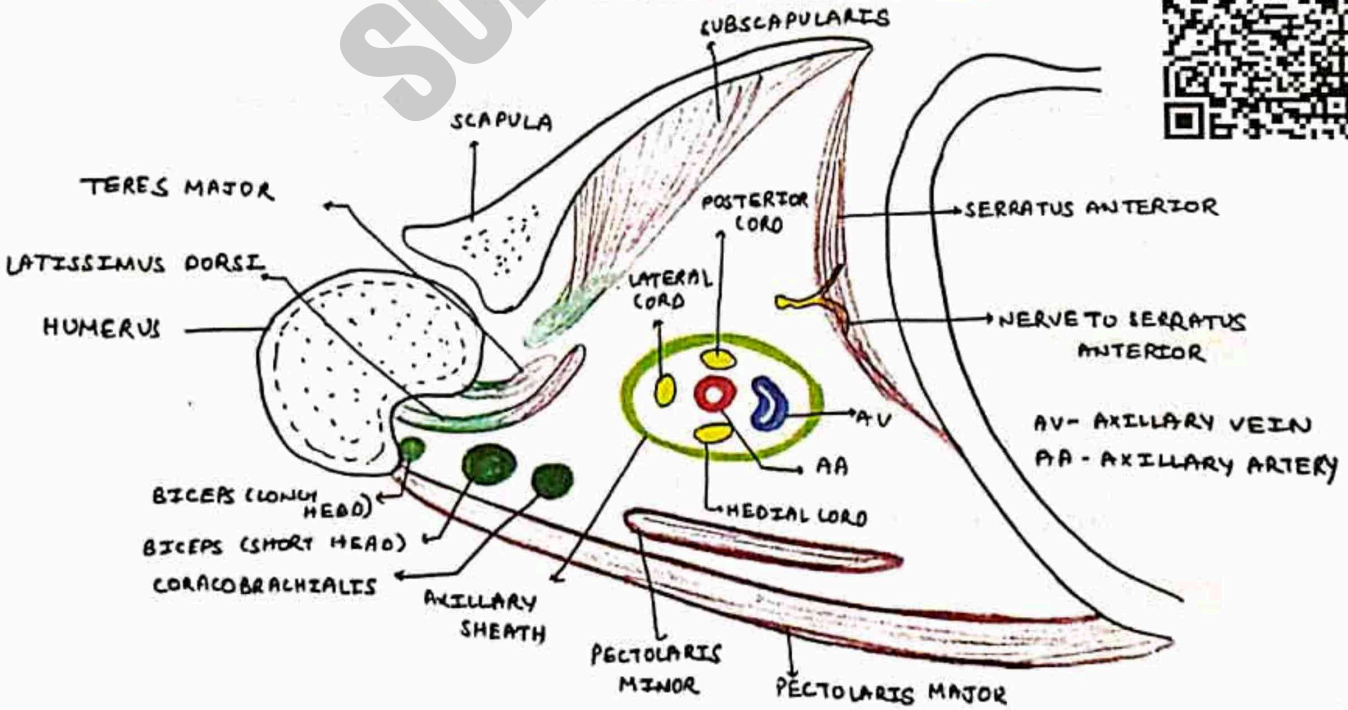


Brachial plexus and its branches

[SS - Suprascapular nerve
US - Upper subscapular nerve
T - Thoraco dorsal nerve
NS - Nerve to subclavius
LS - lower subscapular nerve
DS - Dorsal scapular nerve]

SURAJ KUMAR

AXILLA



¶ DEFINITION :- PYRAMIDAL SPACE, SITUATED - UPPER PART OF THE ARM AND CHEST WALL

BOUNDARIES :- APEX - ANTERIOR COVER BY CLAVICLE
- POSTERIOR BY SUPERIOR BORDER OF SCAPULA
- MEDIAL BY OUTER BORDER OF FIRST RIB

BASE - SKIN , SUPERFICIAL FASCIA

ANTERIOR WALL - PECTORAL MAJOR & MINOR , CLAVIPECTORAL FASCIA

POSTERIOR WALL - SUBSCAPULARIS , TERES MAJOR , LATISSIMUS DORSI

LATERAL WALL - NARROW , CORACOBRACHIALIS & SHORT HEAD OF BICEP BRACHII

CONTENTS :-

1) AXILLARY ARTERY & VEIN BRANCHES
2) INFRACLAVICULAR PART OF BRACHIAL PLEXUS
3) AXILLARY LYMPH NODE
4) LONG THORACIC & INTERCOSTOBRACHIAL NERVE
5) AXILLARY FAT AND AREOLAR TISSUE

SCAN QR WITH QRBOT APP
TO DIRECTLY OPEN VIDEO LINK

SURAJ KUMAR

AXILLARY ARTERY

INTRODUCTION - CONTINUATION OF SUB CLAVIAN ARTERY
-EXTEND - OUTER BORDER OF FIRST RIB TO LOWER BORDER OF TERES MAJOR MS.
-DEVIDED—3 PARTS

PECTORALIS MINOR MUSCLE CROSS ARTERY - DIVIDES INTO 3 PARTS

- 1ST PART - BRANCHES - SUPERIOR THORACIC ARTERY
- 2ND PART - BRANCHES - THORACOCROMIAL ARTERY
  -LATERAL THORACIC ARTERY

* 3RD PART-BRANCHES - SUBSCAPULAR ARTERY
-ANTERIOR CIRCUMFLEX HUMERAL ARTERY
-POSTERIOR CIRCUMFLEX HUMERAL ARTERY

*BRANCHES :-

1.SUPERIOR THORACIS ARTERY -
BRANCH - 1ST PART OF AXILIARY ARTERY
DIRECTION - DOWN, FORWARD & MEDICAL
SUPPLY - PECTORAL MAJOR

2.THORACOCROMIAL ARTERY -
BRANCH - 2ND PART OF AXILLARY ARTERY

3.LATERAL THORACIC ARTERY -

- BRANCH - 2ND PART OF AXILLARY ARTERY
  IT EMERGES - LOWER BORDER OF PECTORALIS MINOR

4.ANTERIOR CIRCUMFLEX HUMERAL ARTERY -

- BRANCH -3RD PART OF AXILLARY ARTERY
  ARISE - LOWER PART OF SUBSCAPULARIS
  SUPPLY -ANASTOMOSIS POSTERIOR CIRCUMFLEXX HUMERAL ARTERY

5.POSTERIOR CIRCUMFLEX HUMERAL ARTERY -

- BRANCH -3RD PART OF AXILLARY ARTERY
  EXTEND - LOWER BORDER OF SUBSCAPULARIS
  SUPPLY -SHOULDER JOINT,DELTOID

6.SUBSCAPULARIS ARTERY -

- BRANCH -3RD PART OF AXILLARY ARTERY
  EXTEND -LOWER BORDER OF SUBSCAPULARIS TO TERMINATE NEAR INFERIOR ANGLE OF SCAPULA.
  SUPPLY - LATISSIMUS DORSI & SERRATUS ANTERIOR MUSCLE.

NOTE :- DIAGRAM OF AXILLARY ARTERY - PAGE NO.005

CUBITAL FOSSA

RAJ KUMAR

INTRODUCTION - TRIANGULAR HOLLOW , SITUATED - FRONT OF ELBOW

***BOUNDARIES :-**

1. LATERAL - MEDIAL BORDER OF BRACHIORADIALIS
2. MEDIAL - LATERAL BORDER OF PRONATOR TERES
3. BASE - IMAGINARY LINE OF EPICHONDYLES
4. APEX - AREA WHERE - BRACHIORADIALIS & PRONATOR TERES CROSSES
5. ROOF - SKIN , SUPERFICIAL FASCIA
   -BICIPITAL APONEUROSIS
6. FLOOR - BRACHIALIS

***CONTENTS :-**

1) MEDIAN NERVE - GIVE BRANCHES - FLEXOR CARPI RADIALIS
   -PALMARIS LONGUS , FLEXOR DIGITORUM SUPERFICIALIS

2) RADIAL NERVE - GIVE - POSTERIOR INTEROSSEI NERVE AND DEEP BRANCH OF RADIAL NERVE

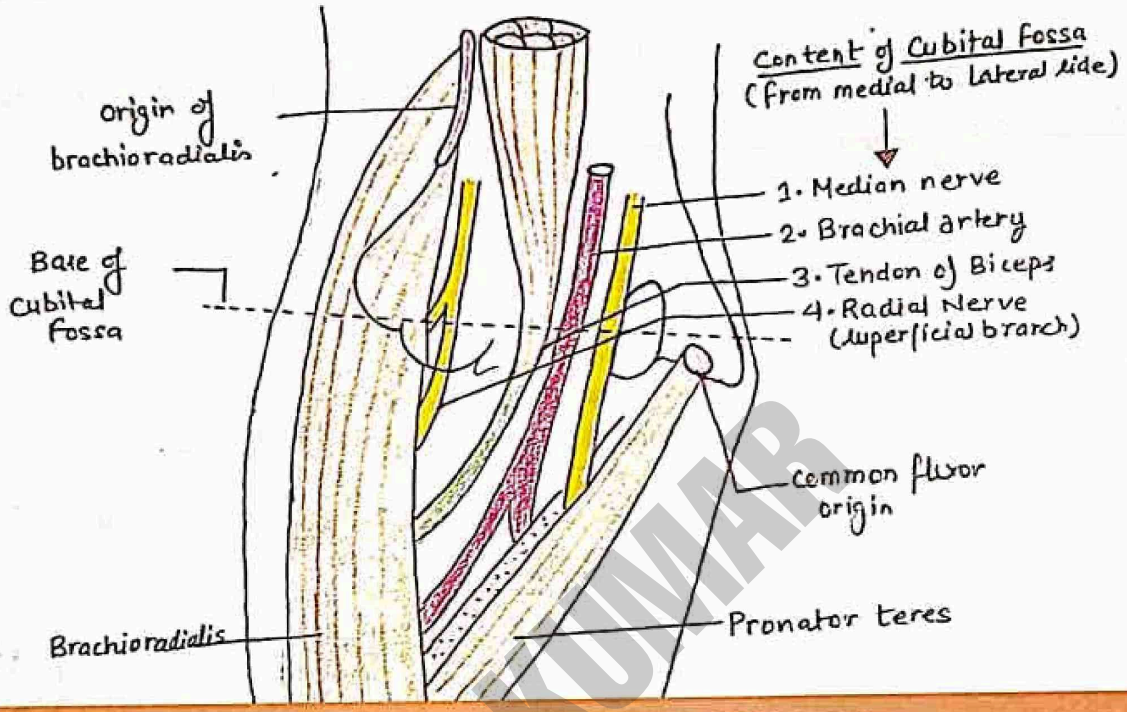
3) BRACHIAL ARTERY - GIVE - RADIAL ARTERY -> RADIAL RECURRENT BRANCH
   -ULNAR ARTERY -> ANTERIOR & POSTERIOR RECURRENT BRANCH , COMMON INTEROSSEOUS BRANCH

4) TENDON OF BICEP BRACHII

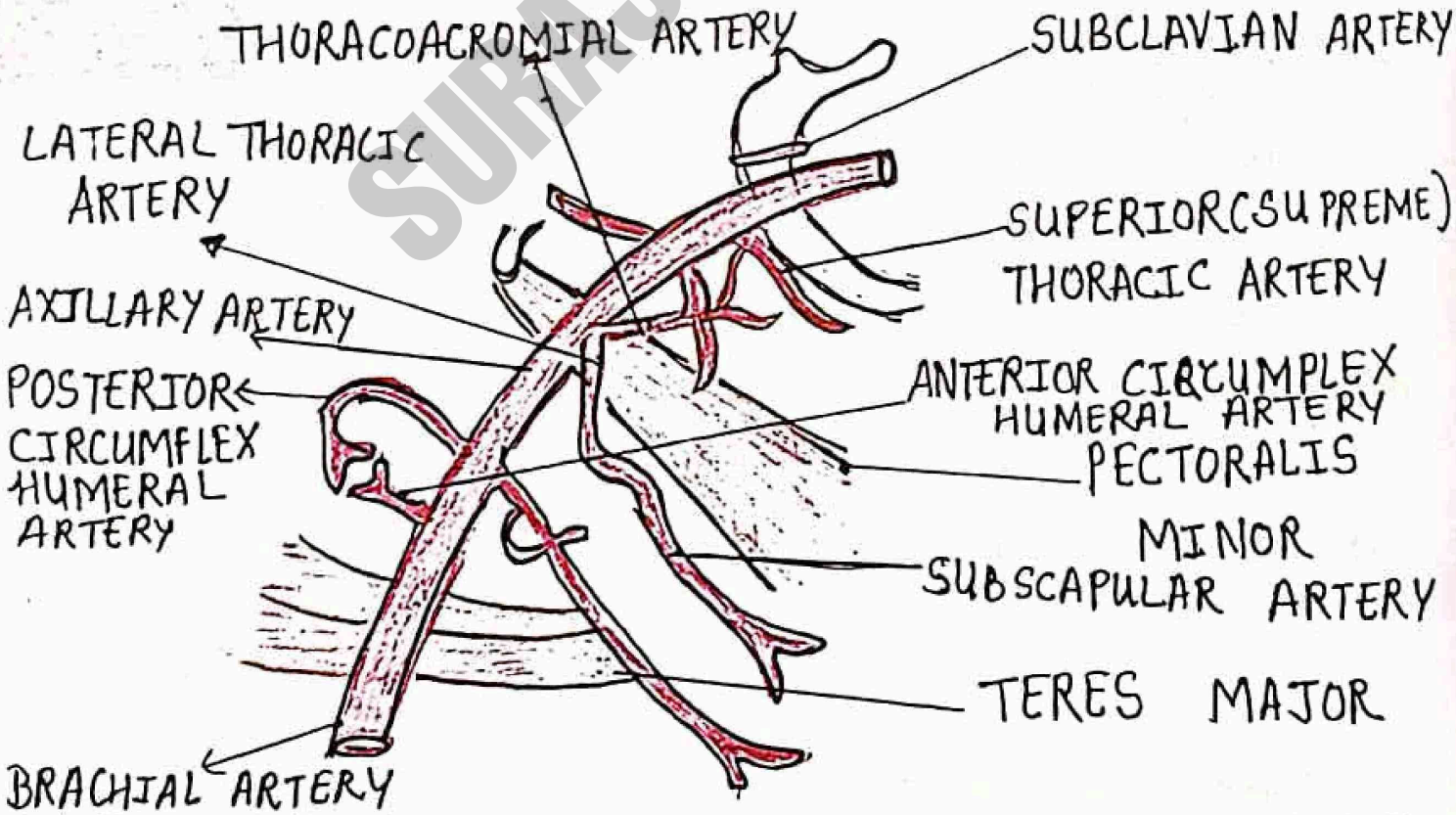
CLINICAL ANATOMY

1. MEDIAN CUBITAL VEIN - VEIN OF CHOICE FOR INTRAVENOUS INJECTION.

2. BLOOD PRESSURE UNIVERSALLY RECORDED - BRACHIAL ARTERY INFRONT OF ELBOW



CONTENT OF CUBITAL FOSSA



BRANCHES OF AXILLARY ARTERY

SURAJ KUMAR

MAMMARY GLAND

BREAST IS MOST IMPORTANT STRUCTURE PRESENT IN PECTORAL REGION.
FOUND IN BOTH SEX BUT RUDIMENTARY IN MALE.
MODIFIED SWEAT GLAND.
*SITUATION - LIE IN SUPERFICIAL FASICA.
DIVIDE INTO 4 QUADRANT - UPPER MEDIAL, UPPER LATERAL, LOWER MEDIAL, LOWER LATERAL

*EXTENSION OF UPPER QUADRANT CALLED AXILLARY TAIL OF SPENCE ->WHICH PASSES THROUGH OPENING IN DEEP FASICA.
*OPENING CALLED FORAMEN OF LANGER.

*EXTENT OF BREAST -
VERTICALLY - 2ND - 6TH RIB
HORIZONTAL - LATERAL BORDER OF STERNUM TO MID AXILLARY LINE
DEEP RELATION - BREAST LIE ON DEEP FASCIA
-DEEPER THERE ARE PART OF 3 MUSCLES - PECTORALIS MAJOR
- SERRATAUS ANTERIOR
- EXTERNAL OBLIQE MUSCLE

*STRUCTURE OF BREAST -
SKIN :- CONICAL POSITION OF NIPPLE ->AT LEVEL->4TH INTERCOASTAL SPACE.
- NIPPLE PIERCED BY 15-20 LECTIFEROUS DUCT.
- IT CONTAIN CIRCULAR & LONGITUDINAL SMOOTH MUSCLE.
- SKIN SURROUNDING BASE OF NIPPLE IS PIGMENTED & FORM CIRCULAR AREA CALLED AREOLA

PARENCHYMA :- MAMMARY GLAND COMPOUND TUBULOALVEOAR GLAND WHICH SECRETES MILK
- IT IS A MODIFIED SWEAT GLAND - GLAND CONSISTS 50-20 LOBES
- EACH LOBE -> CLUSTER OF ALVEOLI -> DRAINED BY LECTIFEROUS DUCT
- LECTIFEROUS DUCT -> CONVERGE -> TOWARD NIPPLE
- LECTIFERAL DUCT NEAR ITS TERMINATION DILATATION -> CALLED LECTIFEROUS SINUS.

STROMA :- FROM SUPPORTIVE FRAMEWORK OF GLAND
- PARTLY FIBROUS PARTLY FATTY

BLOOD SUPPLY :-
1.INTERNAL THORACIC ARTERY
2.LATERAL THORACIC, SUPERFICIAL THORACIC, ACROMIOTHORACIC
3.LATERAL BRANCH -> POSTERIOR INTERCOASTAL ARTERY

*VEINS -> FOLLOW THE ARTERY
1.SUPERFICIAL VEIN -> DRAIN INTERNAL THORACIC VEIN
2.DEEP VEIN -> DRAIN-> AXILLARY & POSTERIOR INTERCOASTAL VEIN

*NERVE SUPPLY :-
SUPPLY BY ANTERIOR AND LATERAL CUTANEOUS BRANCHES OF 4TH & 6TH INTERCOASTAL NERVE.

POINTS TO REMEMBER !

* NERVE DO NOT CONTROL SECRETIION OF MILK

*SECRETION CONTROL BY HORMONE PROLACTIN

SURAJ KUMAR

LYMPHATIC DRAINAGE OF BREAST

*LYMPHATIC DRAINAGE OF BREAST ASSUMED GREAT IMP. TO SURGEON BECAUSE CARCINOMA SPREAD THROUGH LYMPH NODE.

***LYMPH NODE -:**

1. AXILLARY LYMPH NODE - ANT., POST., LAT., CENTRAL, APCAL L.N. (75%)
2. INTERNAL MAMMARY L.N. - (20%)
3. SOME LYMPH (5%) -> FROM BREAST REACH-> * SUPRACLAVICULAR NODE
   * DELTOPECTORAL NODE
   * POST.INTERCOASTAL NODE
   * SUB DIAPHRAGMATIC

***LYMPHATIC VESSELS**

1. SUPERFICAL LYMPHATICS --->DRAIN->SKIN OVER BREAST EXCEPT NIPPLE & AREOLA

2. DEEP LYMPHATIC -->DRAIN->PARENCHYMA OF BREAST

3. LYMPH FROM ANTERIOR & POSTERIOR LYMPH NODE PASS -> CENTRAL & LATERAL LYMPH NODE -->THROUGH-->APICAL GROUP-> -->SUPRACLAVICAL NODES

4. INTERNAL MAMMARY NODE DRAIN LYMPH NOT ONLY FROM INNER HALF OF BREAST BUT ALSO OUTER HALF OF BREAST.

5. PLEXUS OF LYMPHATIC VESSELS PRESENT DEEP TO AREOLA IS SUBAEROLAR PLEXUS OF SAPPEY --->DRAIN--> ANTERIOR LYMPH NODE.

6. DEEP LYMPHATIC OF BREAST -->PASS THROUGH PECTORALIS MAJOR MUSCLE & CLAVIPECTORAL FASICA->APICAL NODE.

SURAJ KUMAR

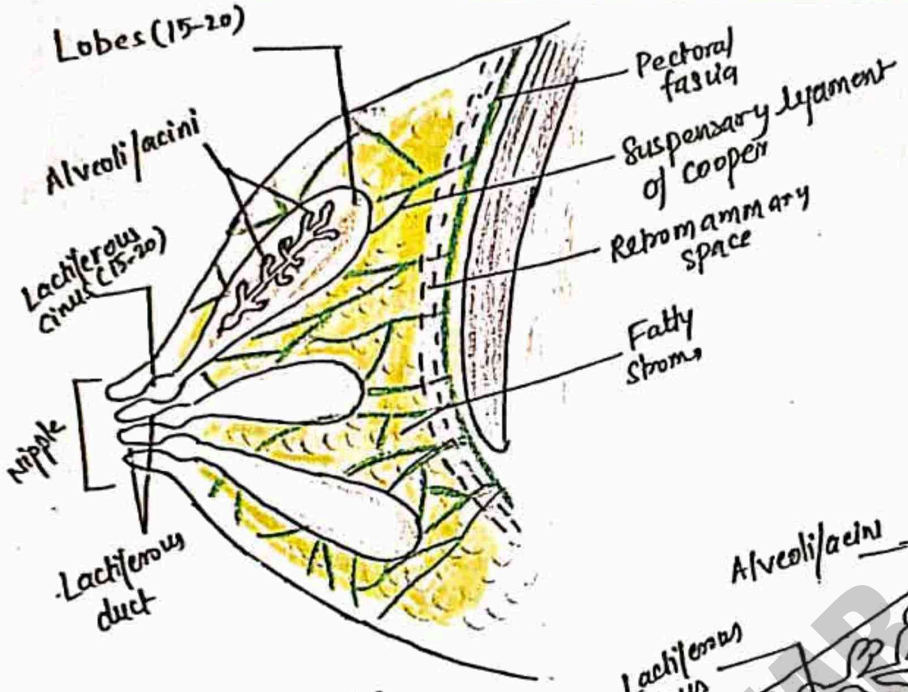
CLINICAL ANATOMY

1. UPPER AND LOWER QUADRANT OF BREAST IS FREQUENT SITE OF CARCINOMA

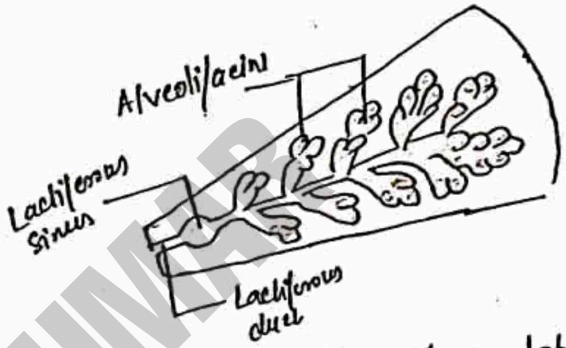
2. CANCER CELL MAY INFILTRATE THE SUSPENSARY LIGAMENT THEN BREAST BECOMES FIX CONTRACTION OF LIGAMENT->CAUSE RETRACTION OR FOLDING OF SKIN.

3. OBSTRUCTION OF SUPERFICIAL LYMPH VESSEL BY CANCER CELL->PRODUCE OEDEMA OF SKIN-->APPEAR GIVING APPEARANCE LIKE SKIN OF ORRANGE (*PEAU D'ORANGE* APPEARANCE).

4. CARCINOMA USUALLY ARISES FROM EPITHILIUM OF LARGE DUCTS.

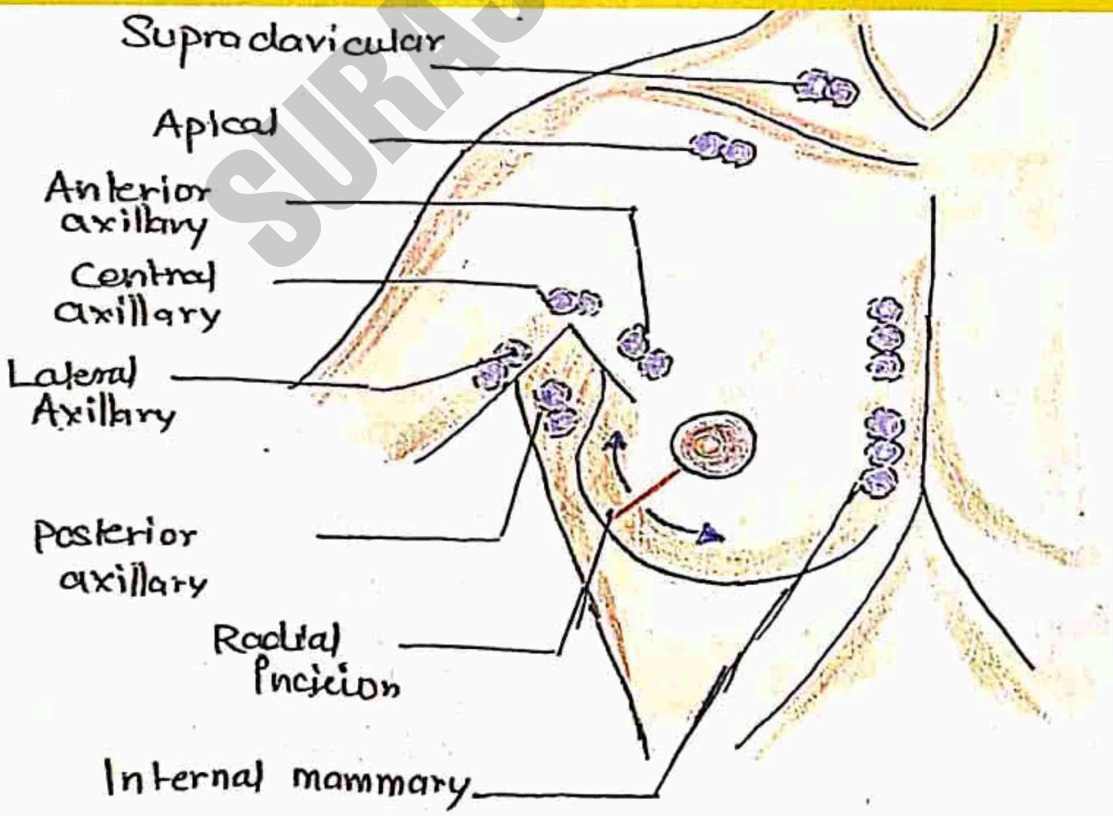


∴ Suspensory ligament of breast / Fitz lobe



∴ Structure of one lobe of mammary gland

STRUCTURE OF BREAST



LYMPH NODE DRAINING BREAST

SURAJ KUMAR

RADIAL NERVE

*THICKEST BRANCH OF BRACHIAL PLEXUS.

➢ ROOT VALUE - VENTRAL RAMI OF C5, C8 & T1

➢ COURSE :-

AXILLA - RADIAL NERVE LIE POSTERIOR TO AXILLARY ARTERY
-THEN ENTER IN LOWER TRIANGULAR SPACE.

RADIAL SULCUS - RADIAL NERVE ENTER THROUGH LOWER TRIANGULAR SPACE INTO RADIAL SULCUS.

CUBITAL FOSSA - NERVE ENTER LATERAL SIDE OF CUBITAL FOSSA.
-TERMNATE INTO SUPERFICIAL & DEEP BRANCH.
DEEP BRANCH-->SUPPLY->EXTENSOR CARPIRADIALIS BREVIS
-->SUPINATOR.
*THEN IT COURSE BETWEEN TWO HEAD OF SUPINATOR MUSCLE -->REACH BACK OF FOREARM.
SUPERFICIAL BRANCH ->LEAVE CUBITAL FOSSA TO ENTER LATERAL SIDE OF FRONT OF FOREARM.

DORSUM OF FOREARM - SUPERFICIAL BRANCH--->DESCEND TILL SNUFF BOX
- TO REACH DORSUM OF HAND--->SUPPLY SKIN OF DORSAL HALF HAND.

➢ BRANCHES :-

1.AXILLA--->MUSCULAR--->LONG HEAD OF TRICEP BRACHI
--->MEDIAL HEAD OF TRICEP BRACHI
--->CUTANEOUS--->POSTERIOR CUTANEOUS NERVE OF ARM

2.RADIAL SULCUS--->MUSCULAR->LATERAL HEAD OF TRICEP BRACHII
-->ANCONEAUS
-->CUTANEOUS-->LATERAL CUTANEOUS NERVE OF ARM
->POSTERIOR CUTANEOUS NERVE OF FOREARM.

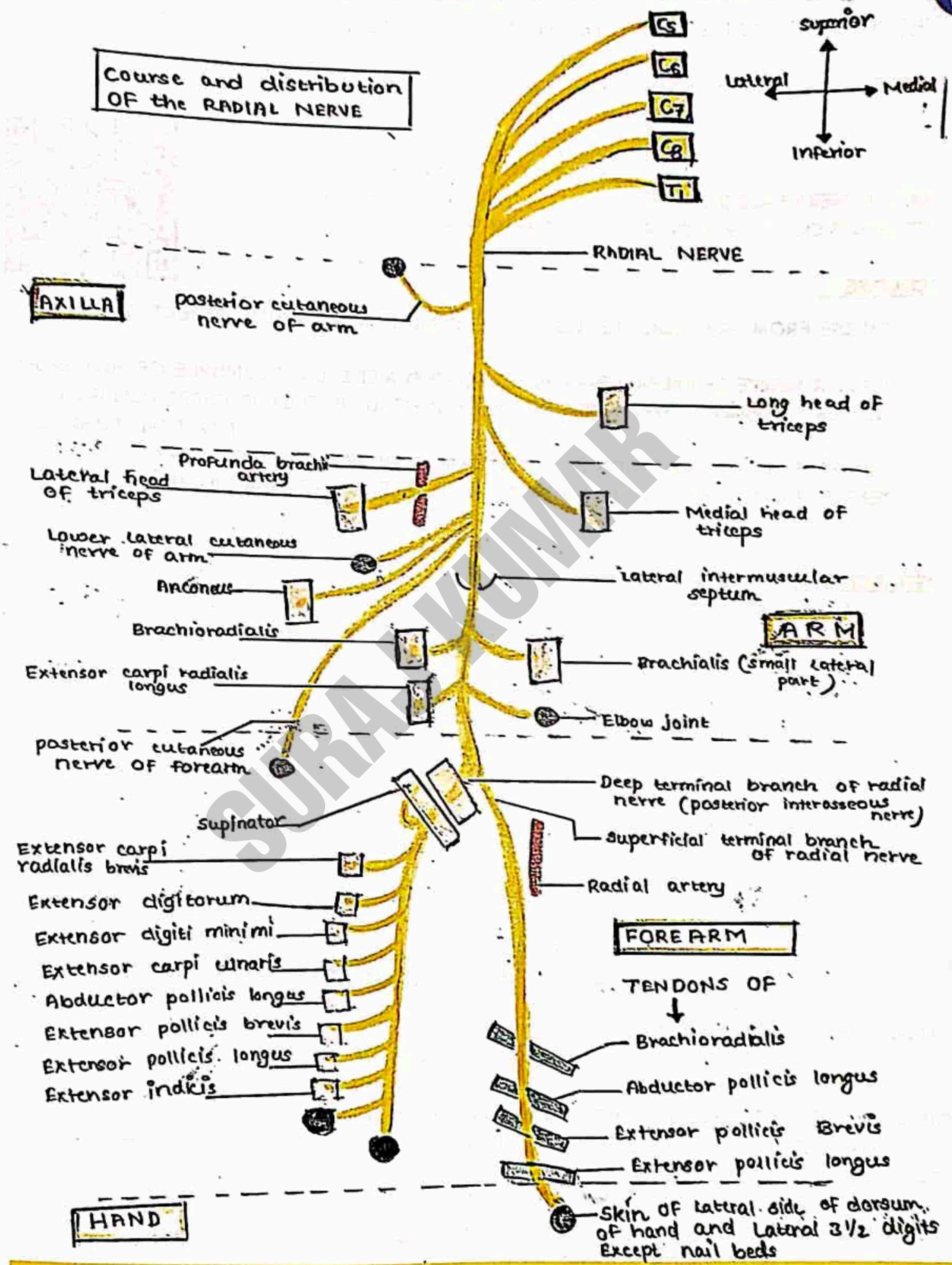
3.LATERAL SIDE OF ARM-->MUSCULAR->BRACHIORADIAL
-->EXTENSOR CARPI RADIALIS LONGUS

4.DEEP BRANCH-->DEEP BRANCH OF RADIAL NERVE
->SUPINATOR MUSCLE
-->EXTENSOR CARPI RADIALIS BREVIS

5.SUPERFICIAL BRANCH-->SUPERFICIAL BRANCH OF RADIAL NERVE
-->CUTANEOUS BRANCHES IN ANATOMICAL SNUFFBOX
(TO SUPPLY LATERAL HALF OF DORSUM OF HAND)

SURAJ KUMAR

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK



BRANCHES OF RADIAL NERVE

SURAJ KUMAR

WRIST DROP-RADIAL NERVE INJURY IN RADIAL GROOVE DUE TO FRACTURE OF SHAFT OF HUMEROUS CAUSE WRIST DROP.
THIS RESULT IN WEAKNESS AND LOSS OF POWER OF EXTENSION IN WRIST.

ULNAR NERVE

*ULNAR NERVE ALSO KNOWN AS MUSICIAN NERVE BECAUSE IT CONTROLS FINE MOVEMENTS OF FINGERS.

*COURSE :-

*ARISE FROM BRACHIAL PLEXUS -->DESCEND DOWN MEDIAL ASPECT OF UPPER ARM.
*ULNAR NERVE IS PALPABLE--->LIES BEHIND MEDIAL EPICONDYLE OF HUMEROUS
*ENTER FOREARM--->PASSING BETWEEN 2 HEAD OF FLEXOR CARPI ULNARIS (CUBITAL TUNNEL)
*IN FOREARM--->LIE ALONG--->LATERAL BORDER OF FLEXOR CARPI ULNARIS.
*ENTER PALM--->PASSING SUPERFICIAL TO FLEXOR RETINACULUM

*BRANCHES:-

1.MUSCULAR--->FLEXOR CARPI ULNARIS
-->FLEXOR DIGITORUM PROFUNDUS (HALF)

2.CUTANEOUS---->PALMER CUTANEOUS BRANCH
(SUPPLIES SKIN OVER HYPOTHENAR MUSCLE)
---->DORSAL CUTANEOUS BRANCH
(SUPPLIES MEDIAL 2HALF FINGER DORSUM OF HAND)

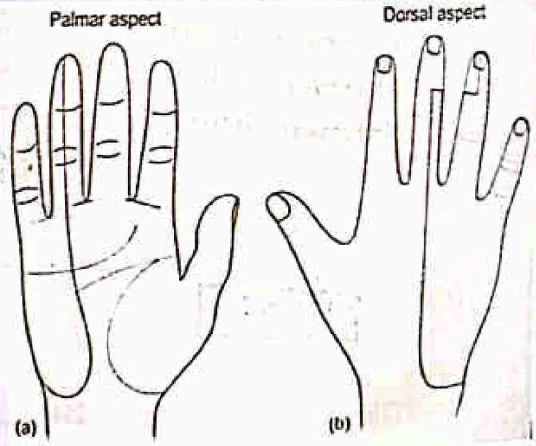
3.PALMAR---->SUPERFICIAL TERMINAL BRANCH->PALMARIS BREVIS
->DIGITAL BRANCH
----->DEEP TERMINAL BRANCH - HYPOTHENAR
- INTEROSSEOUS MUSCLE
-MEDIAL 2 LUMBRICAL

CLINICAL ANATOMY

1.ULNAR NERVE INJURY AT WRIST- PRODUCE ULNAR CLAW HAND

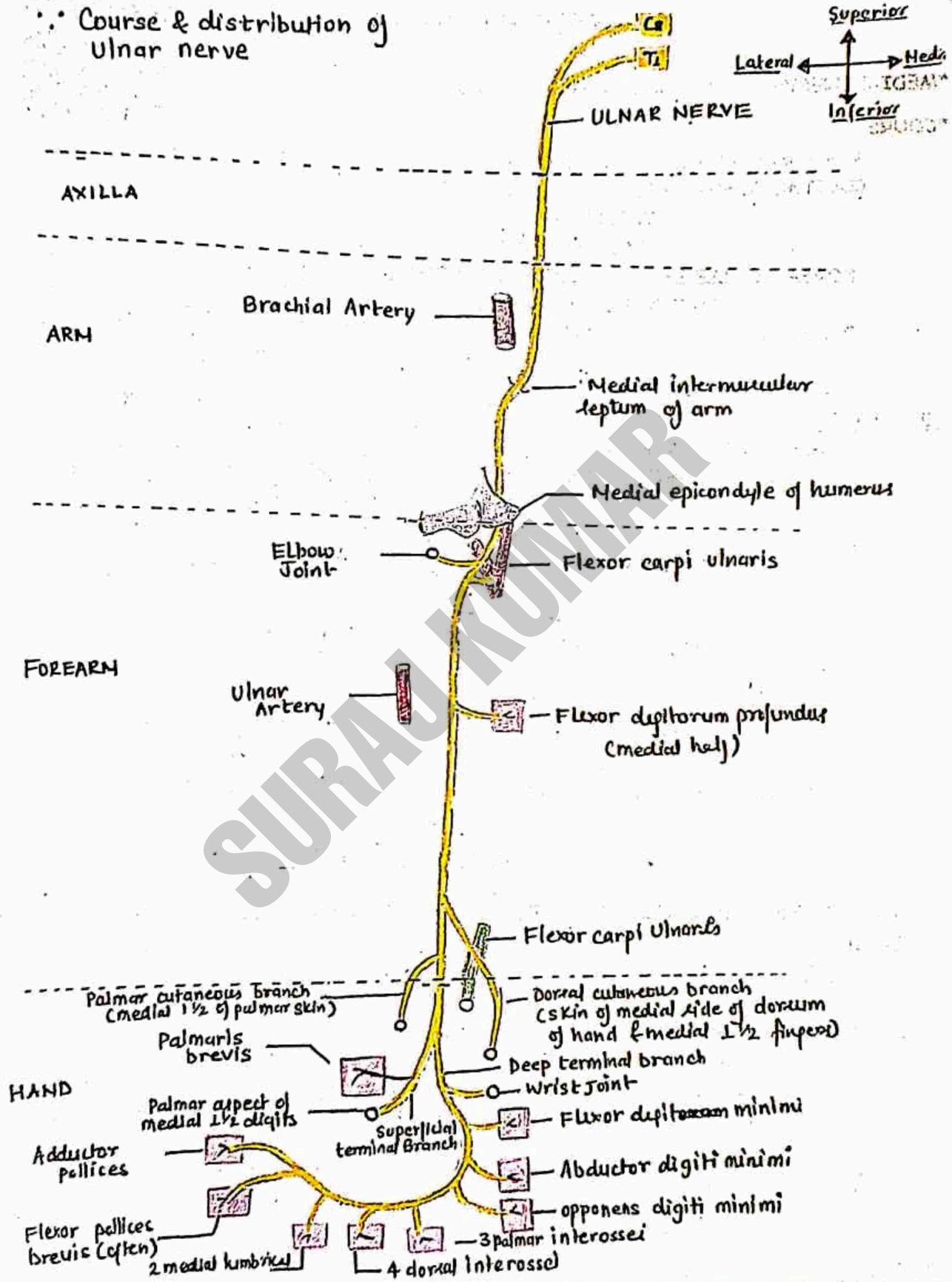
*CAUSE - HYPEREXTENSION AT METACARPOPHALANGEAL JOINTS AND FLEXION AT THE INTERPHALANGEAL JOINTS, INVOLVE-INDEX OR MIDDLE FINGER.

*SENSORY LOSS - MEDIAL 1/3 OF PALM AND MEDIAL 1HALF FINGER.
-MEDIAL HALF OF DORSUM OF HAND.



SUBAJ KUMAR

∴ Course & distribution of Ulnar nerve



BRANCHES OF ULNAR NERVE

SURAJ KUMAR

MEDIAN NERVE

*MEDIAN NERVE IS THE MAIN NERVE OF FOREARM.

*COURSE -

ARM - MEDIAL NERVE LIE MEDIAL TO BRACHIAL ARTERY ENTER CUBITAL FOSSA.

CUBITAL FOSSA - LIE MEDIAL IN CUBITAL FOSSA
- THEN LEAVE FOSSA BY PASSING BETWEEN TWO HEAD OF PRONATOR TERES.

FOREARM - LIE BETWEEN FLEXOR DIGITORUM SUPERFICIALIS AND FLEXOR DIGITORUM PROFUNDUS.

PALM - ENTER IN HAND LIE DEEP TO --->FLEXOR RATINOCULUM THROUGH CARPAL TUNEL.

*BRANCH :-

1.MUSCULAR-ARM-PRONATOR TERES
CUBITAL FOSSA----->FLEXOR CARPI RADIALIS
---->FLEXORDIGITORUM SUPERFICIALIS
-->PALMARIS LONGUS
FOREARM---->FLEXOR DIGITORUM PROFUNDUS
---->PRONATOR QUADRATUS
--->FLEXOR POLLICIS LONGUS
PALM ----> THENAR MUSCLE
--->LATERAL 2 LUMBRICALS

2.CUTANEOUS-PALMARIS CUTANEOUS PALM
(SUPPLY SKIN OVER THENAR MUSCLE)

3.ARTICULAR BRANCHES----->ARE GIVEN TO ELBOW JOINT

4.MUSCULAR BRANCH--->SUPPLY RADIAL AND ULNAR NERVE.

SURAJ KUMAR

CLINICAL ANATOMY

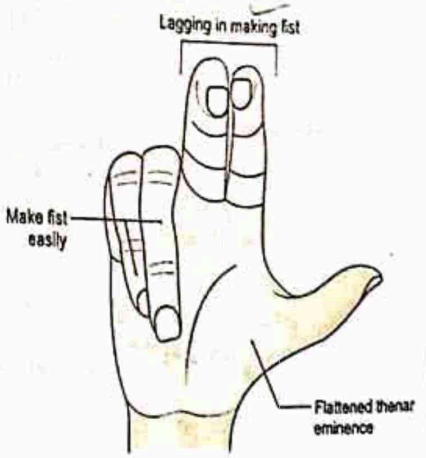
*CARPAL TUNNEL SYNDROME- CAUSED BY COMPRESSION OF MEDIAL NERVE AND CARPI TUNNEL.

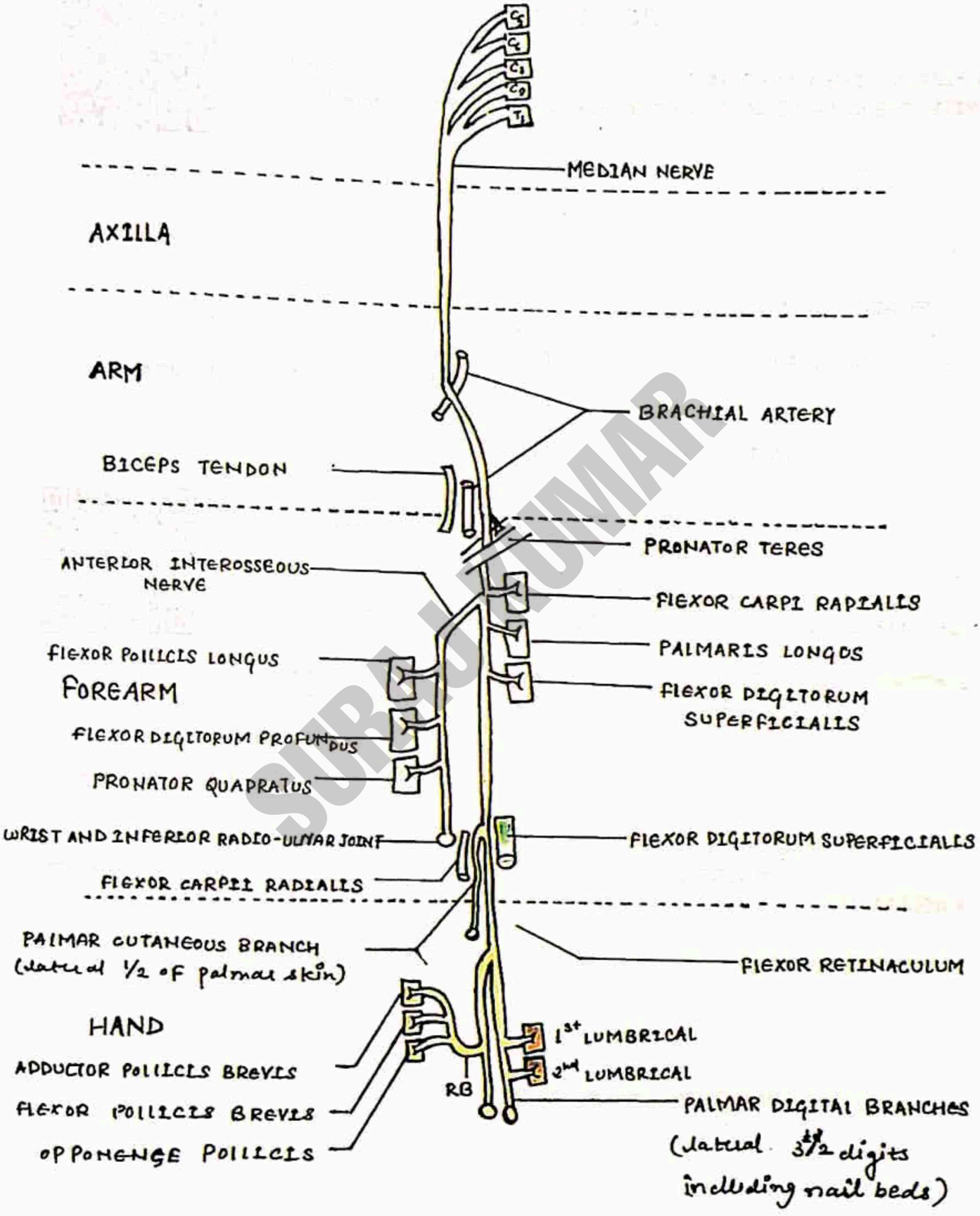
SENSORY CHANGE- LOSS OF SENSATION IN LATERAL 3 HALF DIGITS INCLUDING NAIL BEDS ON DORSUM OF HAND.

MOTOR CHANGE- APE / MONKEY LIKE THUMB DEFORMITY
LOSS OF OPPOSITION OF THUMB, INDEX AND MIDDLE FINGER LAG BEHIND MAKING FIST DUE TO PARALYSIS OF FIRST AND SECOND LUMBRICAL MUSCLE.

VASOMOTOR CHANGE- SKIN AREA WITH SENSORY LOSS WARMER DUE TO ARTERIAL DILATATION.

TROPIC CHANGE- LONG STANDING CAUSE OF PARALYSIS LEAD TO DRYER SCALY SKIN.





SURAJ KUMAR

BRANCHES OF MEDIAN NERVE

ROTATOR CUFF

1.FIBROUS SHEATH→FORM BY 4 FLATTENED TENDONS WHICH BLEND WITH THE CAPSULE OF SHOULDER JOINT AND STRENGTHENED IT.

2.MUSCLE WHICH FORM ROTATOR CUFF ARISE -→SCAPULA
INSERTION-→LESSER & GREATER TUBERCLE HUMEROUS

3.MUSCLE FORM ROTATOR CUFF-→SUBSCAPULARIS, SUPRASPINATUS, INFRASPINATUS, TERES MINOR.

4.CUFF GIVE STRENGTH TO SHOULDER JOINT-→EXCEPT-→INFERIORLY.

5.THUS DISLOCATION OF HUMEROUS OCCUR COMMONLY INFERIOLATERAL.

6.SUBACROMIAL BURSA---→FLUID FILLED SAC. REDUCED FRICTION
-→LARGEST BUSRA OF BODY - BLOW CORACOACROMIAL ARC.

NOTE :- DIAGRAM OF ROTATOR CUFF - PAGE NO.16

FLEXOR RATINACULUM

1.STRONGEST FIBROUS BAND WHICH BRIDGE ANTERIOR CONCAVITY OF CARPUS & CONVERT INTO TUNNEL---→CARPLE TUNNEL.

2.ATTACHMENT - MEDIAL-----→PISIFORM BONE, HOOK OF HAMATE
LATERAL---→TUBERCLE OF SCAPHOID, CREST OF TRAPEZIUM.

MEDIAL SUPERFICIAL SLIP---→PISIFORM BONE
(CONVERT INTO TUNNEL-→PASS-→ULNAR VESSEL & NERVE)

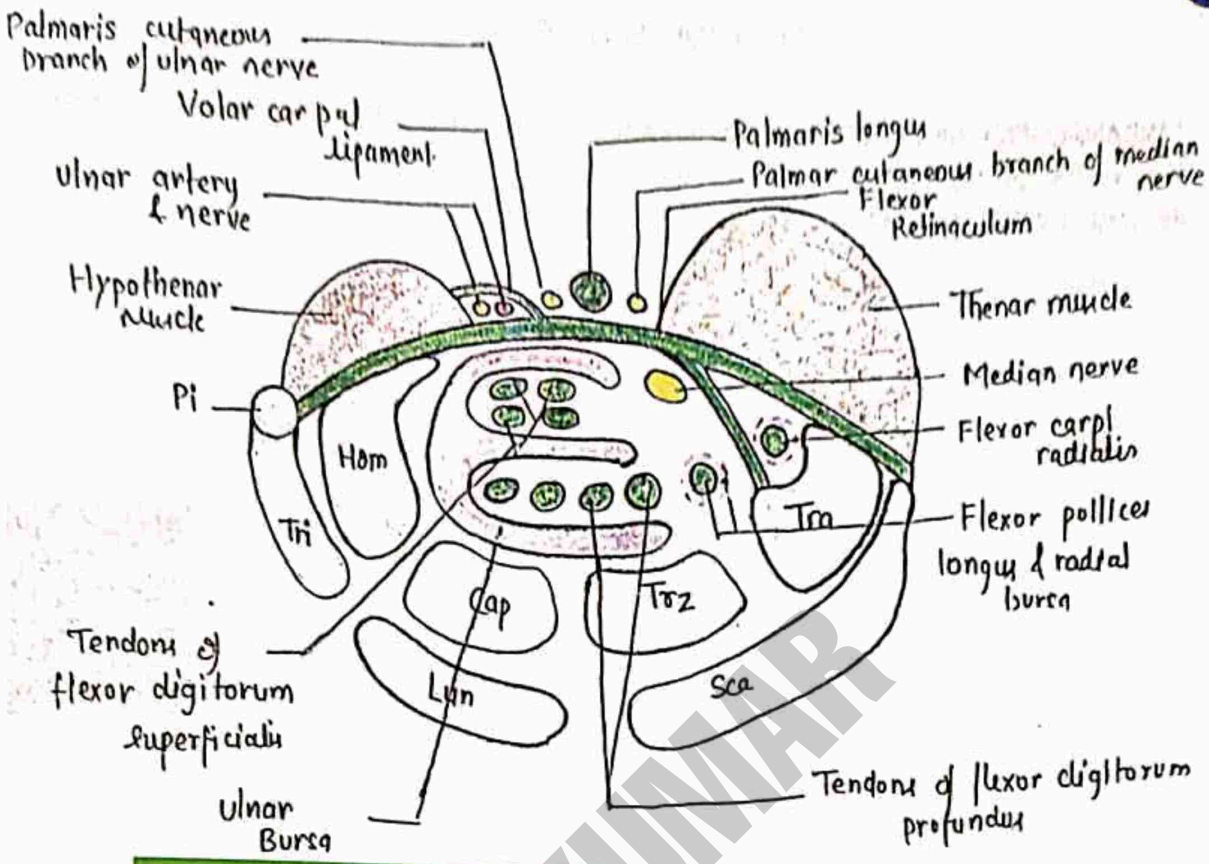
LATERAL DEEP SLIP--→MEDIAL GROOVE ON TRAPEZIUM
(CONVERT INTO TUNNEL--→PASS--→FLEXOR CARPI RADIALIS TENDON)

3.RELATION:-

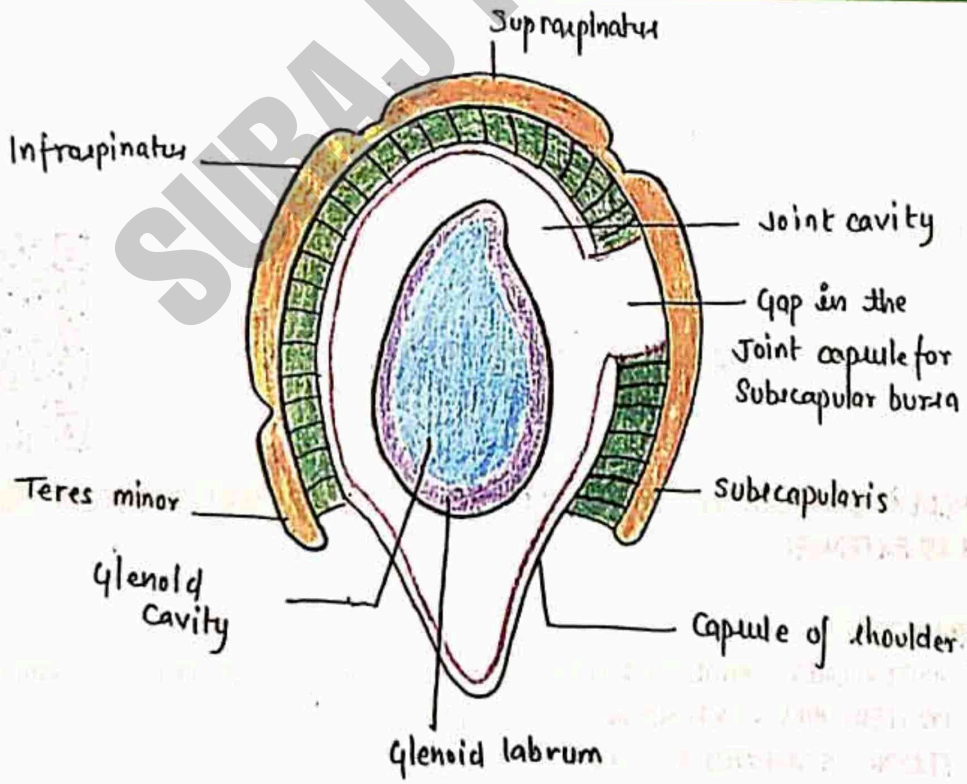
*STRUCTURE PASSES SUPERFICIAL TO FLEXOR RETINACULUM :-
1.PALMARIS CUTANEOUS BRANCH OF MEDIAN NERVE & ULNAR NERVE.
2.PALMARIS LONGUS TENDON.
3.ULNAR NERVE AND VESSEL.

*STRUCTURE PASSES DEEP TO F.R.:-
1.MEDIAN NERVE.
2.ULNAR & RADIAL BURSA.
3.4 TENDON OF FLEXOR DIGITORUM SUPERFICIALIS.
4.4 TENDON OF FLEXOR DIGITORUM PROFUNDUS.
5.TENDON OF FLEXOR POLLICIS LONGUS.

SURAJ KUMAR



FLEXOR RETINOCULUM



∵ Rotator cuff

ROTATOR CUFF

SUBAJ KUMAR

MID PALMAR SPACE

*ARRANGEMENT OF FASCIAE AND THE FASCIAL SEPTA IN THE HAND IN SUCH A WAY MANY SPACES ARE FORMED, SPACES ARE SURGICAL IMPORTANCE BECAUSE THEY MAY BECOME INFECTED- IMPORTANT SPACES ARE AS FOLLOWS :-

A.PALMAR SPACES-->PULP SPACE OF FINGER, MID PALMAR SPACE, THENAR SPACE.

B.DORSAL SPACE----->DORSAL SUBCUTANEOUS AND DORSAL SUBAPONEUROTIC SPACE.

C.THE FOREARM SPACE OF PARONA.

MID PALMAR SPACE

1. SHAPE--->TRIANGULAR

2. SITUATION-->UNDER INNER HALF OF HOLLOW OF PALM

3. EXTEND--->PROXIMAL - FOREARM SPACE OF PARONA
   DISTAL - DISTAL PALMAR CREASE.

4.BOUNDARY-->ANTERIOR - FLEXOR TENDONS OF 3RD,4TH,& 5TH DIGITS.
-2ND, 3RD, & 4TH LUMBRICALS.
- PALMAR APONEUROSIS.

-->POSTERIOR - FASCIA COVERING INTEROSSEI & METACARPALS.
-->LATERAL - INTERMIDIATE PALMAR SEPTUM.
-->MEDIAL - MEDIAL PALMAR SEPTUM.

5.DRAINAGE-->INCISION IN EITHER THE 3RD OR 4TH WEB SPACE.

SURAJ KUMAR

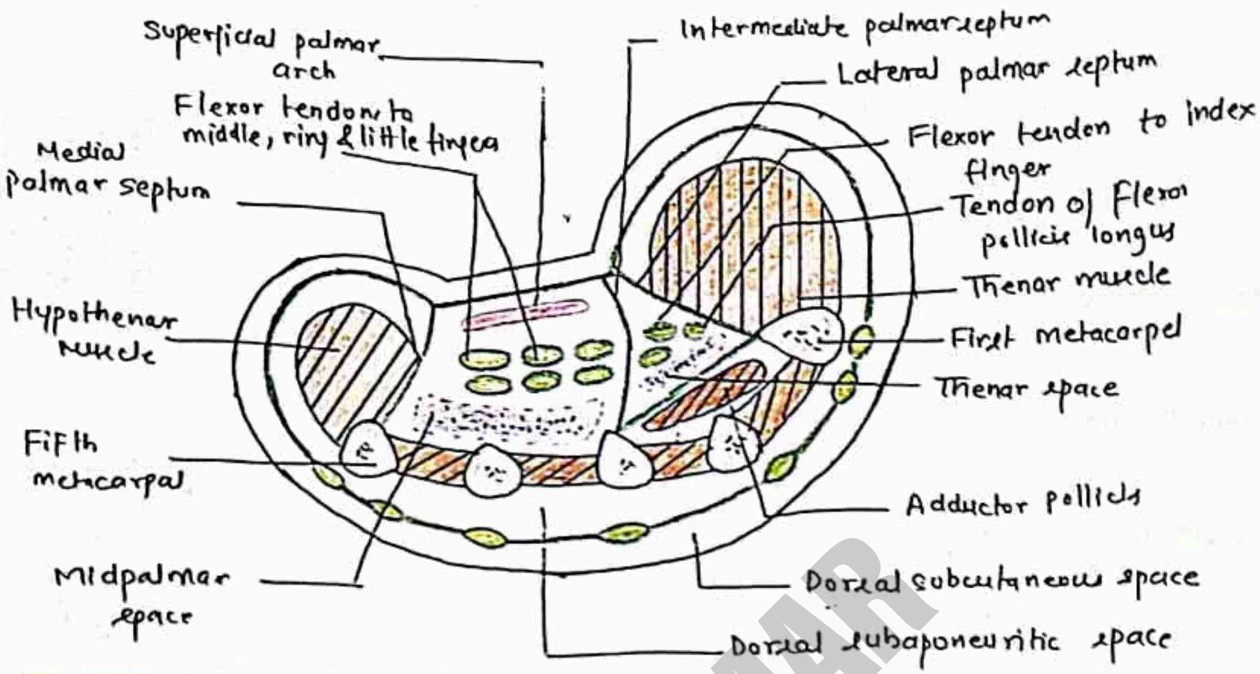
ANATOMICAL SNUFFBOX

*TRIANGULAR DEPRESSION-->POSTERIOR LATERAL SIDE OF WRIST, SEEN WHEN THUMB IS EXTENDED.

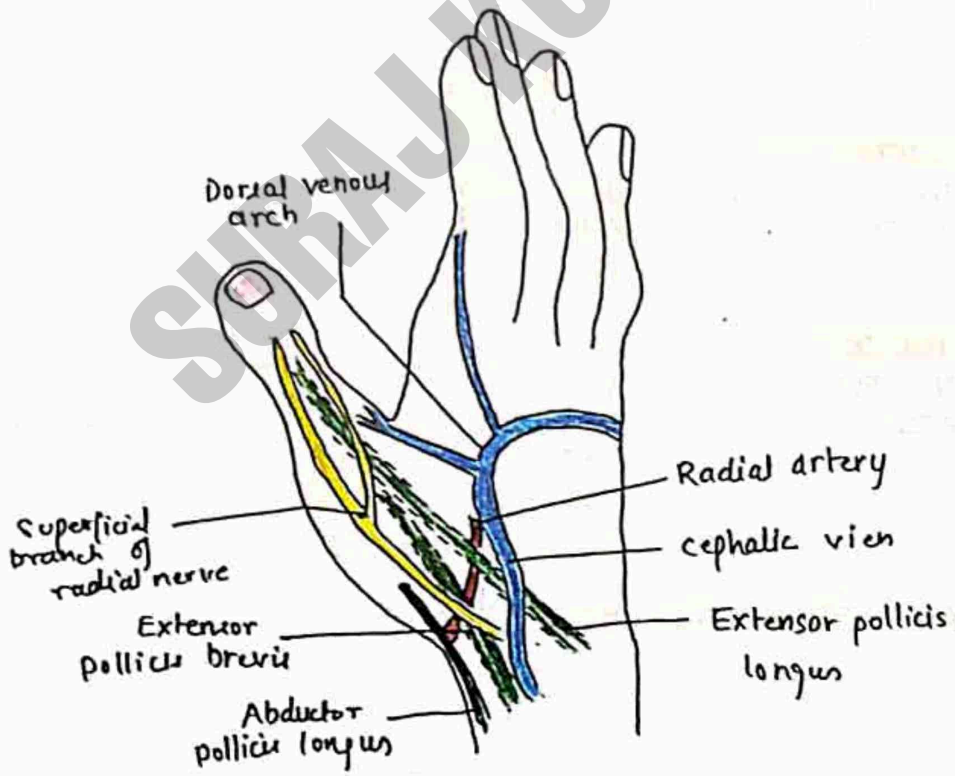
*BOUNDARIES :-

1. ANTERIORLY - ABDUCTOR POLLICIS LONGUS, EXTENSOR POLLICIS BREVIS.
2. POSTERIORLY - EXTENSOR POLLICIS LONGUS.
3. FLOOR - SCAPHOID & TRAPEZIUM.

*CONTENT :- 1.RADIAL ARTERY & RADIAL NERVE
2.CEPHALIC VEIN



MID PALMAR SPACE



∴ Anatomical snuffbox

ANATOMICAL SNUFF BOX

SUBAJ KUMAR

LUMBRICAL MUSCLE

*ORIGIN :-

1ST - LATERAL SIDE OF TENDON OF FLEXOR DIGITORUM PROFUNDUS OF 2ND DIGIT.
2ND - LATERAL SIDE OF SAME TENDON OF 3RD DIGIT.
3RD - ADJACENT SIDES OF SAME TENDONS OF 3RD & 4TH DIGIT.
4TH - ADJACENT SIDES OF SAME TENDONS OF 4TH & 5TH DIGIT.

*INSERTION -

EXTENSOR EXPANSION--->DORSUM OF BASE OF--->DISTAL PHALANGES

*NERVE SUPPLY - 1ST & 2ND -->MEDIAN NERVE
- 3RD & 4TH -->ULNAR NERVE

*ACTION - FLEX METACARPOPHALANGEAL JOINTS
- EXTEND INTERPHALANGEAL JOINT

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK

THENAR EMINENCE

1.ABDUCTOR POLLICIS BREVIS :-

*ORIGIN - TUBERCLE OF SCAPHOID, CREST OF TREPAZIUM, FLEXOR RETINACULUM
*INSERTION - BASE OF PROXIMAL PHALANX OF THUMB
*ACTION - ABDUCTION OF THUMB

2.FLEXOR POLLICIS BREVIS :-

*ORIGIN - FLEXOR RETINACULUM, CREST OF TREPAZIUM & CAPITATE BONES
*INSERTION - BASE OF PROXIMAL PHALANX OF THUMB
*ACTION - FLEXES METACARPOPHALANGEAL JOINT OF THUMB

3.OPPONENS POLLICIS :-

*ORIGIN - FLEXOR RETINACULUM CREST OF TREPAZIUM
*INSERTION - LATERAL HALF OF PALMAR SURFACE OF THE SHAFT OF MATACARPAL BONE OF THUMB
*ACTION - PULLS THUMB MEDIALLY AND FORWARD ACROSS PALM (OPPOSES THUMB TOWARDS THE FINGERS)

4.NERVE SUPPLY OF THENAR MUSCLE :- MEDIAN NERVE

POINTS TO REMEMBER

1.PALMAR INTEROSSEI - ACTION-->ADDUCTION OF FINGERS TOWARDS CENTER.

2.DORSAL INTEROSSEO - ACTION-->ABDUCTION OF FINGER FROM CENTER OF THIRD DIGIT.

SUBAJ KUMAR

SHOULDER JOINT

*TYPE - SYNOVIAL JOINT OF BALL & SOCKET VARIETY.

*ARTICULATING SURFACE :- GLENOID CAVITY OF SCAPULA & HEAD OF HUMERUS.

*WEAK JOINT BECAUSE GLENOID CAVITY IS SMALL & SHALLOW
*STABILITY OF JOINT IS MAINTAIN BY -->CAROCOACROMIAL ARCH
-->ROTATOR CUFF
-->GLENOID LEBRUM (LIP.)

*LIGAMENT :-

1.CAPSULAR LIGAMENT- ATTACHMENT- VERY LOOSE AND PERMIT FREE MOVEMENT
*MEDIALLY- SUPRAGLANOID TENDON
*LATERAL- ANATOMICAL NECK OF HUMERUS
*ANTERIOR- CAPSULE IS REINFORCED BY SUPERIOR, MIDDLE, INFERIOR GLANOHUMERAL LIGAMENT

2.CORACOACROMIAL LIGAMENT :- ATTACHMENT - ROOT OF CORACOID PROCESS TO NECK OF HUMERUS.

3.TRANSVERSE HUMERAL LIGAMENT - BRIDGE UPPER PART OF BICIPITAL GROOVE OF HUMERUS.
-TENDON OF LONG HEAD OF BICEP BRACHII PASS DEEP TO THE LIGAMENT

4.GLANOID LABRUM - FIBRO CARTILAGINOUS RIM COVERS GLENOID CAVITY

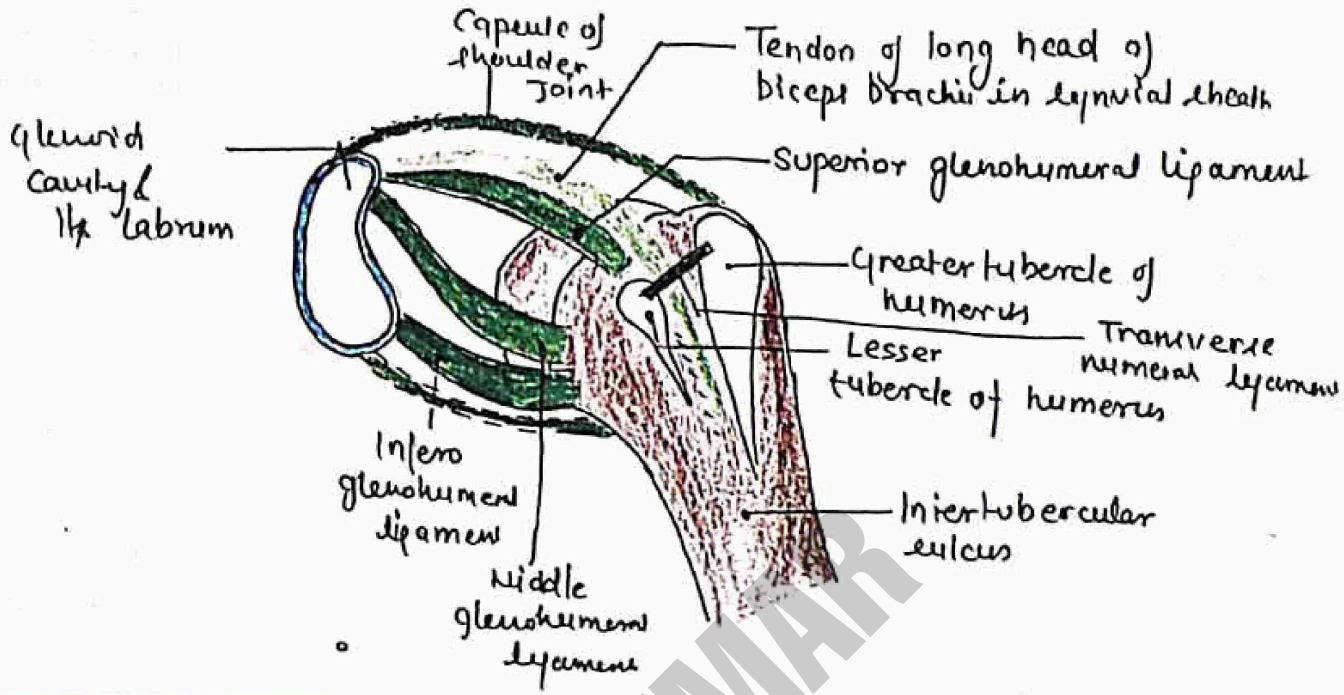
*BURSAE- SUBACROMIAL BURSA, SUBSCAPULARIS BURSA, INFRASPINATOUS BURSA

*RELATION - SUPERIOR - CORACOACROMIAL ARCH, SUBACROMIAL BURSA
-SUPRASPINATOUS & DELTOID
INFERIOR - LONG HEAD OF TRICEP BRACHII
- AXILLARY NERVE
-POSTERIOR CIRCUMFLEX HUMERAL ARTERY
ANTERIOR - SUBSCAPULARIS
-CORACOBRACHIALIS & SHORT HEAD OF BICEP BRACHII
POSTERIOR - INFRASPINATOUS & TERES MINOR

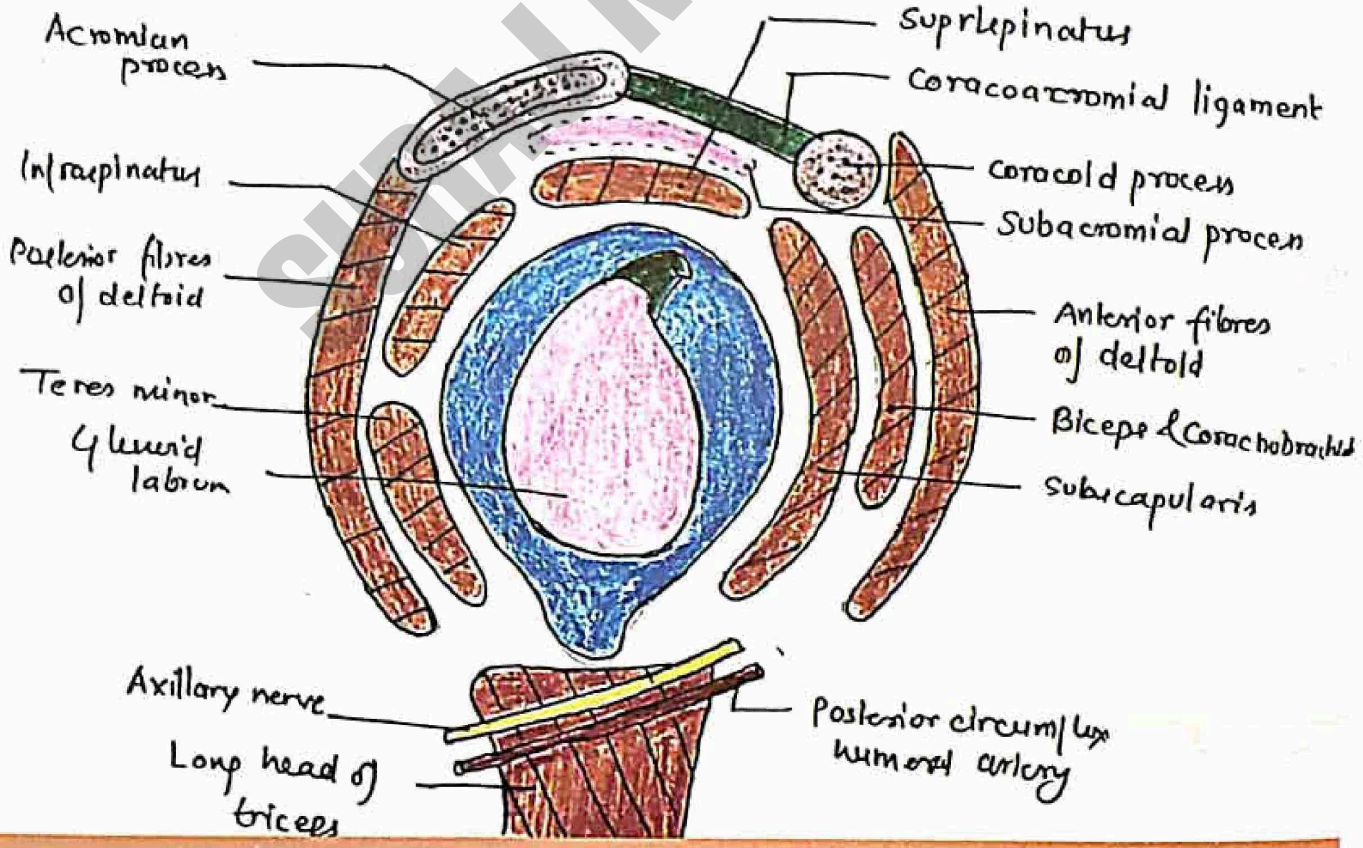
*BLOOD SUPPLY - ANT. & POST. CIRCUMFLEX HMERAL ARTERY
- SUPRASCAPULAR VESSEL & SUBSCAPULAR VESSEL

*NERVE SUPPLY - AXILLARY NERVE SUPPLY
MUSCULOCUTANOUS NERVE
SUPRASCAPULAR NERVE

SUBAJ KUMAR



LIGAMENT OF SHOULDER JOINT



RELATION OF SHOULDER JOINT

SUNIL KUMAR

*MOVEMENT OF SHOULDER JOINT :-

1.FLEXION - ARM MOVE FORWARD & MEDIAL
2.EXTENSION - ARM MOVE LATERAL BACKWARD
3.ADDUCTION - OPPOSITE OF ABDUCTION
4.ABDUCTION - ARM MOVE ANTEROLATERALLY AWAY FROM TRUNK
5.MEDIAL ROTATION - HAND MOVE MEDIAL ACROSS THE CHEST
6.LATERALL ROTATION - HAND MOVE LATERAL OF THE SHOULDER JOINT
7.CIRCUMDUCTION - COMBINATION OF DIFFERENT MOVEMENT

ELBOW JOINT

*TYPE :- SYNOVIAL JOINT OF VARIETY OF HINGE JOINT BETWEEN LOWER END OF HUMEROUS & UPPER END OF RADIUS & ULNA.

*ARTICULATION SURFACE :-

1.UPPER-->CAPITULUM & TROCLEAR CONDYLE OF HUMERUS
-CORNOID FOSSA JUST LIE ABOVE TROCHLEAR
(CORONOID FOSSA FIT CORONOID PROCESS OF ULNA IN FLEXION)
-RADIAL FOSSA JUST ABOVE CAPITULUM
(RADIAL FOSSA FIT RADIAL HEAD OF FLEXION)

2.LOWER----->HEAD OF RADIUS WITH CAPITULUM
-TROCHLEAR NOTCH OF ULNA WITH TROCHLEAR CONDYLE OF HUMERUS

*LIGAMENT :-

1.CAPSULAR LIGAMENT-ATTACHMENT- SUPERIOR - LOWER END OF HUMERUS
INFEROMEDIAL- MARGIN OF TROCHLEAR NOTCH OF ULNA
INFERIOLATERAL- ANGULAR LIGAMENT

2.ULNAR COLLATERAL LIGAMENT- TRIANGULAR IN SHAPE
-APEX-MEDIAL EPICHONDYLE OF HUMERUS
-BASE-ULNA
-ANTERIAR BAND- BELOW CORONOID PROCESS
-POSTERIOR BAND- BELOW OLECRANON PROCESS

3.RADIAL COLLATERAL LIGAMENT-FAN SHAPE
-EXTEND LATERAL EPICHONDYLE-->ANGULAR LIGAMENT

SURAJ KUMAR

*BLOOD SUPPLY - FROM ANATOMOSES AROUND ELBOW JOINT

*NERVE SUPPLY - ULNAR, MEDIAN, RADIAL & MUSCULOCUTANEOUS NERVE

*MOVEMENT - FLEXION -->BRACHIALIS
-->BICEP BRACHII & BRACHIORADIALIS

EXTENSION-->TRICEPS BRACHII
--->ANCHONEUS

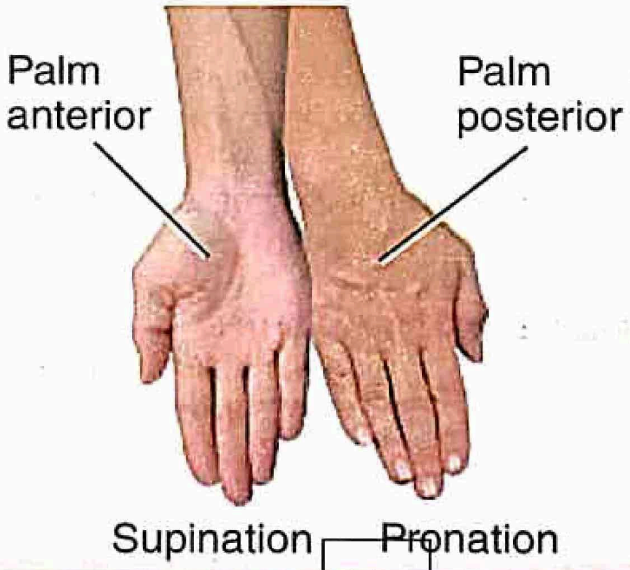
SPINATION & PRONATION

1.SUPINATION & PRONATION ARE ROTATORY MOVEMENT OF FOREARM-->VERTICAL AXIS IN SEMIFLEX ELBOW.

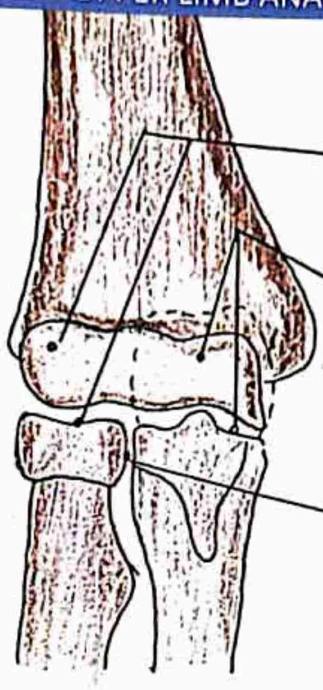
2.SUPINATION-->PALM--->TURN UPWARD.
3.PRONATION----->PALM TURN DOWNWARDS.

4.VERTICAL AXIS MOVEMENT PASS CENTER OF HEAD OF RADIUS ABOVE.
5.THIS AXIS IS NOT STATIONARY BECAUSE -->LOWER END OF ULNA IS NOT FIXED.
6.IT MOVE -->BACKWARD LATERALLY IN PRONATION
-->FORWARD MEDIALLY IN SUPINATION
7.SUPINATION IS POWERFUL BECAUSE IT IS ANTI GRAVITATIONAL
8.SUPINATION MOVEMENT IS RESPONSIBLE FOR ALL SCREWING MOVEMENT OF HAND

9.AROUND 50 DEGREE OF SUPINATIN AND PRONATION ARE GENERALLY TO PERFORM MANY ROUTINE ACTIVITIES
10.PRONATION------>CHEIFLY BY PRONATOR QUADRATUS
----->RAPIDLY MOVEMENT BY PRONATOR TERES
11.SUPINATION----->CAUSE BY SUPINATOR MUSCLE & BICEP BRACHII
---->SLOW SUPINATION---->BY SUPINATR MUSCLE
---->RAPID SUPINATION --->BY BICEP BRACHII



SURAJ KUMAR



Elbow Joint

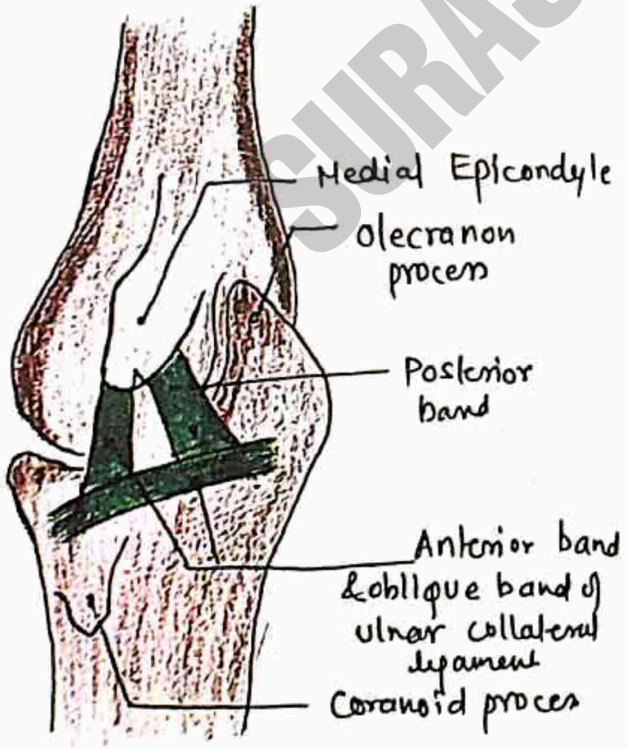
Capitulum of humerus articulates with upper surface of head of radius

Trochlea of humerus articulates with trochlear notch of ulna

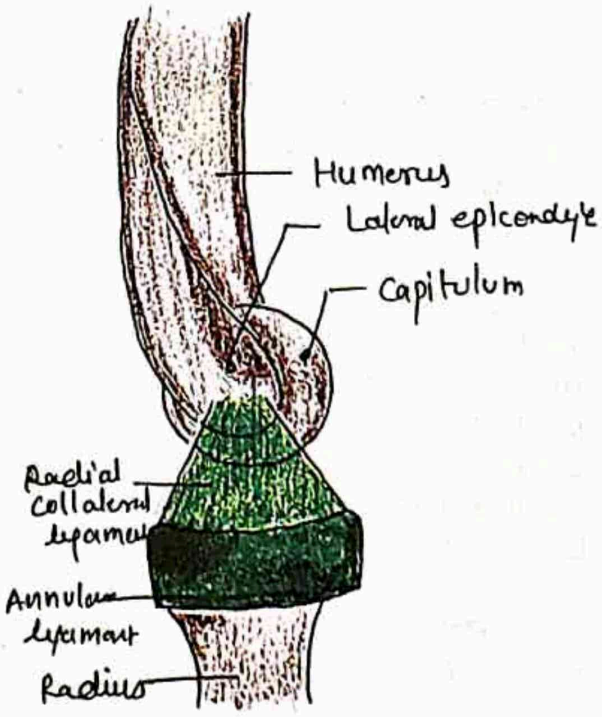
Superior radioulnar joint

The circumference of the head of radius articulates with the radial notch of ulna

∴ The cubital articulations



∴ Ulnar collateral ligament



∴ Radial collateral ligament

ELBOW JOINT

SURAJ KUMAR

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTIONS - 20MARKS

1.DESCRIBE BRACHIAL PLEXUS WITH ITS APPLIED ANATOMY ?

STUDY TIPS FOR MAXIMUM MARKS !!!

1.IN CLINICAL ANATOMY OR APPLIED ANATOMY - YOU HAVE TO WRITE ONLY 3-4 CLINICAL ANATOMY POINTS (ONLY BDC CLINICAL ANATOMY POINTS)

2.IF IT IS MENTION TO WRITE ON SPECIFIC CLINICAL ANATOMY -:

FOR EXAMPLE : WRITE CLINICAL ANATOMY OF ERBS PALSY THEN YOU HAVE TO WRITE ONLY ABOUT ERB PALSY - IN 1-1/2 PAGE.

2.DESCRIBE MEDIAL NERVE UNDER FOLLOWING HEADS -
1.ROOT VALUE 2.COURSE & RELATION 3.DISTRIBUTION 4.CLINICAL ANATOMY

HOW TO WRITE IN UNIVERSITY EXAM (TIME MANAGEMENT)

1.IN MULTIPLE HEADING QUESTION - SPECIFIC MARKS ARE GIVEN TO EACH HEADING

FOR EXAMPLE - 2MARKS FOR ROOT VALUE, 8MARKS FOR COURSE, 5MARKS FOR DISTRIBUTION & 5MARKS FOR CLINICAL ANATOMY.

2.MARKS DISTRIBUTION OF EACH HEADING ARE GIVEN IN PAPER.

3.WRITE ACCORDING TO MARKS DISTRIBUTION.

(ZYADA LIKHNE SE ZYADA MARKS NAHI MILENGE, 2MARKS HAIN TO 2MARKS HI MILENGE - ZYADA LIKHNE KE CHAKKAR ME TIME WASTE MAT KARNA PAPER MISS HO JAAYEGA)

SUBAJ KUMAR

3. A 30-YEAR-OLD MALE PRESENTED TO EMERGENCY DEPARTMENT AFTER A ROAD TRAFFIC ACCIDENT WITH COMPLAINING OF RIGHT ARM PAIN AND INABILITY TO MOVE HIS EXTREMITY. THE FOLLOWING MORNING, HE DEVELOPED CLAWING. NERVE CONDUCTION STUDIES ON THE PERIPHERAL NERVES OF HIS ARM CONFIRMED THE DIAGNOSIS OF INJURY TO THE ULNAR NERVE. KEEPING THIS SCENARIO IN MIND, ANSWER THE FOLLOWING -
QUESTIONS

1) ORIGIN, ROOT VALUE OF THE NERVE 2) COURSE & DISTRIBUTION 3) BRANCHES 4) EXPLAIN REASON FOR DEVELOPING THE CLAW HAND

HOW TO SOLVE CASE STUDY !!!!!!

1.MAINLY NERVES (MEDIAN, RADIAL, ULNAR NERVE) ARE GIVEN IN CASE STUDY.

2.IDENTIFICATION TRICK -
1.IF MENTION INJURY TO SPECIFIC NERVE THEN WRITE ABOUT THAT(FOR EXAMPLE - IN ABOVE CASE STUDY IT MENTION'S ULNAR NERVE INJURY)

2.IF INJURY IS NOT MENTIONED THEN SEE SENSORY LOSS

-IN RADIAL NERVE INJURY SENSORY LOSS - IN DORSAL LATERAL HALF
-IN ULNAR NERVE INJURY SENSORY LOSS IN - MEDIAL HALF OF BOTH DORSAL & PALMAR SIDE
-IN MEDIAN NERVE INJURY SENSORY LOSS IN - LATERAL HALF OF PALMAR SIDE.

(IDENTIFY AARAM SE KARNA, GALAT CASE STUDY IDENTIFY KARNE SE SEEDHE 20MARKS KA LOSS HOGA AUR FAIL.)

4.MID PALMAR SPACE IN HAND ?

HOW TO DRAW A DIAGRAM !!!!!!

1.TRY TO MAKE COLOURFUL DIAGRAM - 1.RED COLOUR - ARTERY
2.BLUE - VEIN
3.BROWN - MUSCLE
4.YELLOW - NERVE
5.GREEN - LIGAMMENT

2.IN LONG QUESTION TRY TO MAKE 2-3 DIAGRAMS ONLY.
(NUMBER OF DIAGRAM = MARKS)
(ZYAADA DIAGRAM BANANE ME TIME WASTE MAT KARNA WARNA NEXT QUESTION MISS HO JAYEGA - TIME MANAGE KARE CHALNA)

3.TRY TO MAKE CLEAR, COLOURFUL & WELL LABELLED DIAGRAM.

5.DESCRIBE MAMMARY GLAND?

HOW TO WRITE IN LONG QUESTIONS !!!!!!

1.IF IN LONG QUESTIONS HEADINGS ARE NOT GIVEN, THEM WRITE IN FOLLOWING HEADINGS
1.DEFINITION
2.EXTERNAL FEATURES
3.BLOOD SUPPLY
4.NERVE SUPPLY
5.LYMPHATIC DRAINAGE
6.CLINICAL ANATOMY

SURAJ KUMAR

6.DESCRIBE SHOULDER JOINT UNDER THE FOLLOWING HEADINGS -
1.TYPE OF JOINT & ARTICULAR SURFACE 2.CAPSULAR LIGAMMENT 3.RELATION
4.APPLIED ANATOMY

7. DESCRIBE ELBOW JOINT UNDER THE FOLLOWING HEADINGS -
1.TYPE OF JOINT & ARTICULAR SURFACE 2.LIGAMENT 3.APPLIED ANATOMY

*SHORT QUESTIONS - 5MARKS

HOW TO WRITE IN SHORT QUESTIONS !!!!!!

1.IN SHORT QUESTIONS WRITE 2-3 PAGES WITH ONE DIAGRAM & 1 CLINICAL ANATOMY - MAXIMUM.
2.MINIMUM 1-2 PAGES WITH DIAGRAM.

(ZYADA LIKHNE ME TIME WASTE MAT KARNA, 5 ME SE 10 MARKS NAHI MILENGE 5 ME 5 HI MILENGE, ZYADA LIKHNE KE CHAKKAR ME PAPER AUR MISS HO JAYEGA)

*SHORTS NOTES :-

1) AXILLA
2) AXILLARY ARTERY
3) CUBITAL FOSSA
4) ROTATOR CUFF
5) FLEXOR RETINACULUM
6) SNUFF BOX
7) LUMBRICAL
8) THENAR MUSCLE
9) SUPINATION & PRONATION
10) FLEXOR RETINACULUM

SURAJ KUMAR

UPPER LIMB ANATOMY IS COMPLETED !!

BEST WAY TO PREDICT FREE FUTURE IS TO CREATE IT

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

THE COPYRIGHT ACT,1957

ALL MATERIAL INCLUDING THEORY, DIAGRAM IS COPYRIGHT REGISTERED .
ANY SCAN AND PHOTOCOPY OF THIS MATERIAL IS NOT PERMITTED BY OWNER

MCQ UPPER LIMB

1) Brachial plexus is formed by:
a ventral ramii of C5 to T1 spinal nerves
b ventral root of C5to T1 spinal nerves
c dorsal ramii of C5 to T1 spinal nerves
d dorsal roots of C5 to T1 spinal nerves

2) All of the following are branches of posterior cord of brachial plexus :
a axillary nerve
b radial nerve
c suprascapular nerve
d thoracodorsal

3) The action of acromial fibres of deltoid are :
a adduction of arm
b abduction of arm
c extension of arm
d flexion of arm

4) all of the following are innervated by musculocutaneous nerve except:
a Coracobrachialis
b biceps brachii
c brachialis
d brachioradialis

5) Which structure does nit lie in anatomical snuff box:
a cephalic vein
b radial artery
c first metacarpal bone
d extensor pollicis longus

6) Following nerve may get injured in downward dislocation of head of humerus
a musculocutaneous
b Axillary
c Radial
d median

7) Following nerve may get injured in fracture of surgical neck of humerus:
a musculocutaneous
b subscapular
c axillary
d radial

8) Fracture of medial epicondyle of humerus may injure :
a median nerve
b ulnar nerve
c ulnar artery
d brachial artery

9) The nerve supply of serratus anterior is:
a axillary nerve
b C4
c thoracodorsal nerve
d long thoracic nerve

10) The most preffered vein for venipuncture is:
a cephalic vein
b basilic vein
c median cubital vein
d brachial vein

11) Which structure lies in the deltopectoral groove:
a basilic vein
b cephalic vein
c axillary vein
d medial pectoral nerve

12) Which nerve passes between ulnar and humeral head of pronator teres
a ulnar nerve
b radial nerve
c median nerve
d anterior interosseous nerve

13) Which nerve passes between ulnar and humeral head of fkexor carpii ulnaris:
a ulnar nerve
b radial nerve
c median nerve
d anterior interosseous nerve

14) All of the following structures pass superficial to flexor retinaculum EXCEPT :
a median nerve
b superficial cutaneous branch of median nerve
c ulnar nerve
d superficial cutaneous branch of ulnar nerve

15) All of the following muscles supplied by median nerve except:
a abductor pollicis brevis
b flexor pollicis
c opponens pollicis
d adductor pollicis

16) Which nerve is involved in carpal tunnel syndrome:
a ulnar nerve
b median nerve
c radial nerve
d anterior interosseous nerve

17 Which lymphatic does not drain the breast:
a subscapular nodes
b infraclavivular nodes
c parasternal nodes
d pectoral nodes

18 Wrist drop occurs due to injury of :
a axillary nerve
b ulnar nerve
c median nerve
d radial nerve

1) A 2)C 3)B 4)D 5)D 6) B 7)C 8)B 9)D 10)C 11)B 12)C 13)A 14)A 15)D 16)B 17)C 18)D

SHARMA SURAJ KUMAR BBS

LOWER LIMB MUSCLES

GLUTEAL REGION

MAJOR MUSCLES
- Gluteus maximus
- Gluteus medius
- Gluteus minimus
- Tensor fasciae latae

MINOR MUSCLES
- Piriformis
- Inferior Gemellus
- Superior Gemellus
- Obturator internus
- Quadratus femoris

THIGH

ANTERIOR COMPARTMENT
- Quadriceps femoris
  - Rectus femoris
  - Vastus lateralis
  - Vastus medialis
  - Vastus intermedius
- Articularis Genu
- Sartorius
- Tensor fasciae Latae

- Inferior Gluteal Nerve
- Superior Gluteal Nerve
- Ventral rami $S_1 S_2$
- Nerve to quadratus femoris
- Nerve to obturator internus
- Quadratus femoris

LATERAL COMPARTMENT
- Peroneus longus
- Peroneus brevis

POSTERIOR COMPARTMENT
SUPERFICIAL
- Gastrocnemius
- Soleus
- Plantaris

DEEP
- Popliteus
- Flexor digitorum longus
- Flexor hallucis longus
- Tibialis posterior

POSTERIOR COMPARTMENT
- Bicep Femoris — short head, long head
- Semitendinosus
- Semimembranosus
- Adductor magnus (ischial part)

MEDIAL COMPARTMENT
ANTERIOR / FIRST LAYER
- Pectineus
- Adductor longus
- Gracilis

MIDDLE / SECOND LAYER
- Adductor brevis

POSTERIOR / THIRD LAYER
- Adductor Magnus
  - Adductor part
  - Hamstring part

- Tibial Sciatic Nerve
- Common Peroneal Sciatic Nerve
- Obturator Nerve
- Femoral Nerve

LEG

MEDIAL COMPARTMENT
Tendons of
- Sartorius
- Gracilis
- Semi tendinosus

ANTERIOR COMPARTMENT
- Tibialis anterior
- Extensor hallucis longus
- Extensor digitorum longus
- Peroneus tertius

- Saphenous Nerve
- Deep peroneal Nerve
- Superficial peroneal Nerve
- Tibial Nerve

FOOT

MUSCLES OF SOLE

1st Layer
- Flexor digitorum brevis
- Abductor hallucis
- Abductor digiti minimi

2nd Layer
- Flexor digitorum accessorius
- 4 lumbricals — 1, 2, 3, 4
- 2 tendon — Flexor digitorum longus, Flexor hallucis longus

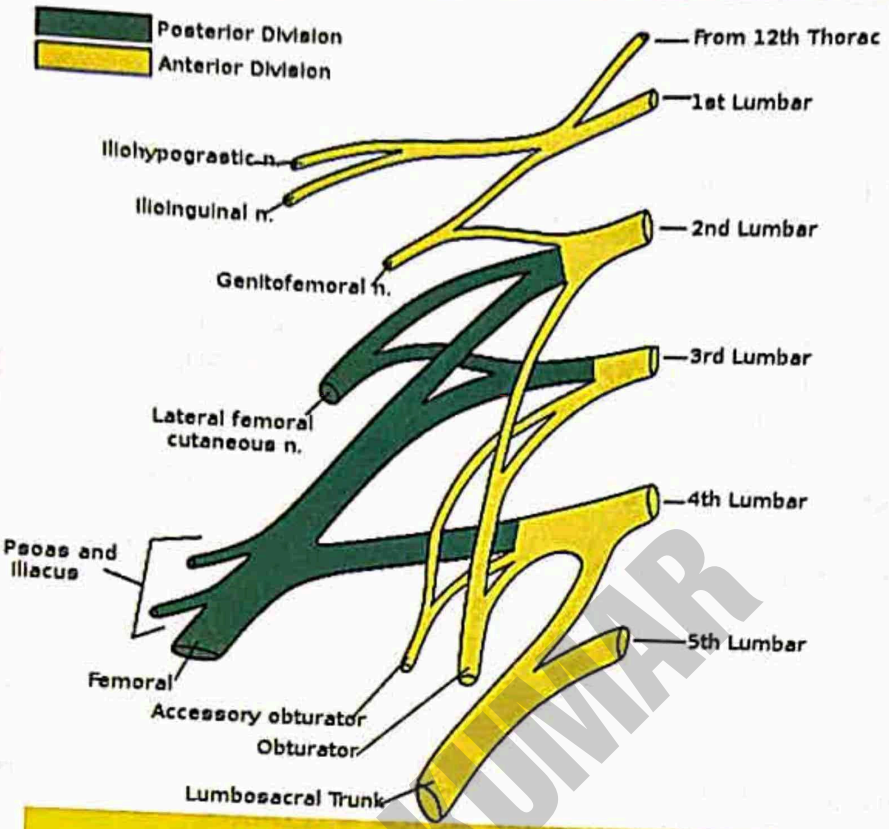
3rd Layer
- Flexor hallucis brevis
- Flexor digiti minimi brevis
- Adductor hallucis

4th Layer
- Interrossei — 3 plantar, 4 dorsal

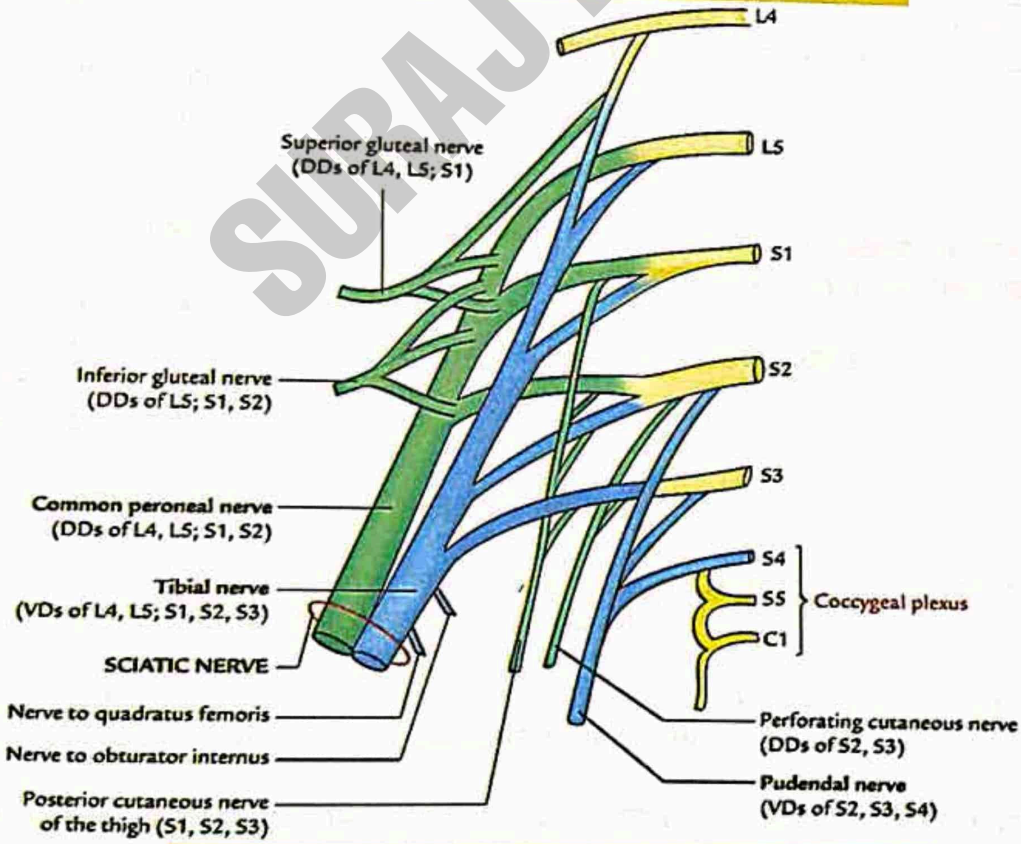
- Tendons of
  - tibialis posterior
  - peroneus longus

- Medial plantar
- Lateral plantar

SURAJ KUMAR



LUMBAR PLEXUS



SACRAL PLEXUS

SURAJ KUMAR

FEMORAL TRIANGLE

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK

*TRIANGULAR DEPRESSION-->INFRONT OF UPPER 1/3RD THIGH BELOW INGUINAL LIGAMENT

*BOUNDARIES :-

1.LATERAL - MEDIAL BORDER SARTORIOUS MUSCLE.
2.MEDIAL - MEDIAL BORDER OF ADDUCTOR LONGUS.
3.BASE - INGUINAL LIGAMENT.
4.APEX - LATERAL & MEDIAL BORDER CROSS.
5.ROOF - SKIN, SUPERFICIAL FASICA & DEEP FASICA.

SUPERFICIAL FASICA---->CONTENT--> SUPERFICIAL INGUNAL L.N.
-> FEMORAL BRANCH OF GENITOFEMORAL NERVE
-> ILIOINGUINAL NERVE
-> GREAT SAPHENOUS VEIN
-> SUPERFICIAL BRANCH OF FEMORAL ARTERY

6.FLOOR - MEDIAL -> ADDUCTOR LONGUS & PECTINEOUS MUSCLE
LATERAL -> PSOAS MAJOR & ILIACUS

*CONTENT :-

1.FEMORAL ARTERY-->IT GIVE 6 BRANCHES - 3 SUPERFICIAL & 3 DEEP BRANCHES
2.FEMORAL VEIN-->MEDIAL TO ARTERY
-->FEMORAL VEIN RECEIVE--->GREAT SAPHENOUS VEIN
--->CIRCUMFLEX VEIN
--->VEIN CORRESPONDING TO BRANCH OF FEMORAL ARTERY

3.FEMORAL SHEATH - ENCLOSE UPPER 4CM OF FEMORAL VESSEL
4.DEEP INGUAL L.N.--->DEEP TO DEEP FASICA
--->MEDIAL TO UPPER PART OF FEMORAL VEIN
->RECEIVED ->LYMPH FROM - SUPERFICIAL INGUINAL L.N.
-DEEP LYMPHATIC OF LOWER LIMB
-FROM GLANS PENIS & CLOTRIES

5.NERVE -

*FEMORAL NERVE - LIE LATERAL TO FEMORAL ARTERY->OUTSIDE FEMORAL SHEATH

*NERVE TO PECTINOUS - ARISE FROM FEMORAL NERVE
-PASS BEHIND FEMORAL SHEATH

*FEMORAL BRANCH OF GENIOFEMORAL NERVE - OCCUUPY LATERAL
-COMPARTMENT TO FEMORAL SHEATH ALONG FEMORAL ARTERY

*LATERAL CUTANEOUS NERVE OF THIGH - CROSS THE LATERAL ANGLE OF

SURAJ KUMAR

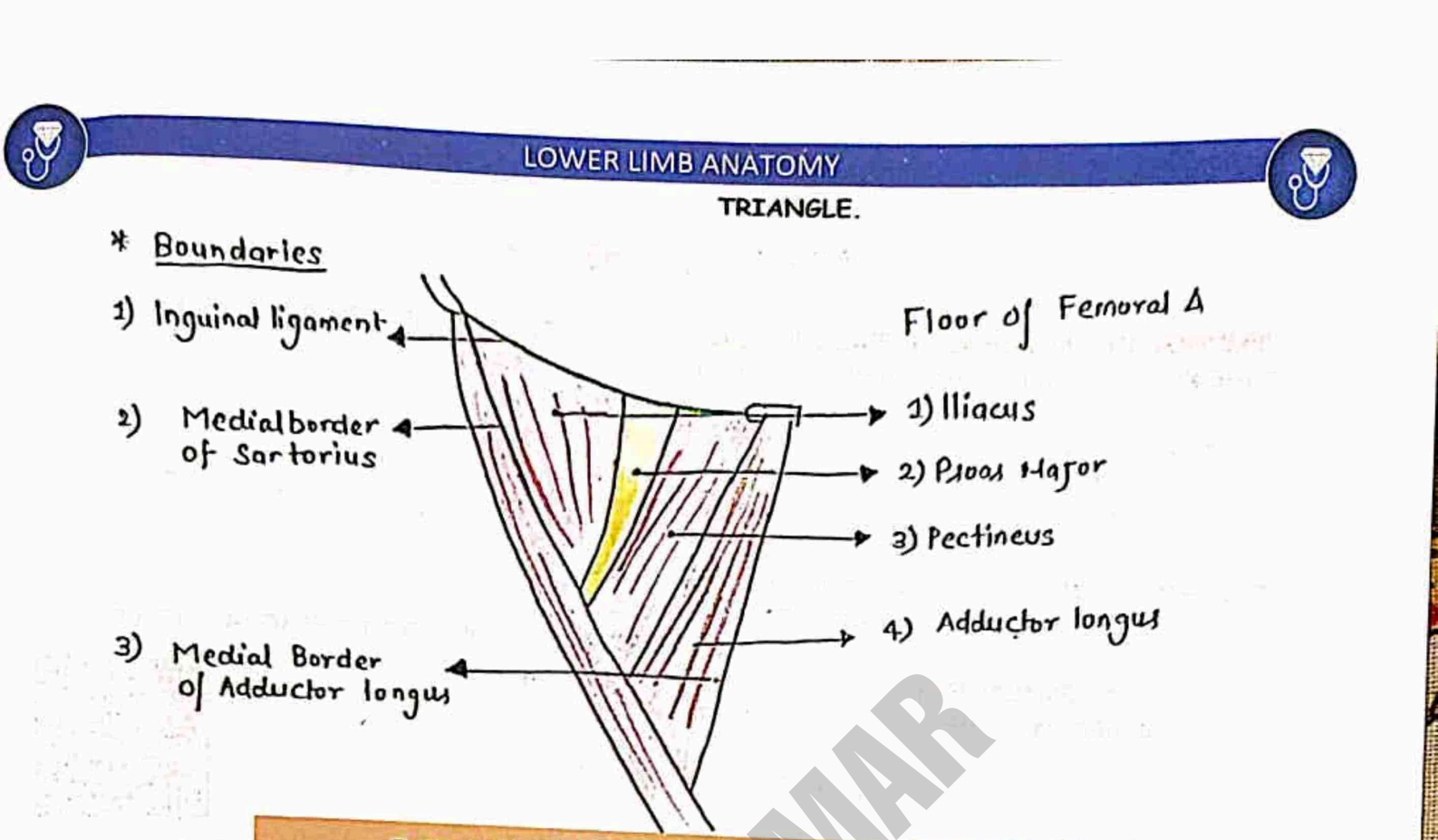
TRIANGLE.

* Boundaries

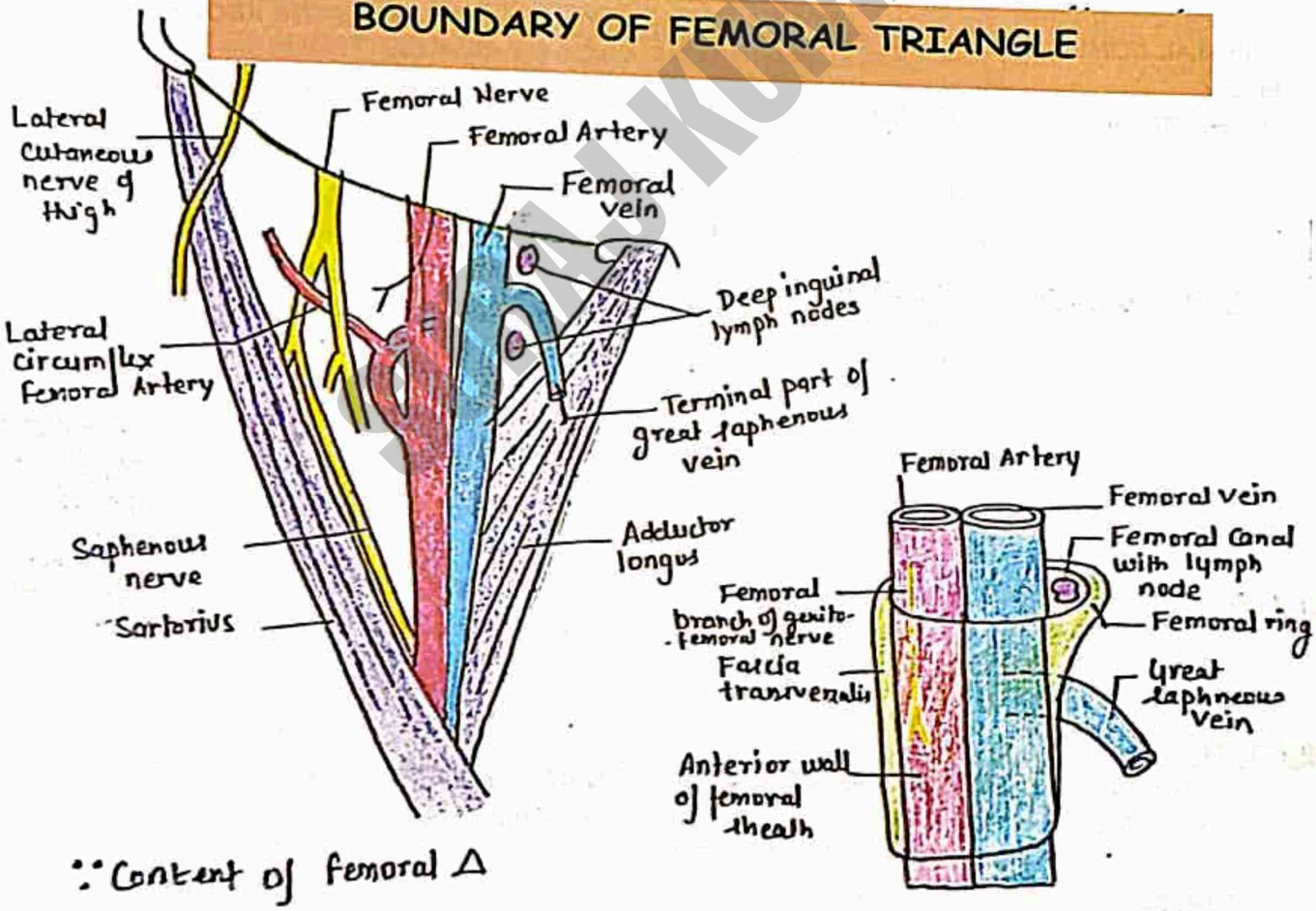
1) Inguinal ligament
2) Medial border of Sartorius
3) Medial Border of Adductor longus

Floor of Femoral Δ

1) Iliacus
2) Psoas Major
3) Pectineus
4) Adductor longus



BOUNDARY OF FEMORAL TRIANGLE



∵ Content of femoral Δ

∵ Femoral canal

CONTENT OF FEMORAL TRIANGLE

ADIL KUMAR

FEMORAL SHEATH

*FUNNEL SHAPE SLEEVE OF FASCIA - ENCLOSING UPPER 4CM OF FEMORAL VESSEL
*SHEATH IS FORM DOWNWARD EXTENSION OF - 2 LAYER OF FASCIA OF ABDOMEN

*ANTERIOR WALL --->FORM--->FASCIA TRANSVERSALIS
*POSTERRIOR WALL--->FORM --->FASCIA ILIACA
*INFERIORLY - SHEATH MERGE WITH CONNECTIVE TISSUE AROUND FEMORAL VESSEL

*SHEATH IS DIVIDED INTO 3 COMPARTMENT BY SEPTA:-
1.LATERAL COMP.-->CONTENT->FEMORAL ARTERY
-->FEMORAL BRANCH OF GENITOFEMORAL NERVE
2.INTERMEDIATE COMP. -->FEMORAL VEIN
3.MEDIAL COMP.--> ALSO KNOWN AS FEMORAL CANAL

FEMORAL CANAL

*MEDIAL COMP.OF FEMORAL SHEATH - CONICAL IN SHAPE - WIDE IN ABOVE, NARROW BLOW
*BASE OR UPPER END OF FEMORAL CANAL ALSO KNOWN AS FEMORAL RING.

CONTENT :- LYMPH NODE & SMALL AMOUNT OF AREOLAR TISSUE

CLINICAL ANATOMY

*FEMORAL HERNIA :-FEMORAL CANAL IS A AREA OF POTENTIAL WEAKNES IN ABDOMINAL WALL-->THROUGH WHICH ABDOMINAL CONTENT MAY BULGE OUT FORM FEMORAL HERNIA

FEMORAL ARTERY

*CHIEF ARTERYOF LOWER LIMB - CONTINUATION OF EXTERNAL ILIAC ARTERY

COURSE - BEGIN - BEHIND INGUINAL LIGAMENT - MID INGUINAL POINT
- PASSES DOWNWARD MEDIALY - 1ST - ENTERING IN FEMORAL TRIANGLE
-THEN ENTER IN ADDUCTOR CANAL - PASSES THROUGH OPENING OF ADDUCTOR MAGNUS ->BECOME POPLITEAL ARTERY

RELATION :-
1.ANTERIOR - SKIN, SUPERFICIAL & DEEP FASCIA, ANTERIOR WALL OF FEMORAL SHEATH
2.POSTERIOR - POSTERIOR WALL OF FEMORAL SHEATH
-PSOAS MAJOR & PECTINEUS
3.MEDIAL - FEMORAL VEIN
4.LATERAL- FEMORAL NERVE

SUBAJ KUMAR

*BRANCHES :- 3 SUPERFICIAL & 3 DEEP

SUPERFICIAL BRANCH - 1.SUPERFICIAL EXTERNAL PUDENDAL ARTERY
2.SUPERFICIAL EPIGASTRIC ARTERY
3.SUPERFICIAL CIRCUMFLEX ILIAC

DEEP BRANCH - 1.PROFUNDA FEMORAL ARTERY
2.DEEP EXTERNAL PUDENDAL ARTERY
3.MUSCULAR BRANCH

FEMORAL NERVE

*CHIEF NERVE OF ANTERIOR COMPARTMENT OF THIGH

*ORIGIN - LARGEST BRANCH OF LUMBER PLEXUS - FORM BY DORSAL DIVISION OF ANTERIOR PRIMARY RAMI (L2, 3,4)

*COURSE - FEMORAL NERVE-→ENTER FEMORAL TRIANGLE BEHIND INGUINAL LIGAMENT LATERAL TO FEMORAL ARTERY
-IN THIGH-→LIE--→BETWEEN ILIACUS & PSOAS OUTSIDE FEMORAL SHEATH.
-AFTER SHORT COURSE--→2.5CM BELOW THE INGUINAL LIGAMENT NERVE DIVIDE INTO ANTERIOR AND POSTERIOR

*BRANCHES :-

MUSCULAR - 1.TRUNK GIVE - BRANCH -→ ILIACUS & PECTINEUS
2.ANTERIOR DIVISION -→SARTORIUS
3.POSTERIOR DIVISION-→RECTUS FEMORIS & 3 VASTI MUSCLE

CUTANEOUS - 1.ANTERIOR DIVISION - GIVE 2 CUTANEOUS BRANCH (MEDIAL & INTERMEDIATE CUTANEOUS NERVE OF THIGH)

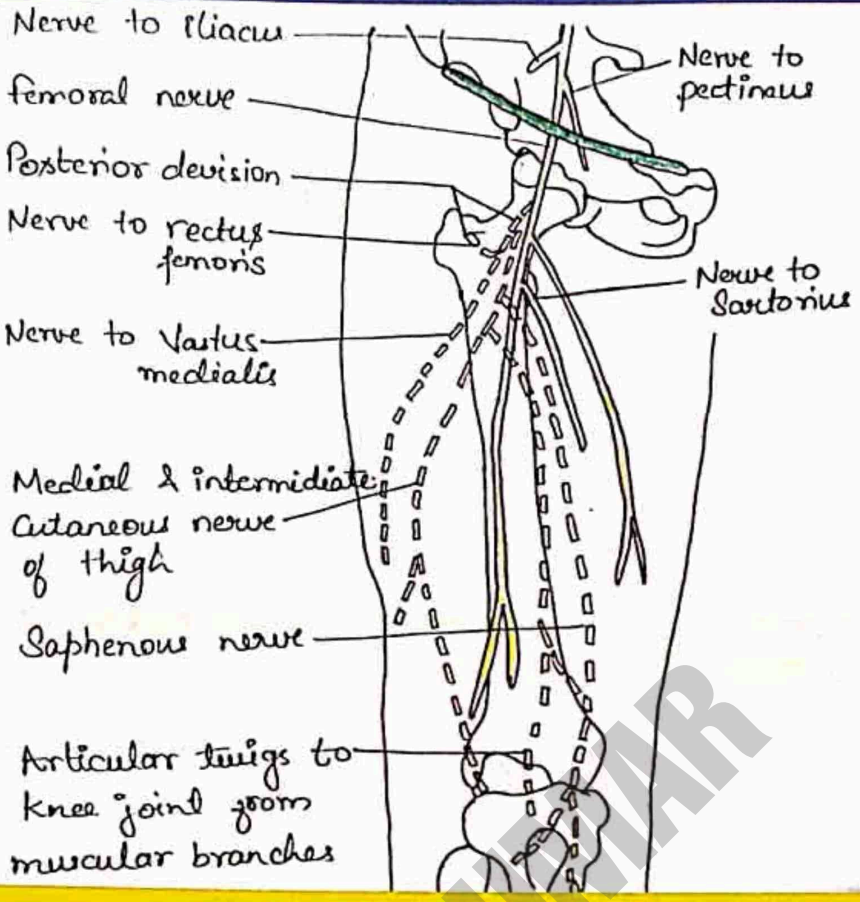
2.POSTERIOR DIVISION - GIVE ONLY ONE CUTANEOUS BRANCH (SAPHANOUS NERVE)

ARTICULAR - ARTICULAR BRANCH TO HIP JOINT & KNEE JOINT
VASCULAR BRANCH TO FEMORAL ARTERY

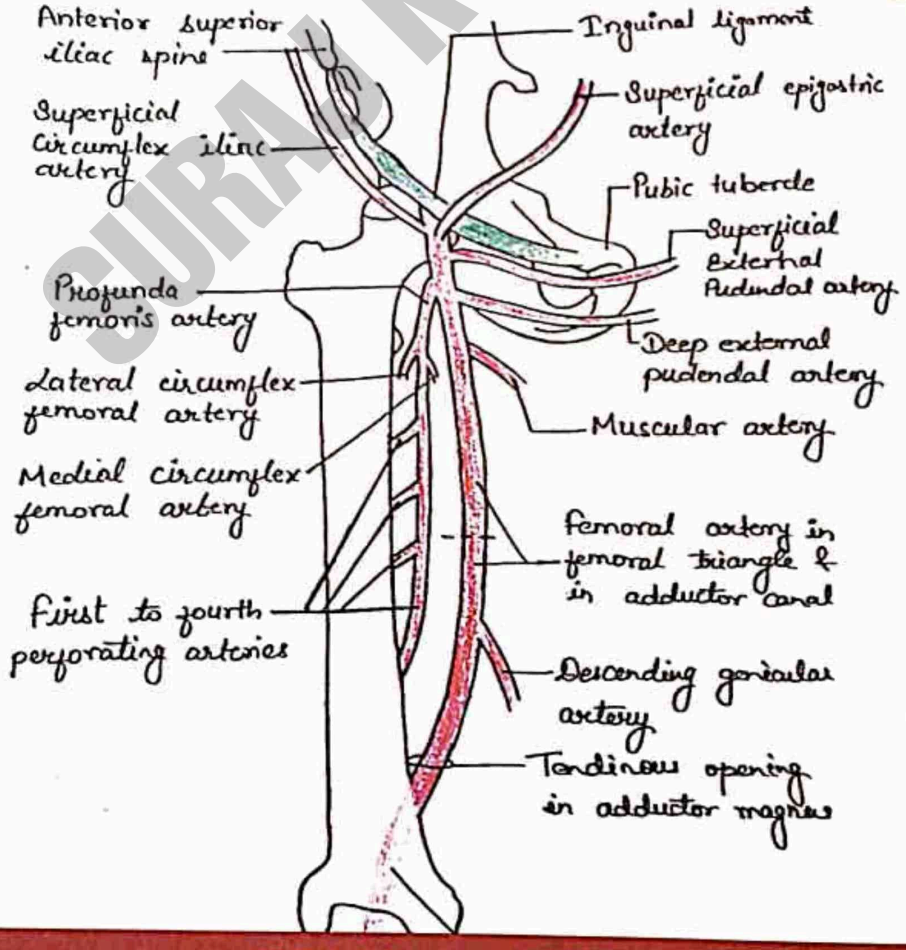
CLINICAL ANATOMY

*FEMORAL NERVE :-FEMORAL NERVE INJURY CAUSE - PARALYSIS OF QUADRICEPS FEMORIS
*SENSORY LOSS - ANTERIOR MEDIAL SIDE OF THIGH &
- MEDIAL SIDE OF LEG

SURAJ KUMAR



FEMORAL NERVE



FEMORAL ARTERY

SURAJ KUMAR

ADDUCTOR CANAL

*INTERMUSCULAR SPACE SITUATED ON MEDIAL SIDE MIDDLE 1/3RD OF THIGH
*EXTEND -APEX OF FEMORAL TRIANGLE→TENDINOUS OPENING OF ADDUCTOR MAGNUS

*BOUNDARIES:-

1.ANTEROLATERAL - VASTUS MEDIALIS

2.POSTEROMEDIAL - ADDUCTOR LONGUS ABOVE
-ADDUCTOR MAGNUS BELOW

3.MEDIAL WALL - SARTORIUS

*SUBSARTORIAL PLEXUS NERVE -→ LIE UNDER SARTORIUS
--→ FORM BY - SAPHENOUS NERVE
- ANT.DIVISION OF OBTURATOR NERVE
- MEDIAL CUTANEOUS NERVE OF THIGH

*CONTENT :-

1.FEMORAL ARTERY--→ENTER CANAL AT APEX OF FEMORAL TRIANGLE
--→GIVE -→MUSCULAR BRANCH AND DESCENDING GENICULAR BRANCH

2.FEMORAL VEIN--→LIE --→POSTERIR TO FEMORAL ARTERY IN UPPER PART
--→LATERAL TO FEMORAL ARTERY IN LOWER PART

3.SAPHENOUS NERVE-→CROSS-→FEMORAL ARTERY ANTERIORLY

4.NERVE TO VASTUS MEDIALIS--→LATERAL TO FEMORAL ARTERY

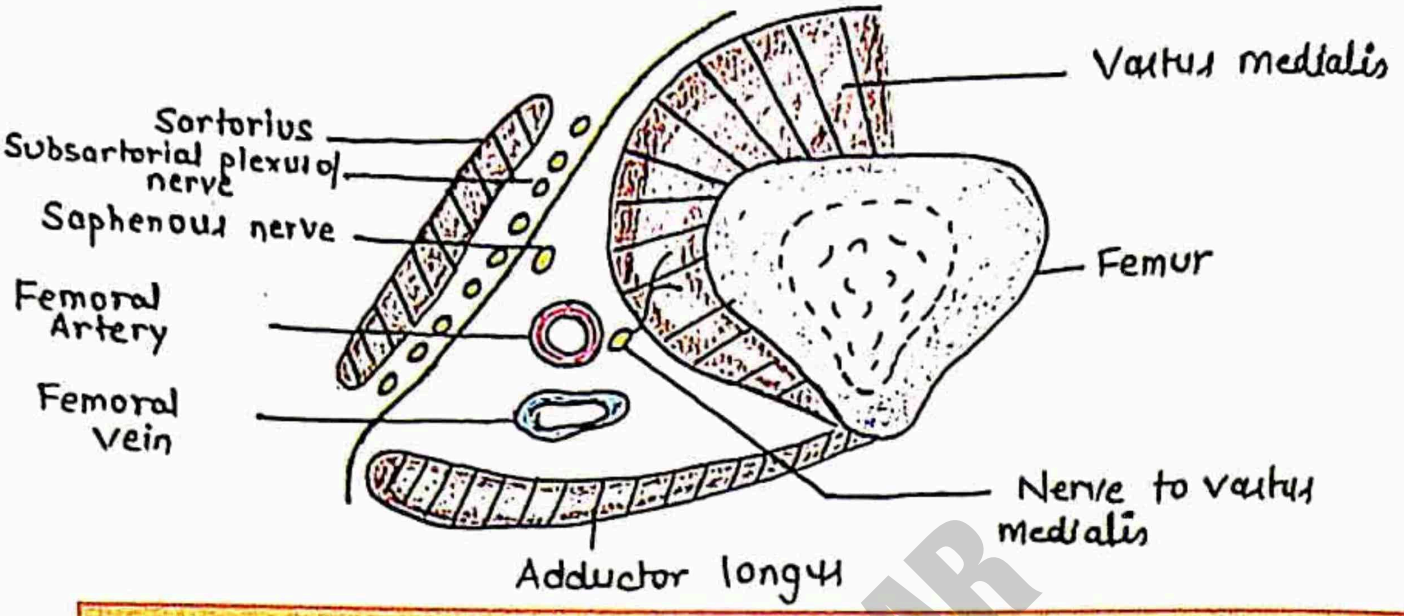
5.BRANCH OF TWO DIVISION OF OBTURATOR NERVE

-→ANTERIOR DIVISION--→EMERGE AT LOWER BORDER OF ADDUCTOR MAGNUS
-->POSTERIOR DIVISION--→ RUN ON ANT. SURFACE OF ADDUCTOR MAGNUS

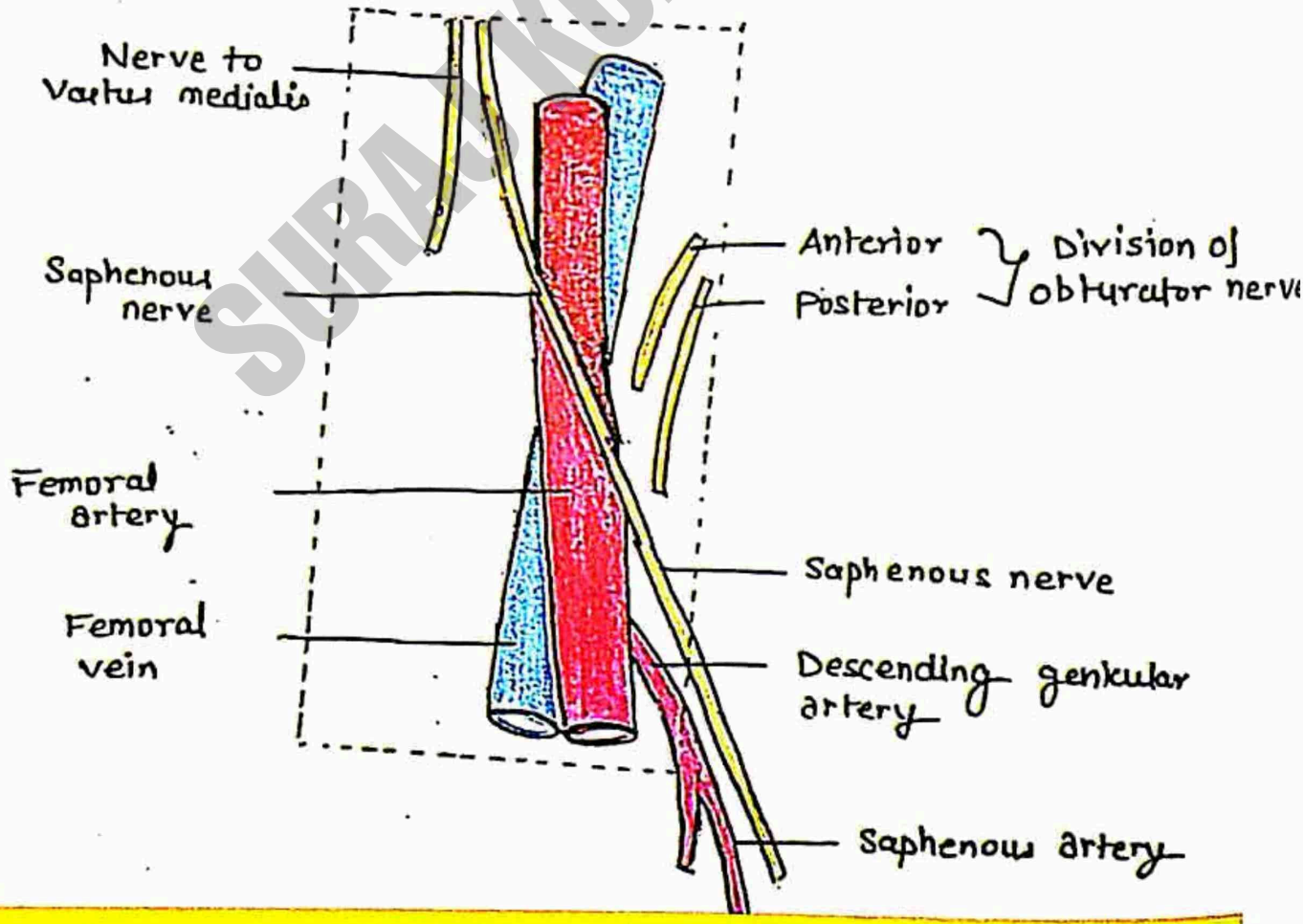
CLINICAL ANATOMY

*A DISEASE OF HIP JOINT MAY CAUSE REFERRED PAIN IN KNEE AND ON THE LATERAL SIDE OF THIGH BECAUSE OF THEIR COMMON NERVE SUPPLY BY THE OBTURATOR NERVE.

SUBAJ KUMAR



BOUNDARIES OF ADDUCTOR CANAL



CONTENT OF ADDUCTOR CANAL

SURAJ KUMAR

POPLITEAL FOSSA

*DIAMOND SHAPE DEPRESSION -->BEHIND KNEE JOINT, LOWER PART OF FEMUR AND UPPER PART OF TIBIA

*BOUNDARIES :-

1.SUPEROLATERAL - BICEP FEMORIS MUSCLE
2.SUPEROMEDIAL - SEMITENDINOSUS MUSCLE
3.INFEROLATERAL - LATERAL HEAD GASTROCNEMIUS
4.INFEROMEDIAL - MEDIAL HEAD OF GASTROCNEMIUS
5.ROOF - DEEP FASICA
-SUPERFICIAL FASICA -->SMALL SAPHENOUS VEIN
-->POSTERIOR CUTANEOUS NERVE OF THIGH
-->POSTERIOR DIVISION OF MEDIAL CUTANEOUS NERVE OF THIGH
-->PERONEAL NERVE
6.FLOOR - POPLITEAL SURFACE OF FEMUR
-CAPSULE OF KNEE JOINT
-POPLITEAL MUSCLE

*CONTENT :- 1.POPLITEAL ARTERY & VEIN
2.TIBIAL NERVE
3.COMMON PERONEAL NERVE
4.GENICULAR BRANCH OF OBURATOR NERVE
5.POPLITEAL LIMPH NODE & FAT SURROUNDING ALL STRUCTURE

SCAN QR WITH QRBOT APP TO DIRECTLY OPEN VIDEO LINK

OBTURATOR NERVE

*CHIEF NERVE OF MEDIAL COMPARTMENT OF THIGH

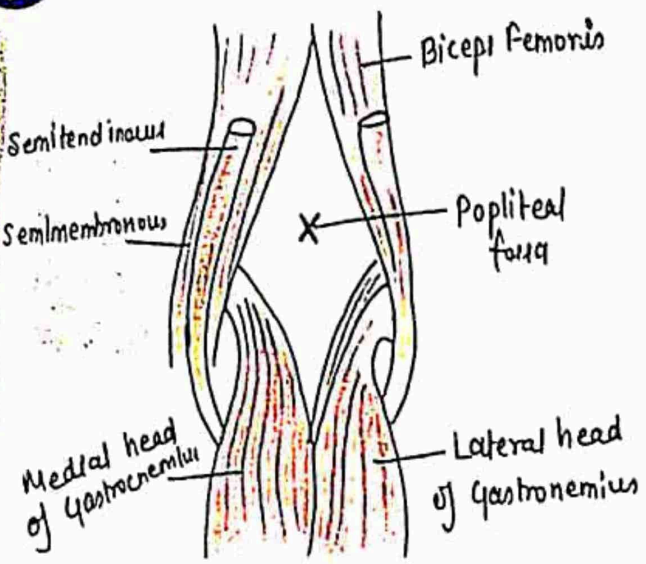
ORIGIN -: BRANCH OF LUMBER PLEXUS -->FORM BY VENTRAL DIVISION OF L2,3,4
-ENTER THE THIGH-->PASSINGG THROUGH OBTURATOR CANAL

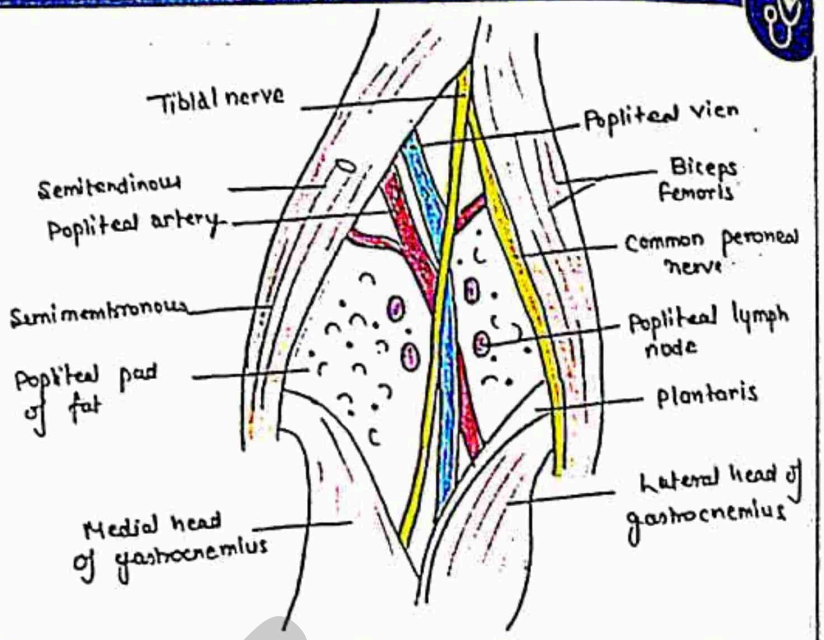
COURSE :-

1.WITH OBTURATOR CANAL -->NERVE DIVIDED INTO ANTERIOR & POSTERIR DIVISION
2.ANTERIOR DIVISION-->PASS DOWNWARD INTO OBTURATOR EXTERNUS
-->THEN DESCEND BEHIND ADDUCTOR LONGUS & PECTINEOUS
-->SUPPLY - PECTINEOUS & GRACILIUS
-ADDUCTOR LONGUS
-ADDUCTOR BRAVIS (IF NOT SUPPLY BY P.D.)

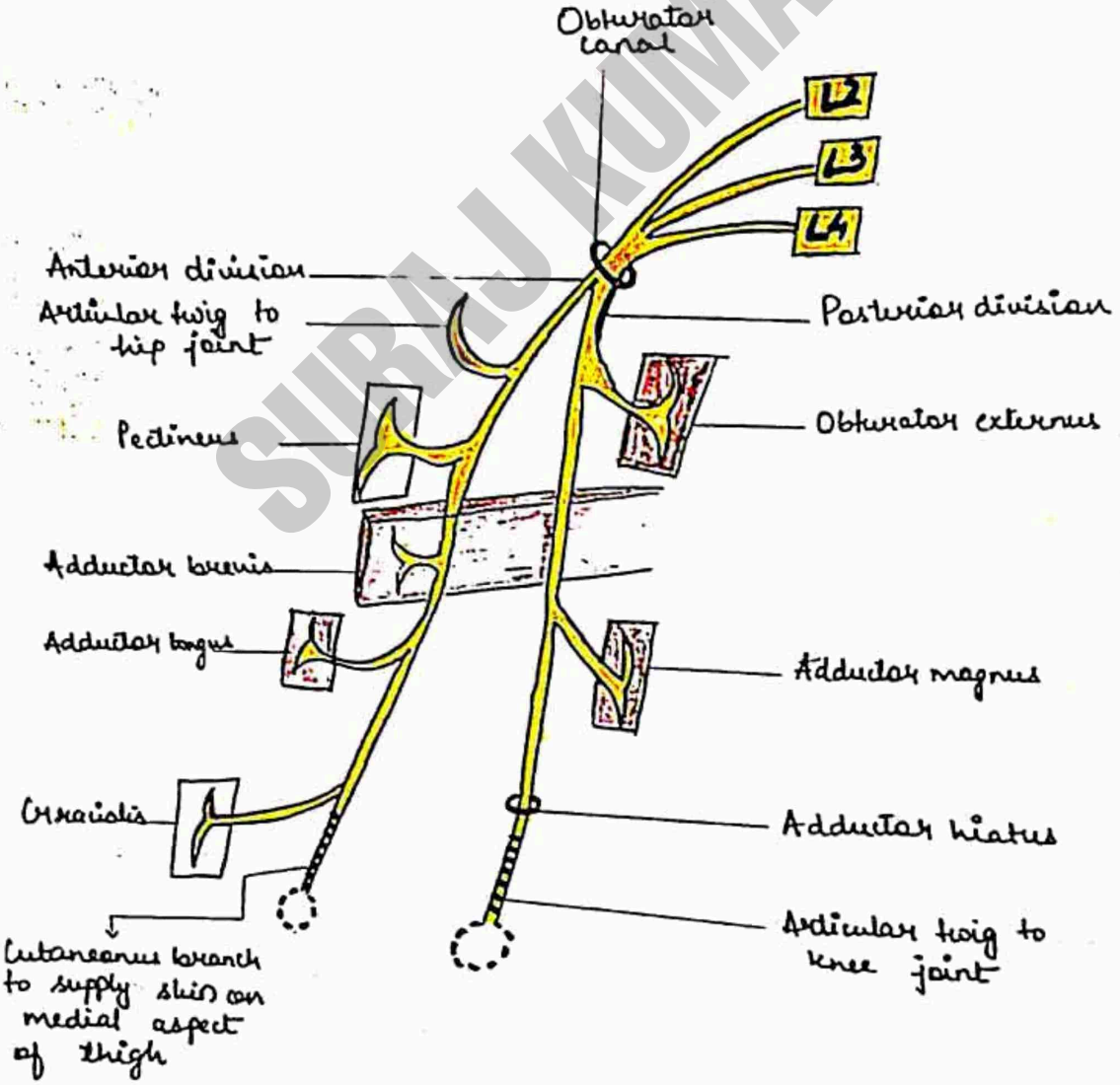
3.POSTERIOR DIVISION --->ENTER THIGH BY PIERCING OBTURATOR EXTERNUS
-->THEN DESCEND BEHIND ADDUCTOR BREVIS
--->SUPPLY - OBTURATOR EXTERNUS
-ADDUCTOR MAGNUS
-ADDUCTOR BREVIS(IF NOT SUPPLY BY ANT.DIVISION)

SUNIL KUMAR





BOUNDARY & CONTENT OF POPLITEAL FOSSA



SURAJ KUMAR

OBTURATOR NERVE

SCIATIC NERVE

*THICKEST NERVE IN THE BODY-→BEGIN IN PELVIS -→TERMINATE AT SUPERIOR ANGLE OF POPLITEAL FOSSA BY DIVIDING - TIBIAL NERVE & COMMON PERONEAL NERVE

*ORIGIN :- LARGEST BRANCH OF SACRAL PLEXUS - ROOT VALUE L4,5 & S1,2,3
-TIBIAL PART - FORM BY - VENTRAL DIVISION OF L4,5 & S1,2,3
-COMMON PERONEAL PART - FORM BY - DORSAL DIVISION OF L4,5 & S1,2

*COURSE :-
1.IN PELVIS - NERVE LIE INFRONT OF-PIRIFORMIS - UNDER COVER OF ITS FASCICA
2.IN GLUTEAL REGION - ENTER THROUGH GREATER SCIATIC FORAMEN
-→ RELATION IN GLUTEAL REGION
1.SUPERFICIAL - GLUTEAL MAXIMUS
2.DEEP - BODY OF ISCHISM
-TENDON OF OBTURATOR INTERNUS WITH GEMELLI
-QUADRATUS FEMORIS
-OBTURATOR EXTERNUS
-ADDUCTOR MAGNUS
3.IN THIGH - ENTER BACK OF THIGH
-RUN VERTICALLY DOWNWARDS UPTO SUPERIOR ANGLE OF POPLITEAL FOSSA
--→ RELATION THIGH :-
1.SUPERFICIAL - CROSS LONG HEAD OF BICEPS FEMORIS
2.DEEP - ADDUCTOR MAGNUS
3.MEDIAL - SEMIMEMBRANOSUS & SEMITENDINOUS
4.LATERAL - BICEPS FEMORIS

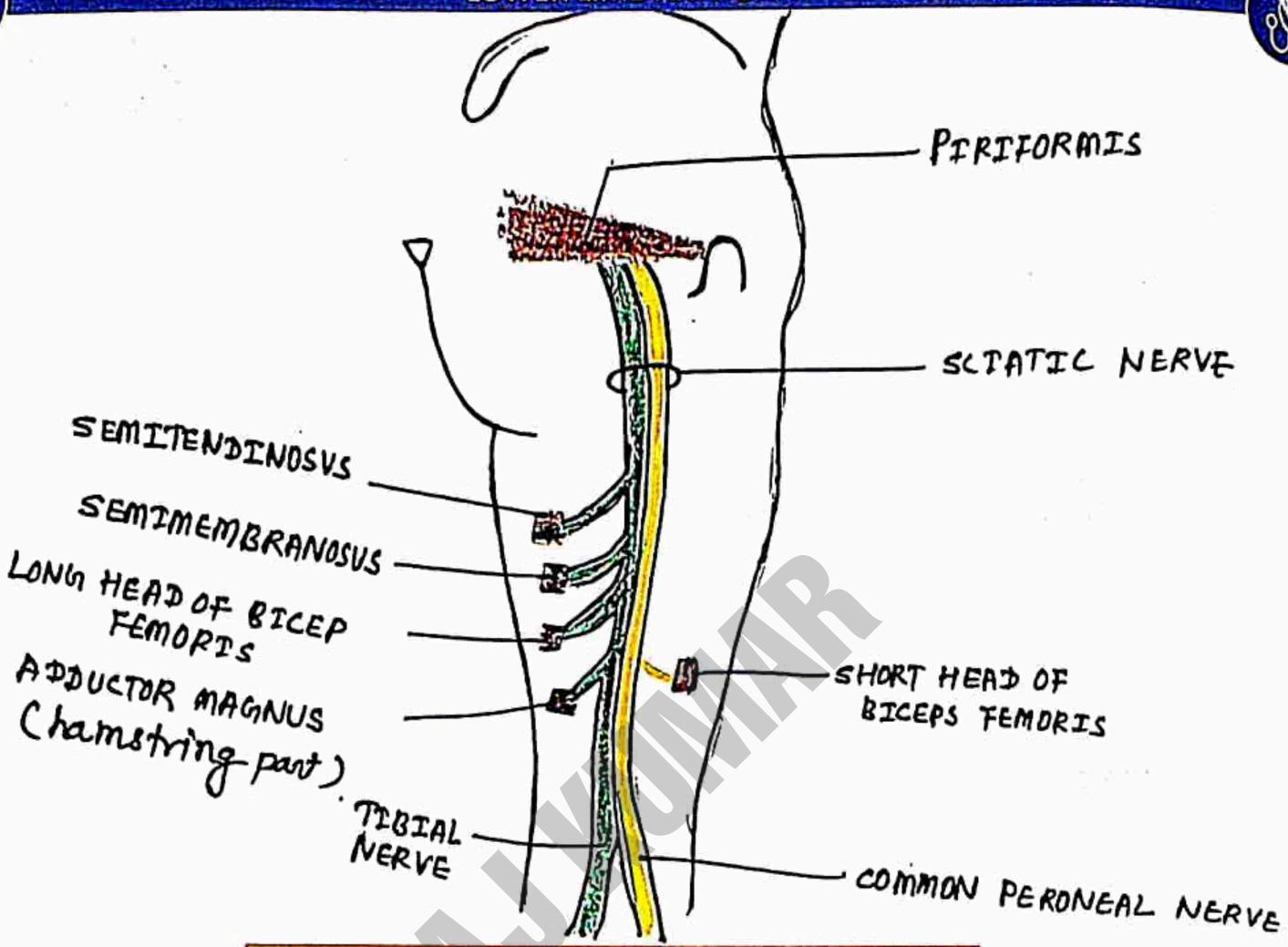
BRANCHES :-
1.ARTICULATING BRANCH OF HIP JOINT
2.MUSCULAR BRANCH OF TIBIAL PART --→SEMITENDINOSUS
- SEMIMEMBRANOSUS
- LONG HEAD OF BICEPS FEMORIS
COMMON PARONEAL PART - SHORT HEAD OF BICEPS FEMORIS

POINTS TO REMEMBER

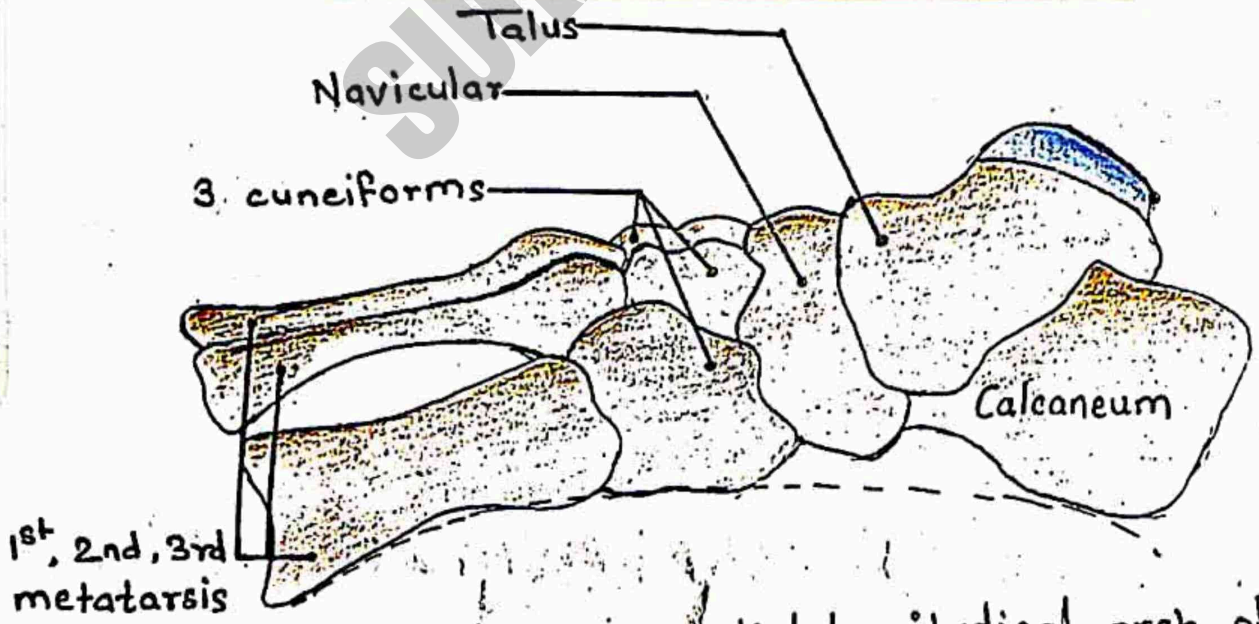
*COMMON PERONEAL NERVE BRANCH -SUPERFICIAL & DEEP PERONEAL NERVE
1.SUPERFICIAL PERONEAL NERVE-→BRANCH--→TIBIALIS ANTERIOR
--→EXTENSOR DIGITORUM LONGUS
--→EXTENSOR HALLUCIS LONGUS
--→PERONEAL TERTIS
2.DEEP PERONEAL NERVE-→BRANCH-→PERONEUS LONGUS & BREVIS

FOOT DROP :- INJURY TO DEEP PERONEAL NERVE RESULT IN LOSS OF POWER OF DORSIFLEXION OF FOOT AS A RESULT FOOT IS PLANTARFLEXED CONDITION IS KNOWN AS FOOT DROP.

SUNIL KUMAR



SCIATIC NERVE



Bones forming the medial longitudinal arch of foot: Medial view. Inset showing big arc of a small circle

ARCHES OF FOOT

SURAJ KUMAR

ARCHES OF FOOT

*ARCHES OF FOOT HELP IN FAST WALKING & JUMPING.

CLASSIFICATION OF ARCHES :-

1.LONGITUDINAL - MEDIAL & LATERAL ARCH
2.TRANSVERSE - ANTERIOR & POSTERIOR

MEDIAL LONGITUDINAL ARCH

ARCH IS CONSIDER HIGHER, MORE MOBILE-→IT CONSIDER AS BIG ARCH OF MORE BONE & MORE JOINT

*ENDS - ANT.END-→HEAD OF 1ST,2ND & 3RD METATARSAL BONE
-POST.END-→MEDIAL TUBERCLE OF CALCANEUM
*SUMMIT :- SUPERIOR ARTICULAR SURFACE OF BODY OF TALUS

*PILLARS :-
ANT.PILLAR - LONG AND WEAK
-FORM BY TALUS, THE NAVICULAR & 3 CUNEIFORM BONES
POST.PILLAR - SHORT AND STRONG
-FORM BY MEDIAL PART OF CALCANEUM
*MAIN JOINT :- TALOCALCANEONAVICULAR JOINT.

LATERAL LONGITUDINAL ARCH

CHARACTERISLY LOW WITH LESS BONE & JOINT & MOBILITY

*END :- ANT.END-→HEAD OF 5TH & 4TH METATARSLE BONE
POST.END-→LATERAL TUBERCLE OF CALCANIUM

*SUMMIT :- SUPERIOR SPACE OF CALCANIUM

*PILLAR :- ANT.PILLAR-→LONG & WEAK
-→FORM BY CUBOID BONE, 4TH & 5TH METATARSAL
POST.PILLAR-→SHORT & STRONG
-→FORM BY LATERAL HALF OF CALCANEUM
*MAIN JOINT :- CALCANEOCUBOIDAL JOINT

TRANSVERSE ARCH (T.A.)

1.ANTERIOR T.A-→FORM BY 5 METATARSAL BONE
-→COMPLETE ARCH BECAUSE HEAD OF 1ST & 2ND METATARSAL COME IN CONTACT WITH GROUND
2.POSTERIOR T.A-→FORM BY GREATER PART OF TARSAL & BASE OF METATARSAL
-→IN COMPLETE ARCH BECAUSE ONLY THE LATERAL END COME IN CONTACT WITH THE GROUND

SURAJ KUMAR

VENOUS DRAINAGE OF LOWER LIMB

*VEIN OF LOWER LIMB CLASSIFIED-→1.SUPERFICIAL- 2.DEEP - 3.PERFORATING VEIN

*DEEP VEIN - MEDIAL PLANTAR, LATERAL PLANTAR, DORSALIS PEDIS , ANTERIOR & POSTERIOR TIBIAL, PERONEAL, POPLITEAL & FEMORAL VEIN
- VALVE ARE MORE NUMEROUS IN DEEP VEIN
- MORE EFFICIENT CHANNEL BECAUSE DRIVING FROCE OF MUSCULAR CONTRACTION

*SUPERFICIAL VEIN - GREATER & SMALL SAPHENOUS VEIN → LIE IN SUPERFICIAL FASCIA
- VALVE ARE MORE IN DISTAL PART
- LARGE PORTION OF BLOOD-→ DRAINAGE - INTO DEEP VEIN THROUGH PERFORATING VEIN

LONG SAPHENOUS VEIN

*FORMATION - UNION OF MEDIAL END OF DORSAL VENOUS ARCH WITH MEDIAL MARGINAL VEIN

*COURSE :-
- PASS UPWARD IN FRONT OF MEDIAL MALLEOLUS
- CROSS LOWER 1/3RD OF MEDIAL SURFACE OF TIBIA TO REACH KNEE JOINT
- TN THING - INCLINE FORWARD TO REACH SAPHENOUS OPENING -→WHERE IT PIERCE CIRBRIFORM FASCIA & OPEN INTO FEMORAL VEIN

*CONNECTION -
- TO DEEP VEIN BY PERFORATING VEIN
  1) 3 MEDIAL PERFORATOR JUST ABOVE ANKLE
  2) 1 PERFORATOR JUST BELOW KEE
  3) ANOTHER ONE IN ADDUCTOR CANA

*TRIBUTARIES -

1.AT COMMENCEMNET - MEDIAL MARGINAL VEIN
2.LEG- COMMUNICATE WITH SMALL SAPHENOUS VEIN & DEEP VEIN
- POSTERIOR ARCH VEIN & VEIN FROM CALF
3.THING - ACCESSARY SAPHENOUS VEIN
- ANTERIOR CUTANEOUS VEIN
4.JUST BEFORE PIERCING THE CRIBRIFORM FASCIA-SUP. EPIIGASTRIC
-SUP.CIRCUMFLEX ILIAC
5.JUST BEFORE TERMINATION - DEEP EXTERNAL PUDENDAL VEIN

SUNIL KUMAR

SMALL SAPHENOUS VEIN

*FORMATION - UNION OF LATERAL END OF DORSAL VENOUS ARCH WITH LATERAL MARGINAL VEIN

*COURSE - IT ENTER THE BACK OF LEG BY PASSING BEHIND LATERAL MALLEOUS TO LOWER PART OF POPLITEAL FOSSA

*DRAIN - LATERAL BORDER OF FOOT - HEEL & BACK OF LEG

PERFORATING VEIN

INDIRECT PERFORATING VEIN -CONECT SUPERFICIAL VEIN WITH DEEP VEIN THROUGH MUSCULAR VEIN.

DIRECT PERFORATING VEIN - CONECT SUPERFICIAL VEIN WITH DIRECTLY WITH DEEP VEIN

1.IN THIGH - ADDUCTOR CANAL PERFORATOR--->CONNECT->GREAT SAPHENOUS WITH FEMORAL VEIN.

2.BELOW THE KNEE - PERFORATOR--->CONNECT--> GREAT SAPHENOUS VEIN WITH POSTERIOR TIBIAL VEIN.

3.IN LEG - LATERAL PERFORATOR---> CONNECT-->SMALL SAPHENOUS VEIN WITH PERONEAL VEIN

- MEDIAL THERE ARE 3 PERFORATORS - CONNECT-->POSTERIOR ARCH VEIN WITH POSTERIOR TIBIAL VEIN.
- MEDIAL 3 PERFORATOR ARE - UPPER MEDIAL PERFORATOR
  -MIDDLE MEDIAL PERFORATOR
  -LOWER MEDIAL PERFORATOR

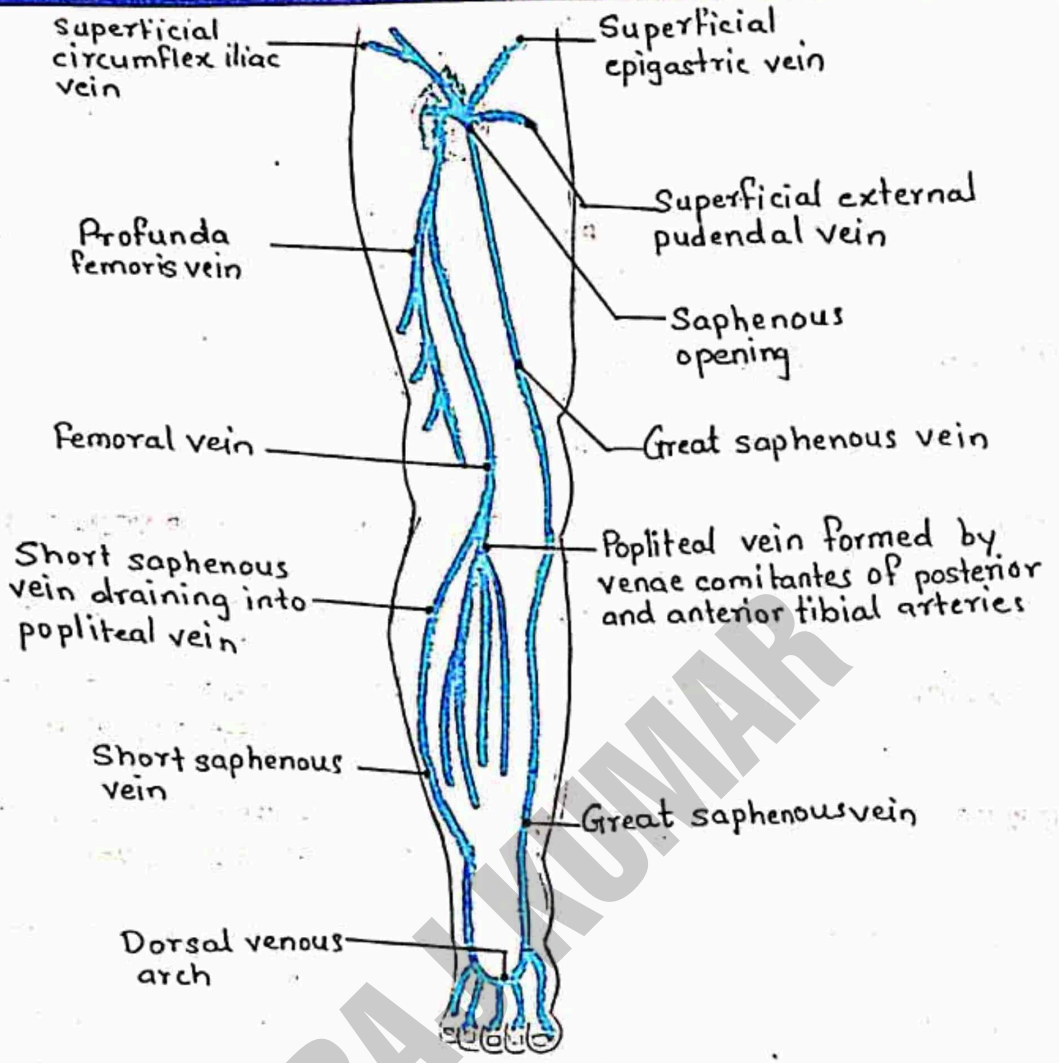
CLINICAL ANATOMY

1.GREAT SAPHENOUS VEIN IS USED FOR BYPASING THE BLOCKED CORONARY ARTERY THE VEIN IS REVERSED SO THAT VALVE DO NOT BLOCK THE PASSAGE OF BLOOD.

2.VARICOSE VEIN - IF THE VALVES IN PERFORATING VEIN OR AT THE TERMINAL OF SUPERFICIAL VEIN BECOME INCOMPETENT --->RESULT IN DILATATION OF SUPERFICIAL VEIN AND GRADUAL DE-GENERATION OF WALL TO PRODUCE VARICOSE VEIN.

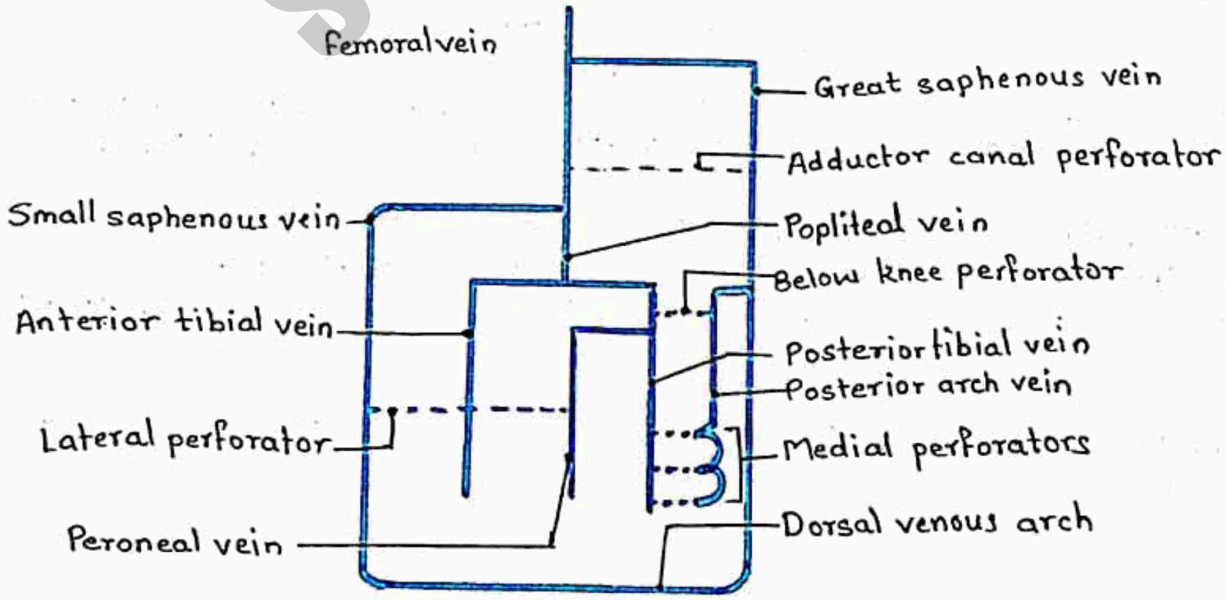
3.VERICOSE VEIN OFTEN OCCUR DURING THIRD TRIMESTER OF PREGNANCY,AS THE ILIAC VEINGET PRSSED DUE TO ENLARGED UTERUS.

SURAJ KUMAR



Scheme to show the arrangement of the veins of lower limb. Popliteal, short saphenous and venae comitantes of posterior tibial artery are on posterior aspect.

SURA KUMAR



VENOUS DRAINAGE OF LOWER LIMB

HAMSTRING MUSCLE

*FEATURES - ORIGIN FROM ISCHIAL TUBEROCITY
- NERVE SUPPLY - TIBIAL PART OF SCIATIC NERVE
- MUSCLE ACT AS FLEXOR KNEE & EXTENSOR OF HIP

1.SEMITENDINOUS :-

*ORIGIN - INFEROMIDIAL IMPRESSION ON THE UPPER PART OF ISCHIAL TUBEROSITY
*INSERTION - UPPER MEDIAL SURFACE OF TIBIA
*NERVE SUPPLY - TIBIAL PART OF SCIATIC NERVE
*ACTION - FLEXION OF KNEE & MEDIAL ROTATION OF LEG, WEEK EXTENSOR OF HIP

2.SEMIMEMBRANOSUS :-

*ORIGIN - SUPEROLATERAL IMPRESSION - ISCHIAL TUBEROSITY
*INSERTION - POSTERIOR SURFACE OF MEDIAL CONDYLE OF TIBIA
*NERVE SUPPLY - TIBIAL PART OF SCIATIC NERVE
*ACTION - FLEXION OF KNEE & MEDIAL ROTATION OF LEG, WEEK EXTENSOR OF HIP

3.BICEPS FEMORIS :-

*ORIGIN - LONG HEAD - INFEROMEDIAL IMPRESSION
SHORT HEAD - ISCHIAL TUBEROSITY
*INSERTION - HEAD OF FIBULA
*NERVE SUPPLY - LONG HEAD - TIBIAL PART OF SCIATIC NERVE
- SHORT HEAD - COMMON PERONEAL NERVE
*ACTION - LATERAL ROTATION OF KNEE - WEEK EXTENSOR OF HIP

4.ADDUCTOR MAGNUS :-

*ORIGIN - LOWER LATERAL PART OF ISCHIAL TUBEROSITY
- INFERIOR RAMUS OF PUBIS
*INSERTION - MEDIAL MARGIN OF GLUTEAL TUBEROSITY
- LINEA ASPERA & MEDIAL SUPRACHONDYLAR LINE
- ADDUCTOR TUBERCLE
*NERVE SUPPLY - DOUBLE NERVE SUPPLY -TIBIAL PART OF SCIATIC NERVE
- OBTURATOR NERVE
*ACTION - EXTENSION OF HIP
- ADDUCTOR PART - CAUSE ADDUCTION OF THIGH

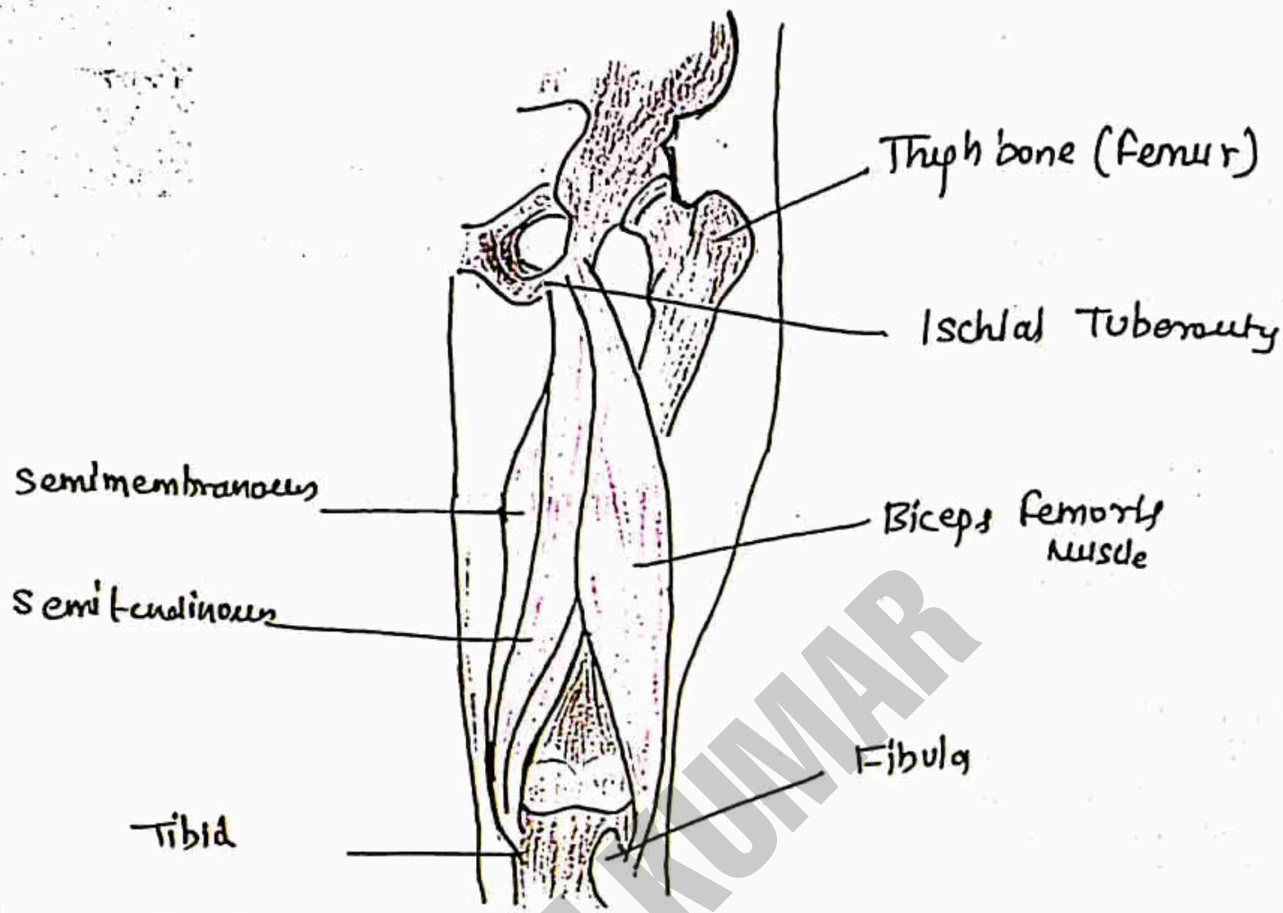
FLEXOR RETINOCULUM

*ATTACHMENT - ANTERIORLY--→ MEDIAL MALLEOLUS
- POSTERIOR & LATERALLY --→ MEDIAL TUBERCLE OF CALCANEUM

*STRUCTURE PASSING DEEP TO FLEXOR RETINOCULUM :-

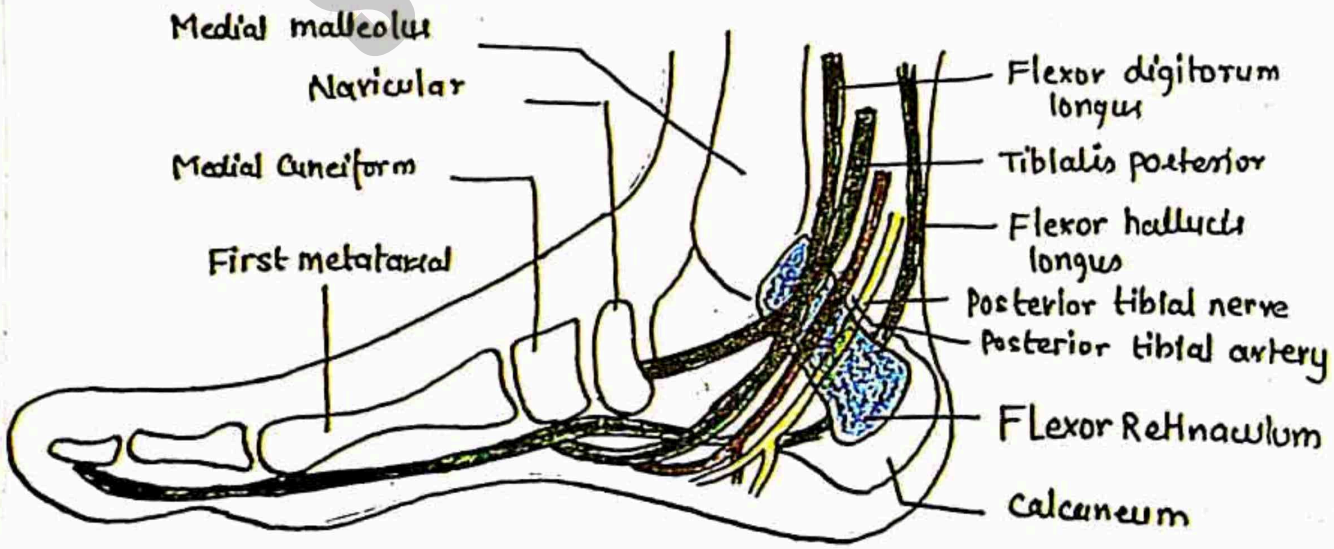
1.TENDON OF TIBIALIS POSTERIOR & FLEXOR DIGITORUM LONGUS
2.POSTERIOR TIBIAL ARTERY & TIBIAL NERVE
3.TENDON OF FLEXOR HALLUCIS LONGUS

BIRAJ KUMAR



HAMSRING MUSCLE

SURAJ KUMAR



FLEXOR RETINACULUM

HIP JOINT

*TYPE - BALL & SOCKET VARIETY OF SYNOVIAL JOINT.

*ARTICULATING SURFACE - HEAD OF FEMUR WITH ACETABULUM OF HIP JOINT.

*HIP JOINT IS UNIQUE HAVING HIGH DEGREE OF STABILITY AS WELL AS MOBILITY.
*STABILITY DEPEND UPON - DEPTH OF ACETABULUM TENSION & STRENGTH OF LIGAMENT.
*STRENGTH OF SURROUNDING MUSCLE
*LENGTH AND OBLIQUITY OF NECK OF FEMUR.

*LIGMANET

1.FIBROUS CAPSULE - ATTACHED ON HIP BONE TO ACETABULUM LABRUM
- ON FEMUR TO THE INTERTROCHANTERIC LINE INFRONT

2.ILIOFEMORAL LIGAMENT - STRONGEST LIGAMENT OF BODY
- TRIANGULAR IN SHAPE
- APEX IS ATTACHED - ANT. INFERIOR ILIAC SPINE
- BASE - INTERTRCHANTERIC LINE.

3.PUBOFEMORAL LIGAMENT - SUPPORT THE JOINT INFEROMEDIALLY
- TRIANGULAR IN SHAPE
- SUPERIOR ATTACH - ILIOPUBIC EMINENCE & OBTURATOR CRUST
-INFERIORLY - ANTEROINFERIOR PART OF CAPSULE
- LOWER BORDER OF ILIOFEMORAL LIGAMENT.

4.ISCHIOFEMORAL LIGAMNET - COVER THE JOINT POSTERIORLY
- FIBRE ARE TWISTED EXTEND FROM ISCHIUM TO ACETABULUM

5.LIGAMENT OF HEAD OF FEMUR - TRIANGULAR IN SHAPE
- APEX ATTACHED - FOVEA CAPITIS
- BASE - MARGIN OF ACETABULUM NOTCH

6.ACETABULAR LABRUM - FIBROCARTILAGINOUS RIM
- ATTACHED TO MARGIN OF ACETABULUM.

*RELATION :-

1. ANTERIORLY - TENDON OF ILIOPSOAS
- FEMORAL ARTERY
- FEMORAL NERVE
- FEMORAL VEIN

SURAJ KUMAR

2.POSTERIORLY - PIRIFORMIS
- SCIATIC NERVE
- OBTURATOR INTERNUS & GEMILLI
- QUADRATUS FEMORIS
- GLUTEUS MAXIMUS

3.SUPERIOR - GLUTEUS MAXIMUS
- GLUTIUS MINIMUS
- GLUTIUS MEDIUS

4.INFERIOR - OBTURATOR EXTERNUS
- HAMSTRING MUSCLE

BLOOD SUPPLY :- OBTURATOR ARTERY & 2 CIRCUMFLEX FEMORAL, 2 GLUTIAL ARTERY

NERVE SUPPLY :- FEMORAL NERVE--->THROUGH NERVE TO RECTUS FEMORIS
-ANTERIOR DIVISION OF OBTURATOR NERVE
-NERVE TO QUADRATUS FEMORIS
-SUPERIOR GLUTEAL NERVE

MOVEMENT :- FLEXION & EXTENSION - TRANSVERSE AXIS
- ADDUCTION & ABDUCTION - ANTERIOR & POSTERIOR AXIS
- MEDIAL & LATERAL ROTATION - VERTICAL AXIS
- CIRCUMDUCTION

LOCKING & UNLOCKING OF KNEE JOINT

1.MECHANISM THAT ALLOW THE KNEE TO REMAIN IN THE POSITION FULL EXTENSION WITHOUT MUSCULAR EFFORT.

2.LOCKING OCCUR -->MEDIAL ROTATION OF FEMUR DURING LAST STAGE OF EXTENSION

3.DIAMETER OF LATERAL FEMORAL CONDYLE IS LESS THAN MEDIAL CONDYLE

4.AS RESULT -->LATERAL CONDYLE FULLY ARTICULATE IN EXTENSION
-->PART OF MEDIAL CONDYLE REMAIN UNUSED
-->AT THIS STAGE LATERAL CONDYLE SERVE AS AXIS & MEDIAL ROTATION OF FEMUR OCCUR
--->SO REMAINING PART OF MEDIAL CONDYLAR SURFACE IS ALSO TAKEN UP.

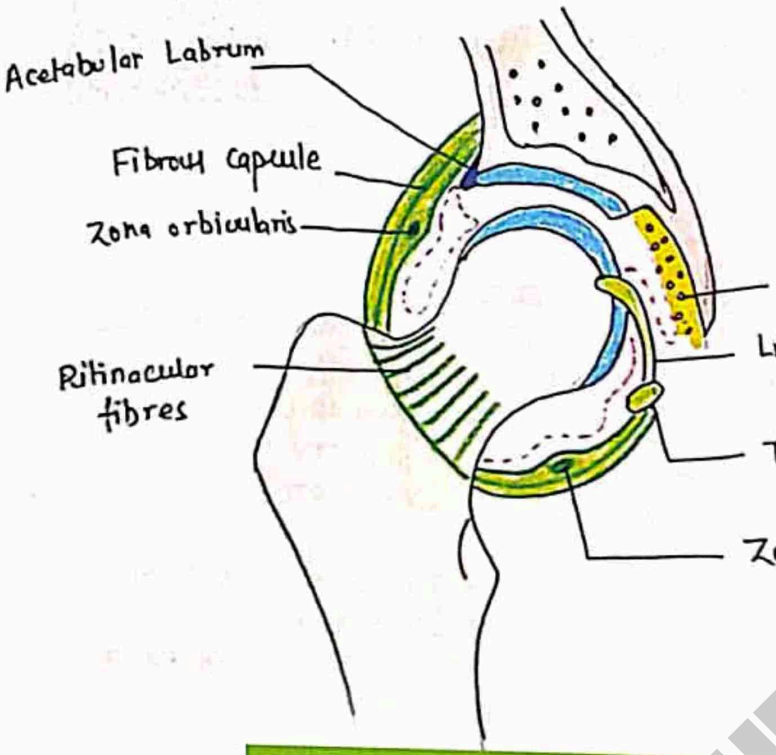
5.LOCKING PRODUCED BY CONTINUED ACTION OF EXTENSOR MUSCLE ->QUADRICEPS FEMORIS

6.UNLOCK IS REVERSE OF LOCKING BY LATERAL ROTATION OF FEMUR

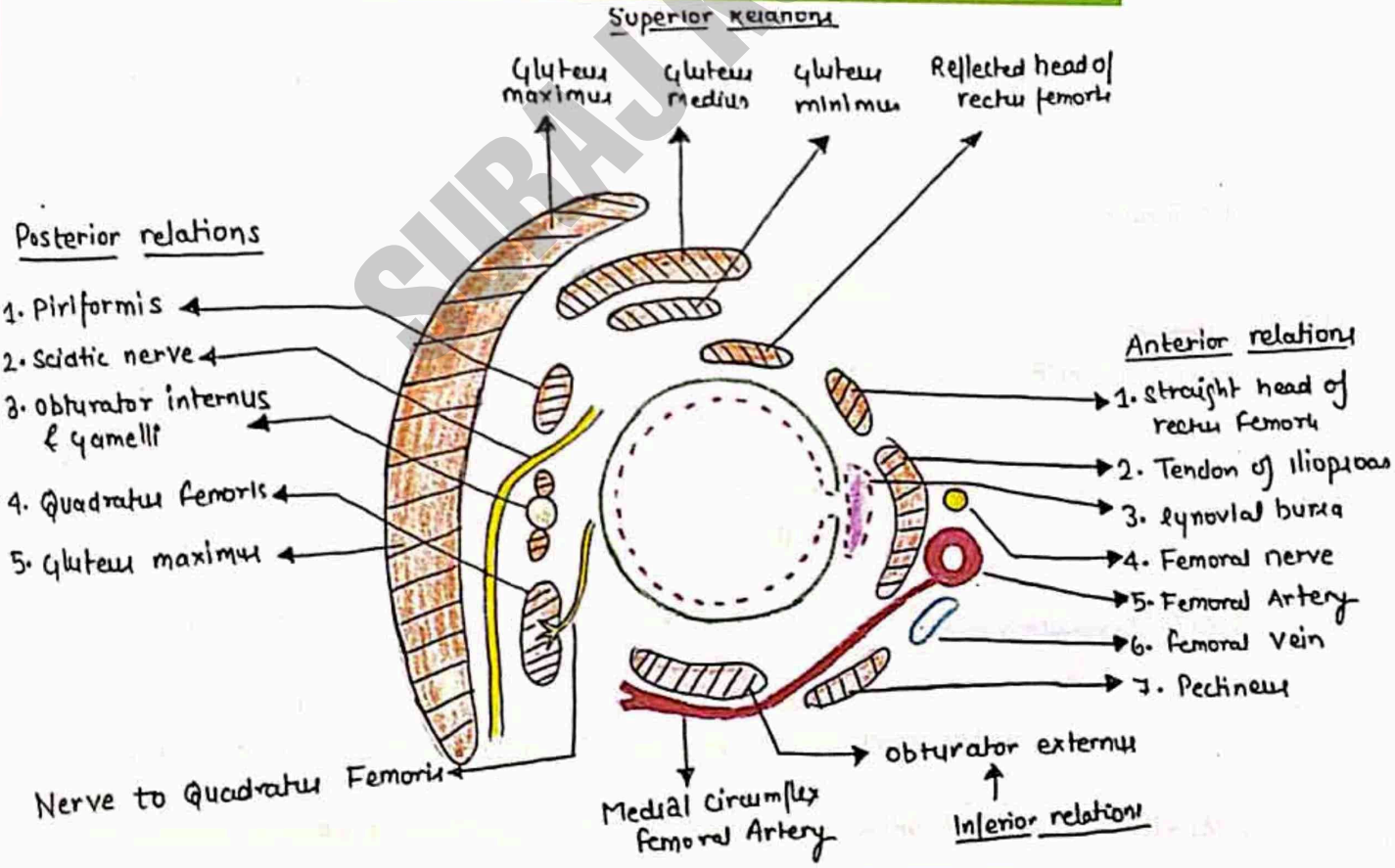
7.UNLOCK CAUSED BY ---->POPLITEUS MUSCLE

SUBHAM KUMAR

∵ Acetabulum of Hip bone



LIGAMENT OF HIP JOINT



RELATION OF HIP JOINT

SURAJ KUMAR

KNEE JOINT

*TYPE :- CONDYLAR SYNOVIAL JOINT
- 2 CONDYLAR JOINT BETWEEN CONDYLE OF FEMUR & TIBIA.
- 1 SADDLE JOINT BETWEEN FEMUR & PATELLA.

*ARTICULAR SURFACE :- CONDYLE OF FEMUR, PATELLA, CONDYLE OF TIBIA

*LIGAMENT :-

1.FIBROUS CAPSULE - THIN AND DEFICIENT ANTERIORLY.
- REPLACE BY QUADRICEP FEMORIS & LIGAMENTUM PATELLAE
- FEMORAL ATTACHMENT - ANT. - DEFICIENT
- POST.- INTERCONDYLAR LINE
- TIBIAL ATTACHMENT - ANT. - TIBIAL TUBEROSITY
- POST. - INTERCONDYLAR RIDGE

2.LIGAMENTUM PATELLAE - THIS IS THE CENTRAL PORTION OF COMMON TENDON OF INSERTION OF QUADRICEPS FEMORIS
-ATTACHMENT-->ABOVE-POSTERIOR SURFACE OF APEX OF PATELLA
BELOW - TIBIAL TUBEROSITY

3.TIBIAL COLLATERAL LIGAMENT - THIS IS A LONG BAND OF GREAT STRENGTH
-ATTACHMENT-->SUPERIOR-MEDIAL EPICONDYLE SHAFT OF FEMUR
-INFERIOR - DEVIDED INTO ANT. AND POST. PART
-ANT.PART-MEDIAL SURFACE OF SHAFT OF TIBIA
-POST.PART-MEDIAL CONDYLE OF TIBIA

4.FIBULAR COLLATERAL - SUPERIOR - ATTACHMENT - LATERAL EPICONDYLE OF FEMUR
-INFERIORLY - HEAD OF FEBULA

5.OBLIQUE POPLITEAL LIGAMENT - EXPANSION FROM TENDON OF SEMIMEMBRANOSUS
-RUN LATERAL WITH POST.SURFACE OF CAPSULE
-ATTACHMENT -->INRECONDYLAR LINE & LATERAL CONDYLE OF FEMUR

6.ARCUATE POPLITEAL LIGAMENT - POSTERIOR EXPANSION FROM SHORT LATERAL LIGAMENT
-EXTEND BACKWARD FROM HEAD OF FIBULA TO POSTERIOR BORDER OF INTERCONDYLAR AREA OF TIBIA.

7.MINISCUS - ARE 2 FIBROCARTILAGINOUS DISC - LATERAL & MEDIAL MINISCUS.

SURAJ KUMAR

8.CRUCIATE LIGAMENT - VERY THICK & STRONG FIBROUS BAND
*ANT.CRUCIATE LIGAMENT - BEGIN FROM ANT.INTRACONDYLAR AREA OF TIBIA TO POSTERIOR PART OF LATERAL CONDYLE OF FEMUR.
*POST.CRUCIATE LIGAMENT - POSTERIOR PART OF INTERCONDYLAR AREA OF TIBIA TO ANT. PART OF MEDIAL CONDYLE OF FEMUR.

9.TRANSVERSE LIGAMENT - CONNECT ANTERIOR END OF LATERAL & MEDIAL MENISCUS

*BLOOD SUPPLY :-

1.5 GENICULAR BRANCH OF POPLITEAL ARTERY
2.DESCENDING GENICULAR BRANCH OF FEMORAL ARTERY
3.DESCENDING BRANCH OF LATERAL CIRCUMFLEX ARTERY
4.2 RECURRENT BRANCH OF ANTERIOR TIBIAL ARTERY
5.CIRCUMFLEX FIBULAR BRANCH OF POSTERIOR TIBIAL ARTERY

*NERVE SUPPLY :-

1.FEMORAL NERVE →BRANCH TO VASTI ESPECIALLY VASTUS MEDIALIS
2.SCIATIC NERVE THROUGH GENICULAR BRANCH OF THE TIBIAL & COMMON PERONEAL NERVES
3.OBTURATOR NERVE--->THROUGH POSTERIOR DIVISION

INVERSION & EVERSION OF FOOT

1.INVERSION - MOVEMENT IN WHICH MEDIAL BORDER OF FOOT IS ELEVATED
- SOLE FACE MEDIALLY.
2.EVERSION - MOVEMENT IN WHICH LATERAL BORDER OF FOOT IS ELEVATED.
- SOLE FACE LATERALLY.

*PERFORM VOLUNTARY ONLY WHEN THE FOOT IS OFF THE GROUND.
*WHEN FOOT IS ON THE GROUND THIS MOVEMENT HELP TO ADJUST THE UNEVEN GROUND.

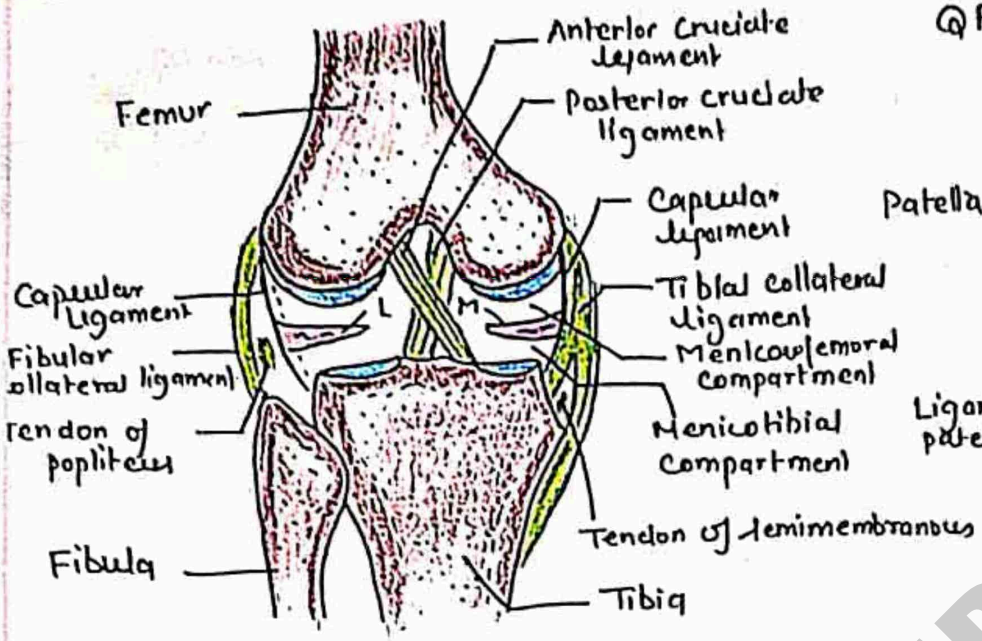
3.INVERSION - ACCOMPANIED BY PLANTAR FLEXION OF FOOT
- ADDUCTION OF FORE FOOT.
4.EVERSION - ACCOMPANIED BY DORSI FLEXION OF FOOT
- ABDUCTION OF FORE FOOT.

DORSIFLEXION & PLANTAR FLEXION

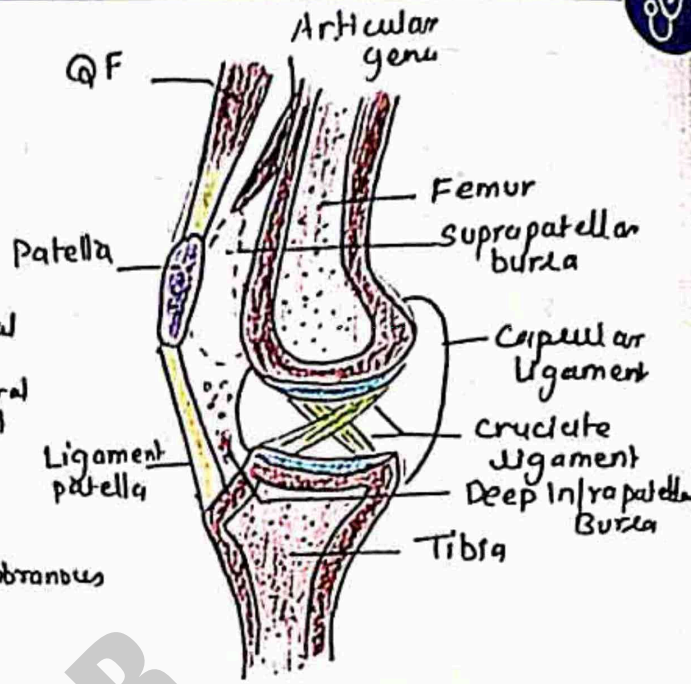
1.DORSIFLEXION → FORE-FOOT IS RAISED
→ANGLE BETWEEN FRONT OF LEG & DORSUM OF THE FOOT IS DIMINISHED

2.PLANTAR FLEXION→ FORE-FOOT IS DEPRESSED
→ANGLE BETWEEN LEG & FOOT IS INCREASED.

SURAJ KUMAR

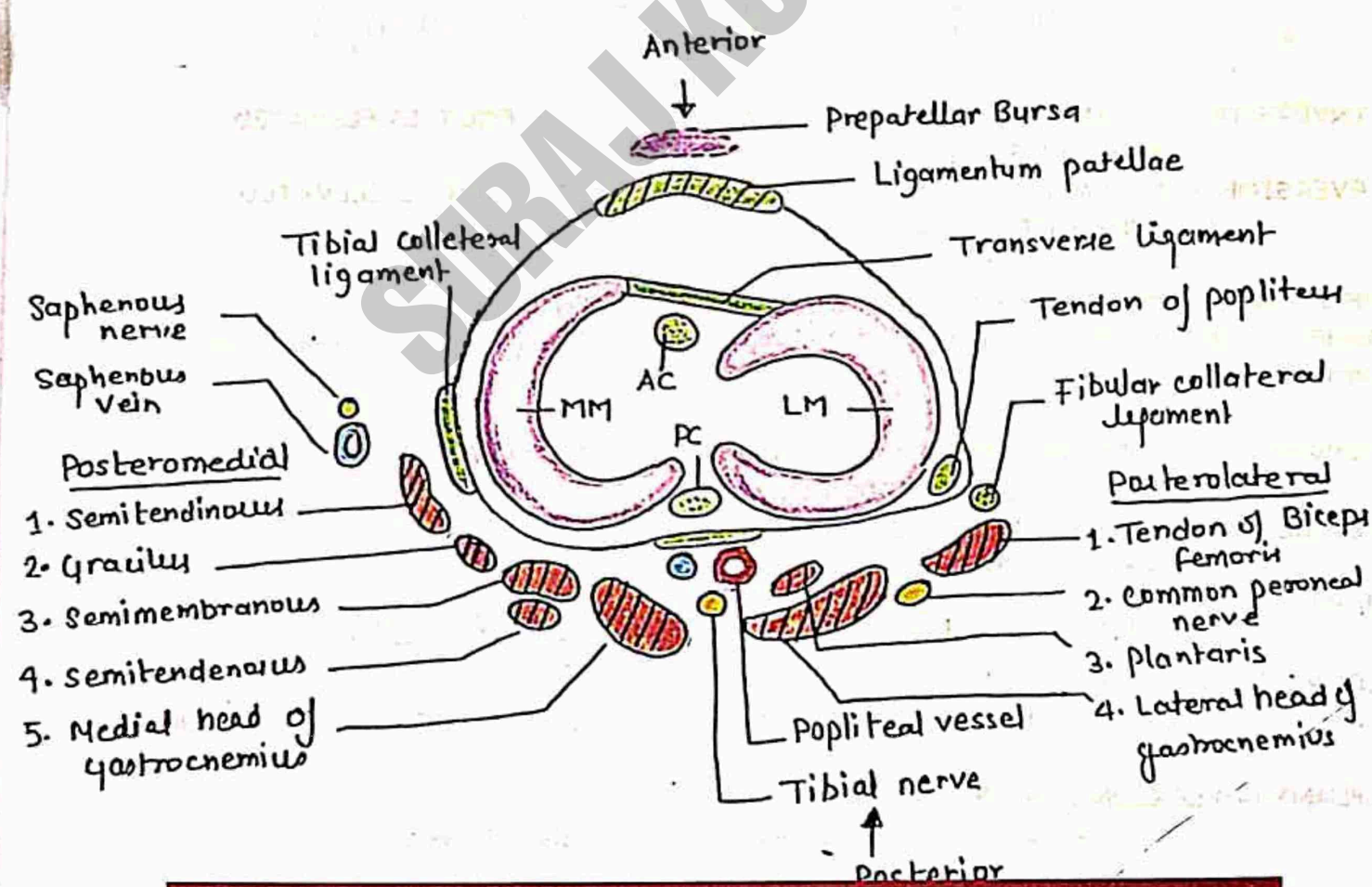


∴ Knee Joint – coronal section



∴ Knee joint – Sagittal section

TIBIAL & FIBULAR COLLATERAL LIGAMENT OF KNEE JOINT



TRANSVERSE SECTION THROUGH KNEE JOINT

SURAJ KUMAR

STUDY TIPS FOR MAXIMUM MARKS

1.HOW TO WRITE IN UNIVERSITY EXAM AND HOW TO DRAW DIAGRAMS ALREADY MENTION IN UPPER LIMB (PAGE NUMBER - 25)

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTIONS - 20 MARKS

1) DESCRIBE THE ROOT VALUE. COURSE, RELATIONS BRANCHES AND DISTRIBUTION AND APPLIED ANATOMY OF OBTURATOR NERVE ?

2) DESCRIBE THE TYPE LIGAMENTS, RELATION, MUSELES PRODUCING THE MOVEMENTS AND APPLIED ANATOMY OF HIP JOINT ?

3) FEMORAL TRIANGLE AND FEMORAL SHEATH & CANAL WITH CLINICAL ANATOMY

4) DESCRIBE VENOUS DRAINAGE OF LOWER LIMBS ITS APPLIED ASPECT ?

5) DESCRIBE THE TYPE LIGAMENTS , MOVEMENTS AND MUSELES PRODUCING THE MOVEMENTS AND APPLIED ANATOMY OF HIP JOINT ?

*SHORTS QUESTION - 5 MARKS

1)FEMORAL ARTERY
2)FEMORAL NERVE
3)ADDUCTOR CANAL
4)POPLITEAL FOOSA
5)ARCHES OF FOOT
6)HAMSTRING MUSCLE
7)LOCKING & UNLOCKING OF KNEE
8) FLEXOR RETINACULUM OF FOOT

SURAJ KUMAR

LOWER LIMB ANATOMY IS COMPLETED !!

STOP DOUBTING YOURSELF , WORK HARD AND MAKE IT HAPPEN

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

THE COPYRIGHT ACT,1957

ALL MATERIAL INCLUDING THEORY, DIAGRAM IS COPYRIGHT REGISTERED .
ANY SCAN AND PHOTOCOPY OF THIS MATERIAL IS NOT PERMITTED BY OWNER

MCQ LOWER LIMB

1) Which of the following does not form boundary of femoral triangle :
a Inguinal ligament
b medial border of adductor longus
c lateral border of adductor longus
d medial border of sartorius

2) All of the following are content of femoral triangle except :
a femoral artery
b femoral nerve
c deep inguinal lymph nodes
d great saphenous vein

3) Femoral nerve supplies all of the following except :
a adductor longus
b pectineus
c iliacus
d sartorius

4) All of the following have dual nerve supply except :
a Pectineus
b biceps femoris
c quadriceps femoris
d adductor magnus

5) The following muscle is flexor of both hip & knee joint:
a rectus femoris
b biceps femoris
c gracilis
d sartorius

6) Anterior division of femoral nerve supplies the following muscle :
a iliacus
b pectineus
c rectus femoris
d sartorius

7) Anterior division of obturator nerve supplies all the following except :
a obturator externus
b pectineus
c gracilis
d adductor longus

8) Medial boundary of femoral ring is formed by:
a Inguinal ligament
b pectineus muscle
c lacunar ligament
d femoral vein

9) Which muscle of the thigh is preffered to give intramuscular injection in children :
a sartorius
b vastus lateralis
c vastus medialis
d rectus femoris

10) All of the following muscle are inserted on greater trochanter EXCEPT :
a obturator externus
b piriformis
c gluteus medius
d gluteus maximus

11) All the following are supplied by posterior division of obturator nerve EXCEPT :
a adductor brevis
b adductor magnus
c obturator internus
d obturator externus

12) Which structure lies midway between anterior superio iliac spine and pubic symphysis :
a femoral artery
b femoral vein
c femoral nerve
d deep inguinal ring

13) Which of the following ligament prevents hypertensio hip:
a Inguinal ligament
b Iliofemoral ligament
c pubofemoral ligament
d Ischiofemoral ligament

14) All the following are lateral rotators of thigh except :
a gluteus medius
b obturator internus
c sartorius
d quadratus femoris

15) The following are part of hamstrings EXCEPT :
a Semitendinosus
b Semimembranosus
c biceps femoris
d tensor fascia lata

16) The superficial inguinal lymph nodes drain all the following except :
a lower part of ant Abdominal wall
b thigh
c scrotum
d glans penis

17) Saphenous nerve is a branch of:
a femoral nerve
b obturator nerve
c tibial nerve
d common peroneal nerve

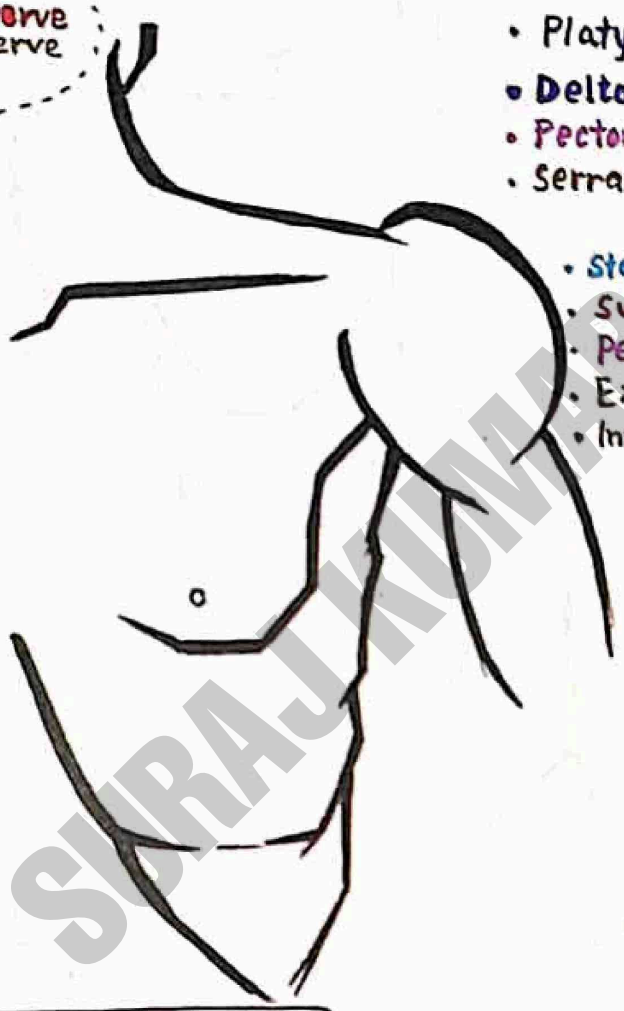
18) The deep inguinal lymph node of Rosenmuller drains the:
a Testis
b scrotum
c glans penis
d ductus deferens

1) C 2)D 3)A 4)B 5)D 6)D 7)A 8)C 9)B 10)D 11)C 12)A 13)B 14)A 15)D 16)D 17)A 18)C

SOHAIL KUMAR BBS
SURAJ KUMAR

THORAX MUSCLES

- Facial Nerve
- Axillary Nerve
- Lateral & Medial Pectoral Nerve
- Long thoracic Nerve
- Accessory Nerve
- Subclavian Nerve
- Intercostal Nerve



ANTERIOR AXIO-APPENDICULAR

SUPERFICIAL

- Platysma
- Deltoid
- Pectoralis Major
- Serratus Anterior

DEEP

- Sternocleido mastoid
- Subclavius
- Pectoralis Minor
- External intercostal Muscle
- Internal intercostal Muscle

POSTERIOR AXIO - APPENDICULAR

SUPERFICIAL

- Deltoid
- Trapezius
- Rhomboid Major
- Rhomboid Minor
- Latissimus dorsi

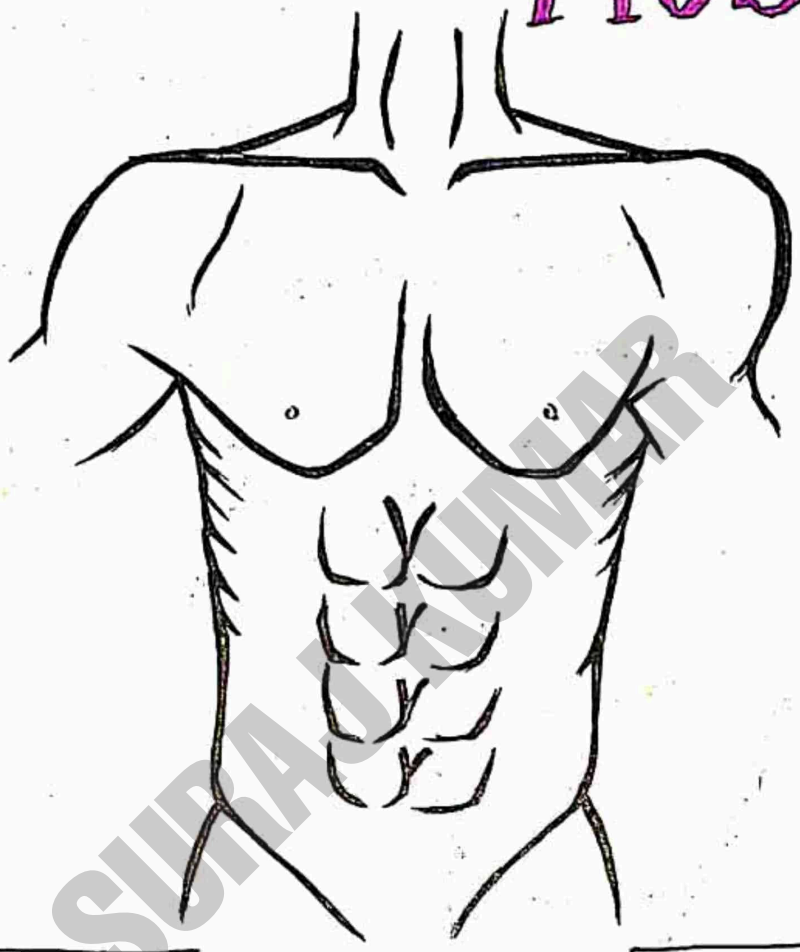
DEEP

- Supraspinatus
- Infraspinatus
- Teres minor
- Teres major
- Lerator scapulae

- Dorsal scapular Nerve
- Thoracodorsal Nerve
- Suprascapular Nerve
- Lower subscapular Nerve

SURAJ KUMAR

ABDOMINAL MUSCLES



SURAJ KUMAR

ANTERIOR WALL

- External Oblique
- Internal Oblique
- Trans-verse Abdominis
- Rectus Abdominis

- Anterior primary rami of lower 6 thoracic nerves (t_7-12)
- Lower 5 intercostal & subcostal nerves (t_7-12)
- Anterior primary rami of lower 6 thoracic nerves (t_7-12) & 1st lumbrical (L_1)

POSTERIOR WALL

- Quadratus lumborum
- Psoas minor
- Psoas major
- Iliacus

- Ventral rami of T_{12} & L_{1-4} lumbar spinal nerves
- Ventral rami of L_{1-4} spinal nerves
- Branch of L_1 spinal nerve
- Femoral nerve (L_{2-4})

PLEURA

*PLEURA IS A SEROUS MEMBRANE - LINED BY MESOTHELIUM.
*2 PLEURAL SAC, 1 ON EITHER SIDE OF THE MEDIASTINUM.
*OUTER LAYER IS PARIETAL PLEURA.
*INNER LAYER IS VISCERAL OR PULMONARY PLEURA.
*2 PLEURA CONTINUE WITH EACH OTHER AROUND HILIUM OF LUNGS BETWEEN PLEURAL CAVITY.

COMPARISON OF VISCERAL & PARIETAL PLEURA

DEVELOPMENT	SPLANOPLEURIC MESODERM	SOMATOPLEURIC MESODERM
POSITION	LINES SURFACE OF LUNG INCLUDING TISSUES	LINE THORACIC WALL, MEDIASTINUM & DIAPHRAM.
NERVE SUPPLY	SYMPATHETIC NERVES FROM T2-T5 GANGLIA, PARASYMPATHETIC FROM VAGUS NERVE	THORACIC NERVES & PHRENIC NERVES
SENSITIVITY	INSENSITIVE TO PAIN	SENSITIVE TO PAIN
BLOOD SUPPLY	BRONCHIAL VESSELS	INTERCOASTAL & PERICARDIACOPHRENIC VESSELS
LYMPHATIC DRAINAGE	TRACHEOBRONCHIAL LYMPH NODE	INTERCOASTAL LYMPH NODE

VISCERAL PLEURA :- COVER SURFACE & FISSURES OF LUNGS EXCEPT HILUM WHERE IT CONTINUE WITH PARIETAL PLEURA.
-FIRMLY ADHERENT TO THE LUNGS CAN NOT BE SEPARATED.

PARIETAL PLEURA :- THICKER THAN VISERAL PLEURA SUB DIVIDED INTO 4 PARTS - 1.COSTAL, 2.DIAPHRAGMATIC, 3.MEDIASTINAL, 4.CERVICAL

SURFACE MARKING OF PLEURA :- (PARIETAL)

1.CERVICAL PLEURA - REPRESENT BY CURVED LINE. - ABOUT 2.5CM ABOVE CLAVICLE.

2.ANTERIOR MARGIN OF COSTOMEDIASTINAL -
- RIGHT SIDE - EXTEND - STERNOCLAVICULAR JOINT TO STERNUM ANGLE.
- VERTICALLY DOWNWARD TO XIPHISTERNAL POINT
- LEFT SIDE - FOLLOW THE SAME COURSE UP TO 4TH COSTOCARTILAGE LEVEL.
- THEN ARCH DOWN TO 6TH COSTOCARTILAGE.

3.POSTERIOR MARGIN - PASS 2CM LATERAL TO 12 THORACIC SPINE TO 2CM LATERAL TO 4TH CERVICLE SPINE.

4.INFERIOR MARGIN - CROSS 8TH RIB IN MID CLAVICULAR LINE - 10TH RIB IN MID AXILLARY LINE

SURAJ KUMAR

SURFACE MAKING OF PLEURA :-(VISCERAL)

*APEX - 2.5CM ABOVE MEDIAL 2/3 CLAVICLE.

1.ANTERIOR BORDE OF VISCERAL PLEURA -

-RIGHT SIDE :- CORRESPOUND CLOSELY WITH COASTO MEDIASTINAL LINE.

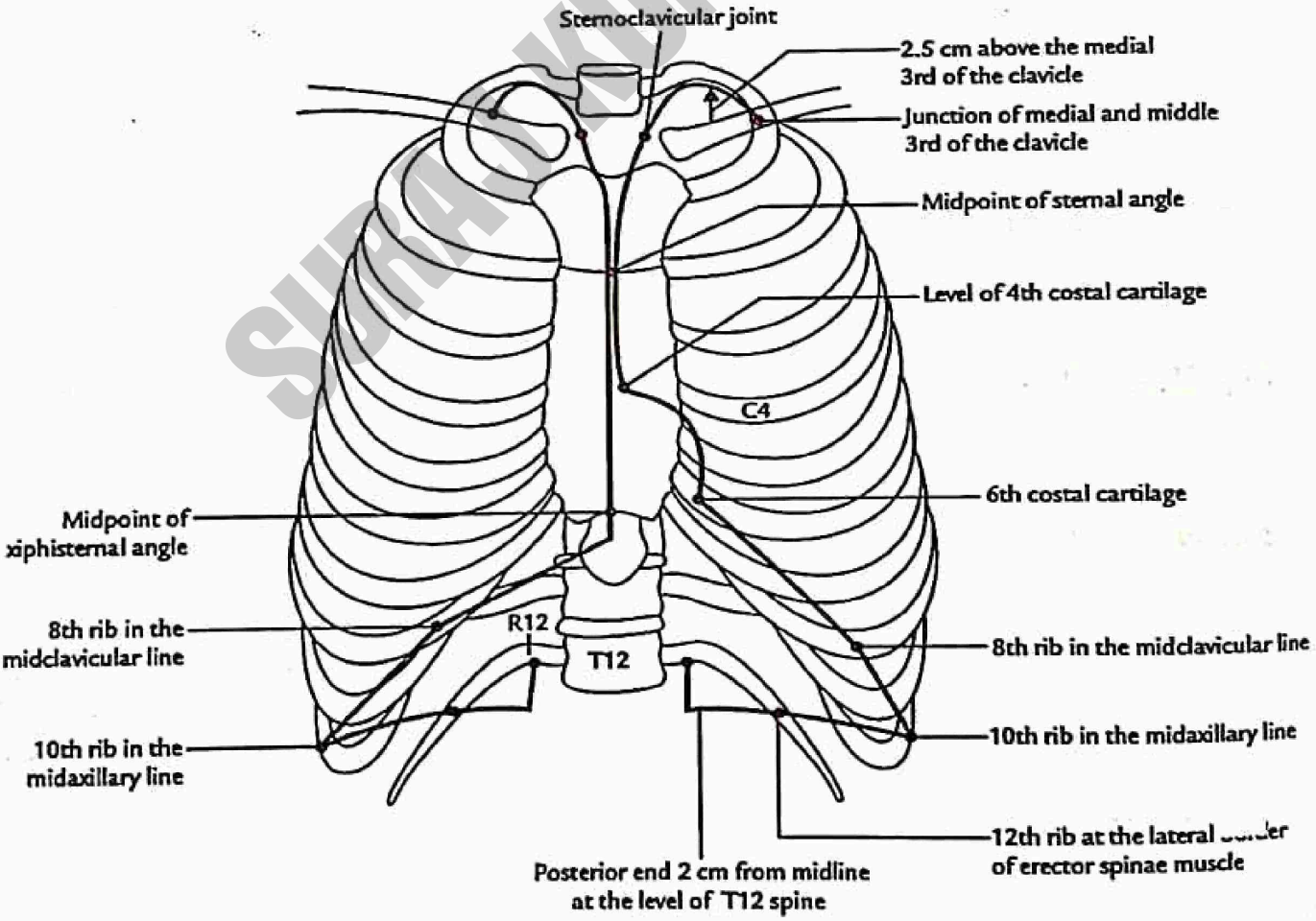
-OBTAIN BY JOINING - STERNOCLAVICULO JOINT & STERNAL ANGLE

-MEDIAL PLANE JUST ABOVE XIPHISTERNAL JOINT.

-LEFT SIDE :- CORRESPOUND TO COASTOMEDIASTINAL LINE UP TO LEVEL OF 4TH COSTOCARTILAGE

- LOWER PART REPRESENT A CARDIAC NOTCH
- FROM 4TH COASTOCARTILAGE IT PASS - LATERAL 3.5CM FROM STERNAL MARGIN
- THEN CURVE DOWN MEDIALLY TO REACH 6TH COASTOCATILAGE
- AREA OF CARDIAC NOTCH IS DULL ON PERCUSSION CALLED AREA OF SUPERFICIAL CARDIAC DULLNESS.

2.LOWER BORDER- CROSS 6TH RIB IN MIDCLAVICULAR LINE, 8TH RIB IN MID AXILLARY LINE.



SURFACE MARKING OF PARIETAL PLEURA

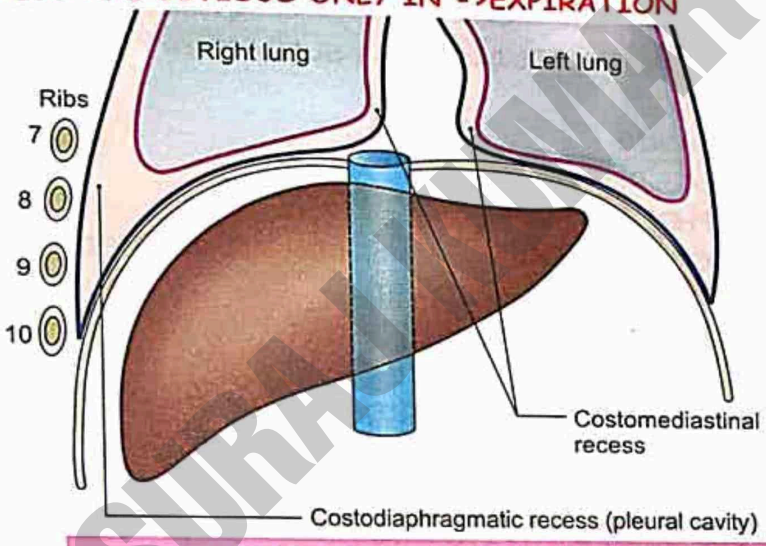
SURAJ KUMAR

RECESSES OF PLEURA

*THERE ARE 2 RECESSES OF PARIETAL PLEURA ACT →RESERVE SPACE FOR LUNGS TO EXPAND DURING INSPIRATION

1) COSTOMEDIASTINAL RECESS-
- *LIE ANTERIORLY ,BEHIND STERNUM AND COSTAL CARTILAGE
- *BETWEEN COSTAL AMD MEDIASTINAL PLEURA
- *PARTICULARLY IN RELATION →CARDIAC NOTCH OF LEFT LUNG
- *FILLED UP BY -→ANTERIOR MARGIN OF LUNGS EVEN DURING QUIET BREATHING
- *ONLY OBVIOUS-→REGION OF CARDIAC NOTCH OF LUNG

2) COSTODIAPHRAGMATIC RRECESS-
- *LIE INFERIORLY BETWEEN COSTAL AND DIAPHRAGMATIC PLEURA
- *VERTICALLY 5CM AND EXTEND FROM 8TH →TO 10TH RIB IN MIDAXILLARY LINE
- *FILLED BY -→LUNGS EXPEND INTO RECESSES DURING INSPIRATION
- *SO THIS RECESS ARE OBVIOUS ONLY IN ->EXPIRATION



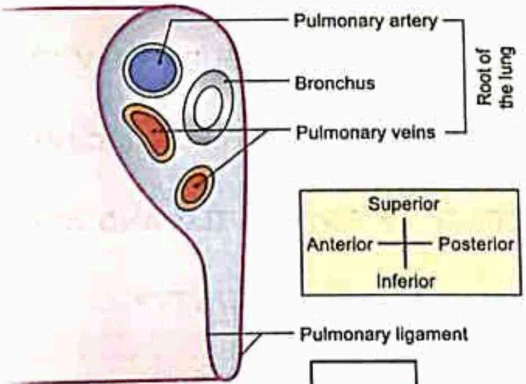
SCAN QR WITH QRBOT APP
TO DIRECTLY OPEN VIDEO LINK

PULMONARY LIGAMENT

1.PARITEAL PLEURA- SURROUND ROOT OF LUNGS-→EXTEND DOWNWARD BEYOUND ROOT AS FOLD CALLED PULMONARY LIGAMENT

2.FOLD CATAIN -→THIN LAYER OF LOOSE AREOLAR TISSUE WITH FEW LYMPHATICS

3.PROVIDE - DEAD SPACE--→INTO PULMONARY VEIN EXPAND DURING -→INCREASE IN VENOUS RETURN IN EXCERCISE



SURAJ KUMAR

FISSURES & LOBES OF LUNGS

*RIGHT LUNG-→ DIVIDED INTO 3 LOBES - UPPER, MIDDLE, & LOWER BY OBLIQUE FISSURE AND HORIONTAL FISSURE

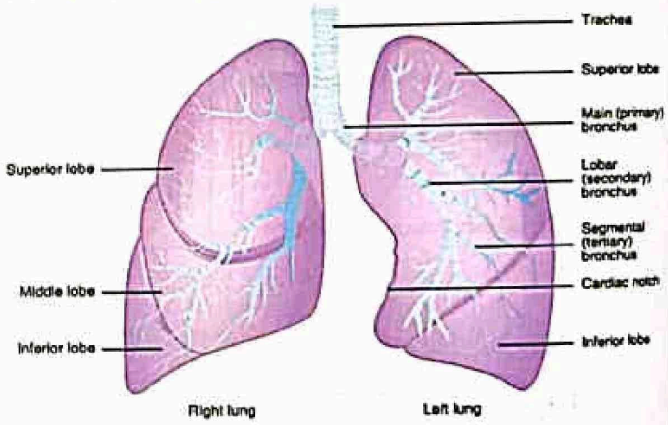
*LEFT LUNG-→ DIVIDED INTO 2 LOBES - UPPER & LOWER LOBE BY OBLIQUE FISSURE

*OBLIQUE FISSSURE - CUT INTO WHOLE THICKNESS OF LUNGS, EXPECT AT HILUM
-DUE TO OBLIQUE PLANE OF FISSURE -→LOWER LOBE IS MORE POSTERIOR
-→UPPER & MIDDLE LOBE MORE ANTERIOR

*HORIZONTAL FISSURE - PASSES FROM ANTERIOR BORDER UP TO OBLIQUE FISSURE
-SEPARATE WEDGE SHAPED MIDDLE LOBE FROM UPPER LOBE
-RUN HORIZONTALLY AT LEVEL OF FOURTH COSTAL CARTILAGE
-TONGUE SHAPE PROJECTION OF LEFT LING BELOW CARDIAC NOTCH CALLED LINGULA

IMPRESSION ON MEDIASTINAL SURFACE :-

1.RIGHT LUNGS - OESOPHAGUS
- TRACHEA - RIGHT VAGUS NERVE
- RIGHT PHRENIC NERVE
- SUPERIOR VENA CAVA_-→ ARCH OF AZYGOS VEIN
- RIGHT BRACHIOCEPHALIC VEIN
- INFERIOR VENA CANA
- HIULM
- IMPRESSION OF RIGHT ATRIUM



2.LEFT LUNGS - OESPHAGUS
- THORACIC DUCT
-TRACHEA
- LEFT PHRENIC NERVE
- LEFT VAGUS MERVE
- PULMONARY TRUNK
- HILUM
- IMPRESSION OF LEFT VENTICLE

CLINICALANATOMY

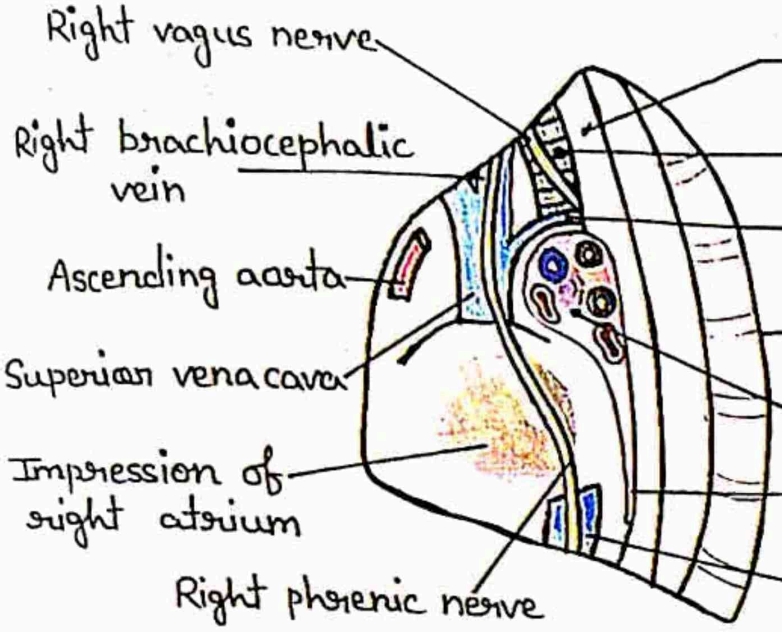
1.PNEUMOTHORAX- PRESENCE OF AIR IN THE PLEURAL CAVITY

2.HAEMOTHORAX- PRESENCE OF BLOOD IN THE PLEURAL CAVITY

3.HYDROPNEUMOTHORAX- PRESENCE OF BOTH FUILD AND AIR CAVITY IN PLEURAL CAVITY

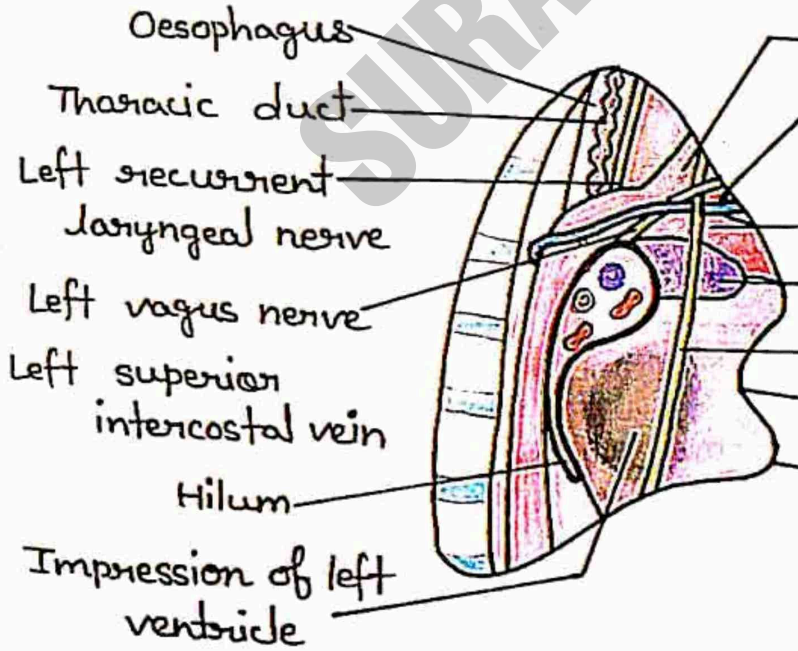
4.EMPYEMA- PRESENCE OF PUS IN THE PLEURAL CAVITY

SURAJ KUMAR



IMPRESSION IN MEDIASTINUM SURFACE OF RIGHT LUNGS

Superior
Anterior — Posterior
Inferior



IMPRESSION IN MEDIASTINUM SURFACE OF LEFT LUNGS

Superior
Posterior — Anterior
Inferior

SURAJ KUMAR

ROOTS OF LUNGS

*SHORT, BROAD PEDICLE WHICH CONNECT MEDIAL SURFACE OF LUNGS TO MEDIASTINUM

*FROM BY STRUCTURE WHICH EITHER ENTER OR COME OUT OF LUNGS AT HILUM.

*LIE OPPOSITE THE BODY OF 5TH 6TH & 7TH THOARCIC VERTEBRAE

*CONTENTS - PRINCIPAL BRONCHUS ON LEFT SIDE
- EPARTERIAL & HYPARTERIAL BRONCHI ON RIGHT SIDE
- 1 PULMONARY ARTERY
- 2 PULOMONARY VEIN - SUPERIOR & INFERIOR
- BRONCHIAL ATERIES
- BRONCHIAL VEINS
- ANTERIOR & POSTERIOR PULMONARY PLEXUSES OF NERVE
- LYMPHATICS OF LUNGS

*ARRANGEMENT OF STRUCTURE IN ROOT :-

* ANT. TO POST. - SIMILAR IN BOTH SIDES - SUPERIOR PULMONARY VEIN
- PULMONARY ARTERY
- BRONCHUS

*ABOVE DOWNWARD - DIFFERENT IN 2 SIDES

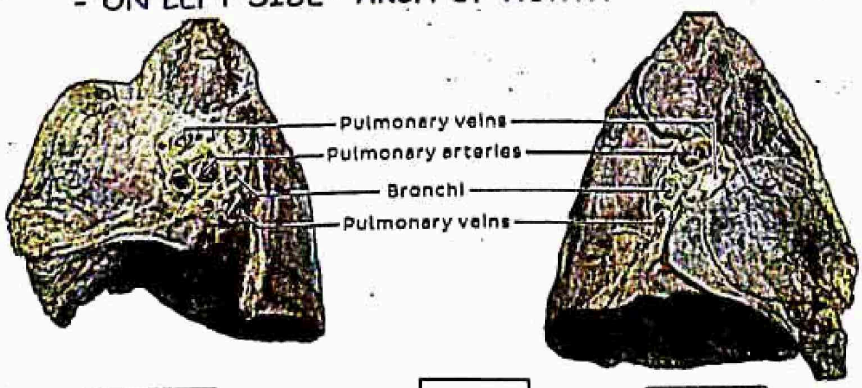
1) RIGHT SIDE - EPARTERIAL BRONCHUS
-PULMONARY ARTERY
-HYPARTERIAL BRONCHUS
-INFERIOR PULMONARY VEIN

2) LEFT SIDE - PULMONARY ARTERY
- BRONCHUS
- INFERIOR PULMONARY VEIN

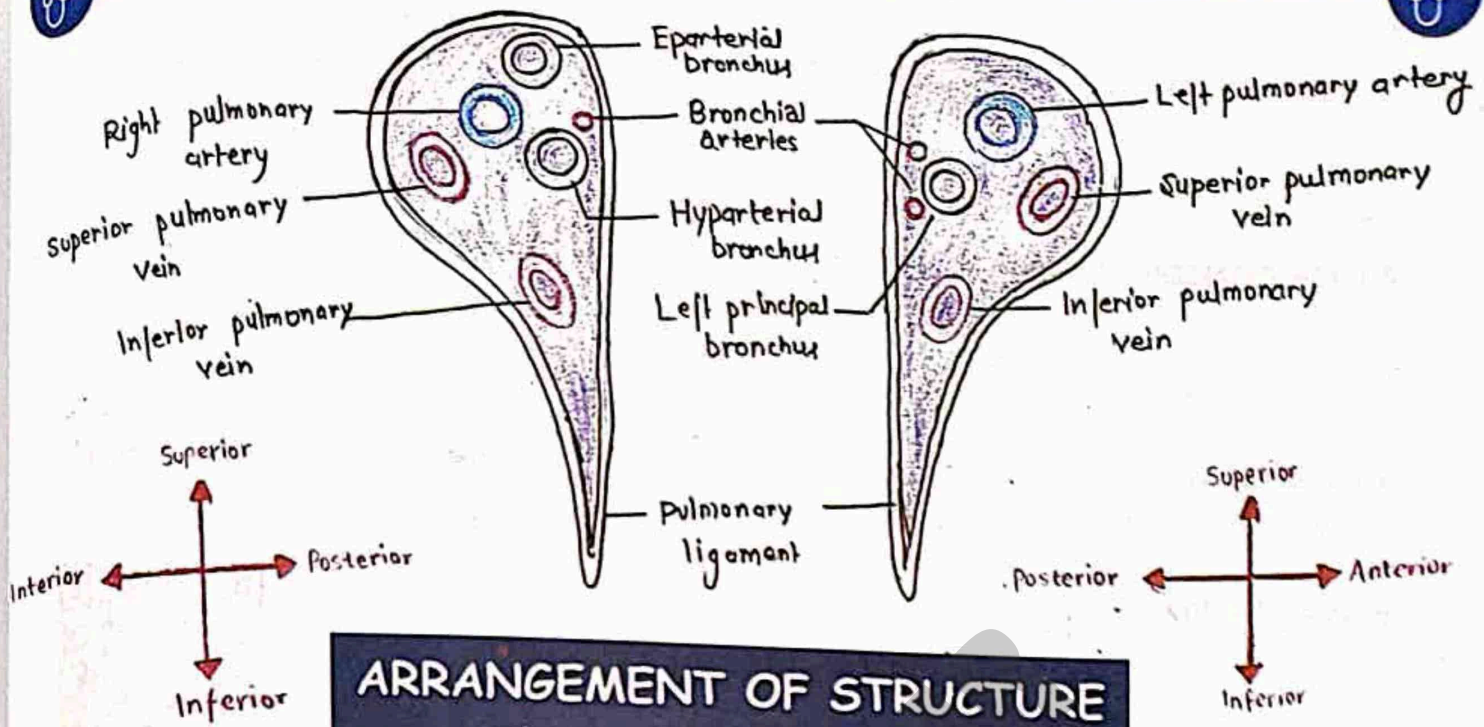
*RELATONS :- ANTERIOR - COMMON ON 2 SIDES - PHRENIC NERVE
- PERICARDIACOPHRENIC VESSELS
- ON RIGHT SIDE - SUPERIOR VENA CAVA

POSTERIOR - COMMON ON 2 SIDES - VAGUS NERVE
- ON LEFT SIDE - DESCENDING THORACIC AORTA

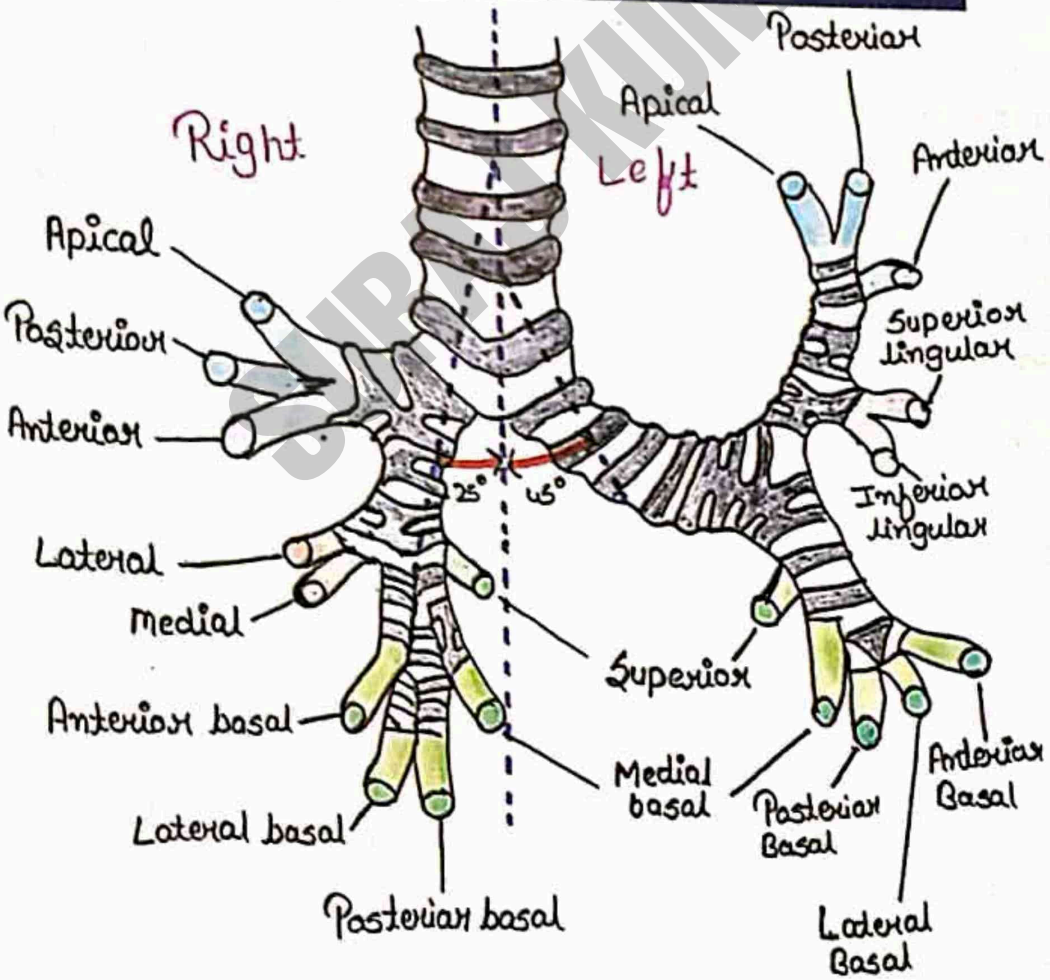
SUPERIOR - ON RIGHT SIDE - TERMINAL PART OF AZYGOS VEIN
- ON LEFT SIDE- ARCH OF AORTA



SURAJ KUMAR



ARRANGEMENT OF STRUCTURE IN ROOT OF LUNGS



BRONCHOPULMONARY SEGMENTS

BRONCHIAL TREE & BRONCHOPULMONARY SEGMENTS

1.TRACHEA DIVIDES INTO 2 PRIMARY PRINCIPAL BRONCHI -→RIGHT & LFT BRONCHUS

2.EACH PRINCIPAL BRONCHUS ENTER LUNGS -→THROUGH HILIUM

3.DIVIDES INTO SECONDARY LOBAR BRONCH-→ONE FOR EACH LOEBS -→ 3 LOBAR BRONCHI - RIGHT SIDE 2 LOBAR BRONCHI - LEFT SIDE

4.EACH LOBAR BRONCHUS DIVIDE INTO TERTIARY -→ 1 FOR EACH BRONCHOPULMONARY SEGMENTS --→ WHICH ARE 10 ON RIGHT AND LEFT SIDE

5.SMALLER BRANCHES ARE CALLED RESPIRATORY BRONCHIOLES

BRONCHOPULMONARY SEGMENTS -

*10 SEGEMNTS IN RIGHT SIDE AND 10 ON LEFT SIDE

***FEATURES :-**

1. WELL DEFINED ANATOMIC , FUNCTIONAL AND SURGICAL SECTOR OF LUNGS
2. EACH ONE IS AERATED BY TERTIARY BRONCHUS
3. EACH SEGMENTS IS PYRAMIDAL IN SPACE -→APEX -→TOWARD ROOT OF LUNGS
4. EACH SEGMENTS HAS SEGMENTAL BRONCHUS & ARTERY AND AUTONOMIC NERVE
5. SEGMENTAL VENULES LIE-→IN CONNECTIVE TISSUE BETWEEN ADJACENT PULMONARY UNITS
6. DURING SEGMENTAL RESCENTION, THE SURGEON WORK ALONG SEGMENTAL VEIN TO ISOLATE A PARTICULAR SEGMEMT

***RELATION TO PULMONARY ARTERY** - BRANCHES OF PULMONARY ARTERY ACCOMPANY THE BRONCHI

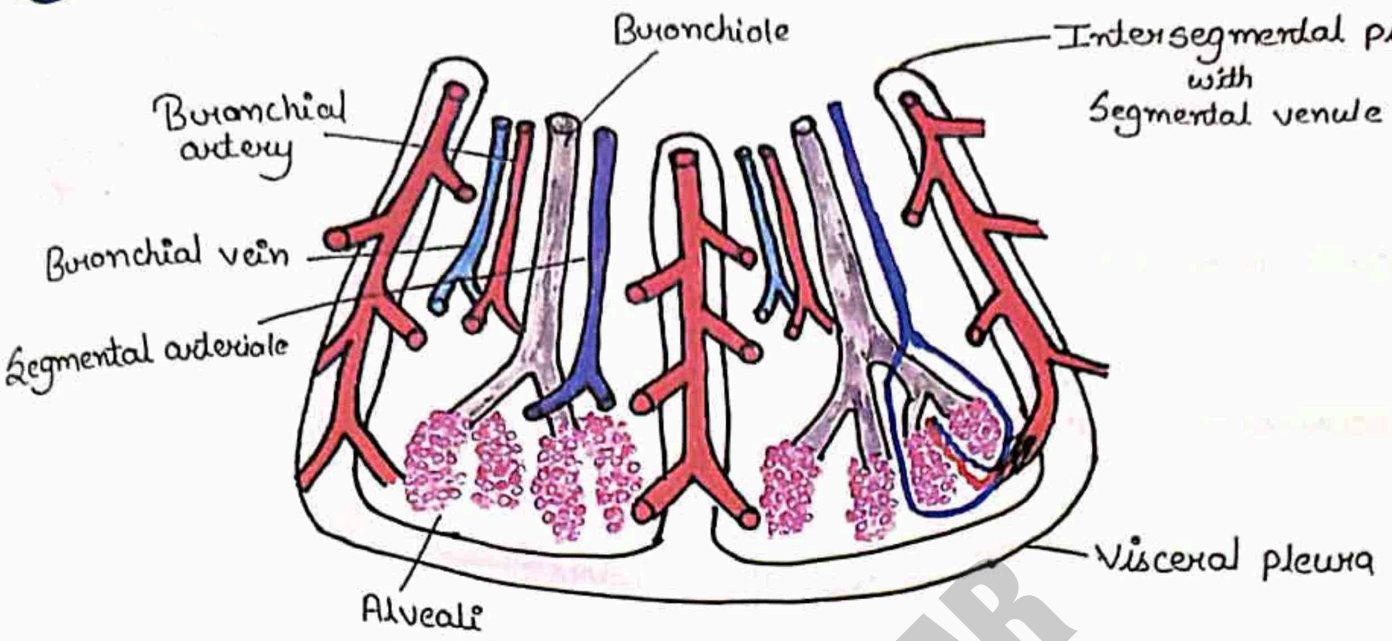
***RELATION TO PULMONARY VEIN** - PULMONARY VEIN DO NOT ACCOMPANY THE BRONCHI NOR PULMONARY ARTERIES
-RUN IN THE INTERSEGMENTAL PLANES

CLINICAL ANATOMY

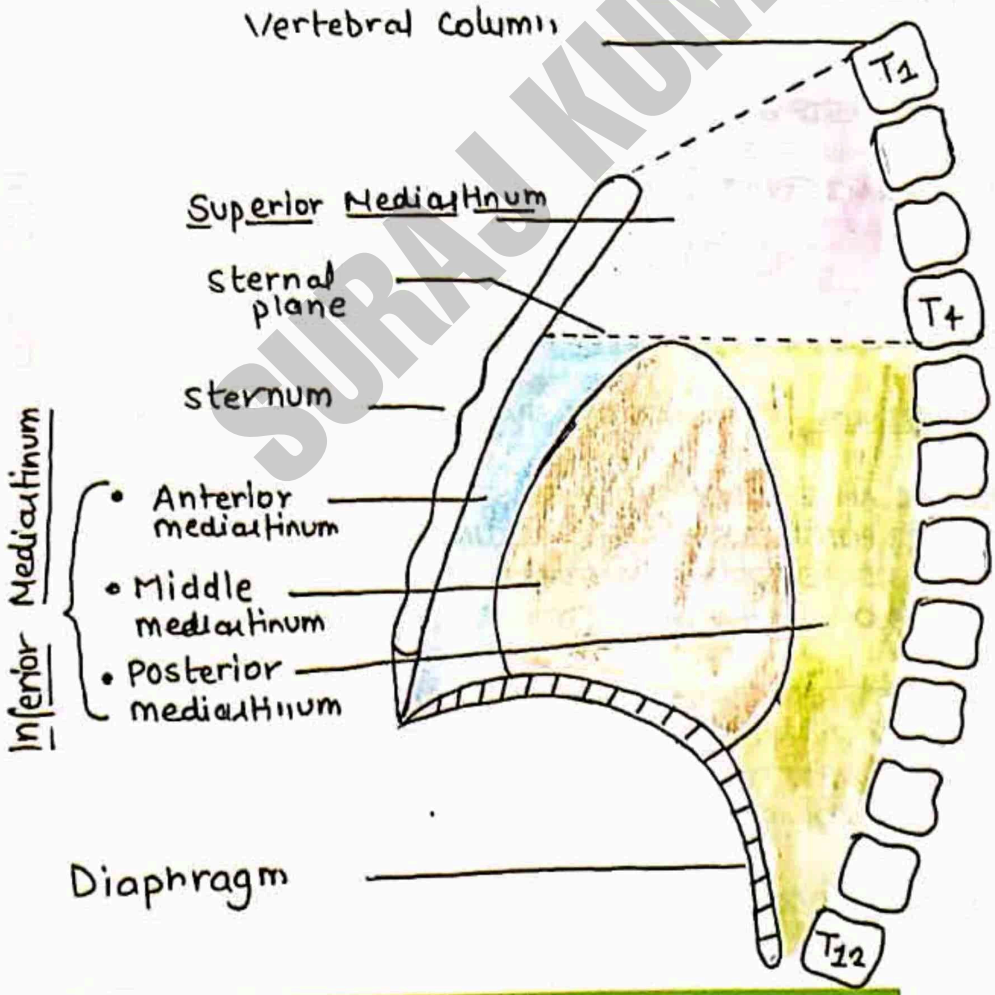
1.USUALLY, THE INFECTION OF A BRONCHOPULMONARY SEGMENT REMAINS RESTRICTED TO IT, ALTHOUGH TUBERCULOSIS & BRONCHOGENIC CARCINOMA MAY SPREAD FROM ONE SEGMENT TO ANOTHER.

2.KNOWLEDGE OF THE DETAILED ANATOMYOF THE BRONCHIAL TREE HELPS CONSIDERABLY IN :- SEGMENTARY RESECTION.

SUBAL KUMAR



BRONCHOPULMONARY SEGMENTS



MEDIASTINUM

SURAJ KUMAR

SUPERIOR MEDIASTINUM

*MEDIASTINUM - MIDDLE SPACE IN THORACIC CAVITY BETWEEN LUNGS
*MOST IMP CONTENT IS HEART ENCLOSE IN PERCARDIUM

*DIVISION :-SUPERIOR & INFERIOR MEDIASTINUM

*SUPERIOR MEDIASTINUM - SEPARATED FROM INFERIORLY BY IMAGINARY PLANE PASSING THROUGH STERNAL ANGLE ANTERIORLY & THORACIC VERTEBRAE POSTERIORLY

*BOUNDARIES :
- ANTERIORLY - MANUBRIUM STERNI
- POSTERIORLY - UPPER THORACIC 4 VERTEBRAE
- SUPERIORLY - PLANE OF THORACIC INLET
- INFERIORLY - AN IMAGINARY LINE
- ON EACH SIDE - MEDIASTINAL PLEURA

*CONTENTS :-
1. TRACHEA & OESOPHAGUS
2. MUSCLE - STERNOHYOID & STERNOTHYROID
3. ARTERIES - ARCH OF AORTA
   - BRACHIOCEPHALIC ARTERY
   - LEFT COMMON CAROTID ARTERY & SUBCLAVIAN ARTERY
4. VEIN - LETF & RIGHT BRACHIOCEPHALIC VEIN
5. NERVE - VAGUS , PHRENIC & CARDIAC NERVE OF BOTH SIDE
6. THORACIC DUCT & LYMPH NODES

INFERIOR MEDIASTINUM

*DIVIDED INTO - ANTERIOR, MIDDLE , POSTERIOR MEDIASTINUM

*ANTERIOR MEDIASTINUM - VERY NARROW SPACE INFRONT OF PERICARDIUM

*BOUNDARIES :-
1. ANTERIORLY - BODY OF STERNUM
2. POSTERIORLY - PERCARDIUM
3. SUPERIORLY - IMAGINARY LINE
4. ON EACH SIDE - MEDIASTINAL PLEURA

*CONTENTS :-
1. STERNOPERICARDIAL LIGAMENT
2. LYMPH NODE
3. LOWEST PART OF THYMUS
4. AREOLAR TISSUE

*MIDDLE MEDIASTINUM :- OCCUPIED BY PERICARDIUM & ITS CONTENTS.

SUNIL KUMAR

*BOUNDARIES :- 1.ANTERIORLY - STERNOPERICARDIAL LIGAMENT
2.POSTERIORLY - OESOPHAGUS
-DESCENDING THORACIC AORTA.
3.ON EACH SIDE - MEDIASTINAL PLEURA

*CONTENT :- 1.HEART ENCLOSED IN PERICARDIUM
2.ASCENDING AORTA
3.PULMONARY TRUNK
4.2 PULMONARY ARTERIES
5.VEINS - LOWER PART OF SUPERIOR VENACAVA
- TERMINAL PART OF AZYGUS VEIN
- RIGHT AND LEFT PULMONARY VEIN
6.NERVES - PHRENIC
- DEEP CARDIAC PLEXUS
7.LYMPH NODES
8.BIFURCATION OF TRACHEA

*POSTERIOR MEDIASTINUM :-

*BOUNDARIES :- ANTERIORLY - PERICARDIUM
- BIFURCATION OF TRACHEA
- PULMONARY VESSELS
- POSTERIOR PART OF THE UPPER SURFACE O THE DIAPHRAM.
-POSTERIORLY - LOWER 8 THORACIC VERTEBRAE & INTERVENING DISCS.
-ON EACH SIDE - MEDIASTINAL PLEURA.

*CONTENT :- 1.OESOPHAGUS
2.ARTERIES - DESCENDING THORACIC AORTA & ITS BRANCHES
3.VEINS - 1.AZYGOS VEIN
2.HEMIAZYGOS VEIN
3.ACCESSORY HEMIAZYGOS VEIN
4.NERVES - 1.VAGI
2.SPLANCHNIC NERVES (GREATER, LESSER)
3.8 THORACIC GANGLIA OF THE SYMPATHETIC CHAIN
5.LYMPH NODE

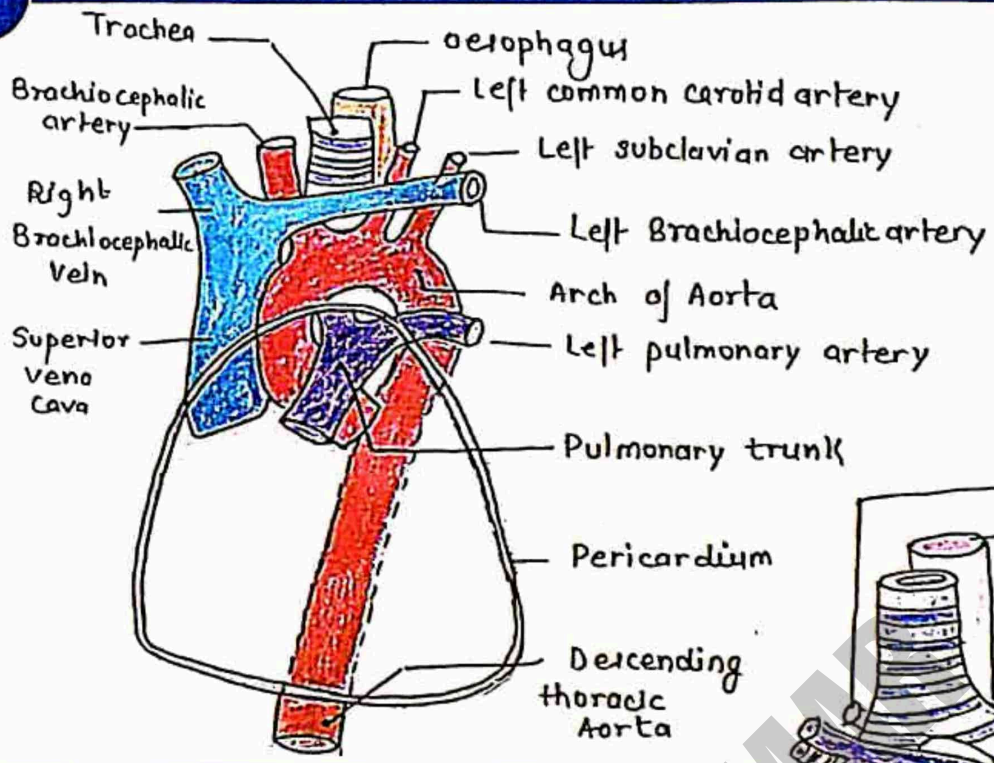
CLINICAL ANATOMY

1.MEDIASTINAL SYNDROME:-COMPRESSION OF MEDIASTINAL STRUCTURES BY ANY TUMOUR GIVES RISE TO A GROUP OF SYMPTOMS KNOWN AS MEDIASTINAL SYNDROME.

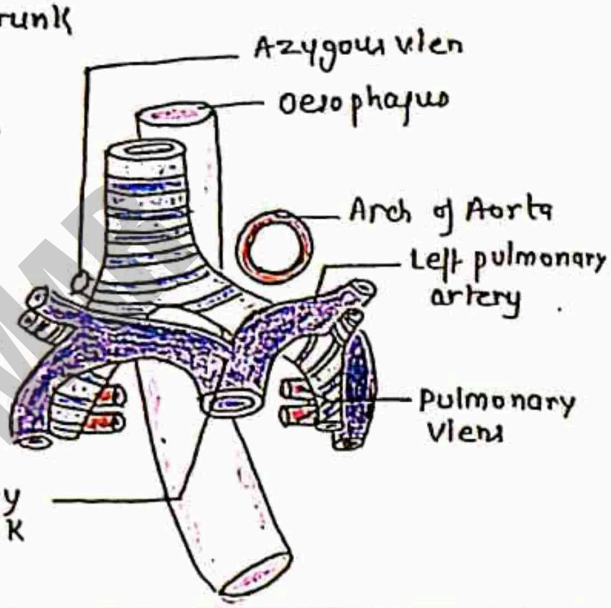
THE COMMON SYMPTOMS ARE AS FOLLOWS -

A. OBSTRUCTION OF SUPERIOR VENA CAVA GIVES RISE TO ENGORGEMENT OF VEINS IN THE UPPER HALF OF THE BODY.
B. PRESSURE OVER THE TRACHEA CAUSES DYSPNOEA, AND COUGH.
C. PRESSURE ON OESOPHAGUS CAUSES DYSPHAGIA.

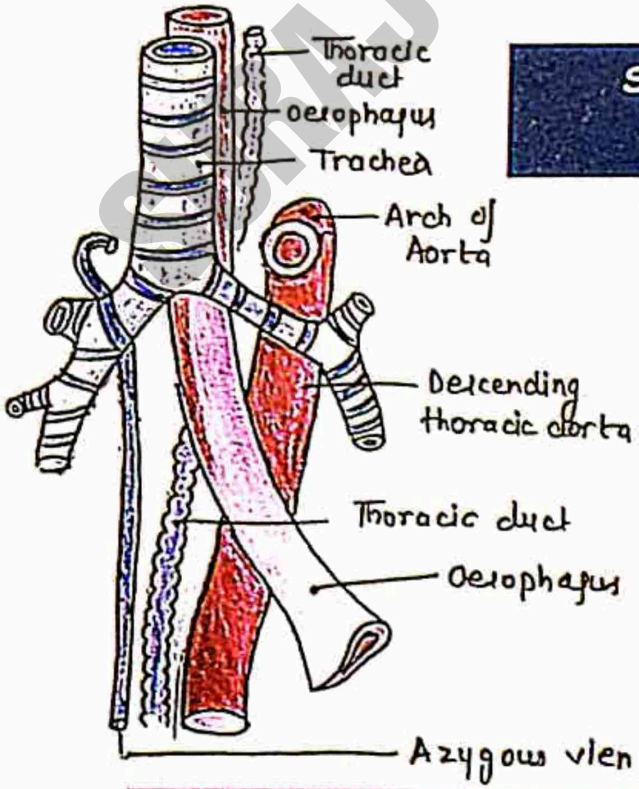
SURA KUMAR



STRUCTURE IN SUPERIOR MEDIASTINUM



STRUCTURE IN MIDDLE MEDIASTINUM



STRUCTURE IN POSTERIOR MEDIASTINUM

PERICARDIUM

*FIBROSEROUS SAC WHICH ENCLOSED HEART & ROOT OF GREAT VESSEL
*CONSISTS OF FIBROUS OF PERICARDIUM & SEROUS PERICARDIUM

1) FIBROUS PERICARDIUM :-
- CONICAL SAC MADE OF FIBROUS TISSUE
- PARIETAL LAYER OF SEROUS PERICARDIUM ATTACHED TO ITS DEEP SURFACE

*FEATURE :-
1.APEX IS BLUNT AND LIE AT LEVEL OF STERNAL ANGLE - FUSED WITH GREAT VESSEL.
2.BASE IS BROAD AND INSEPARABLY BLENDED WITH CENTRAL TENDON OF DIAPHRAGM
3.ANTERIOR CONNECTED TO STERNUM BY SUP. & INF. STERNOPERICARDIAL LIGAMENT
4.POSTERIOR RELATED TO PRINCIPAL BRONCHI, OESOPAGUS WITH NERVE PLEXUS
5.EACH SIDE BY →MEDIASTINAL PLEURA.

BLOOD SUPPLY :- FIBROUS & PARIETAL PARICARDIA-->SUPPLY BY INTERNAL THORACIC, MUSCULOPHRENIC ARTERY
VEIN DRAIN INTO CORRESPONDING VEIN.

NERVE SUPPLY :- FIBROUS AND PARIETAL ARE SUPPLIED BY -→PHRENIC NERVE
-SENSITIVE TO PAIN

SEROUS PERICARDIUM :-
-SEROUS PERICARDIUM IS THIN, DOUBLE LAYERED SEROUS MEMBRANE
-OUTER LAYER - PARIETAL PERICARDIUM FUSED - WITH FIBROUS PERICARDIUM
-INNER LAYER - UISCERAL PERICARDIUM FUSED TO HEART

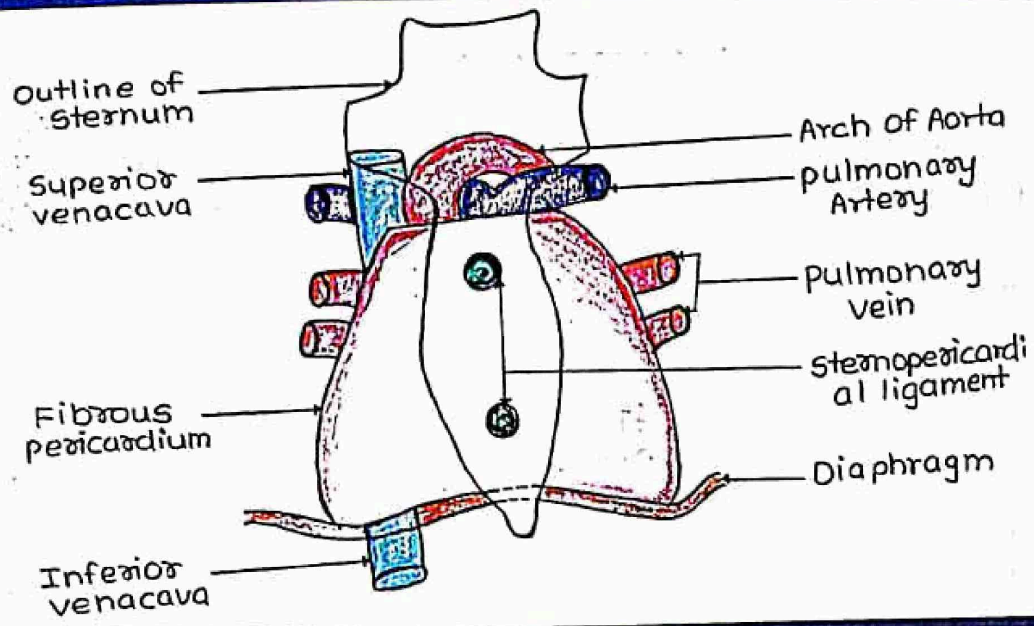
PERICARDIAL CAVITY-POTENTIAL SPACE - BETWEEN PARIETAL & VISCERAL PERCARDIUM
- CONTAIN ONLY THIN FILM OF SEROUS FLUID

2) SINUSES OF PERICARIUM :-
1.EPICARDIUM AT ROOT GREAT VESSEL ARRANGED IN FORM OF TWO TYPE -→
-->ARTERIAL TUBE ENCLOSE -->ASCENDING AORTA & PULMONARY TRUNK
--> VENOUS TUBE ENCLOSE-->VENACAVA & PULMONARY TRUNK
2.PASSAGE BETWEEN TWO TUBES--->TRANSVERSE SINUS
3.DURING DEVELOPMENT TO BEGIN-→VEIN OF HEART CROWDED TOGETHER
4.HEART INCREASE IN SIZE & VEIN SEPARATE OUT
5.PERICARDIAL REFLECTION SURROUND ALL OF THEM & FORM OBLIOUS PERICARDIAL SINUS.

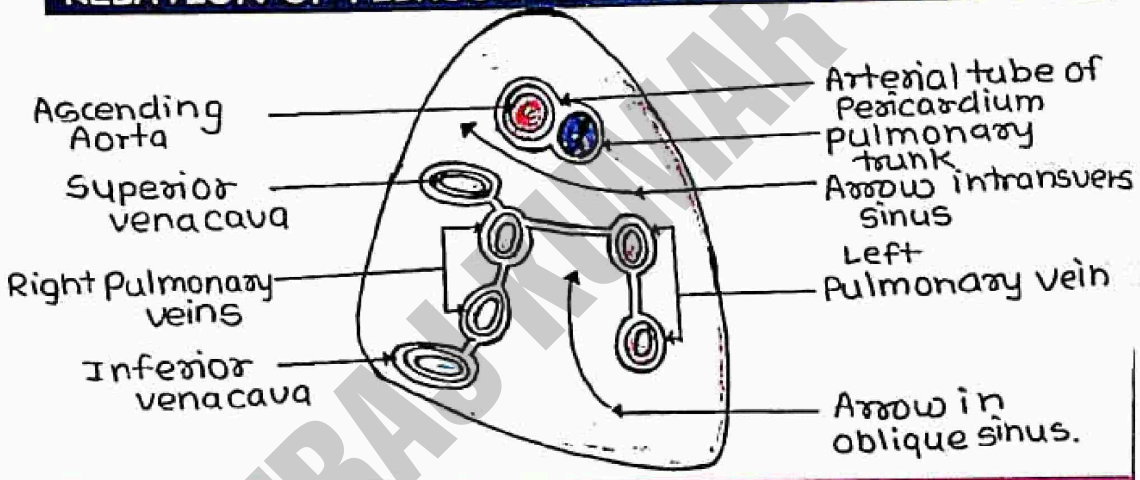
TRANSVERE SINUS :-BOUNDED - ANTERIORLY - ASCENDING AORTA,PULMONARY TRUNK
-POSTERIORLY - SUPERIOR VENACAVA

OBLIQUE SINUS :- BOUNDED - ANTERIORLY-LEFT ATRIUM
POSTERIORLY-PARIETAL PERICARDIUM OESOPHAGUS

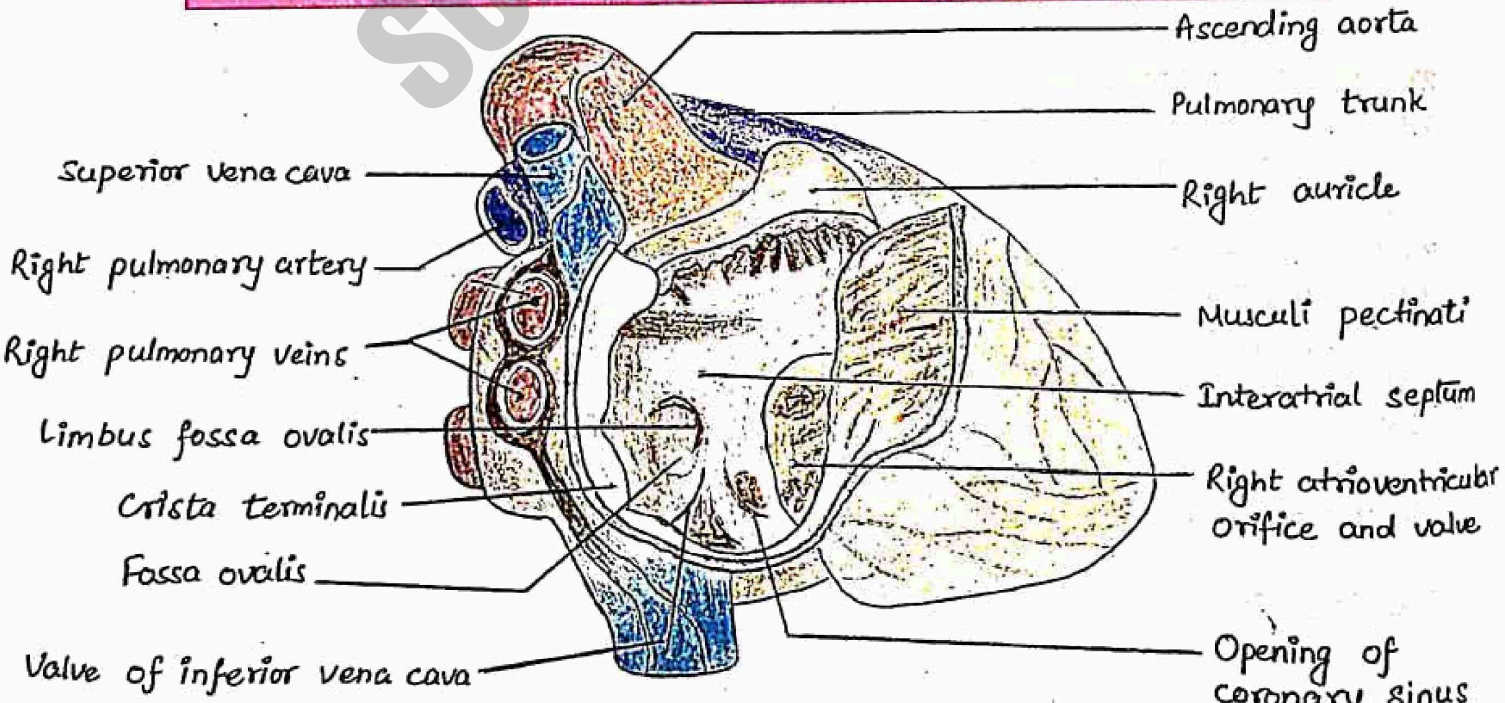
SUBAL KUMAR



RELATION OF FIBROUS PERCARDIUM TO STURNUM



SINUS OF PERICARDIUM



INTERIOR OF RIGHT ATRIUM

SUBAS KUMAR

RIGHT ATRIUM

*RIGHT UPPER CHAMBER OF HEART RECEIVE VENOUS BLOOD FROM BODY
*PUMP INTO RIGHT VENTRICLE THROUGH TRICUSPID VALVE

EXTERNAL FEATURE -

1. CHAMNER IS ELONGATED VERTICALLY, RECEIVING HE SUP. VENA CAVA AT THE UPPER END & INF. VENA CAVA AT THE LOWER END.

2. ALONG RIGHT BORDER OF ATRIUM-→ SHALLOW VENTRICAL GROOVE WHICH PASSES FROM ---→ SUPERIOR VENA CAVA TO INFERIOR VENA CAVA IS CALLED SULCUS TERMINALIS

3. IT PRODUCED BY INTERNAL MUSCULAR RIDGE CALLE CRISTA TERMINALIS

4. UPPER PART OF SULCUS CONTAINED S.A NODE

5. ANTRIOVENTRICULAR GROOVE SEPARATED RIGHT ATRIUM FROM RIGHT VENTRICLE
INLET OF RIGHT ATRIUM - 1. SUPERIOR VENA CAVA
2. INFERIOR VENA CAVA
3. CORONARY SINUS
4. ANTERIOR CARDIAC VEINS

INTERNAL FEATURES :- DIVIDED INTO 3 PARTS

1.SMOOTH PART - DEVELOPMENT - DRIVED FROM RIGHT HORN SINUS VENOSUS
- MOST OF TRIBUTARIES EXCEPT ANT. CARDIAC VEINS OPEN INTO IT
1.SUPERIOR VENA CAVA
2.INFERIOR VENA CAVA - EUSTACHIAN VALVE
3.CORONARY SINUS - THEBESIAN VALVE
4. VENAE CORDIS MINIMAE

2.ROUGH ANT.PART - DEVELOPMENT - DERIVED FROM PRIMITIVE ATRIAL CHAMBER
- IT PRESENT A SERIES OF MUSCULAR RIDGE CALLED MUSCULI PECTINATI

3.INTERATRIAL SEPTUM - DEVELOPMENT - DRIVED FROM PRIMITIVE ATRIAL CHAMBER
- IT PRESENT THE FOSSA OVALIS A SHALLOW DEPRESSION IN LOWER PART
- ANNULUS OVALIS & FOSSA OVALIS THE PROMINENT MARGIN FOSSA OVALIS

NOTE :- DIAGRAM OF RIGHT ATRIUM - PAGE NO.65

SURAJ KUMAR

ARTERIAL SUPPLY OF HEART

*HEART IS SUPPLY BY 2 CORONARY ARTERIRES -→ARISING FROM ASCENDING AORTA

RIGHT CORONARY ARTERY

POSITION - SMALLER THAN LEFT CORONARY ARTERY
- ARISE FROM ANT. AORTIC SINUS

COURSE - FIRST PASS FORWARD BETWEEN ROOT OF PULMONARY TRUNK & RIGHT AURICLE
- THEN RUN DOWNWARD - RIGHT ANTERIOR CORONARY SULCUS
- WINDS AROUND THE INFERIOR BORDER TO REACH - DIAPHRAGMATIC SURFACE OF HEART
- TERMINATE BY ANASTOMOSIS WITH LEFT CORONARY ARTERY

BRANCHES :-

1. ARTERIAL BRANCH - ANT., POST. & LATERAL
2. RIGHT CONUS ARTERY - FROM CIRCLE AROUND PULMONARY TRUNK CALLED 'ANNULUS OF VIEUSSENS'
3. VENTRICLAR BRANCH- ANT. & POST GROUP
4. RIGHT MARGINAL ARTERY
5. POSTERIOR INTERVENTRICULAR BRANCH - LIE IN POST. INTERVENTRICULAR GROOVE

AREA OF DISTRIBUTION :- RIGHT ATRIUM
- VENTRICLE - GREAT PART OF RIGHT VENTRICLE - EXCEPT AREA ADJOING INTERVENTRICULAR GROOVE
- SMALL PART OF LEFT VENTRICLE
- WHOLE CONDUCTING SYSTEM OF HEART EXCEPT A PART OF LEFT BRANCH OF AV BUNDLE.

LEFT CORONARY ARTERY

***POSITION** - ARISE FROM LEFT AORTIC SINUS

***COURSE** - FIRST RUN FORWARD TO LEFT END EMERGE BETWEEN PULMONARY TRUNK
-HERE IT GIVE THE INTERVENTRICULAR BRANCH WHICH RUN DOWNWARD IN GROOVE OF SAME NAME
-FUTHER CONTINUATION OF LCA IS CALLED CIRCUMFLEX ARTERY
-IT WIND AROUND LEFT BORDER OF HEART & CONTINUE IN LEFT CORONARY SULCUS
-TERMINATE BY ANASTOMOSIS WITH RCA

SURAJ KUMAR

BRANCHES :- 1.ANT. INTERVENTRICULAR BRANCH - LARGEST BRANCH
2.CIRCUMFLEX BRANCH IS TERMINATE PART OF LCA
3.ANT.POST VENTRICULAR BRANCH
4.ARTERIAL BRANCH - ANT.POST. LAT. GROOVE

AREA OF DISTRIBUTION :- 1.LEFT ATRIUM
2.VENTRICLE - GREATER PART OF LEFT VENTRICLE EXCEPT AREA ADJOINING POSTERIOR INTERVENTRICULAR GROOVE & SMALL PART OF RIGHT VENTRICLE OR ANT.PART OF INTERVENTRICULAR SEPTUM.
3.ANT.PART OF INTERVENTRICULAR SEPTUM.
4.LEFT BRANCH OF AV BUNDLE.

CARDIAC DOMINANCE :-
1.ARTERY GIVING POSTERIOR INTERVENTRICULAR BRANCH IS DOMINANT ARTERY
2.IN 90% RIGHT CORONARY ARTERY GIVE POSTERIOR INTERVENTRICULAR ARTERY
-→SUCH HEART CALLED RIGHT DOMINANCE
3.IN 10% OF CIRCUMFLEX ARTERY CONTINUATION OF LEFT CORONARY ARTERY
4.PROVIDE POSTERIOR INTERVENTRICULAR BRANCH SUCH HEART CALLED LEFT DOMINANCE HEART.

VEINS OF HEART

*THESE ARE GREAT CARDIAC VEIN, MIDDLE CARDIAC, RIGHT MARGINAL VEIN POSTERIOR VEIN OF LEFT VENTRICLE, ANTERIOR CARDIAC VEIN, VENAE CORDIS MINIMAE.
*ALL VEIN EXCEPT ---→ANT.CARDIAC VEIN OPEN INTO -→CORONARY SINUS
*CORONARY SINUS OPENS INTO --→RIGHT ATRIUM.

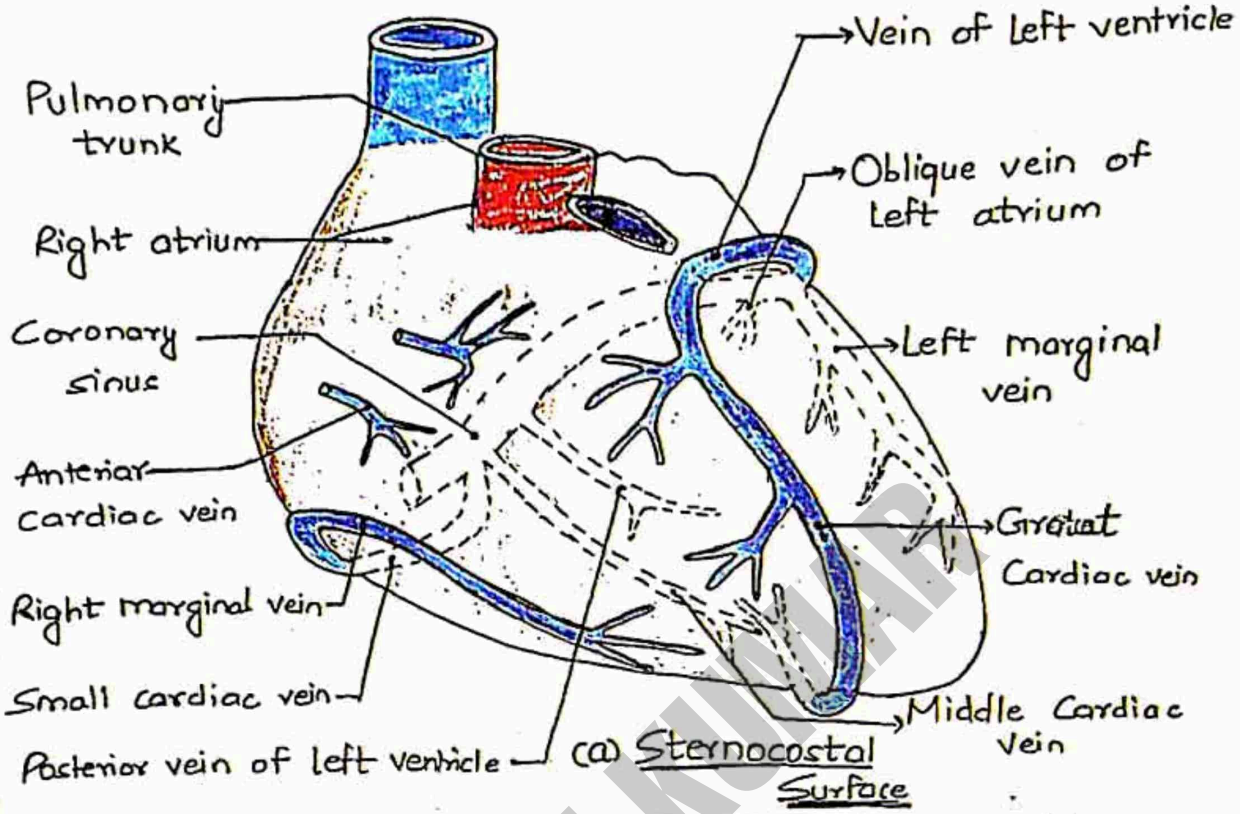
1.CORONARY SINUS :- LARGEST VEIN OF HEART
-SITUATED INTO LEFT POSTERIOR CORONARY SULCUS END INTO RIGHT ATRIUM

-TRIBUTRIES OF CORONARY SINUS -
1.GREAT CARDIAC VEIN - ACCOMPANY FIRST ANT.VENTRICULAR ARTERY THAN LEFT CORONARY ARTERY
2.MIDDLE CARDIAC VEIN - ACCOMPANY - POSTERIOR INTERVENTRICULAR ARTERY
3.SMALL CARDIAC VVEIN - ACCOMPANY - RIGHT CORONARY ARTERY
4.RIGHT MARGINAL VEIN - ACCOMPANY - MARGINAL BRANCH OF RIGHT CORONARY ARTERY
5.POST.VEIN OF LEFT VENTRICLE - RUN ON DIAPHRAGMATIC SURFACE ON LEFT VENTRICLE
6.OBLIQUE VEIN OF LEFT ATRIUM - RUNNING OF POST.SURFACE LEFT ATRIUM

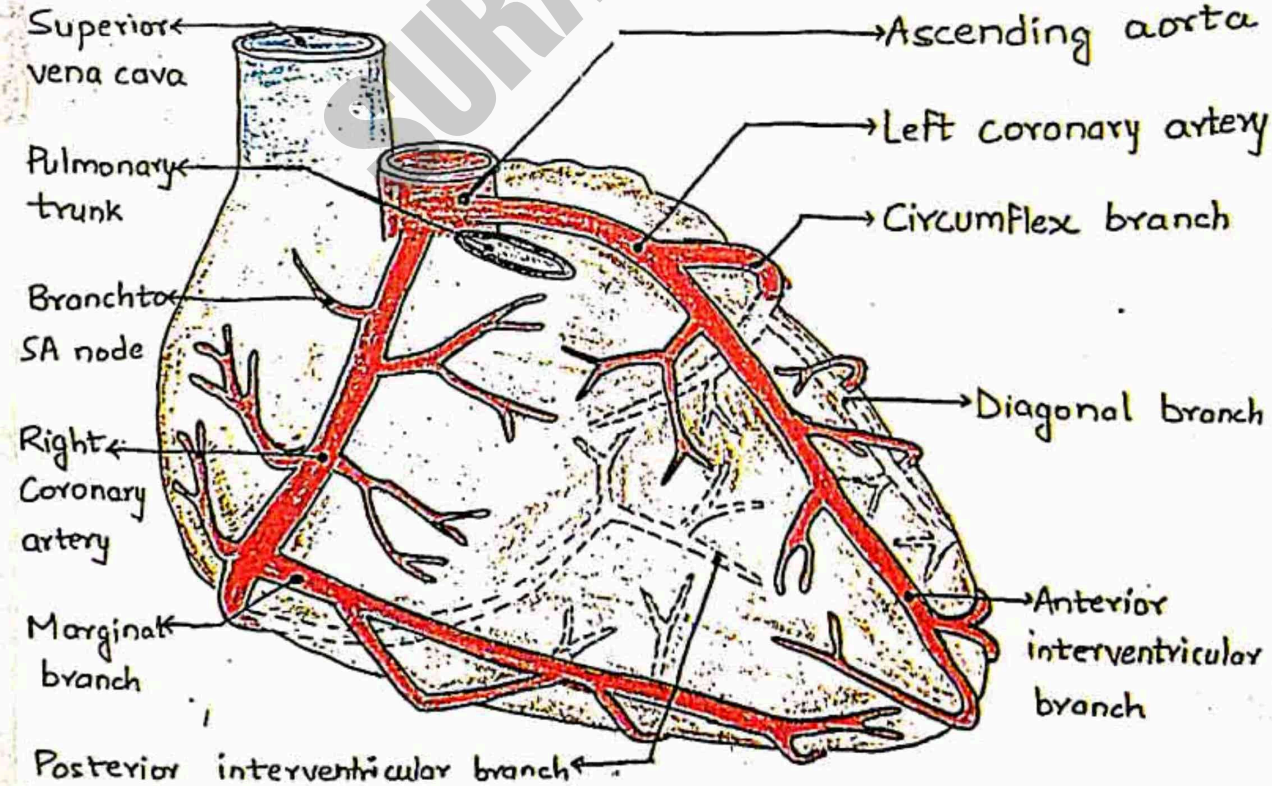
2.ANT.CARDIAC VEIN :- 3 TO 4 SMALL VEIN WHICH RUN PARALLEL TO EACH OTHER ON THE ANT.WALL OF RIGHT VENTRICLE.
-OPEN DIRECTLY INTO RIGHT ATRIUM.

RAJ KUMAR

3. VENA CORDIS MINIMAE :- NUMEROUS SMALL VALVE LESS VEIN PRESENT IN ALL FOUR CHAMBER OF HEART --→ DIRECTLY OPEN RIGHT ATRIUM.



VEINS OF HEART



ARTERY OF HEART

SURAJ KUMAR

AZYGOS VEIN

*THE AZYGOS TERM MEANS UNPAIR.
*AZYGOS VEIN DRAIN THORACIC WALL & THE UPPER LUMBAR REGION.
*FROM IMPORTANT CHANNEL CONNECTING IVC.
*VEIN OCCUPIES UPPER PART OF POST. WALL OF ABDOMINAL WALL & THE POST.MEDIASTINUM.

*FORMATION :- FORM BY UNION OF LUMBAR AZYGOS
- RIGHT SUB COASTAL
- ASCENDING LUMBAR VEIN.

*TRIBUTARIES :-

1.RIGHT SUPERIOR INTERCOASTAL VEIN-→ FORM BY UNION OF 2ND, 3RD,4TH POST.INTERCOASTAL VEIN.
2.5TH TO 8TH POST.INTERCOASTAL VEIN.
3.HEMIAZYGOS---→AT LEVEL OF LOWER BORDER OF 8 THORACIC VERTEBRAE.
4.ACCESSORY HEMIAZYGOS VEIN - AT THE LEVEL OF UPPER BORDER OF 8TH THORACIC VERTEBRA
5.RIGHT BRONCHIAL VEIN, NEAR THE TERMINAL END OF AZYGOS VEIN.
6.SEVERAL OESOPHAGEAL, MEDIASTINAL, PERICARDIAL VEINS.

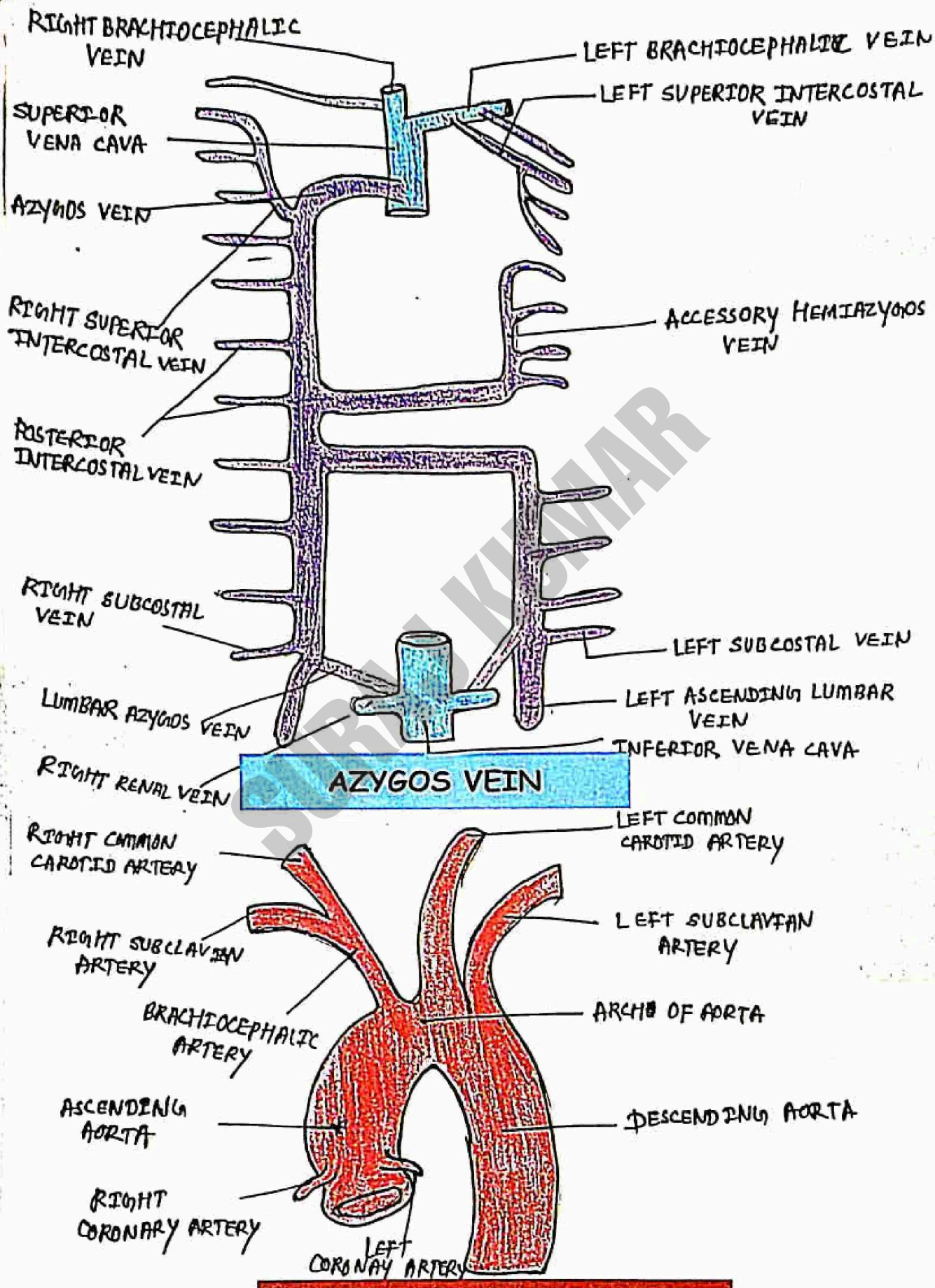
ARCH OF AORTA

*CONTINUATION OF ASCENDING AORTA
*SITUATED IN MEDIASTINUM.

*COURSE :-
1.BEGIN BEHIND THE UPPER BORDER OF THE 2ND RIGHT STERNOCHONDRAL JOINT.
2.RUN UPWARD, BACKWARD & BIFURACTION OF TRACHEA.
3.IT THUS ARCHES OVER ROOT OF LUNG.
4.ENDS AT LOWER BORDER OF BODY OF 4TH THORACIC VERTEBRAE.

*BRANCHES :-
- BRACHIOCEPHALIC ARTERY ---→DIVIDES RIGHT COMMON CAROTID RIGHT SUBCLAVIAN ARTERIES
- LEFT COMMON CAROTID ARTERY
- LEFT SUBCLAVIAN ARTERY

SURAJ KUMAR


RIGHT BRACHIOCEPHALIC VEIN
LEFT BRACHIOCEPHALIC VEIN
LEFT SUPERIOR INTERCOSTAL VEIN
SUPERIOR VENA CAVA
AZYGOS VEIN
RIGHT SUPERIOR INTERCOSTAL VEIN
ACCESSORY HEMIAZYGOS VEIN
POSTERIOR INTERCOSTAL VEIN
RIGHT SUBCOSTAL VEIN
LEFT SUBCOSTAL VEIN
LUMBAR AZYGOS VEIN
LEFT ASCENDING LUMBAR VEIN
INFERIOR VENA CAVA
RIGHT RENAL VEIN
AZYGOS VEIN
LEFT COMMON CAROTID ARTERY
RIGHT COMMON CAROTID ARTERY
LEFT SUBCLAVIAN ARTERY
RIGHT SUBCLAVIAN ARTERY
ARCH OF AORTA
BRACHIOCEPHALIC ARTERY
ASCENDING AORTA
DESCENDING AORTA
RIGHT CORONARY ARTERY
LEFT CORONARY ARTERY
ARCH OF AORTA

SURYA KUMAR

THORACIC DUCT

*LARGEST LYMPHATIC VESSEL OF THE BODY
*EXTEND FROM UPPER PART OF ABDOMEN TO LOWER PART OF NECK
*CROSS POST.BORDER SUP.PART OF MEDIASTINUM

*COURSE :-

1. BEGINS AS A CONTINUATION OF THE UPPER END OF THE CISTERNA CHYLI NEAR LOWER BORDER -→12 THORACIC VERTEBRAE
2. ENTER THORAX THROUGH AORTIC OPENING OF DIAPHRAM
3. THEN ASCEND THROUGH POST.MEDIASTINUM FROM LEVEL OF 12TH - 5TH THORACIC VERTEBRAE.
4. THEN IT COURSE THROUGH SUP.MEDIASTINUM
5. IN NECK - ARCH LAT. AT LEVEL OF 7TH CERVICAL VERTEBRAE
6. AND BY OPENING INTO ANGLE OF JUNCTION BETWEEN LEFT SUBCLAVIAN & LEFT INTERNAL JUGULAR VEIN.

*RELATION :-

-AT AORTIC OPENING OF DIAPHRAM - ANTERIORLY-DIAPHRAM
-POSTERIORLY-VERTEBRAL COLUMN
-TO THE RIGHT - AZYGOS VEIN
-TO THE LEFT - AORTA

-IN THE POST.MEDIASTINUM - ANTERIORLY - DIAPHRAGM
-OESOPHAGUS
-RIGHT PLEURAL RECESS
-POSTERIORLY - VERTEBRAL COLUMN
-RIGHT POST.INTERCOASTAL ARTERIES
TO THE RIGHT - AZYGOS VEIN
TO THE LEFT - DESCENDING AORTA

-IN SUP.MEDIASTINUM - ANTERIORLY - ARCH OF AORTA
-POSTERIORLY - VERTEBRAL COLUMN
-TO THE RIGHT - OESOPHAGUS
-TO THE LEFT - PLEURA

-IN THE NECK - THE THORACIC DUCT FORMS AN ARCH RISING ABOUT 3-4CM ABOVE THE CLAVICLE
-ANTERIORLY - LEFT COMMON CAROTID ARTERY
- LEFT VAGUS
- INTERNAL JUGULAR VEIN
-POSTERIORLY - VERTEBRAL ARTERY
- SYMPATHETIC TRUNK
- LEFT PHRENIC NERVE

SURAJ KUMAR

*TRIBUTARIES :- THORACIC DUCT RECEIVE LYMPH FROM BOTH HALF OF BODY BELOW DIAPHRAM & LEFT HALF ABOVE ABDOMEN.
- IN THORAX THORACIC DUCT RECEIVE - LYMPH VESSEL FROM POST.MEDIASTINAL NODE & SMALL INTERCOASTAL NODE.
- LEFT JUGULAR TRUNK, LEFT SUBCLAVIAN TRUNK END IN THORACIC DUCT
- LEFT BRONCHOMEDIASTINAL TRUNK - DRAIN LYMPH FROM LEFT HALF OF THORAX & END IN THORACIC DUCT .

RIGHT HALF ABOVE DIAPHRAGM :-
-RIGHT SIDE THERE IS RIGHT LYMPHATIC DUCT.-→INTO WHICH RIGHT BRONCHIOMEDIASTINAL, RIGHT JUGULAR, RIGHT SUBCLAVIAN LYMPH DRAIN.
-RIGHT LYMPHATIC TRUNK END IN RIGHT BRONCHIOCEPHALIC VEIN.

NOTE :- DIAGRAM OF THORACIC DUCT - PAGE NO.74

INTERCOSTAL NERVE

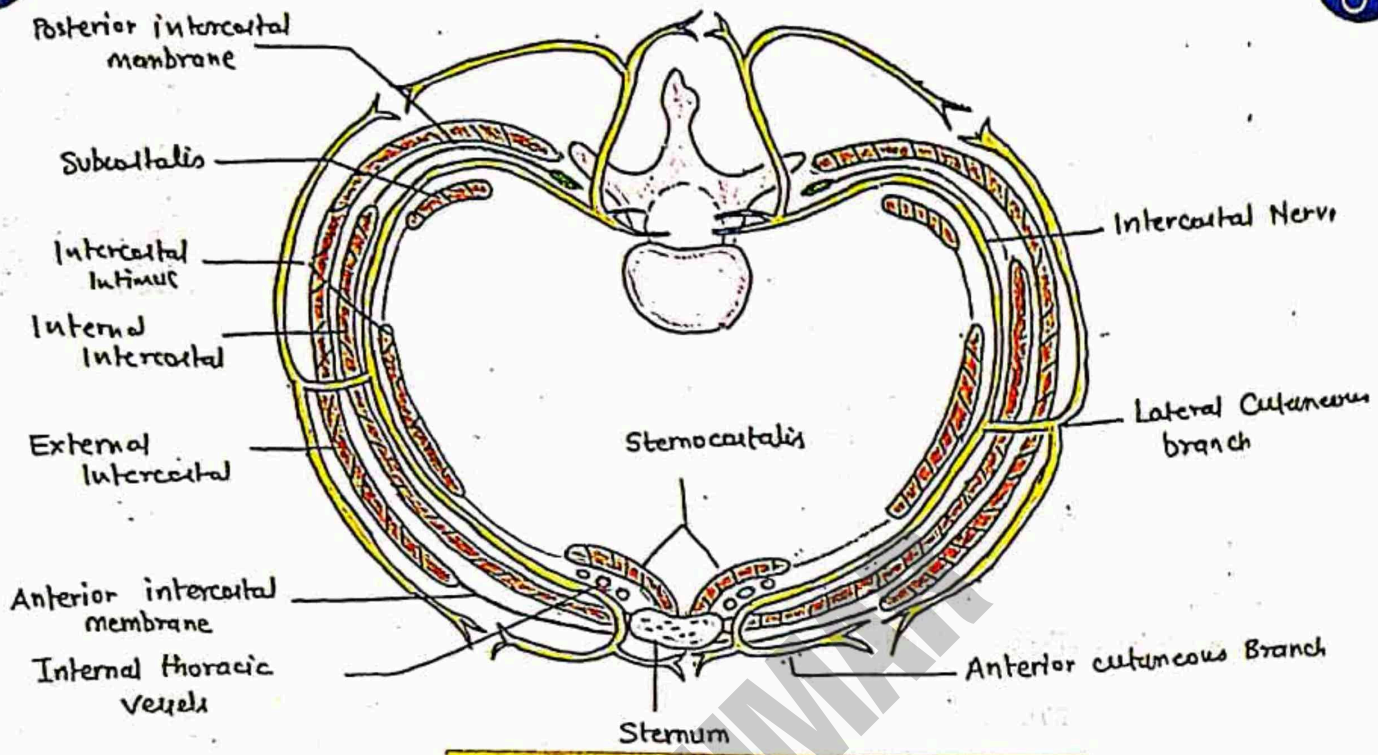
*ALL INTERCOSTAL NERVE - SUPPLY INTERCOSTAL MUSCLE

INTERCOSTAL NERVE :-
-INTERCOASTAL NERVE ARE ANT.PRIMARY RAMI OF THORACIC 1-12 SPINAL NERVE
-THE ANT.PRIMARY RAMI OF 12 THORACIC NERVE - FORM SUBCOASTAL NERVE.
-IN ADDITIONAL TO SUPPLY INTERCOASTAL SPACE -
-UPPER 2 INTERCOASTAL NERVE - ALSO SUPPLY UPPER LIMB MUSCLE
-LOWER 5 - SUPPLY ABDOMINAL WALL ALSO CALLED THORACO ABDOMINAL NERVE
-3-6 SUPPLY ONLY THORACIC WALL THEY ARE CALLED TYPICAL INTERCOASTAL NERVE.

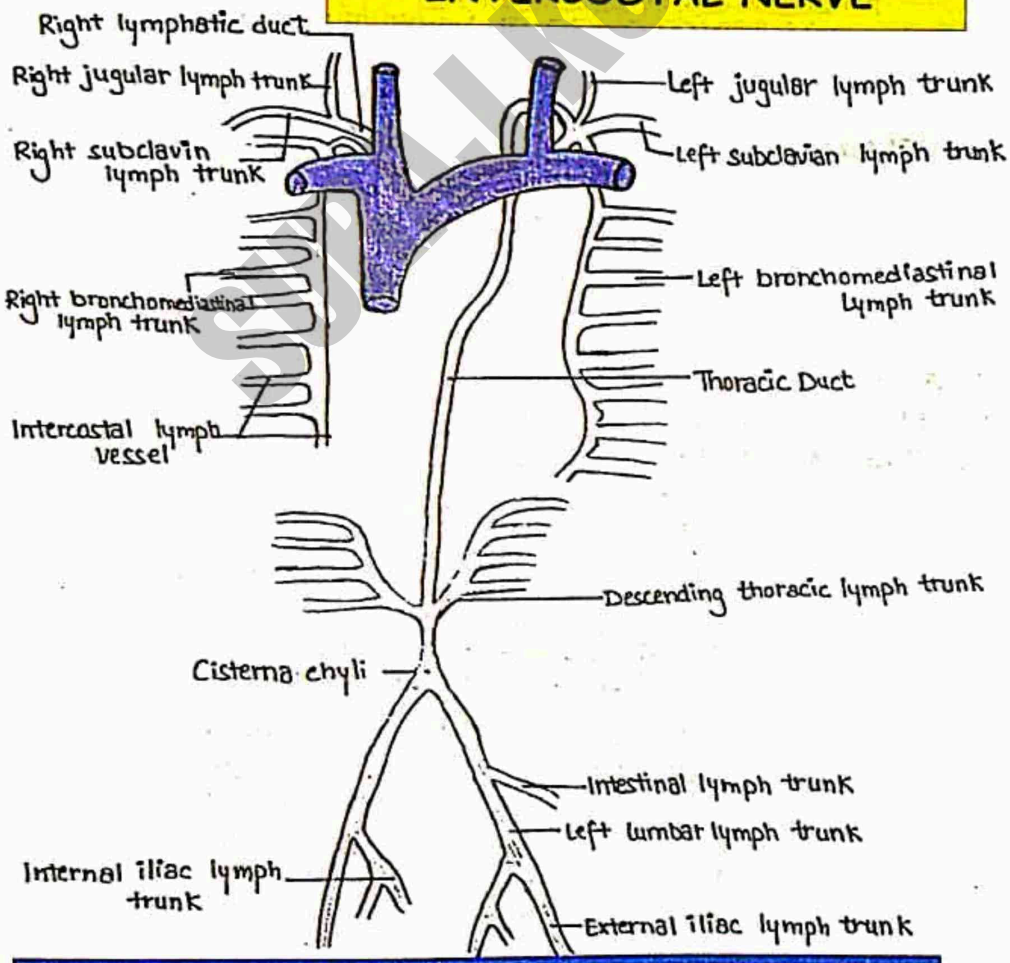
*COURSE :- INTERCOASTAL NERVE RUN IN COASTAL GROOVE-→END NEAR STERNUM

*BRANCHES :- MUSCULAR BRANCH - NUMEROUS MUSCULAR BRANCH SUPPLY INTERCOASTAL MUSCLE - TRANSVERSUS THORACIS
-SENSORY BRANCH - MAIN BRANCH & COLLATERAL BRANCH ALSO SUPPLY PARIETIAL PLEURA & PERIOSTEUM OF RIB
-LAT.CUTANEOUS BRANCH ARISE-NEAR ANGLE OF RIB WHERE IT PAIR INTERCOASTAL MUSCLE & DISTRIBUTED TO THE SKIN.
-ANT.CUTANEOUS BRANCH - EMERGE ON THE SIDE OF STERNUM -SUPPLY OVERLYING SKIN.
-COMMUNICATING BRANCH - EACH NERVE IS CONNECTED TO THORACIC SYMPATHETIC GANGLION BY A DISTAL PLACE WHITE & PROXIMAL GREY RAMUS COMMUNICANS.

SURAJ KUMAR



INTERCOSTAL NERVE



TRIBUTARIES OF THORACIC DUCT

SUBHAN

INTERCOASTAL SPACE

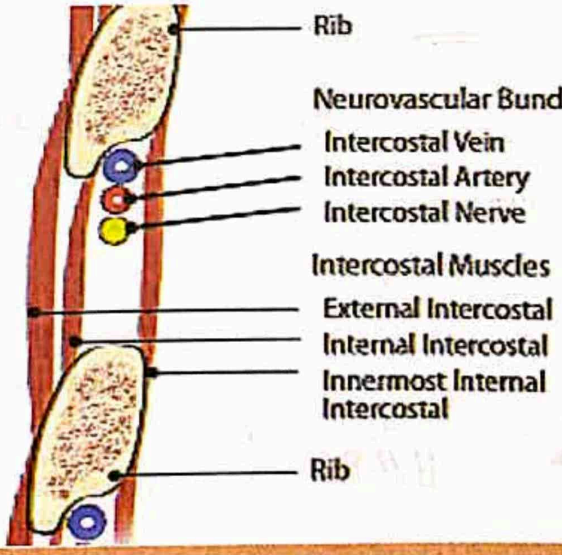
*GAP BETWEEN RIBS ARE CALLED INTERCOASTAL SPACE
- FILL BY INTERCOASTAL MUSCLE & CONTAIN INTERCOASTAL NERVE & VESSEL-
*9-INTERCOASTAL SPACE - ANTERIORLY
*11 INTERCOASTAL SPACE POSTERIORLY

INTERCOASTAL MUSCLE - 1.EXTERNAL INTERCOASTAL MUSCLE
2.INTERNAL INTERCOASTAL MUSCLE
3.TRANSVERSUS THORACIS - DIVIDED INTO 3 PARTS- SUBCOSTALIS, INTERCOSTALIS INTIMI, STERNOCOSTALIS

*EXTEND
1.EXTERNAL INTERCOASTAL - POST. - TUBERCLE OF RIB
- ANT. - COSTOCHONDRAL JUNCTION
*ANTERIORLY - BETWEEN COSTOCHONDRAL JUNCTION AND STERNUM
REPLACED BY EXTERNAL INTERCOASTAL MEMBRANE
*POSTERIORLY - END OF MUSCLE IS CONTINUOUS WITH POSTERIOR FIBRES
→SUPERIOR COSTO TRANSVERSE LIGAMENT
*DIRECTION OF FIBRE - DOWNWARD, FORWARD & MEDIALLY IN FRONT

2.INTERNAL INTERCOASTAL - ANT. - LATERAL BORDER OF STERNUM-->CONTINOUS -->SUPERIOR COSTO LIGAMENT
- POST. - ANGLE OF RIB BEYOND POSTERIOR INTERCOASTAL MEMBRANE
*DIRECTION OF FIBRE - DOWNWARD, BACKWARD, LATERALLY

3.SUBCOSTALIS - POSTERIOR PART OF LOWER INTERCOASTAL SPACE
4. INTERCOSTALIS - MIDDLE 2/4 OF ALL INTERCOSTAL SPACE
5.STERNOCOSTALIS - ANTERIOR PART OF UPPER INTERCOSTAL SPACE



SECTION THROUGH INTERCOSTAL SPACE

SURAJ KUMAR

INLET OF THORAX

*NARROW UPPER END OF THORAX - CONTINUOUS WITH THE NECK, KIDNEY SHAPED.

*BOUNDARIES - ANT. - UPPER BORDER OF MANUBRIUM STERNI
-POST. - SUPERIOR SURFACE OF 1ST THORACIC VERTEBRA
-EACH SIDE ON 1ST RIB

*PLANE OF INLET IS DIRECTED DOWNWARDS AND FORWARD WITH AN OBLIQUITY OF 45 DEGREE.

*PARTITION AT INLET OF THORAX :-

1.PARTITION IN 2 HALVES----→RIGHT & LEFT --→WITH CLEFT BETWEEN
2.EACH HALF IS COVERED BY FASCIA - SIBSON FASCIA OR SUPRAPLEURAL MEMBRANE
3.MEMBRANE - TRIANGULAR IN SHAPE
-APEX ----→ATTACHED TO TIP OF TRANSVERSE PROCESS OF 7TH CERVICAL VERTEBRA
-BASE ----→INNER BORDER OF 1ST RIB
4.SIBSON FASCIA--→IS FLATTENED TENDON OF SCALENUS MINIMUS

STRUCTURE PASSING THROUGH THE INLET OF THORAX

1.VISCERA - TRACHEA
- OESOPHAGUS
- APICES OF LUNGS WITH PLEURA
- THYMUS

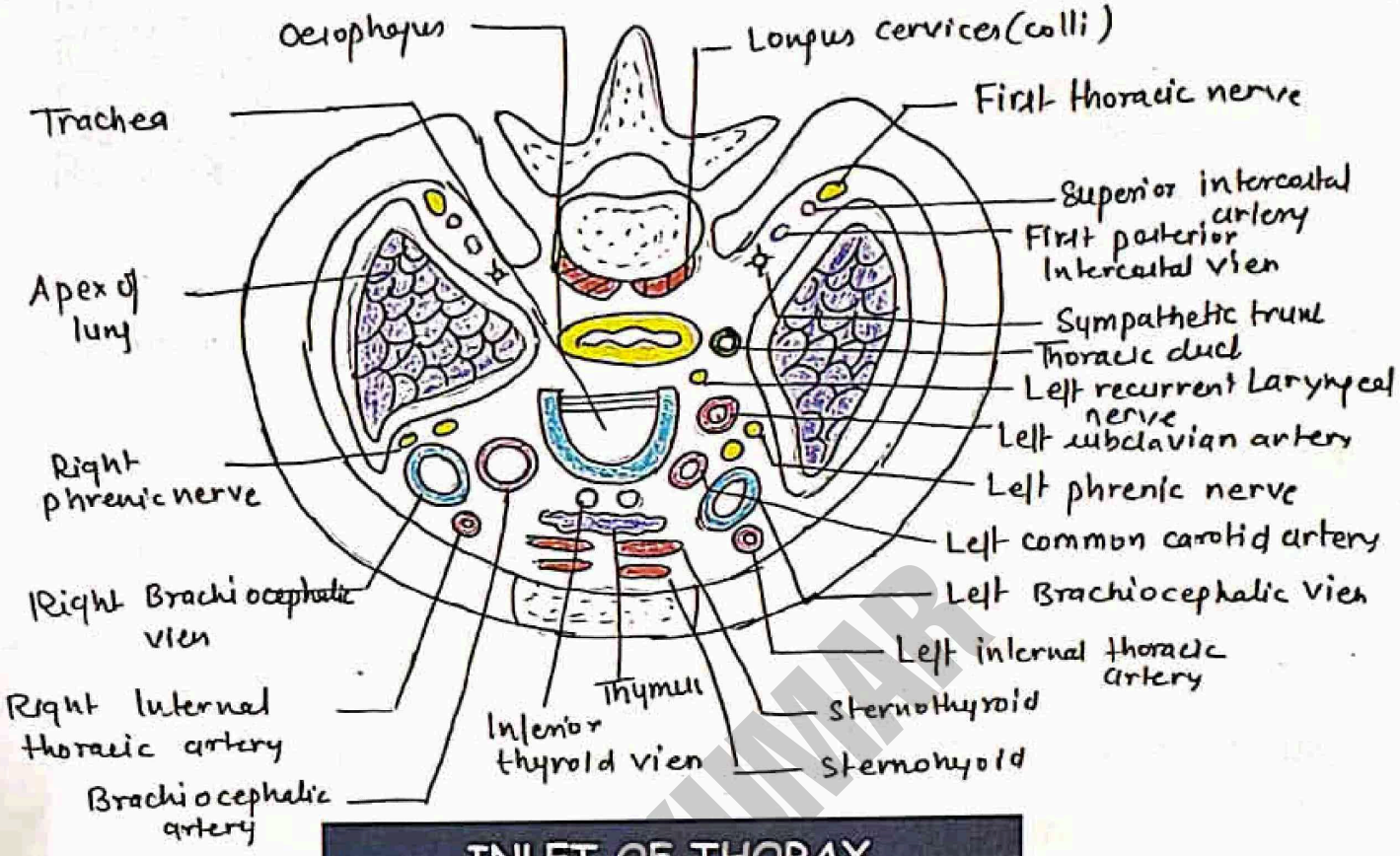
2.LARGE VESSELS - BRACHIOCEPHALIC ARTERY ON RIGHT SIDE
- LEFT COMMON CAROTID ARTERY -→ON LEFT SIDE
- LEFT SUBCLAVIAN ARTERY--→ON LEFT SIDE
- RIGHT & LEFT BROCHIOCEPHALIC VEIN

3.SMALLER VESSELS - RIGHT & LEFT INTERNAL THARACIC ARTERY
- RIGHT & LEFT SUPERIOR INTERCOSTAL ARTERIES
- RIGHT & LEFT FIRST POSTERIOR INTERCOSTAL VEINS
- INFERIOR THYROID VEINS

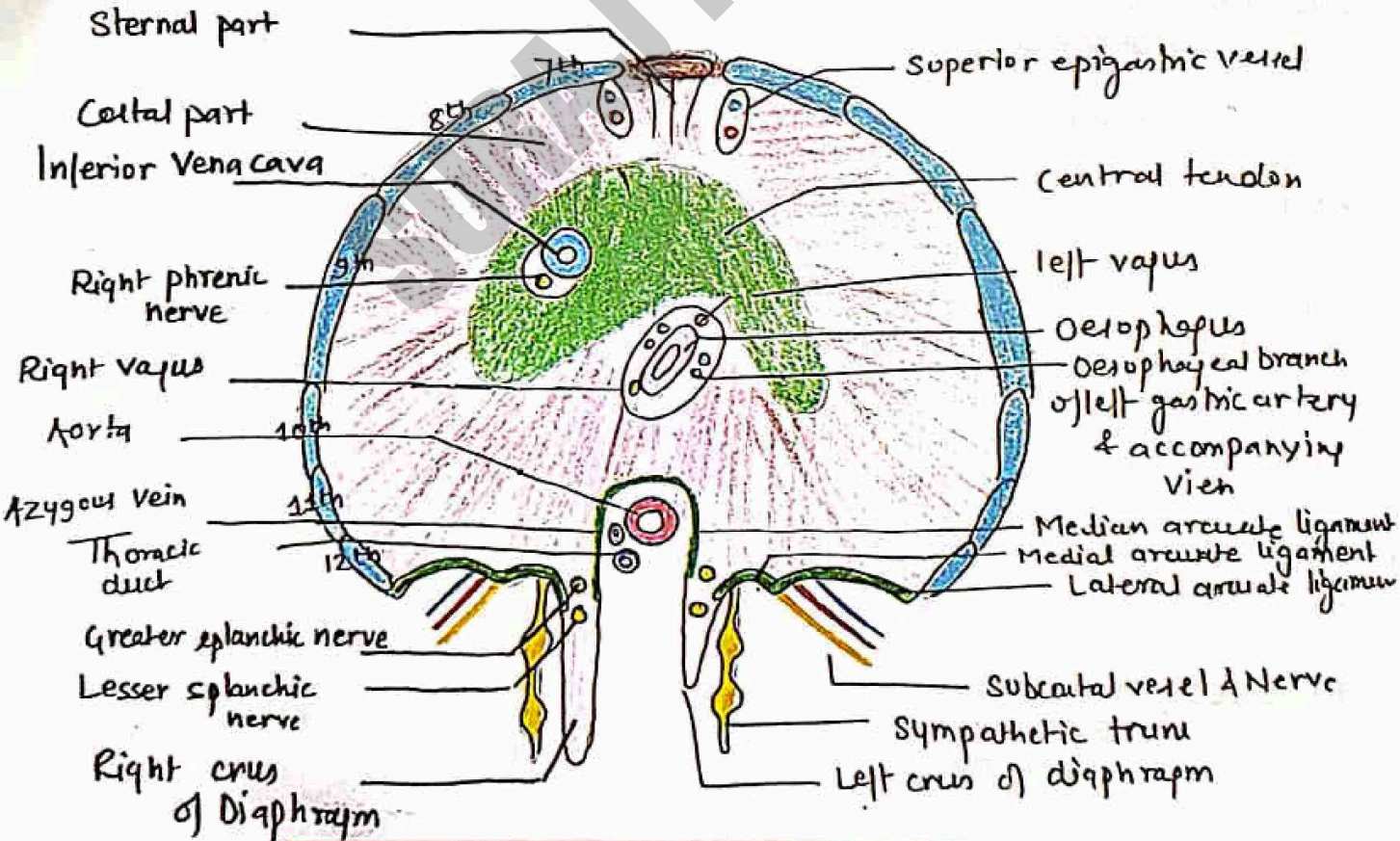
4.NERVES - RIGHT & LEFT PHRENIC NERVE, VAGUS NERVE, SYMPATHETIC TRUNKS

5.MUSCLES - STERNOHYOID
- STERNOTHYROID
- LONGUS COLLI

SURAJ KUMAR



INLET OF THORAX



OUTLET OF THORAX

STUDY TIPS FOR MAXIMUM MARKS

1.HOW TO WRITE IN UNIVERSITY EXAM AND HOW TO DRAW DIAGRAMS ALREADY MENTION IN UPPER LIMB (PAGE NUMBER - 25)

PREVIOUS YEAR UNIVERSITY QUESTIONS

*LONG QUESTIONS - 20 MARKS

1.DESCRIBE BRONCHOPULMONARY SEGMENTS WITH ITS APPLIED

2.DESCRIBE BLOOD SUPPLY OF HEART

3.DESCRIBE RIGHT ATRIUM UNDER FOLLOWING HEADINGS
1.EXTERNAL FEATURES 2.INTERNAL FEATURES 3.DEVELOPMENT

*SHORT NOTE - 5MARKS
1.ROOT OF LUNGS
2.RECESSES OF PLEURA
3.SUPERIOR MEDIA STINUM
4.INFERIOR MEDIA STINUM
5.AZYGUS VEIN
6.THORACIC DUCT
7.INTERCOASTAL NERVE
8.INTERCOASTAL SPACE
9.DRAW THE DIAGRAM OF INLET OF THORAX
10.DRAW THE DIAGRAM OF OUTLET OF THORAX

SUBAJ KUMAR

THORAX ANATOMY IS COMPLETED !!

WORK HARD TO HAVE A GOOD RESULT

ALL THE BEST FOR YOUR EXAMS

DR.JOHARI

THE COPYRIGHT ACT,1957

ALL MATERIAL INCLUDING THEORY, DIAGRAM IS COPYRIGHT REGISTERED .
ANY SCAN AND PHOTOCOPY OF THIS MATERIAL IS NOT PERMITTED BY OWNER

MCQ THORAX

1) Manubriosternal joint is a :
a synovial joint
b primary cartilaginous joint
c secondary cartilaginous joint
d syndesmosis

2) The inferior margin of pleura in the midaxillary line extends up to:
a 10th rib
b 9th rib
c 8th rib
d 7th rib

3) The pleura is supplied by all the following arteries except:
a intercostal arteries
b internal thoracic artery
c pulmonary artery
d bronchial arteries

4) The anterior most structure at the hilum of lung is:
a Pulmonary vein
b bronchial artery
c bronchus
d pulmonary artery

5) The structure that arches above the hilum of right lung is:
a superior venacava
b arch of aorta
c brachiocephalic vein
d azygous vein

6) Hypertrophy of following structures may compress oesophagus :
a left atrium
b right atrium
c left ventricle
d right ventricle

7) All the following are in right atrium except :
a azygous vein
b SVC
c IVC
d coronary sinus

8) Superior mediastinum shows all the following except:
a trachea
b ascending aorta
c arch of aorta
d vagus nerve

9) The venacaval opening foramen in the diaphragm lies at the level of:
a T8
b T12
c T10
d C7

10) The most superficial structure in the thoracic inlet is :
a vagus nerve
b thoracic duct
c superior venacava
d left subclavian artery

11) The most superficial structure in the thoracic inlet is :
a vagus nerve
b superior venacava
c right subclavian artery
d thoracic duct

12) which passes through the diaphragm with the oesophagus:
a azygous vein
b right vagus
c thoracic duct
d phrenic nerve

13) Which muscle is not used in forced expiration :
a transversus abdominis
b rectus abdominis
c diaphragm
d external oblique

14) Accessory muscle of inspiration includes all except:
a latissmus dorsi
b scalene muscle
c sternocleidomastoid
d quadratus lumborum

15) the posterior mediastinum contains all except :
a thoracic aorta
b oesophagus
c azygous vein
d sympathetic trunk

16) The oesophageal opening in the diaphragm transmits:
a azygous vein
b thoracic duct
c vagus nerve
d right phrenic nerve

17) Which doesn't drain into cardiac sinus :
a great cardiac vein
b anterior cardiac vein
c small cardiac vein
d oblique vein if rt atrium

18) Which muscle is not used in inspiration :
a transversus thoracis
b lattismus dorsi
c erector spinae
d pectoralis major

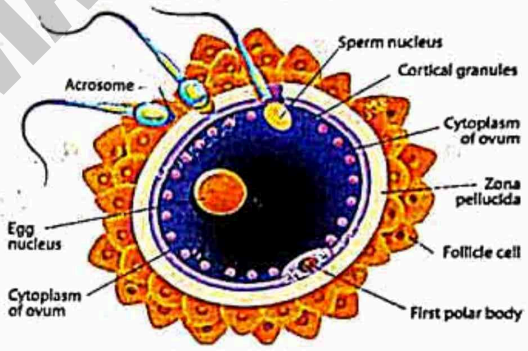
1)C 2)A 3)C 4)A 5)D 6)A 7)A 8)B 9)A 10)C 11)B 12)B 13)C 14)A 15)D 16)C 17)B 18)A

SOHAN SURAJ KUMAR MBBS

FIRST WEEK OF DEVELOPMENT

*In the first week - fertilization takes place in ampulla of fallopian tube
*zona pellucida - a membrane to prevent implantation and polyspermy.

The sperm binds to zona pellucida
↓
Triggers the release if acrosomal enzyme called - Acrosomal reaction
(Acrosin, acid phosphatase, hyaluronidase - acrosomal enzyme)
↓
The sperm penetrates the zona pellucida
↓
The cortical granules release its secretion called - Cortical reaction
↓
The cortical enzymes reach the zona
↓
Zona changes permeability of sperms called - zona reaction
↓
Fusion of male and female pronucleus
↓
Forms zygote
↓
undergoes 1st cleavage - 2 cells stage
↓
and cleavage - 4 cells stage
↓
3rd cleavage - 8 cells stage.



TIME AND SITE OF EVENTS

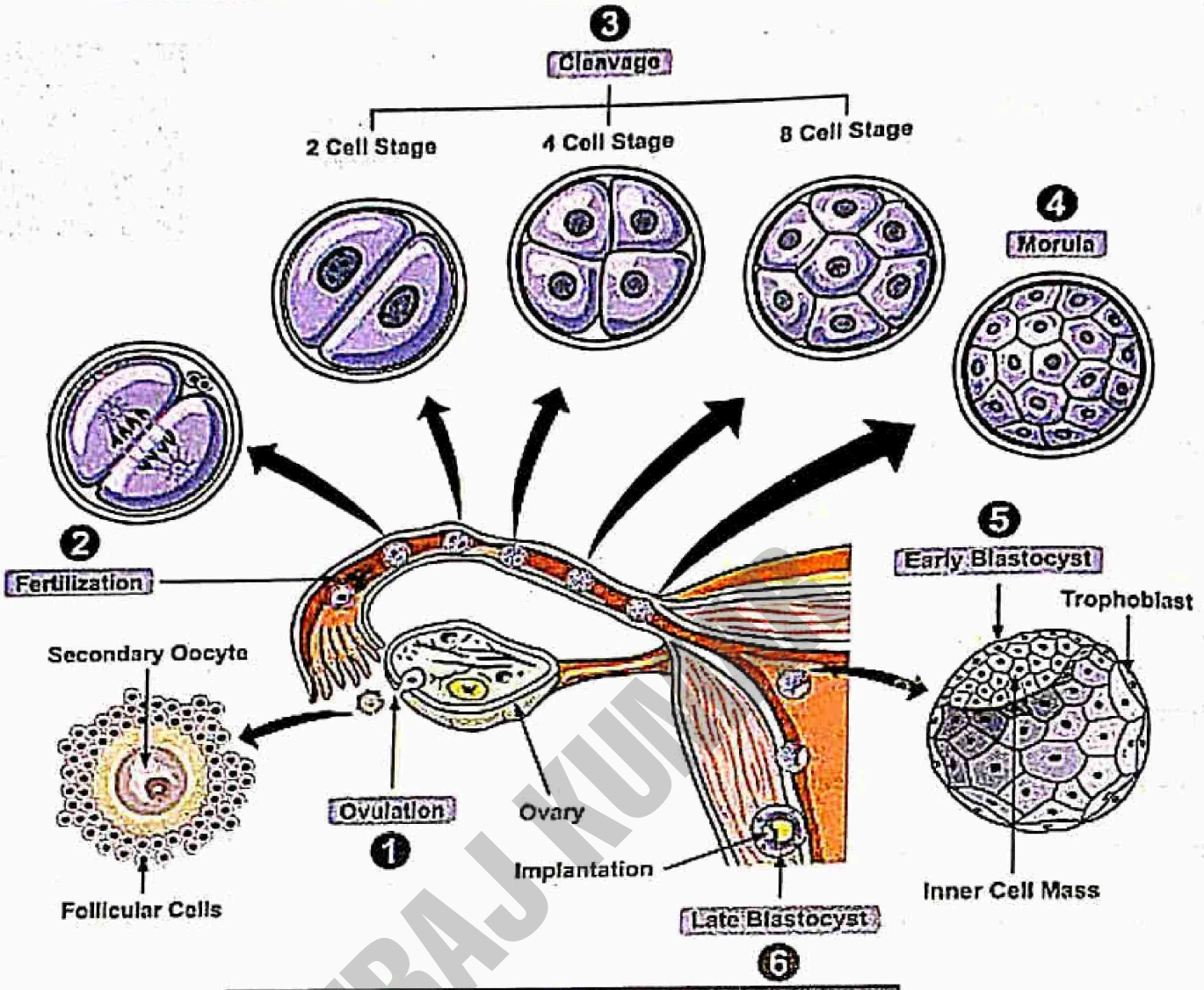
1.Fertilization- takes place within 12 to 24 hours of ovulation.
2.One day after fertilization- 2 cell stage.
3.Third day after fertilization- 8 to 16 cells stage [morula].
4.Fourth day after fertilization- 32 cells [advanced morula]--->Enters uterine cavity.

5. uterine fluid enters and advanced morula forms blastocyst.

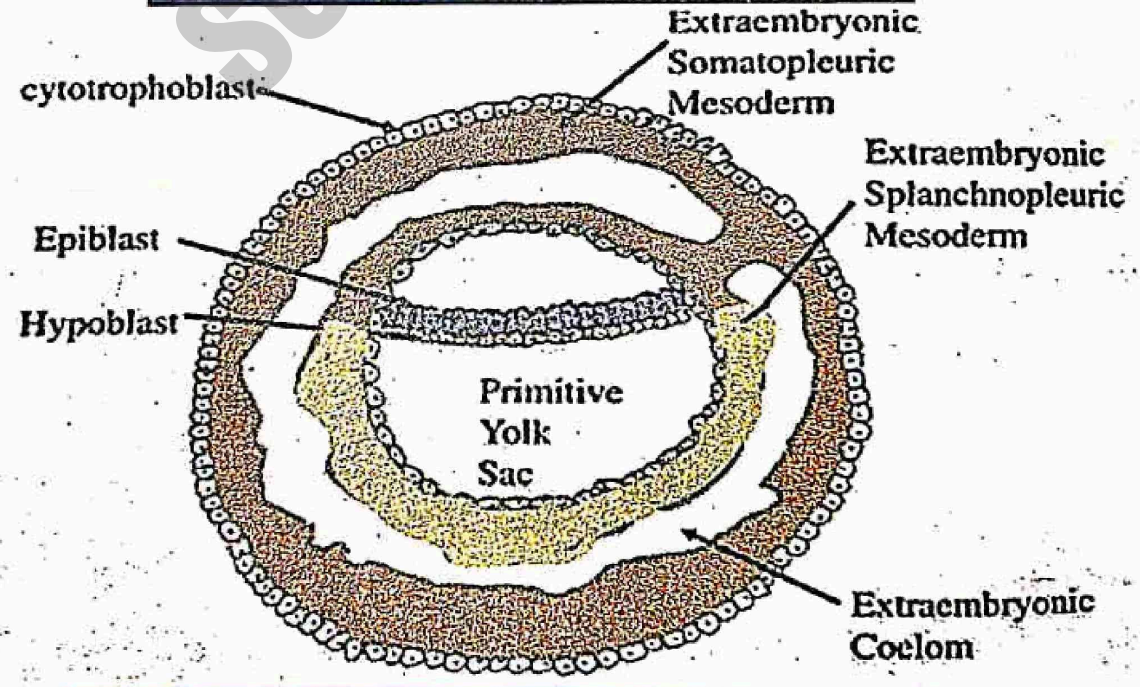
6.At the time of formation - blastocyst contains 58 cells:
*53 cells in outer cell mass forms trophoblast - forms placenta.
*5 cells in inner cell mass forms embryoblast-forms embryo proper

7. Blastocyst is formed on 4 1/2 days after fertilization.
8. 5th day after fertilization zona pellucida disappears.
9. 6th to 7th after fertilization implantation occurs.

SURAJ KUMAR



FIRST WEEK OF DEVELOPMENT



SECOND WEEK OF DEVELOPMENT

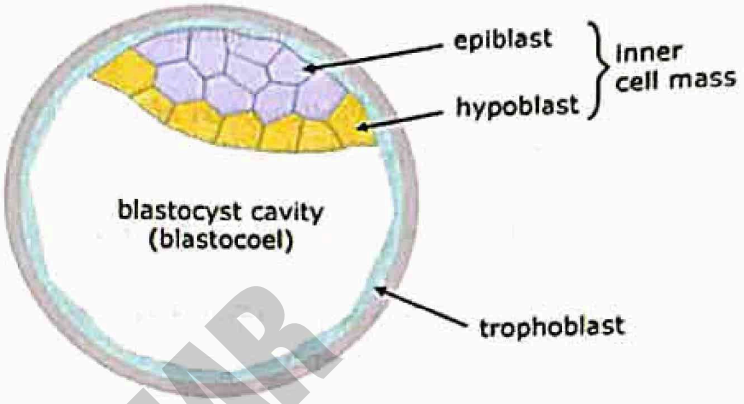
SUBAJ KUMAR

SECOND WEEK OF DEVELOPMENT

*Development of embryoblast and trophoblast :-

-The inner cell mass - forms embryoblast.
-The embryoblast forms two layers - Columnar cells & cuboidal cell

1.Columnar cells
↓
-Epiblast cells
↓
Formed on dorsal end
↓
Encloses amniotic cavity



2.Cuboidal cells -→ Hypoblast cell
-Hypoblast cells
↓
Formed on ventral end -→forms Exocoelom membrane
↓
It is flattened cell---→Encloses the primary yolk sac

*Formation of secondary yolk sac and umbilical cord :-

-The small cavity formed in the Eem joins to form a larger cavity.
-The undivided part of Eem (extra embryonic mesoderm) - cranial to amniotic cavity
↓
forms connecting stalk -→which becomes umbilical cord.

-The cavity in Eem divides it into:
↓
Splanchnopleuric layer of Eem- surrounding the yolk sac makes the yolk sac smaller in size
Called the secondary yolk sac.

*Summary of second week development: -

1.Two layers of inner cell mass - Epiblast
-Hypoblast.

2. Two layers of outer cell mass - Cytotrophoblast
-Syncytiotrophoblast.

3. Amnion and chorion are formed.
4.Two layers of eem somatic and visceral layer.
5.Yolk sac divided into primary and secondary yolk sac.

SURAJ KUMAR

THIRD WEEK OF DEVELOPMENT

*GASTRULATION & NEURULATION OCCUR IN 3RD WEEK
- GASTRULATION - FORMATION OF GERM LAYER
- NEURULATION - FORMATION OF NEURAL TUBE

*GASTRULATION :-

*Prochordal plate: columnar hypoblast cell
-From Buccopharyngeal membrane- ruptures & forms oral cavity

1. Head end :-
-Primitive streak: Proliferation of epiblast.
-Primitive streak indicates the beginning of gastrulation.

2. Tail end :-
-End of and week/beginning of 3rd week

1.Ist germ layer to develop -→Endoderm
2. 2nd germ layer: mesoderm
↑
derived from primitive streak
↑
derived from epiblast

3.Last germ layer: Ectoderm
-The remains of epiblast after the formation of intraembryonic mesoderm.
-All germ layers derived from epiblast > yolk sac

NOTOCHORD DEVELOPMENT

*Introduction →midline structure
- Develop lying Cranial end Primitive streak & Caudal end prochordal plate

* Stage -

1. Cranial end primitive streak -→thick-→called Primitive knot

2. Depression appear-→center of primitive knot called primitive pit

3. Cell of primitive knot multiply-→passing between ectoderm & endoderm to prochordal plate called notochordal process

4. Cavity of promotive pot extend into -→ notochordal process called notochordal canal

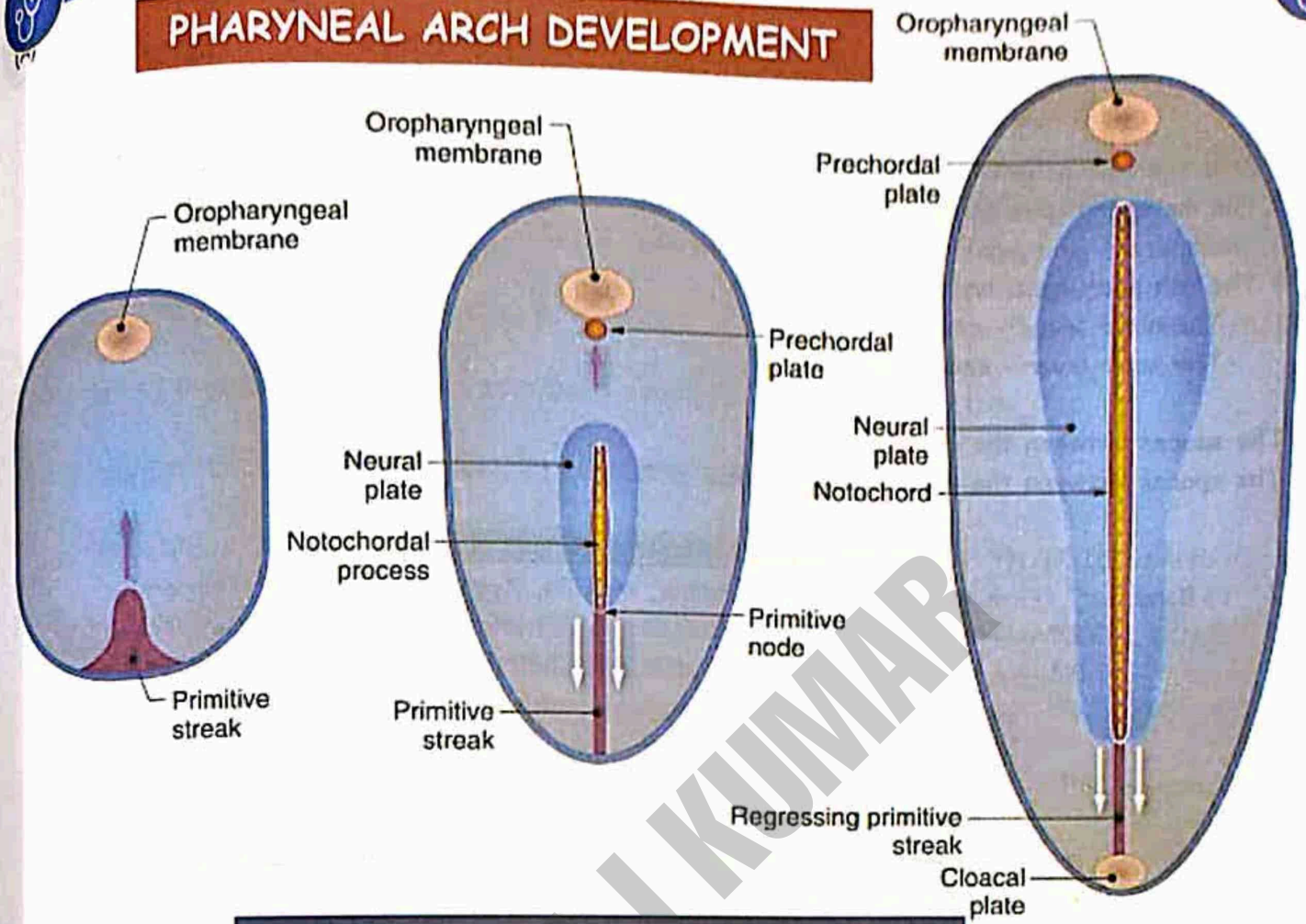
5. Cell of notochordal canal -→intercalated in cell of endoderm and break down ,this stage form communication of amniotic cavity & yolk sac

6. Wall of canal flatted -→notochordal plate

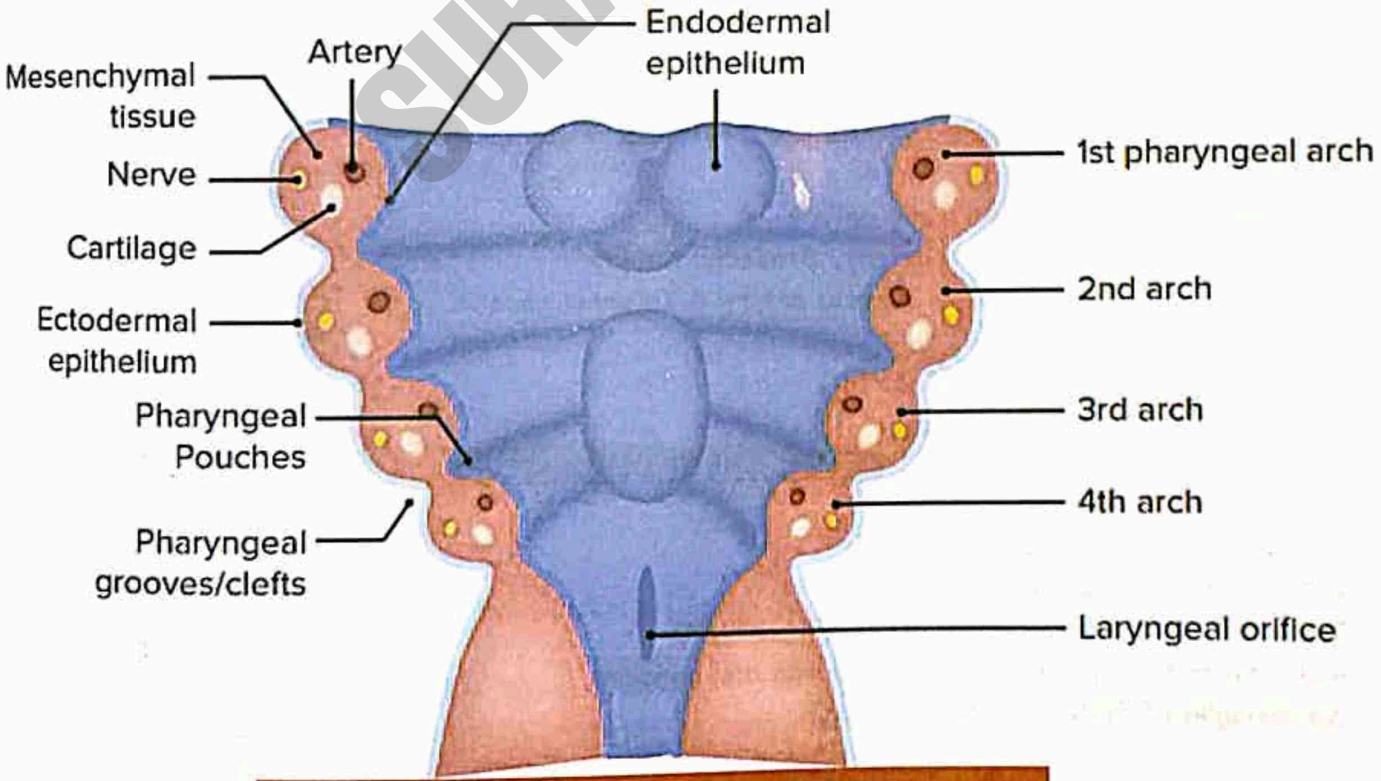
7. Cell proliferation-→flat to curve -→notochord

SURAJ KUMAR

PHARYNEAL ARCH DEVELOPMENT



NOTOCHORD DEVELOPMENT



PHARYNEAL ARCH DEVELOPMENT

SURAJ KUMAR

PHARYNEAL ARCH DEVELOPMENT

1.It is the mesenchymal condensation.
2.The mesenchyme is derived from the NCC, paraxial & lateral mesoderm.
3.There are 5 pharyngeal arches - I, II, III, IV, V, VI.
4.The 5th pharyngeal arch disappears :-
-The outer layer - ectoderm
-The inner layer - endoderm

The spaces between the ectoderm - cleft.
The spaces between the endoderm - pouches

PHARYNGEAL ARCH DERIVATIVES

1. Bone - 1st ---→ Malleus, Incus, Mandible, Maxilla, Zygomatic, Palatine, Temporal
2nd ---→Stapes, Styloid, lesser cornu of hyoid, Upper body of hyoid
3rd ---→Greater cornu of hyoid & lower half of body of hyoid
4th ---→Thyroid cartilage, Arytenoid cartilage, cuneiform cartilage

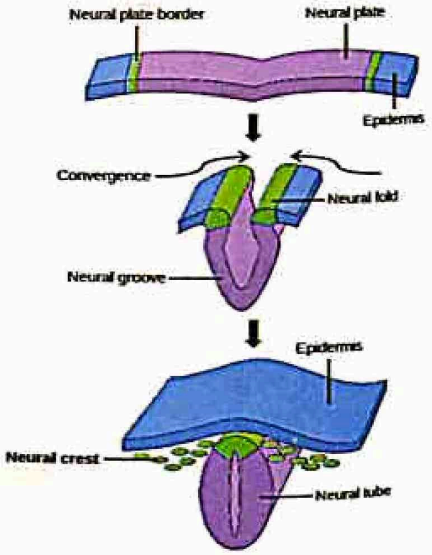
2.Muscle- 1st ---→Muscle of Mastication, Tensor tympani, Tensor plate, Anterior belly Digastric muscle
2nd --→Stapedius, Stylohyoid, Posterior belly of Digastric, Facial muscle
3rd---→Stylopharyngeus
4th --→larynx muscle

3.Nerve - 1st ----→ Trigeminal nerve - maxillary & mandibular
2nd ----→ Facial Nerve
3rd ----→ Glossopharyngeal nerve
4th ----→ Superior laryngeal branch of vagus

4.Artery- 1st-----→ Maxillary artery
2nd ----→ Hyoid artery, Stapedial artery
3rd----→ Common carotid artery & Internal carotid artery
4th ----→Arch of aorta

NEURAL CREST DERIVATIVES

1.Neuron out side CNS
2.Dorsal root of ganglion
3.Sensory ganglion of cranial nerve 5th , 7th ,10
4.Sympathetic & parasympathetic neuron and ganglion
5.Neuroglia of PNS---→ satellite cell



SURAJ KUMAR

THYROID DEVELOPMENT

1. THYROID GLAND--→ DEVELOP FROM --→ THYROGLOSSAL DUCT

2. PARAFOLLICULAR CELL --→DERIVED FROM -→CAUDAL PHARYNGEAL COMPLEX

3. MIDLINE OF MANDIBULAR ACHES-→SEPARATED BY MIDLINE SWELLING -→ TUBERCULUM IMPAR → BEHIND DEPRESSED SURFACE→ TO FORM THYROGLOSSAL DUCT

4. SITE OF ORIGIN OF DIVERTICULUM -→CALLED FORAMEN CAECUM

5. DIVERTICULUM GROW DOWN -→ MIDLINE INTO NECK

6. PROLIFERATION OF DIVERTICULUM CELL -→ FORM 2 LOBES OF THYROID GLAND

7. THYROID COME--→ INTIMATE RELATIONSHIP WITH CAUDAL PHARYNGEAL COMPLEX----→FUSE TO FORM→ PARAFOLLICULAR CELL

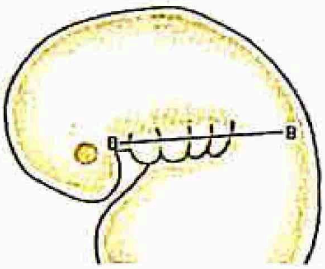
TONGUE DEVELOPMENT

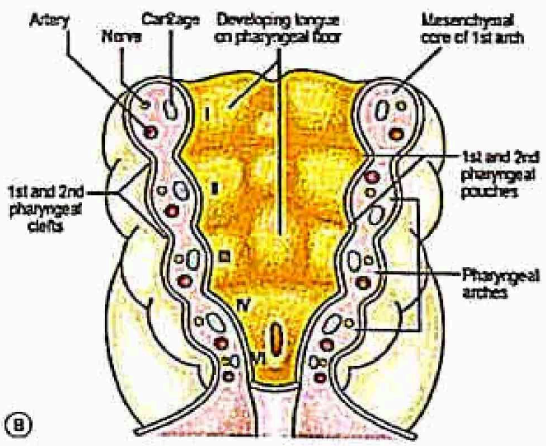
*Development of tongue

1. Epithelium of the tongue : - Anterior 2/3rd- surface ectoderm.
-Posterior 1/3rd- endoderm.

2. All the muscles are derived from myotome of occipital somites [Except palatoglossus CIV pharyngeal arch].

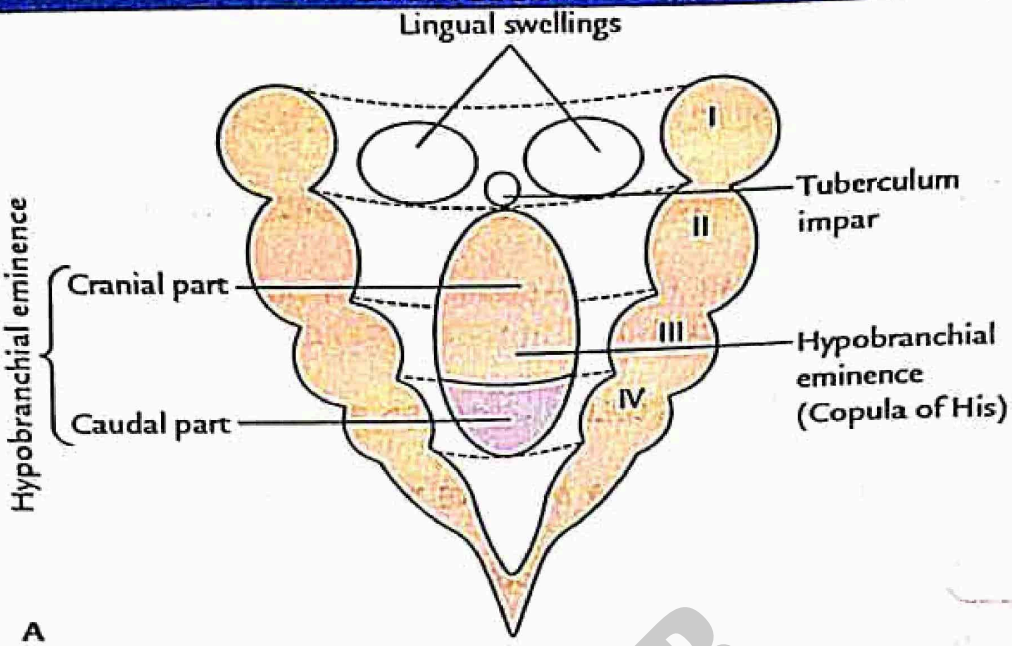
3. Nerve supply of the tongue
-Anterior 2/3rd tongue : mandibular nerve-lingual nerve.
-Posterior 1/3rd tongue : 9th nerve/glossopharyngeal nerve.
-Posterior most part of tongue:10 nerve.



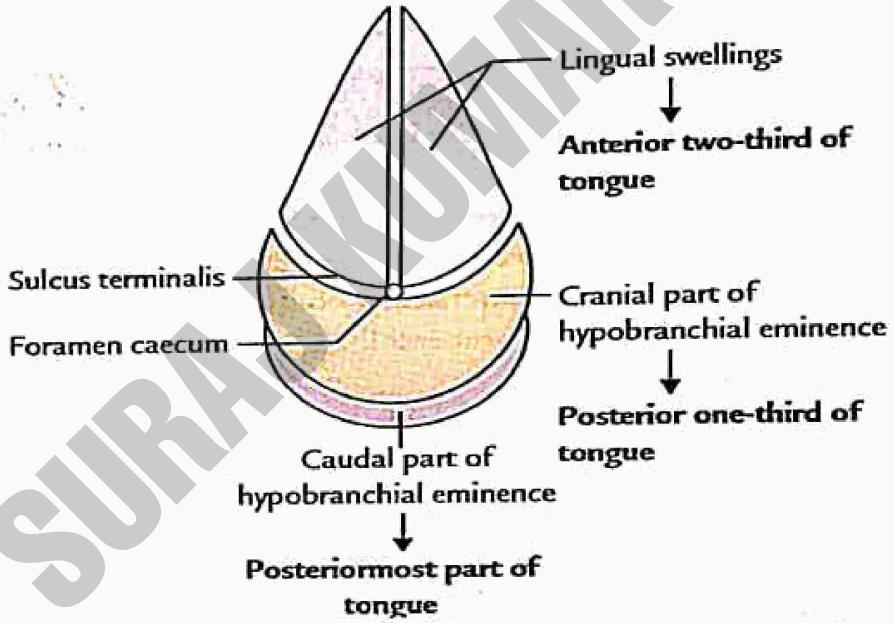


*On the floor of Ist arch there are three swellings:-
1. Two lingual swellings
2. One tuberculum impar
↓
Forms anterior 2/3rd of tongue
3. Floor of II, III, IV arches - hypobranchial eminence
4. The III arch mesoderm of hypobranchial eminence forms
↓
posterior 1/3rd of the tongue.
5. The IV arch mesoderm of hypobranchial eminence forms
↓
Posterior most part of the tongue and epiglottis.
6. The II arch mesoderm does not contribute to tongue development
↓
Because mesoderm of III arch overgrows 11 & fuse with anterior a/3rd of tongue

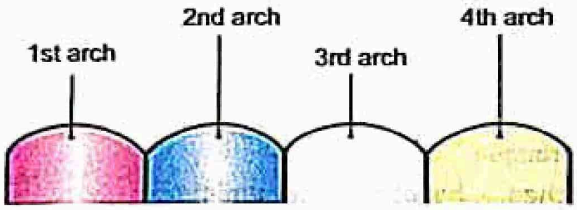
SURAJ KUMAR

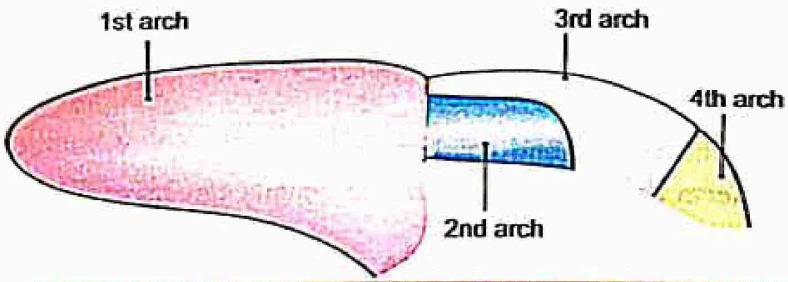


A



B





TONGUE OF DEVELOPMENT

SURAJ KUMAR

HEART TUBE FORMATION

1.Bulbous cordis :-
   a. Proximal 1/3 will form the rough part of the right ventricle
   b. Conus part will form infundibulum of both right ventricle and left ventricle
   c. Spiral septum [derived from neural crest cell] is developed in truncus arteriosus and divides it into aorta and pulmonary trunk

2.Primitive ventricle: develops into rough part of left ventricle

3.Bulbo ventricular sulcus: filled with septum which forms Interventricular septum

4.Primitive atrium: will form the rough part of both right and left atrium

5.Sinus venosus:
   a. Body absorbed into right atrium along with right horn
   b. Left horn obliterates and forms coronary sinus
   c. Right common cardinal vein will form superior vena cava
   d. Right vitelline vein will form inferior vena cava (cranial part)

*LOOPING OF HEART - Due to dextro looping atrial chamber goes above and behind, ventricle chamber move front and below

INTERVENTRICULAR SEPTUM FORMATION

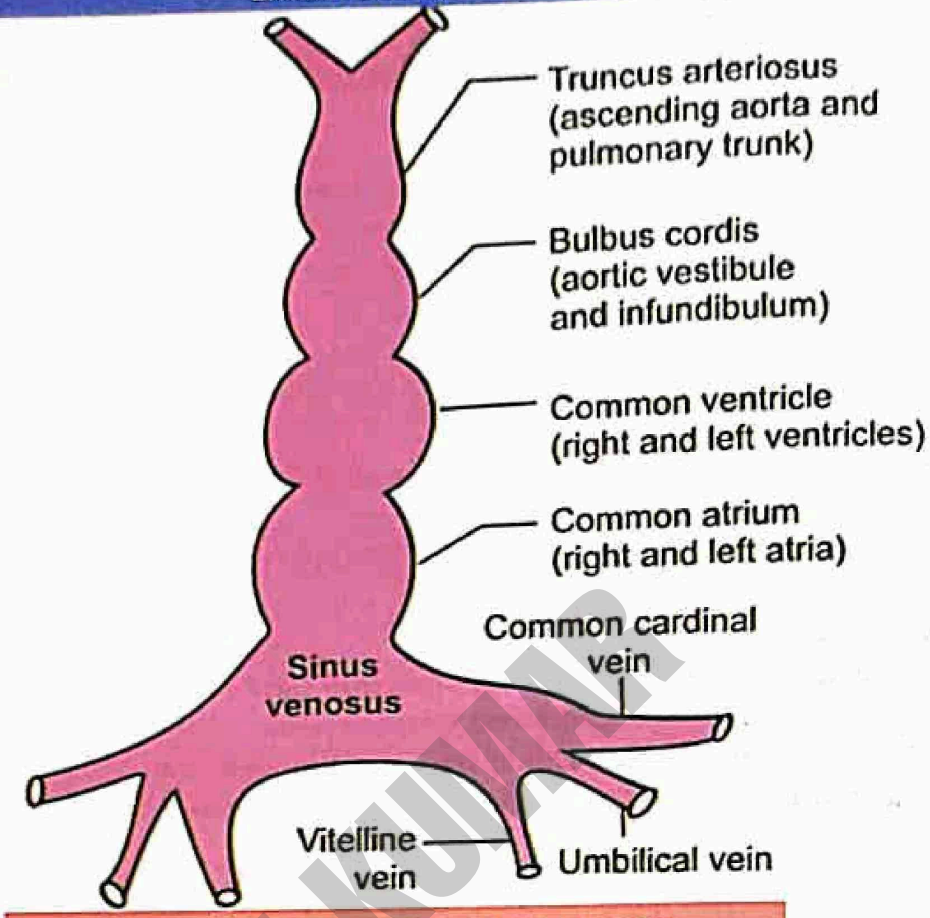
1) Muscular part: from the muscular proliferation of Bulb ventricular cavity
2) Membranous part: derived from proliferation of Av cushion
3) Bulbar part: derived from spiral septum
   - Proximal 1/3 of bulbous cordis (RV) + Primitive ventricle (LV)
   ↓
   Common ventricular chamber

RIGHT ATRIUM DEVELOPMENT

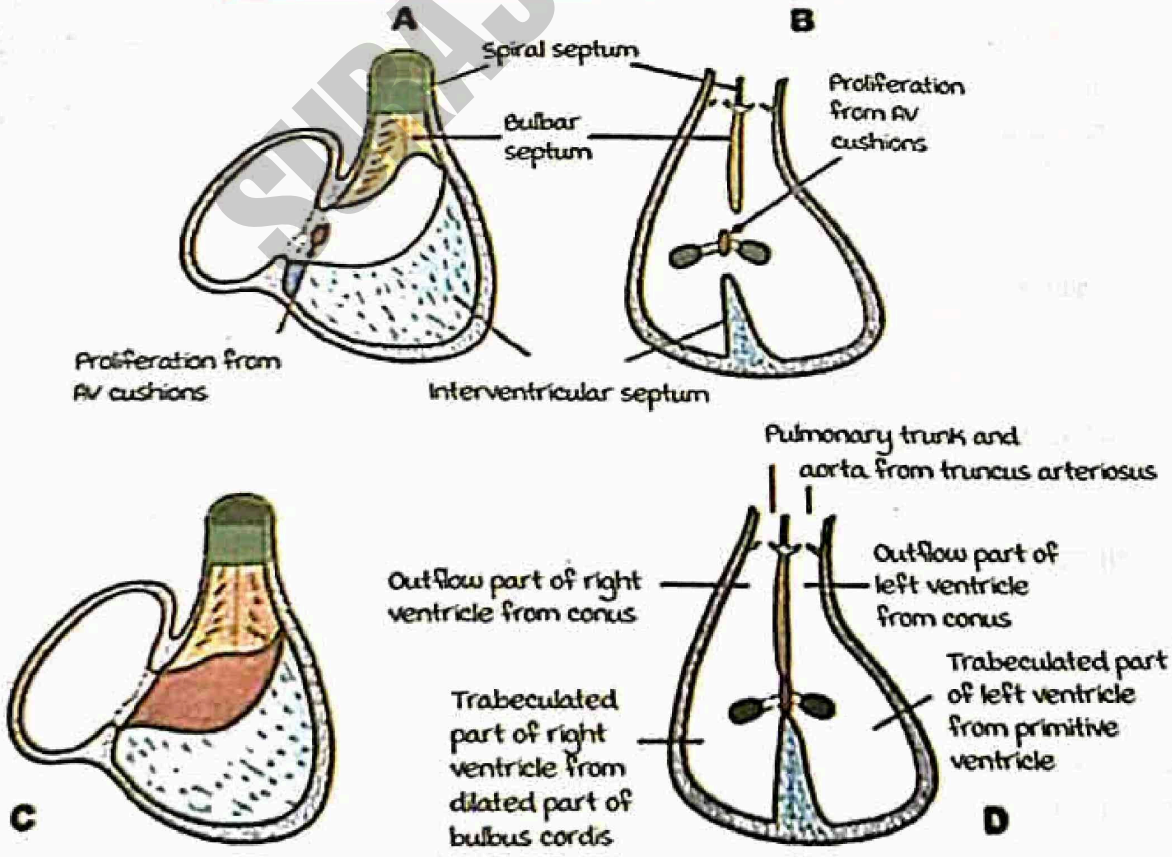
*RIGHT ATRIUM -→ 3 SOURSE
1.RIGHT HALF OF PRIMITIVE ATRIUM - FROM ROUGH TRABECULATED PART -IN FRONT CRISTA TERMINALIS
2.ABORPTION OF RIGHT HORN OF SINUS VENOSUS - SMOOTH PART BEGIND CRISTA TERMINALIS
3.ABORPTION OF RIGHT HALF OF AV CANAL FORM- MOST VENTRAL -→SMOOTH PART

*ABSORPTION OF SINUS VENOUS - CORONARY SINUS & VENA CAVA ARE SEEN
*RIGHT VENOUS VALVE --→BECOME - CRISTA TERMINALIS, INFERIOR VENA CAVA
*LEFT VENOUS VALVE --→ FUSED WITH ATRIAL SEPTUM

SURAJ KUMAR



HEART TUBE FORMATION



INTERVENTRICULAR SEPTUM FORMATION

SURAJ KUMAR

DEVELOPMENT OF FACE

*Face: Derived from - Frontonasal process
-medial & lateral process
-maxillary process
-mandibular process
-Lateral nasal process

1.Frontonasal process - forms forehead.
2.medial Nasal Process - fuse with each other
↓
Forms philtrum and primitive palate.
3. Lateral Nasal process - forms ala of the nose.
4.maxillary process - upper lip and cheek.
5.mandibular process - lower lip and chin.
6.The maxillary process fuses with MNP and LNP

DEVELOPMENT OF SOFT PALATE

*PALATE CONSIST OF 2 PART - PRIMARY PALATE & SECONDARY PALATE
*PRIMARY PALATE DEVELOPS FROM -→ FRONTONASAL PROCESS
*SECONDARY PALATE DEVELOPS FROM-→ MAXILLARY PROCESS

*DEVELOPMENT OF PALATE BEGINS IN - 6TH WEEK
*DEVELOPMENT OF PALATE END IN - 12TH WEEK

*PALATE DEVELOP IN 2 STAGES-
1.DEVELOPMENT OF PIRMARY PALATE
2.DEVELOPMENT OF SECONDARY PALATE

*DEVELOPMENT OF PRIMARY PALATE - FORM BY FUSION OF -→2 MEDAIL PROCESS OF FRONTONASAL PROCESS

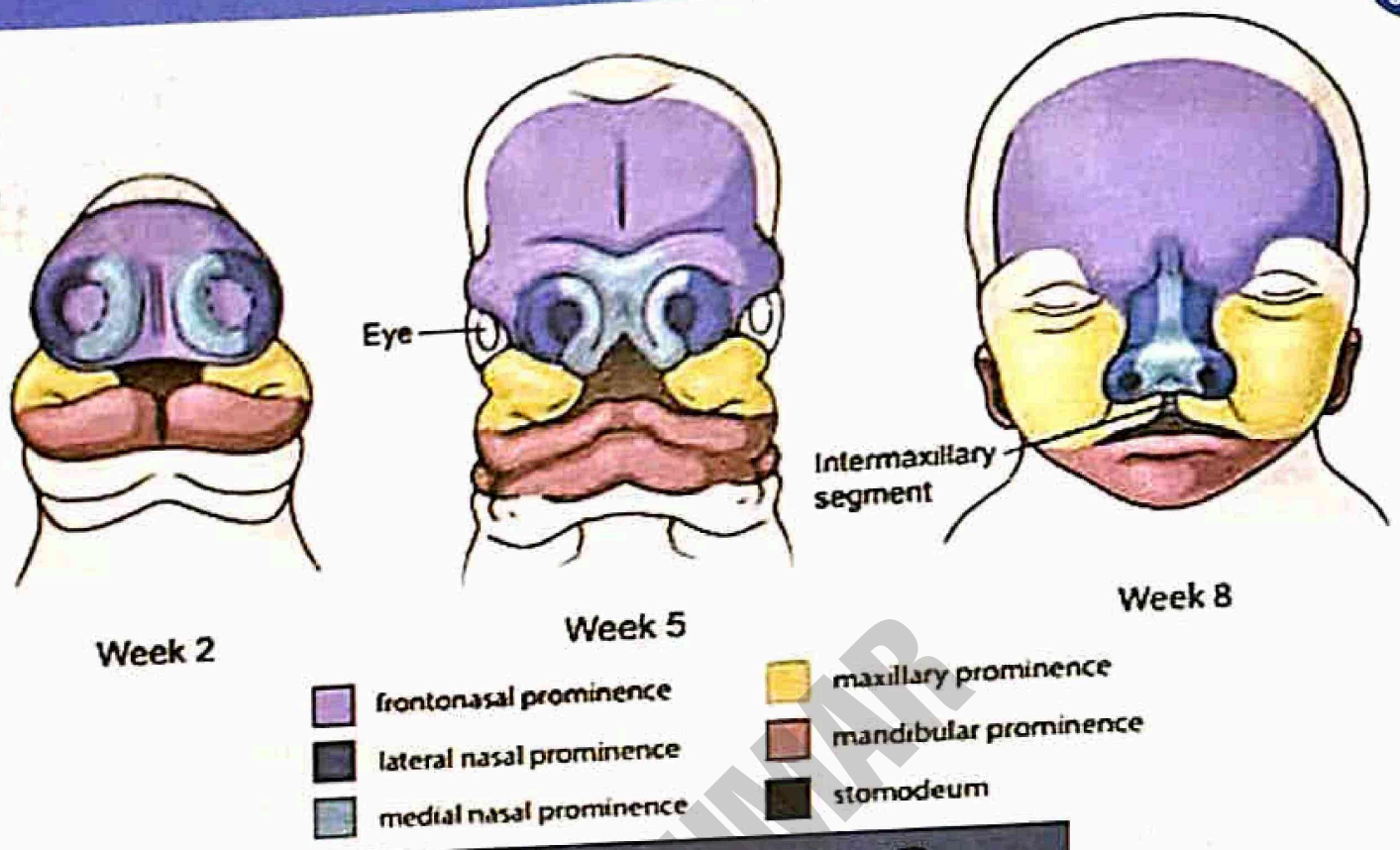
*DEVELOPMENT OF SECONDARY PALATE - FORM BY FUSION OF 2 SHELVE LIKE OUTGROWTH CALLED PALATINE SHELVES - APPEAR IN 6TH WEEK

*INTIALLY - GROWTH - DOWNAWARD, MEDIALLY AND BELOW TONGUE

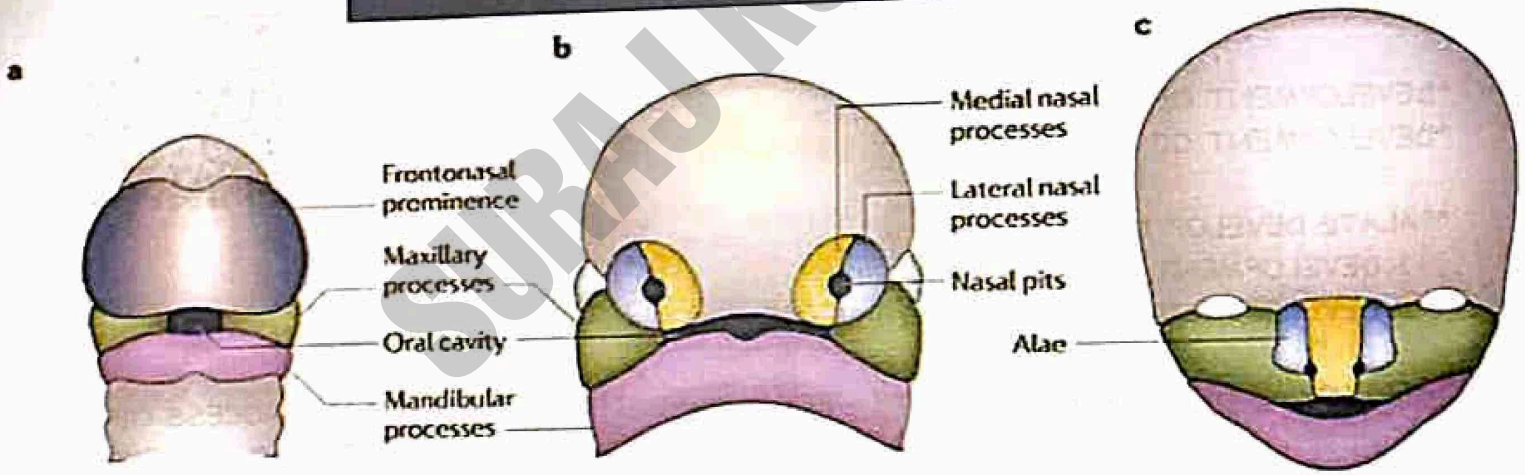
*7TH AND 8TH WEEK - ASSUME HORIZONTAL POSTION ABOVE TONGUE & FUSE TO FORM SECONDARY PALATE

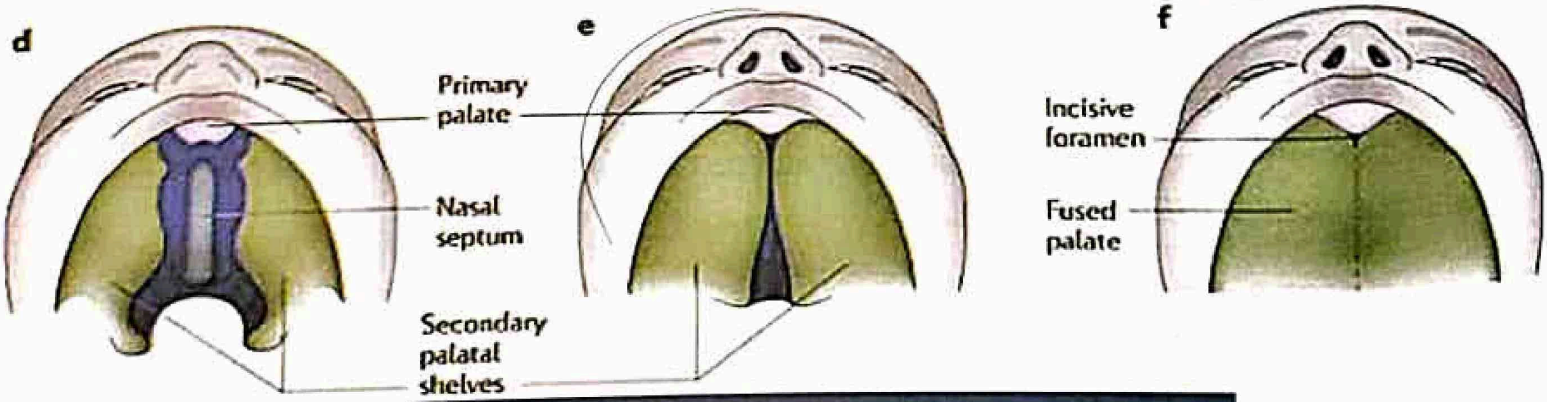
*POSTERIOR PART OF THESE--→ FAIL RO OSSIFY AND FORM - SOFT PALATE

SURAJ KUMAR



DEVELOPMENT OF FACE





DEVELOPMENT OF SOFT PALATE

SUBAJ KUMAR

DEVELOPMENT OF PANCREAS

*Pancreas Development from - Two buds : Dorsal pancreatic bud
Ventral pancreatic bud

*Ventral bud moves dorsally, lies below the dorsal bud in - 7th week

1.ventral bud - part of head of pancreas + uncinate process
2.Dorsal bud- remaining entire pancreas

*PANCREATIC DUCT -
1.Main pancreatic duct derived from distal part of dorsal bud & whole of ventral bud
2.Accessory Pancreatic Duct - derived from - proximal part of dorsal bud

*PARENCHYMA DEVELOPS - from - surrounding mesoderm

DEVELOPMENT OF DUODENUM

*Duodenum --->Derived from foregut + midgut
*Duodenum Supply - celiac trunk & superior mesenteric artery
*During rotation of stomach, duodenum comes to right side, forms a C - shaped loop
*Mesoduodenum suspends duodenum from Posterior abdominal wall.
↓
Disappears- duodenum becomes retroperitoneal.

* Duodenum retroperitoneal except the 1st proximal part of duodenum

DEVELOPMENT OF LIVER

*Development of liver, From a sources-liver bud and septum transversum
1.Liver bud- In foregut tube, Forms parenchyma of liver & biliary system
2.Septum transversum-Forms connective tissue of liver, capsule, sinusoids, Kupffer cells.

ROTATION OF MIDGUT

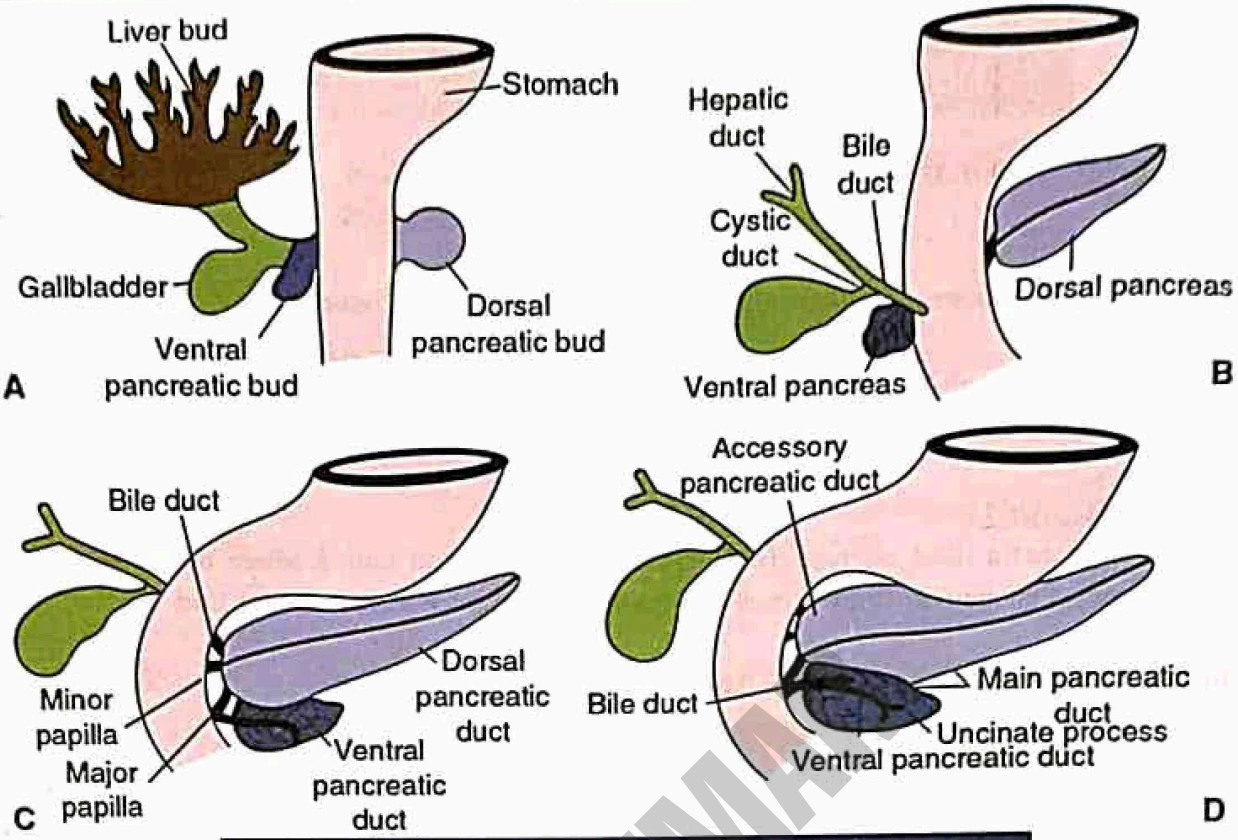
*During reduction, midgut rotates 3 times, each time by 90° in counter clockwise/anti-clockwise manner.

1.1st rotation: - Pre arterial segment - right side.
-Post arterial segment - left side.
2. 2ND rotation:-Pre arterial segment - elongates and enters abdominal cavity and rotates by 90°anti-clockwise.

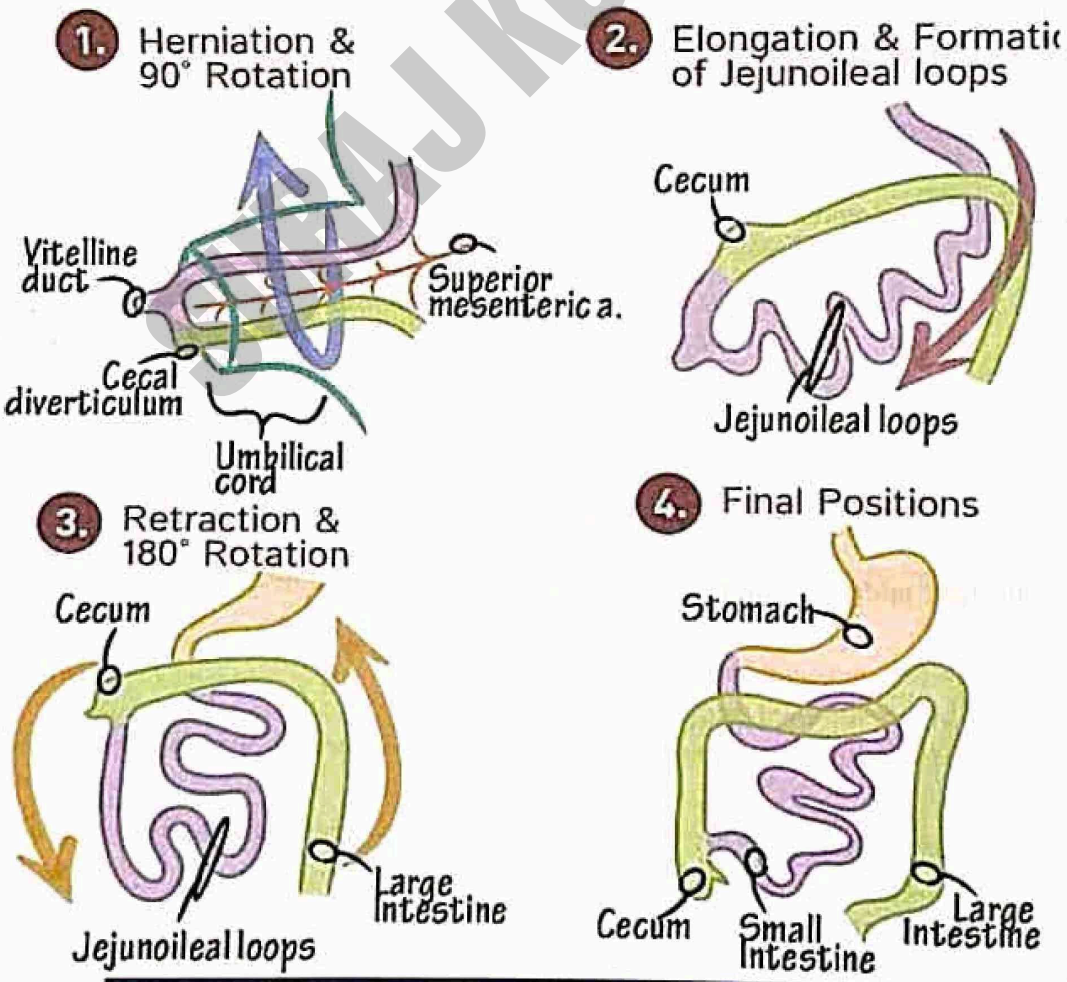
*Pre arterial segment comes to lie behind the superior mesenteric artery.

3.3rd rotation: -Post arterial segment - enters abdominal cavity and rotates by 90°anti-clockwise.

SURAJ KUMAR



DEVELOPMENT OF PANCREAS



ROTATION OF MIDGUT

SIRAJ KUMAR

DEVELOPEMNT OF KIDNEY

*Nephrogenic cord – forms ---→Kidney, mesonephric duct & Paramesonephric duct.

1. At 4th week – pronephros appears, functions as the kidney & disappears by end of 4th week.
2. By the end of 4th week – mesonephros appears, functions as interim Kidney up to a months.
3. By 5th week – metanephros appear and starts its function by 3rd month.
4. A diverticulum arises from the mesonephric duct called ureteric bud
   ↓
   which forms metanephric blastema.

*The kidneys are derived from two sources:-

1. Excreting component – derived from metanephros.
   - Metanephric blastema – contact with ampulla → under go differentiation form nephron

2. Collecting component – derived from ureteric bud.
   - uretic bud grow toward metanephric blastema → become ampulla
   -ampulla –→ form pelvis of kidney – then form minor & major calyx and collecting tubules

*The kidneys become functional from 12th week.

*Ascent of kidney :- Initial in sacral region –→ kidney ascend to lumbar region

*Rotation of kidney – Hilum – first anterior, later hilum come medially

DEVELOPEMNT OF HINDGUT

*Development of hindgut

1. Diverticulum arising from hindgut is called allantois.
2. Cloaca is a part of hindgut distal to allantois.
3. The urorectal septum – separates the cloaca and cloacal membrane.
4. The cloaca divides into – primitive urogenital sinus
   -Primitive rectum.
5. The cloacal membrane divides into – anal membrane & urogenital membrane.
6. The primitive rectum forms – rectum, Anal canal up to pectinate line

*Derivation of upper anal canal :

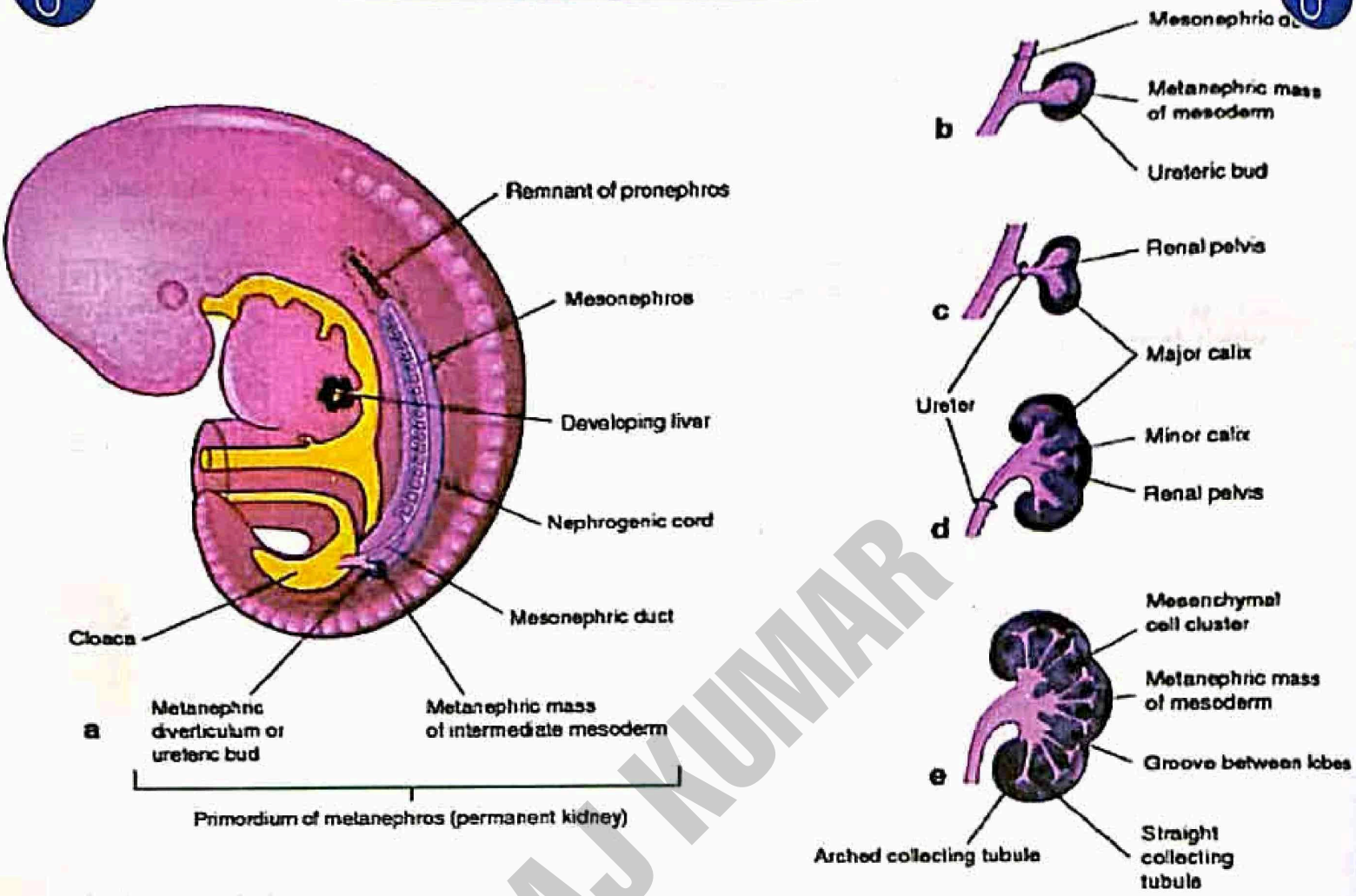
- Endoderm
  ↓
  Gut ---→ Hindgut –→ Cloaca –→ Primitive rectum –→ upper part of anal canal

*Derivation of lower part of anal canal :

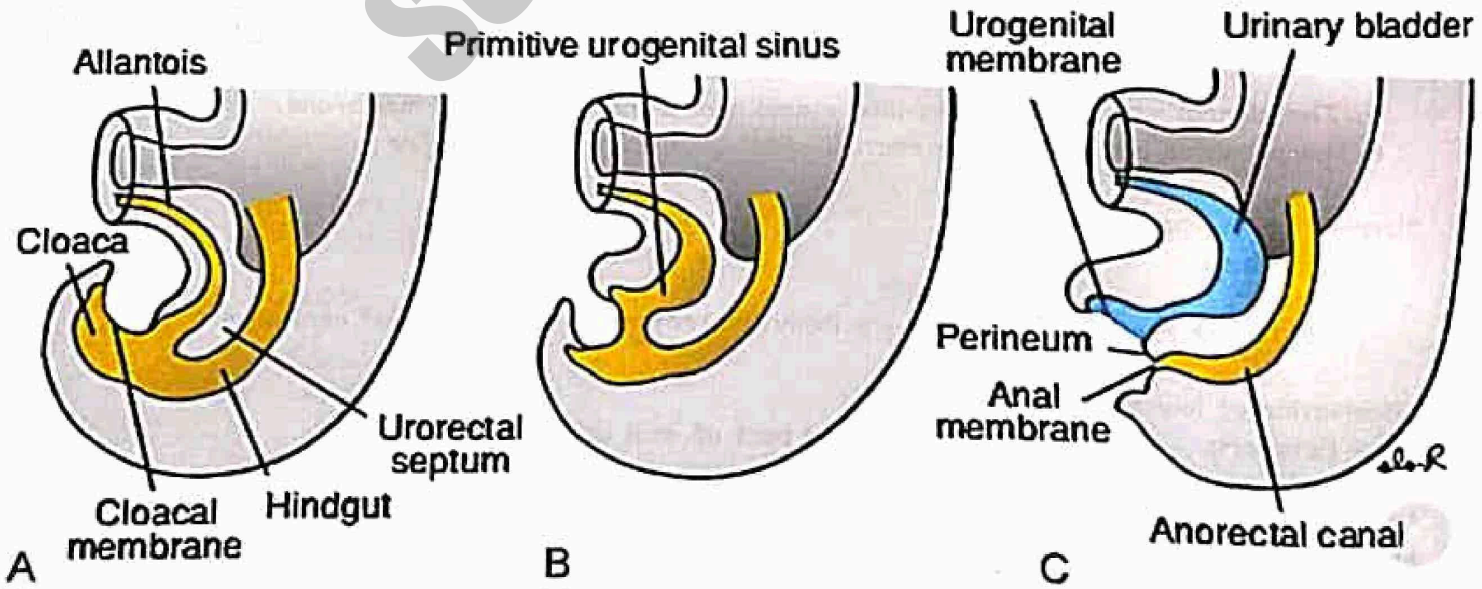
- Ectoderm --→ Proctodeum --→lower part of anal canal

*The anal membrane is formed at the junction of upper and lower anal canal.
*It ruptures by the 9th week therefore, upper and lower anal canal communicate.
*The pectinate line is formed at the site of formation of anal membrane.

SURAJ KUMAR



DEVELOPEMNT OF KIDNEY



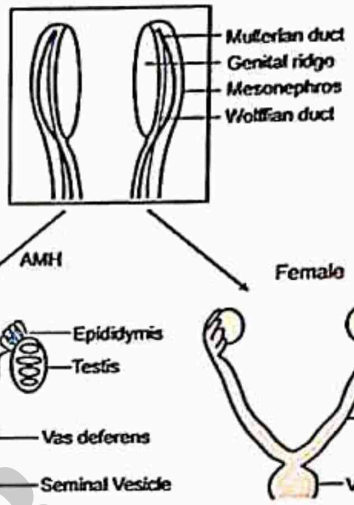
DEVELOPEMNT OF HINDGUT

SURAJ KUMAR

DEVELOPEMNT OF GONADS

*Genotype that is xx/xy is determined at the time of fertilization itself

*Phenotypic differentiation will start only at 7th week
. until 6th week gonads are indifferent gonads
. Primordial germ cells reach the gonads at 5th week



*Gonads:
. Testis develops at 7th week
Ovary develops at 8th - 10th week

*External genitalia:
. In males: 14th week
. In females: 11th week

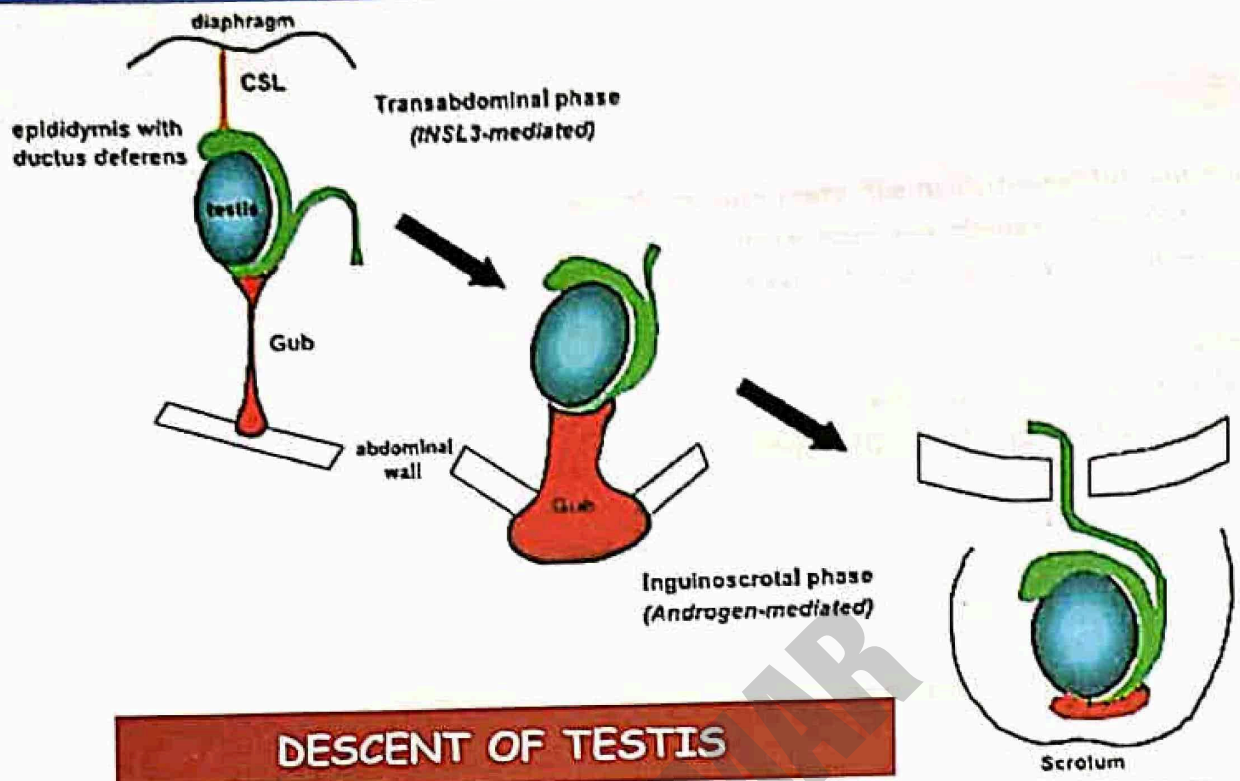
*Sex determination can be done at 15th week (illegal to reveal)

Structures	Male derivative	Female derivative
1. Gonadal ridge	-Testis	-Ovary
2. Genital Tubercle	-Penis	-Clitoris
3. Genital swelling	-Scrotum	-Labia majora
4. Mesonephros	-Efferent tubules	-Epoophoron
5. Mesonephric duct	-Appendix of epididymis vas deferens Epididymis Ejaculatory duct Seminal vesicle	-Appendix of ovary
6. Paramesonephric duct	- Appendix of Testis	-Fallopian tube uterus vagina (upper part)

SURAJ KUMAR

DESCENT OF TESTIS

1. Descend to reach scrotum.
2. Each testis begin to descent during second month of intrauterine life
3. Reach iliac fossa by 3RD month
   - Deep inguinal ring - 4th to 6th month
   -Inguinal canal 7th month
   -Superficial inguinal ring → 8th month
   -Bottom of scrotum by 9th month
4. Extension of peritoneal cavity - Called processus vaginalis-→Preceds the descent of tests into the scrotum
5. Processus vaginalis close above the testis
6. Doescent doesn't occur after one year of age



DESCENT OF TESTIS

STUDY TIPS FOR MAXIMUM MARKS

1) IN EMBRYOLOGY ONLY SHORT NOTES ARE ASKED

2) IN LONG QUESTION ,IT IS USUALLY ASKED AS SUB HEADING
(FOR EXAMPLE - DEVELOPMENT OF RIGHT ATRIUM)

3) ONLY WRITE MAIN CONTENT WITH DIAGRAMS, ABOUT 1/2 OR 1 PAGE

SURAJ KUMAR

PREVIOUS YEAR UNIVERSITY QUESTIONS

*SHORT NOTES - GENERAL EMBRYOLOGY

1.NOTOCORD DEVELOPMENT(VIMP)
2.PHARYNGEAL ARCHES DERIVATIVES

*SYSTEMIC EMBRYOLOGY -

1.DEVELOPMENT OF TONGUE
2. DEVELOPMENT OF THYROID
3. DEVELOPMENT OF SOFT PALATE
4. DEVELOPMENT OF PANCREAS
5. DEVELOPMENT OF DEUDENUM
6. DEVELOPMENT OF ANAL CANAL(HIND GUT)
7. DEVELOPMENT OF RIGHT ATRIUM
8. DEVELOPMENT OF INTERVENTRICULAR SEPTUM
9.ROTATION OF MID GUT
10.DESCENT OF TEST

GENERAL ANATOMY

EPIPHYSIS

- The ends and tips of a bone which ossify from secondary centre are called epiphysis.
- **Classification:**

According to number of epiphysis -

SIMPLE
- ends of long bones develop from many epiphysis, which fuse independently with shaft.

e.g. femur.

COMPOUND
- ends of bones develop from many centre which unite to form single epiphysis which in turn fuses with shaft.

e.g. humerus.

Based on function -

PRESSURE
- Articular
- takes part in weight transmission.

e.g. head of Humerus.

TRACTION
- Non articular
- provide attachment to one or more tendon which exert traction on epiphysis.

e.g. Trochanter of femur.

ATAVISTIC
- phylogenitically independent bone which in man becomes fused to another.

e.g. Coracoid process of scapula.

ABERRANT
- not always present.

e.g. at head of first metacarpal.

COMPOUND
- in some bones 2-3 smaller epiphysis join to form compound epiphysis before joins a shaft.

e.g., upper end of humerus.

SURAJ KUMAR

BONES

- Bone is one-third connective tissue.
- It is impregnated which Calcium constituting remaining two-third.

- **Classification -**

According to Shape -

- **LONG BONES** - has elongated shaft (diaphysis) and two expanded ends (epiphysis) which are smooth & articular.
  - Typical long bones - two secondary epiphysis.
  - Miniature/short long bone - one epiphysis.
  - Modified long bones - no medullary cavity.

- **SHORT BONES** – Shape is usually cuboid / scaphoid.
- **FLAT BONES** – form boundaries of certain body cavities.
- **IRREGULAR BONES** – Hip bone.
- **PNEUMATIC BONES** – Irregular bone with large air space.
- **SESAMOID BONES** – Bony nodule embedded in tendon / joint capsules.
- **ACCESSORY BONES** – Not always present. May occur as ununited epiphysis developed from extra centre of ossification.

Developmental Classification –

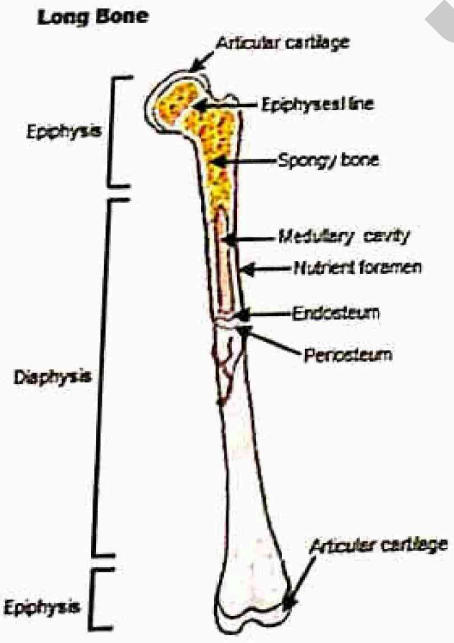
- **MEMBRANE (DERMAL) BONES** – Ossify in membrane.
- **CARTILAGINOUS BONES** – Ossify in cartilage.
- **MEMBRANO-CARTILAGINOOS BONES** – Ossify partly in membrane and partly in cartilage.
- **SOMATIC BONES** – Most bones are somatic.
- **VISCERAL BONES** – few and develop from pharyngeal arch.

Regional Classification

- **AXIAL SKELETON** – includes skull, vertebrae, thoracic cage.
- **APPENDICULAR SKELETON** – includes bone of limb

Structural Classification

- **MACROSCOPICALLY**
  - **COMPACT** – bone is dense, texture is ivory but extremely porus.
  - **CANCELLOUS** – Spony or trabaecular bone is open in texture and is made up of meshwork of trabeculae.
- **MICROSCOPICALLY**
  - **LAMELLAR** – most of mature bones are composed of thin plates of bony tissue called lamallae.
  - **WOVEN** – Seen in fetal bone, fracture repair or cancer bone.
  - **FIBROUS** – young foetal bone.
  - **DENTINE**, **CEMENT** – Occur in teeth.



SURAJ KUMAR

BLOOD SUPPLY of LONG BONE

- Arterial Supply is derived from -

NUTRIENT ARTERY -

enters cortex through nutrient foramen → divides into ascending and descending branch → at end, bend to form hairpin loop.

- Supplies metaphysis, inner 2/3rd cortex, medullary cavity.
- direction of nutrient foramen is away from growing end, i.e., towards elbow away from knee.
- any blood infection causes emboli can block nutrient artery at bend and cause osteomylitis.

PERIOSTEAL ARTERY -

- numerous beneath muscular and ligamentous attachments.
- enter Volkmann's canal to supply outer 1/3rd cortex.

EPIPHYSEAL ARTERY -

- derived from periarticular vascular arcades found on non-articula bony surface.

METAPHYSEAL ARTERY -

- derived from neighbouring systemic vessels.
- pass directly into metaphysis and reinforce metaphysial branch from primary nutrient artery.

LONG SHORT BONE

- Nutrient artery → enters middle of shaft → divide to form a plexus.
- major part of bone supplied by periosteal artery.

SHORT BONE

- Supplied by numerous periosteal arteries which enter their non-articular surface.

SURAJ KUMAR

VERTEBRA

- Body – Supplied by anterior posterior vessels
- Vertebral arch – Supplied by large vessels entering base of transverse process.

RIBS

- Nutrient artery – enter just beyond tubercle.
- Periosteal artery

VENOUS DRAINAGE –

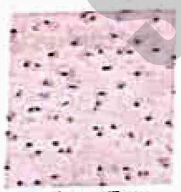
- Veins are large and numerous – Cancellous bone.
- Accompany artery in Volkmann's canal – Compact bone.

CARTILAGE

- Cartilage is a connective tissue composed of cells and fibres embedded in a firm, gel-like matrix which is rich in a mucopolysaccharide.

- Classification → ELASTIC
  - HYALINE
  - FIBROCARTILAGE

→ **HYALINE** – • located in articular cartilage of long bones, epiphyseal plates, nasal cartilage, thyroid, trachae, bronchi, coastal cartilage.


Hyaline cartilage

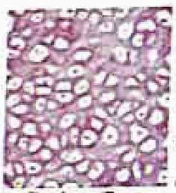
- Bluish white in colour.
- Shiny or translucent.
- Very thin fibres.
- flexible
- perichondrium is present.

→ **FIBROCARTILAGE** – • located in intervertebral disc, Temporomandibular joint, sternoclavicular joint.

Fibrocartilage

- Glistening white in colour.
- Opaque appearance.
- Numerous white fibres.
- perichondrium is absent.

→ **ELASTIC** – • located in pinna, external auditory meatus, eustachian tube, epiglottis.


Elastic cartilage

- yellowish in colour.
- opaque appearance.
- numerous yellow fibres.
- Most flexible.
- perichondrium is present.

SURAJ KUMAR

JOINT

- It is a junction between two or more cartilage.
- Device to permit movement.

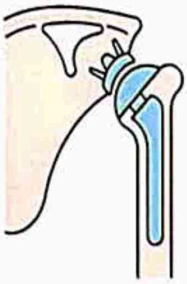
Classification –

Structural Classification -

- **FIBROUS JOINTS** – Immovable or permit slight movement
  - SUTURES – present only in skull.
  - SYNDESMOSIS – fibrous union b/w bones.
  - GOMPHOSIS – peg and socket junction between tooth and socket.
- **CARTILAGINOUS JOINTS** – bones are joined by plates of cartilage.
  - PRIMARY / SYNCHONDROSIS – temporary in nature as after a certain age cartilage plate is replaced by bone.
  - SECONDARY / SYMPHYSES – permanent and persist throughout life.
- **SYNOVIAL JOINTS** – most evolved, most mobile.
  - BALL AND SOCKET JOINT
  - SELLAR / SADDLE JOINT
  - CONDYLAR / BICONDYLAR JOINT
  - ELLIPSOID JOINT
  - HINGE JOINT
  - PIVOT JOINT
  - PLANE JOINT

ELBOW JOINT

KNEE JOINT



SHOULDER JOINT

HIP JOINT

Functional Classification

- SYNARTHROSES – Immovable.
- AMPHIARTHROSES – Slightly movable.
- DIARTHROSES – freely movable.

ANKLE JOINT

Regional Classification

- SKULL TYPE – Immovable.
- VERTEBRAL TYPE – Slightly movable.
- LIMB TYPE – freely movable.

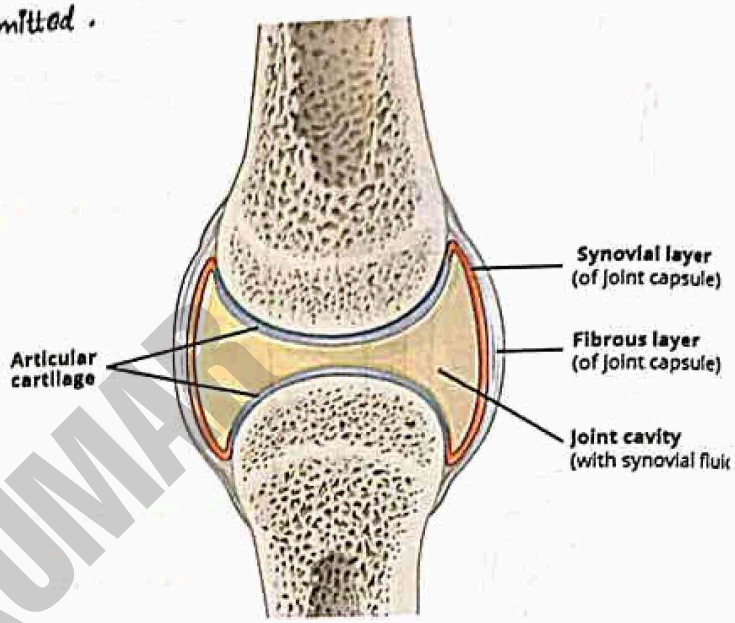
According to number of Articulating Bones

- SIMPLE JOINT – when only two bones articulate.
- COMPOUND JOINT – more than two bones articulate in capsul
- COMPLEX JOINT – when joint cavity is divided by an intra-articular disc.

SURAJ KUMAR

SYNOVIAL JOINT

- most evolved and most mobile.
- **Characters –**
  - Articular surface are covered with hyaline cartilage.
  - Synovial fluid is filled between articular surfaces.
  - Joint cavity is surrounded by articular capsule.
  - Varying degree of movement are always permitted.



- **Classification**

1. PLANE SYNOVIAL JOINT
- Articular surfaces are more or less flat.
- permit gliding movement in various directions

EXAMPLE –
- Intercarpal joints.
- Intertarsal joints.
- Carpometacarpal (except first).
- Tarsometatarsal
- Intermetatarsal.

2. HINGE JOINT
- Articular surface are pulley shaped.
- Movements are permitted in one plane.

EXAMPLE –
- Elbow joint.
- Ankle joint.

3. PIVOT JOINT
- Comparises a central bony pivot surrounded by osteoligamentous ring.
- Movement permitted in one plane.

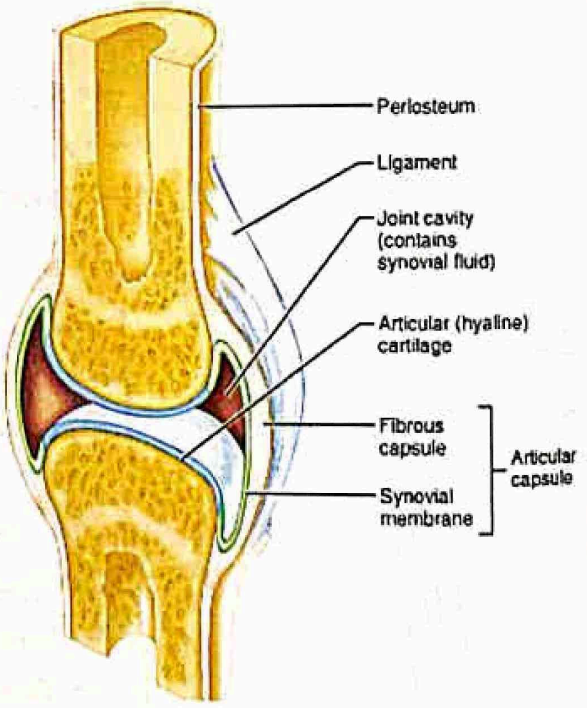
EXAMPLE –
- Median atlantoaxial joint.
- Superior and Inferior Radioulnar joint.

4. CONDYLAR JOINT
- Articular surface includes two distinct condyles.

EXAMPLE –
- Knee joint.
- TMJ.

5. ELLIPSOID JOINTS
- Articular surface include an oval.
- free movements are permitted.

EXAMPLES –
- wrist joint.
- Atlanto-axial joint.



6. SADDLE JOINT
- Articular surface are reciprocally concavoconvex.
- Movement are similar to those permitted by an ellipsoid joint.

EXAMPLE –
- Sternoclavicular joint.
- first carpometacarpal joint.

7. BALL-SOCKET JOINT
- Articular surface include a globular head filling into a cup-shaped socket.
- Movement occur around indefinite axis.

EXAMPLE –
- Shoulder joint.
- Hip joint.

SURAJ KUMAR

KARYOTYPING

- also called 'IDIOGRAM'.
- standard arrangement of photographed metaphase spread of chromosomes in decreasing length is called 'KARYOTYPING'.
- chromosomes are identified on the basis of :-
  - length of arm of chromosome.
  - position of centromere of chromosome.

Procedure -

- done by culture of lymphocyte in medium of phytohaemagglutinin (PHA).

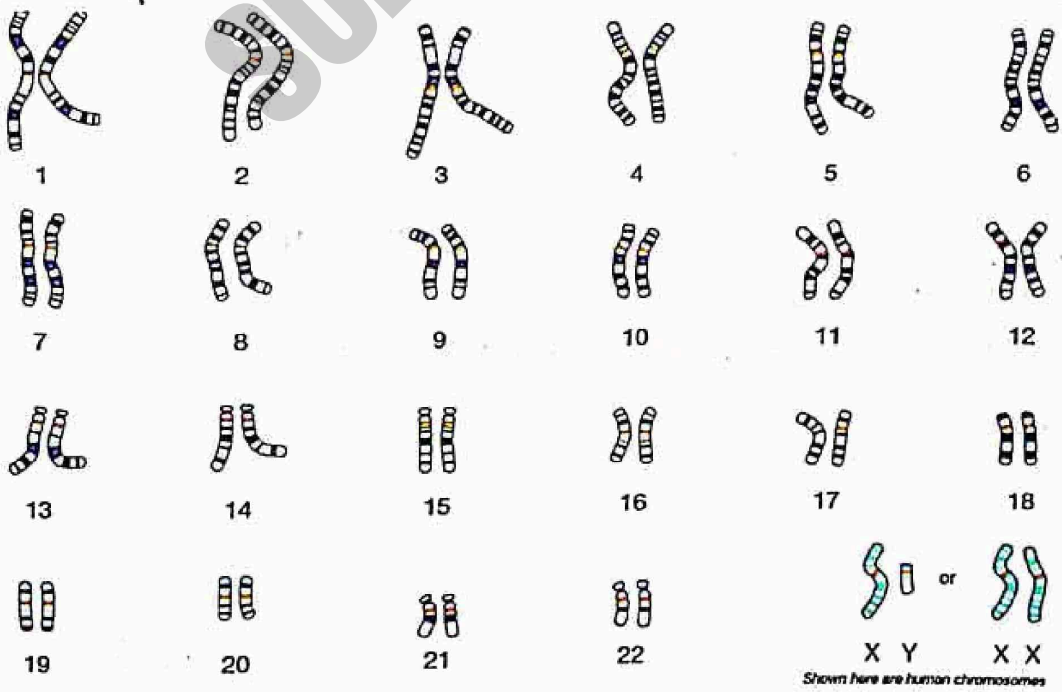
Cell division → Arrested at metaphase by action of cholchicine → Chromosome are spread, counted & photographed under microscope.

image are cut & arranged in pair in decreasing order of length. → chromosomes are numbered 1-22 in decreasing order of length.

Karyotyping helps to -

1. Identify pattern of abnormal chromosome.
2. Determine the sex.

- the length of pair number 1 is long & 22 is short.
- Individual chromosomes can be recognized with quinacrine band and Giemsa band.



Shown here are human chromosomes

SURAJ KUMAR

DOWNS SYNDROME

- Most common congenital anomaly due to numerical abbreation first described by Dr. Down.
- There is trisomy of 21st chromosome, i.e., 4 xx or 47XY.
- Commonly seen in early primigrivida or infants born to female suffering from viral infection during pregnancy.

Clinical features –
- Mentally retarded.
- poor growth & short stature.
- widely spaced great toe.

facial features
- moon face / short head / epicanthal fold / narrow palate / protruding tongue / flat nasal bridge / small ears.

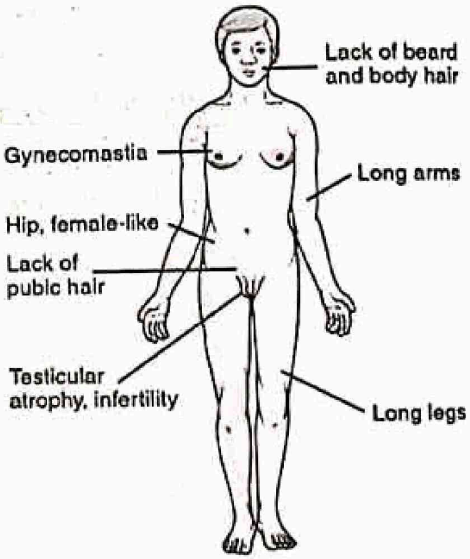
hand features
- short and broad hand / single characteristic crease called siminean crease.

Prenatal diagnosis –
- Aminocentesis
- Increased Inhibin A level.

- also called mongolism due to characterstic mongolian face of patient.

KLINEFELTER'S SYNDROME

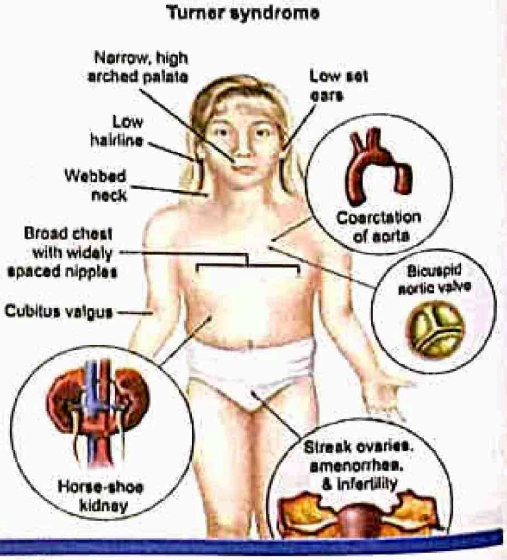
- It is a sex-chromosome abnormality with an extra X chromosome in male.
- Also called XXY syndrome.
- Karyotype is 47XXY.
- Barr body is seen in afflicted males.
- Occurs due to Meiosis II nondisjunction.



Clinical features –
- small degree of mental retardation.
- small testis leading to infertility.
- normal penis and scrotum.
- gynaecomastia
- scanty pubic and facial hairs.

TURNER'S SYNDROME

- sex chromosomal abnormality with partial or complete loss of X chromosome.
- Karyotype is 45X chromosome.
- No Barr body is seen in buccal smear as one chromosome is absent.



Clinical features:
- short stature.
- webbed neck.
- low set ears.
- normal / slightly retarded intelligence.
- Cubitus valgus, i.e., increased carrying angle at elbow.
- Amenorrhoea.
- widely placed nipples.
- Rudimentary ovary / streak gonads.

Treatment:
- Replacement therapy.
- IVF for sterile females.

SUBAJ KUMAR

SKULL ANATOMY

*Study of Skull done from Outside or Inside

*Exterior View - Norma Verticalis (Superior aspect)
- -Norma Occipitalis (Posterior)
- -Norma frontalis (Anterior)
- - Norma lateralis (lateral)
- -Norma Basalis (Inferior)

NORMA VERTICALIS

→oval
→ Wider Posteriorly than anteriorly.
→ Bones 1. Frontal Bone -->Squamous Part
2. Parietal Bone -> PAIRED
3. Occipital Bone → Squamous Part

→ Sutures 1. Coronal Suture - B/W Frontal and 2 Parietal Bonus
2. Sagittal Suture - B/W 2 Parietal Bones
3. lamdoid Suture - B/w occipital and 2 Parietal Bones

SCAN QR WITH ORBOT APP TO DIRECTLY OPEN VIDEO LINK

* Metopic Suture in 3%-8%. Cases Present in Median Plane meet

→ Other Features -
1. Bregma→ Point Where Sagittal and coronal sutures
2. Parietal Eminence - Area of Maximum Convexity in Parietal Bone
3. Vertex → highest Point on skull
4. Parietal Foramen → Present near Sagittal Suture transmits emissary Vein
5. Obelion→ Point B/W 2 Parietal foramen

NORMA OCCIPITALIS

*Norma Occipitalis → Convex Upwards and sides and Flattened Below

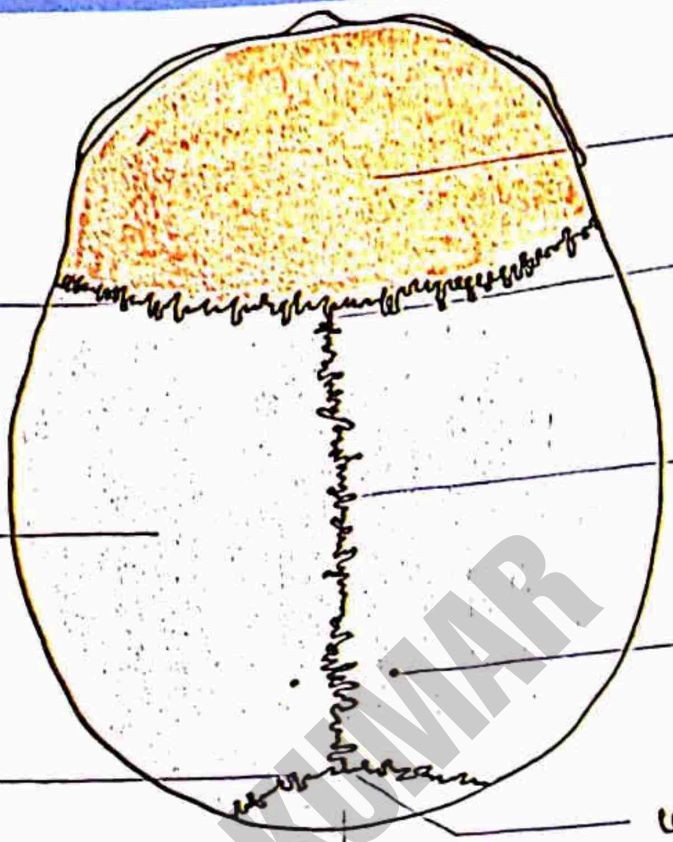
*Bones - 1.Parietal Bones - Posterior Portions of Parietal Bone Superiorly One on each side
2.Occipital Bone - Upper Part of Occipital Bone Present inferiorly.
3.Mastoid Part of temporal Bone - Present inferolaterally on each side

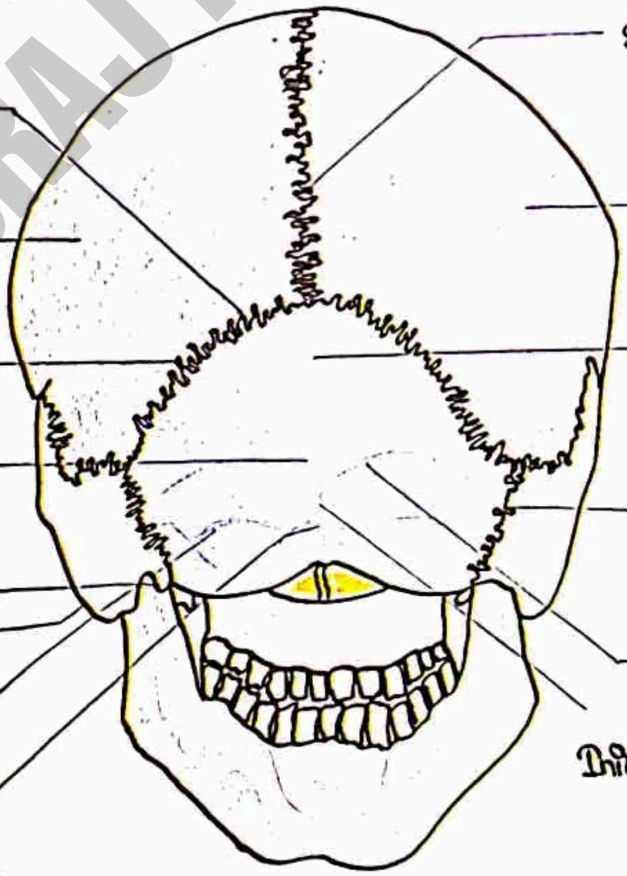
*Sutures-lambdoid suture •Parietomastoid Suture

*Other features - tamda
- External Occipital protubrence
- Superior nuchal lines
- inferior nuchal lines
- highest nuchal lines (some Cases)

SUBAJ KUMAR

NORMA VERTICALIS

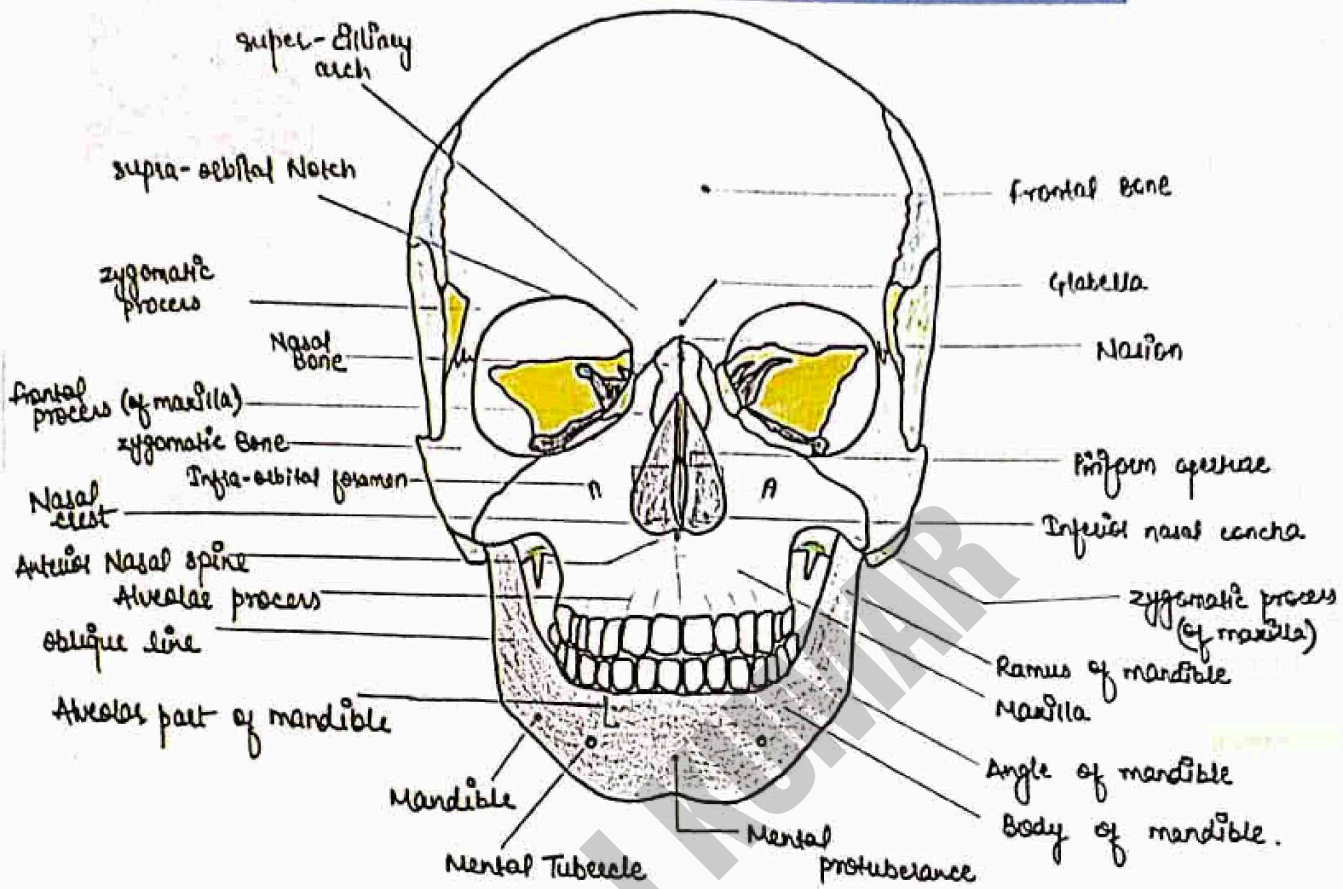


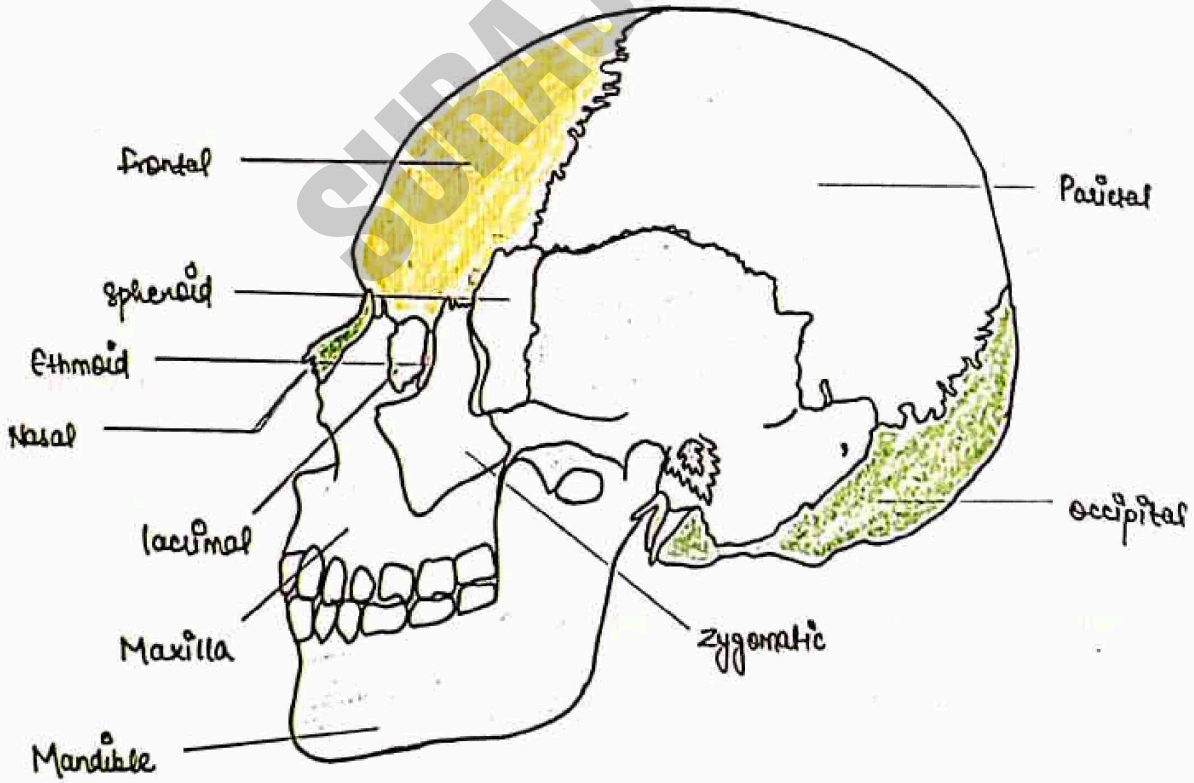


NORMA OCCIPITALIS

SURAJ KUMAR

NORMA FRONTALIS





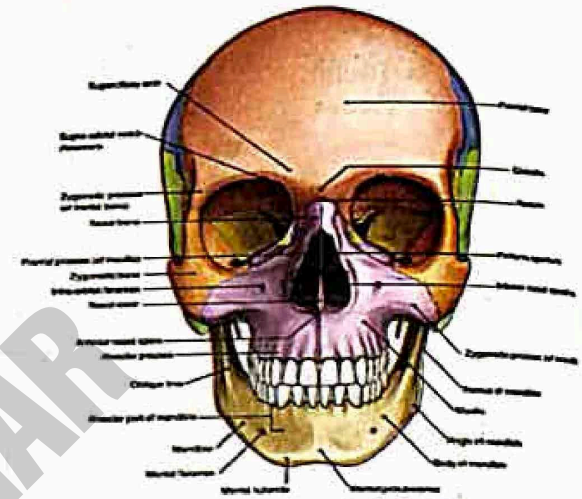
NORMA LATERALIS

SURAJ KUMAR

NORMA FRONTALIS

*Norma Frontalis -

*Bones - 1. Frontal Bone
2. Nasal Bone
3. Maxilla
4. Zygomatic Bone (Right and left)
5. Mandible



*Sutures - 1. Frontonasal 3. Frontomaxillary
2.Internasal 4. Zygomaticomaxillary

*Anterior aspect of skull Presents -

1. Frontal Region formed By Squamous Part of frontal Bone
2. Orbital openings
3. Prominences of the cheek formed By Zygomatic Bones
4. Bony External nose and Anterior Nasal Aperture.
5. Upper and lower Jaw Bearing teeth.

*foramen : 1. Supraorbital Foramen /Notch - Supraorbital nerve and Vessels
- Frontal diploic Vein.

2. Superior orbital Fissure - lacrimal artery and nerve
-Frontal nerve
·Trochlear nerve
·Abducent nerve
· Nasociliary nerve
*Superior and inferior division of oculomotor nerve
*Sympathetic Roots of Ciliary ganglions

3. Inferior orbital Fissure - Maxillary nerve
· Infraorbital Vessels
-Zygomatic nerve
·Vein Connecting inferior ophthalmic Vein with Pterygoid Venous Plexus.

4. Infraorbital Foramen - Infra Orbital nerve and artery → Continuation of maxillary artery
-Infraorbital vein

5. Zygomatico-Orbital foramen - Zygomatic Branch of maxillary nerve.

6. Mental Foramen.- Mental nerve and Vessels.

SURAJ KUMAR

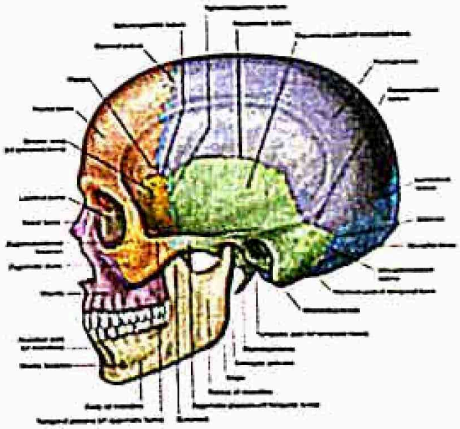
NORMA LATERALIS

*Bones - Frontal - Mandible - temporal
Parietal -Sphenoid - zygomatic
-Occipital - Maxilla - nasal

*Suture- -Coronal Suture
-Parietosquamosal suture - B/W Parietal and Squamous Part of temporal Bone
-lamdoid Suture

Other features -

1. Temporal line - Commences at the frontal Process of zygomatic Bone arches Upwards and Backwards across the Parietal Bone where it Splits in Superior and inferior Lines
2. Zygomatic arch → horizontal Bar of Bone form by temporal Process of zygomatic Bone and zygomatic Process of temporal Bone.
3. External auditory Meatus
4. Suprameatal triangle - Small depression posterosuperior to the External auditory meatus
5. Mastoid Process of temporal Bone
6. Anterior - Meeting Point of Parietomastoid, occipitomastoid and lambdoid Suture.
7. Styloid Process of temporal Bone
8. Temporal fossa and infratemporal Fossa
9. Pterion - Region in anterior Part of temporal fossa where 4 bones meet to form H Shaped structure
   1. Frontal
   2. Squamous temporal
   3. Parietal
   4. Greater Wing of sphenoid



***Foramen and Fissures of Norma lateralis**

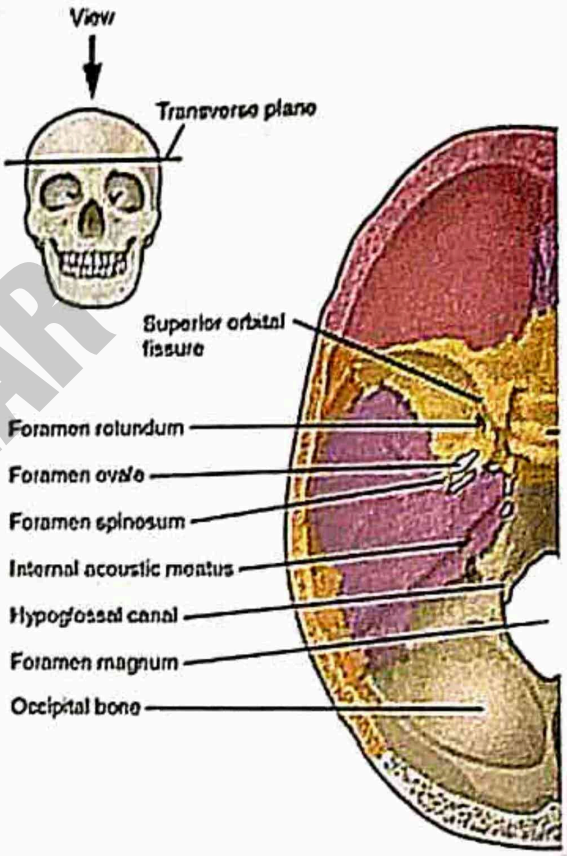
1. Zygomatico temporal foramen - Zygomatico temporal artery and nerve Passes
2. Sphenopalatine foramen - Nasopalatine nerves & Sphenopalatine Vessels.
3. Pterygomaxillary fissure - 3rd part of Maxillary artery
   - Maxillary Nerve
4. Greater Palatine Canal - Palatine nerves and Vessels.

SURAJ KUMAR

NORMA BASALIS

*Foramina and Structure Passing through : -

1. Incisive foramina - greater palatine Vessel
   -nasopalatine nerve
2. Greater palatine foramen- greater palatine nerves and Vessels
3. Lesser Palatine foramina.- Middle and Posterior palatine nerve
4. Foramen ovale - Mandibular nerve
   -Accessory middle meningeal artery
   -lesser Petrosal nerve
   -Emissary Vein
5. Foramen Rotendum - Maxillary nerve
   -Emissary Vein
6. Foramen Spinosum - Middle meningeal artery
   -Middle meningeal Vein
   -Nerve Spinous
7. Foramen Lacerum -Meningeal→
   Ascending pharyngeal Artery Branch
   -Emissary Vein
   -internal carotid artery
   -greater Petrosal nerve



8. Carotid canal - Infernal carotid artery and vein
   Emissary Vein.
9. Juglar Foramen - Anterior Part-Inferior Petrosal sinus
   -Middle part- 9th 10th 11th Cranial nerve
   -Posterior - Sigmoid sinus
10. Stylomastoid foramen - 7th Cranial nerve
    Stylomastoid artery
11. Foramen Magnum - Anterior Compartment
    Apical ligament of Dens
    Upper Band of Cruciform ligament of atlas
    Membrana tectoria
    - Posterior Compartment
    - Medulla oblangata
    -Two posterior spinal artery
    ·1x anterior Spinal artery
    · 2x Vertebral arteries
    - Sympathetic plexus
    -Spinal Root of two accessory
12. Hypoglossal Canal - 12th Cranial nerve
    - Meningeal Branch of ascending Pharyngeal artery

SURAJ KUMAR

CLAVICLE

• "Collar bone" • Medially - Sternum + 1st rib cartilage • Laterally - Acromion process

FEATURES
- Only long bone lying horizontally
- No medullary cavity
- Membrocartilaginous ossification
- Maybe pierced through cutaneous nerve "INTERMEDIATE SUPRACLAVICULAR NERVE"
- Subcutaneous throughout its extent
- Only long bone with 2 primary centres.
- First to ossify (B/w 5th-6th wk of IUL)
- Last to ossify (25 years)

PARTS 2 ends + Shaft

ENDS
- Medial / Sternal
  - Quadrilateral
  - with clavicular notch of manubrium sterni
- Lateral / Acromial
  - Flattened
  - with acromion process margin

SHAFT
- curved
- Medial 2/3rd round & convex forward
- Lateral 1/3rd flattened & concave forward
- Inferior surface Subclavian groove in middle third

LATERAL / ACROMIAL END
- Flattened above downward
- Oval facet articulate with medial margin of acromion and form 'Acromioclavicular Joint'
- Lateral end gives attachment to fibrous capsule of acromioclavicular joint.

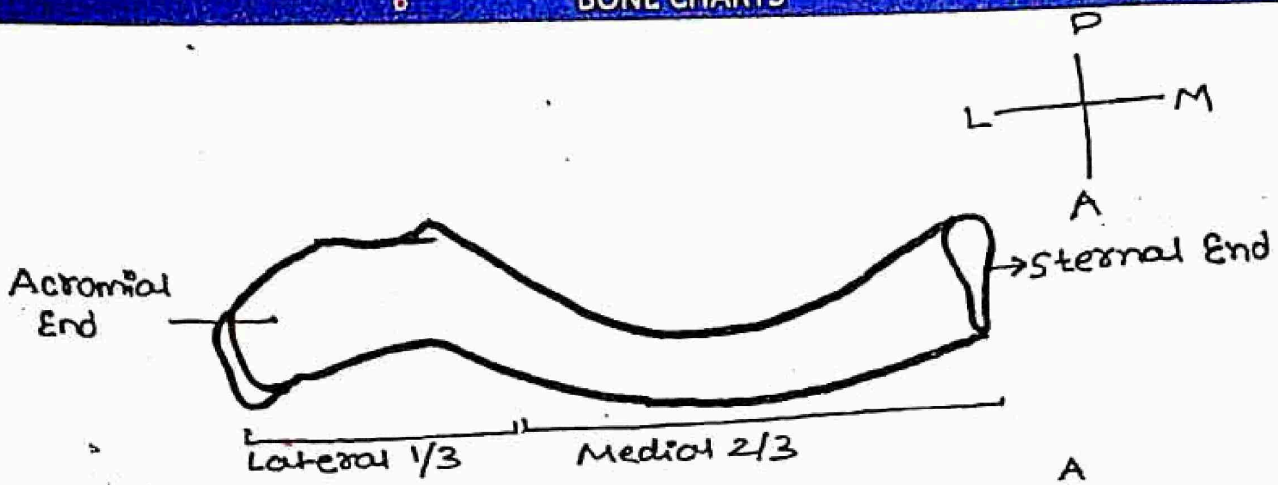
MEDIAL / STERNAL END
- Saddle shaped articular surface with clavicular notch of manubrium sterni to form 'Sternoclavicular Joint'
- Gives attachment to
  - Fibrous capsule
  - Articular disc
  - Interclavicular ligament

SHAFT

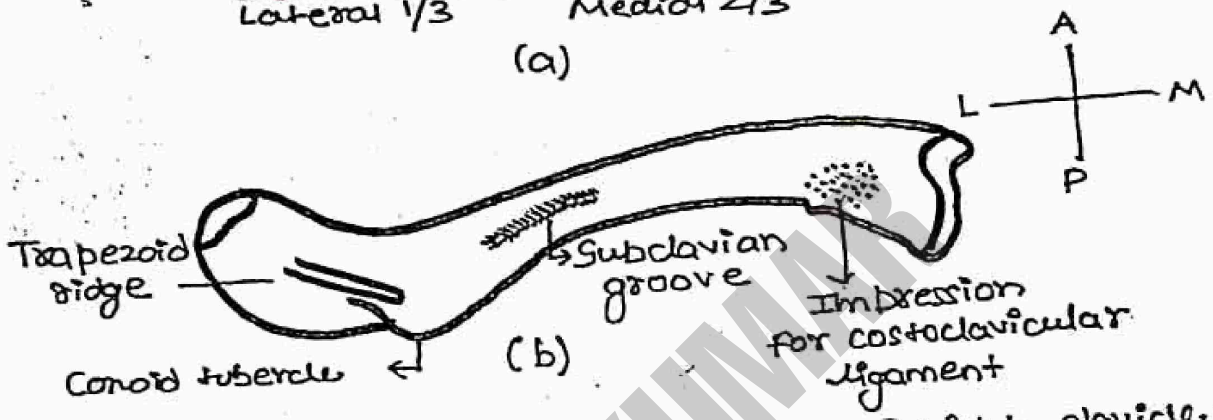
- LATERAL 1/3rd :- • Flattened • 2 surface and 2 borders
  - ⇒ SURFACE
    - SUPERIOR • Subcutaneous b/w attachment of deltoid and trapezius.
    - INFERIOR
      1. Conoid Tubercle • Near posterior border at junction of lateral 1/4th and Medial 3/4th of clavicle
      2. Trapezoid ridge • Extend forward & laterally from conoid tubercle

      → Attachment to 'Coracoclavicular ligament'
  - ⇒ BORDER
    - ANTERIOR • Concave forward • Origin to deltoid muscle • Deltoid tubercle maybe present
    - POSTERIOR • Convex forward • Insertion to trapezius muscle
- MEDIAL 2/3rd :- • Cylindrical • Has 4 surface
  - ⇒ SURFACE
    - SUPERIOR • Origin of clavicular head of sternocleidomastoid muscle
    - INFERIOR • Attachment of costoclavicular ligament to oval impression • Insertion of subclavius muscle into subclavian groove. • Attachment of clavipectoral fascia to margin of subclavian groove • Presence of nutrient foramen on late subclavian groove
    - ANTERIOR • Convex forward • Origin to clavicular head of pectoralis major muscle.
    - POSTERIOR • Concave backward • Origin to sternohyoid muscle. • Lateral part - forms anterior boundary of cervico-axillary canal.
      - Trunks of brachial plexus
      - 3rd part of subclavian artery.

SURAJ KUMAR

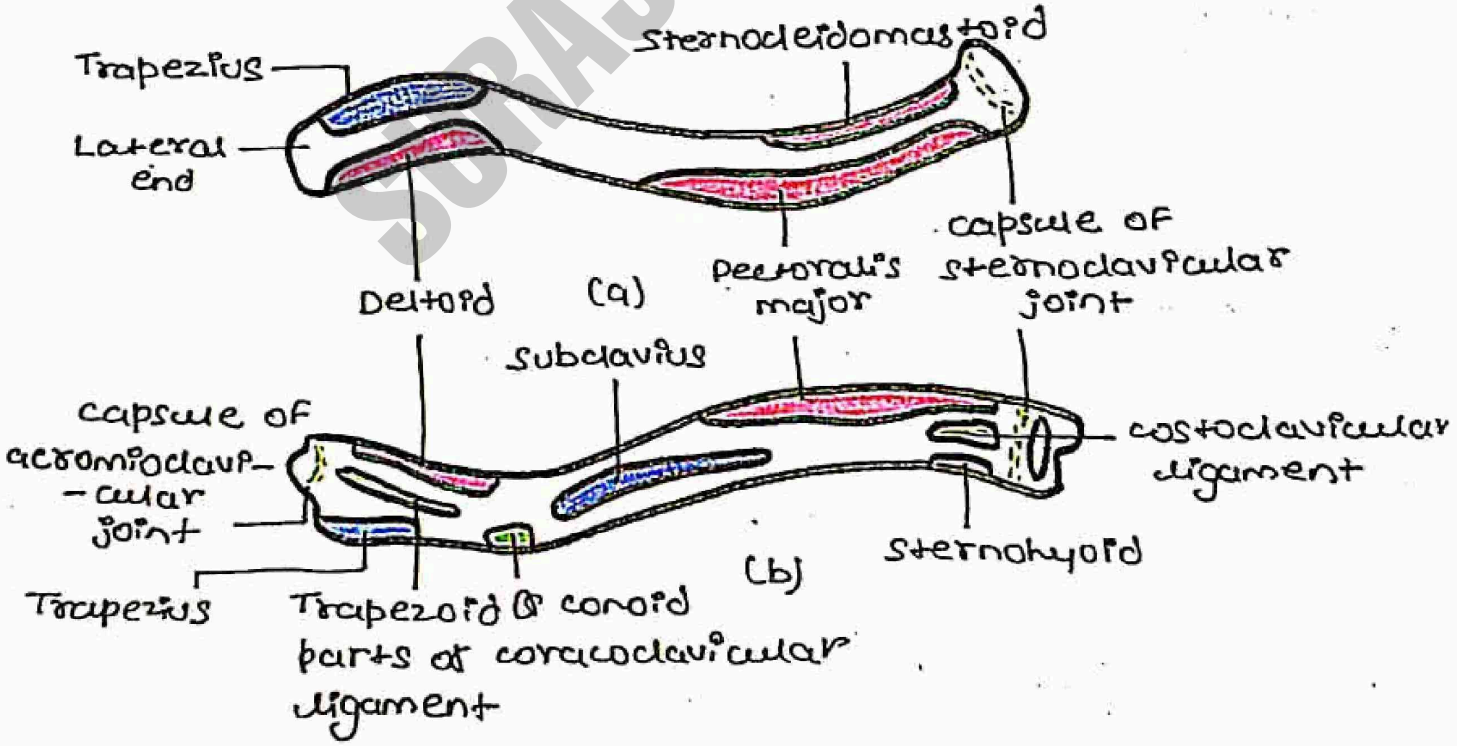


(a)



(b)

Figure showing general features of right clavicle
(a) Superior aspect (b) Inferior aspect



(a) (b)

Figure showing attachments of right clavicle
(a) Superior aspect (b) Inferior aspect

SURAJ KUMAR

SCAPULA

: "Shoulder blade" • large, flattened, triangular bone • highly mobile • Upper part of postero lateral aspect of thorax, against 2nd - 7th ribs

PARTS 4 parts
- Body
- Processes spinous, acromion, coracoid

Body 2 surface + 3 borders + 3 angles

SURFACE

COSTAL SURFACE
(Subscapular fossa)
- Concave & directed medially & forward
- 3 longitudinal ridge — attachment to subscapularis muscle — arising from medial 2/3rd except near subscapular bursa.

DORSAL SURFACE
- convex
- Presents spinous process (shelf-like)
  - supraspinatous fossa (1/3rd)
  - infraspinatus fossa (2/3rd)
- Spinoglenoid notch — suprascapular nerve & vessels
- Origin of supraspinatus and infraspinatous muscle from medial 2/3rd of supraspinatous fossa
- Origin of teres minor muscle from upper 2/3rd of dorsal surface of lateral border. Circumflex scapular artery interrupt its origin.
- Origin of teres major muscle from lower 1/3rd of dorsal surface of lateral border & inferior angle.
- Origin of latissimus dorsi muscle

BORDERS

SUPERIOR BORDER
- shortest border
- Suprascapular notch on coracoid process — into suprascapular foramen by superior transverse ligament.
- Above the ligament → suprascapular artery
  Below the ligament → suprascapular nerve
- Origin of inferior belly of omohyoid near suprascapular notch

LATERAL BORDER
- Thickest border
- Presence of infraglenoid tubercle at upper end
- Origin of long head of triceps from this tubercle.

MEDIAL BORDER
- Thin and angled at root of spine
- Insertion of serratus anterior on costal surface.
- Insertion of levator scapulae on dorsal aspect.
- Insertion of rhomboideus minor on dorsal aspect
- Insertion of rhomboideus major on dorsal aspect.

ANGLES

- INFERIOR → lies over 7th rib or 7th intercostal space.
- SUPERIOR → Junction of superior and medial borders • lies over 2nd rib.
- LATERAL (Head of scapula) → Glenoid cavity + head of humerus → 'Glenohumeral Joint' (Shoulder Joint)
  → Glenoid labrum → margin of glenoid cavity to deepen its concavity.
  → Capsule of shoulder joint → margin of glenoid cavity, proximal to attachment of glenoid labrum.
  → Origin of long head of biceps brachii from supraglenoid tubercle. Origin is intracapsular.

SURAJ KUMAR

PROCESSES

SPINOUS PROCESS :- (Spine of Scapula)

- Junction of Upper 1/3rd and Lower 2/3rd.
- Upper Supraspinous fossa & Lower Infraspinous fossa.
- Has 2 surface & 3 borders.

- **SURFACE**
  - **SUPERIOR**
    - Lower boundary of supraspinous fossa
    - Origin to supraspinatus.
  - **INFERIOR**
    - Upper boundary of infraspinous fossa
    - Origin to infraspinatus.
- **BORDERS**
  - **ANTERIOR**
    - Attached to dorsal surface
  - **LATERAL**
    - Spinoglenoid notch — through which passes suprascapular nerve and vessels.
  - **POSTERIOR**
    - 'Crest of spine'
    - Insertion of trapezius on the upper lip of crest of spine.
    - Origin of posterior fibers of deltoid from lower border of crest of spine.

ACROMION PROCESS :- (Acromion)

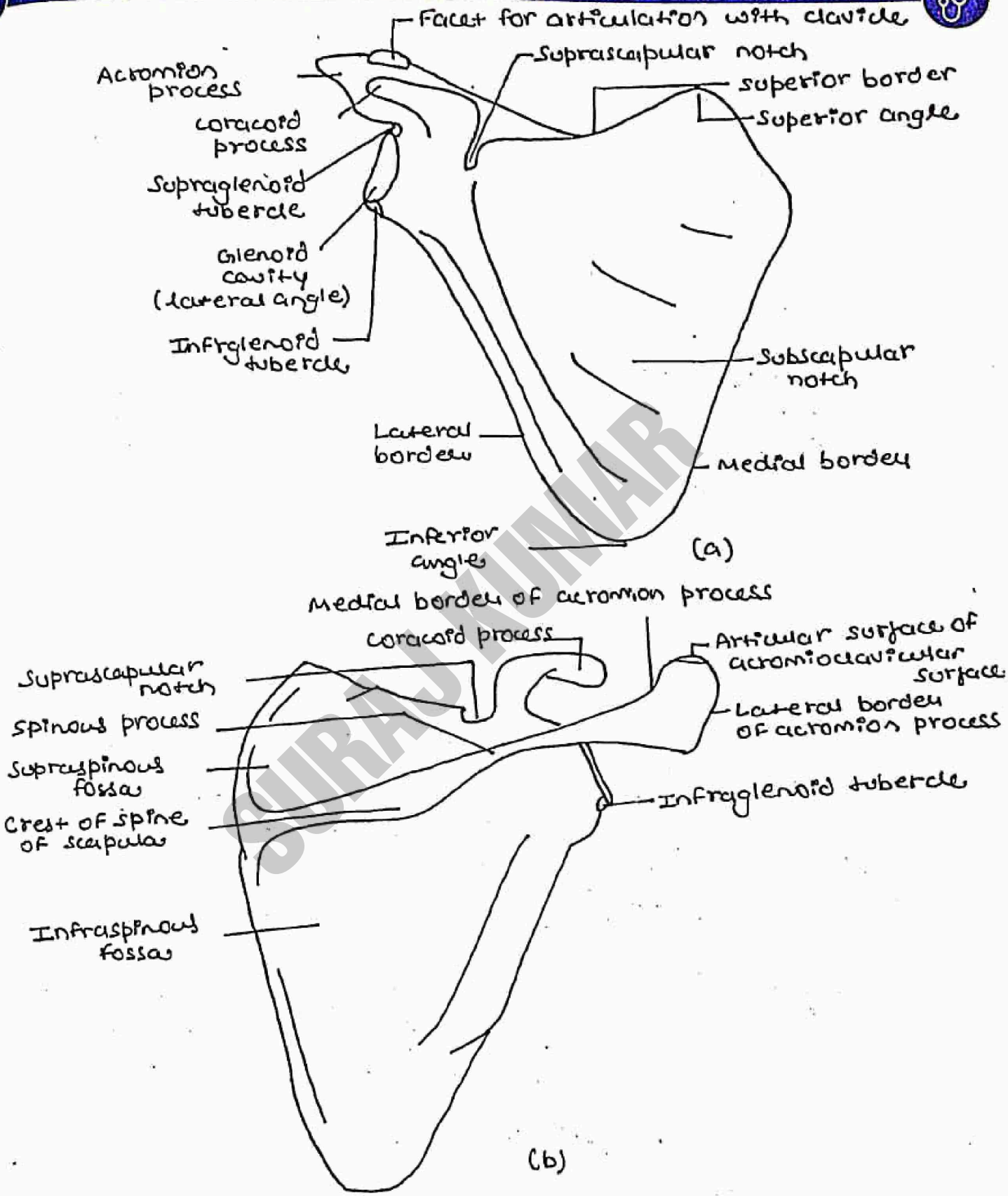
- Projects forward almost at right angle from lateral end of spine & overhangs glenoid cavity.
- Superior surface — subcutaneous.
- Has tip, 2 borders, 2 surface.

- **SURFACE**
  - **SUPERIOR**
    - Rough & subcutaneous
  - **INFERIOR**
    - Smooth
    - Related to 'Subacromial bursa'.
- **BORDERS**
  - **MEDIAL**
    - Continues to form upper lip of crest of spine
    - Insertion to trapezius muscle
    - Circular facet articulate with lateral end of clavicle to form 'Acromioclavicular Joint'.
  - **LATERAL**
    - Continues to form lower lip of crest of spine
    - Origin to intermediate fibers of deltoid muscle.
- Attachment of coracoacromial ligament to the tip of acromion.
- Acromial angle — junction of lateral border of acromion and lateral border of crest of spine of scapula.

CORACOID PROCESS :-

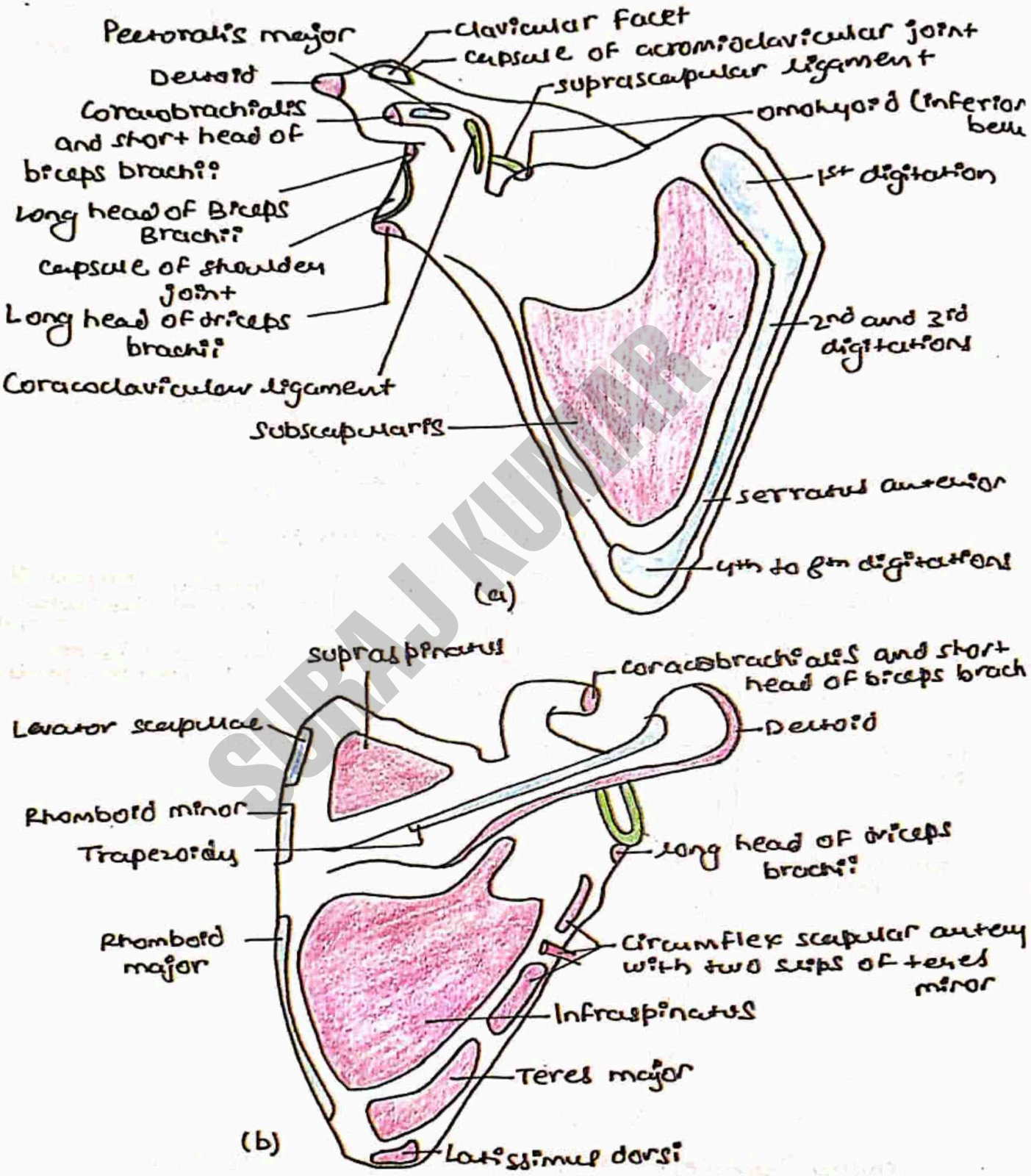
- From upper part of head of scapula
- Bent sharply to project forward and slightly laterally.
- Origin of short head of biceps brachii and coracobrachialis from tip by a common tendon.
- Insertion of pectoralis minor muscle on medial border of upper surface.
- Attachment of coracoacromial ligament to its lateral border.
- Coracoclavicular ligament
  - conoid part (rhomboid ligament) — knuckle.
  - trapezoid part (rhomboid ligament) — ridge on superior aspect b/w pectoralis minor muscle and coracoacromial ligament.
- Coracohumeral ligament — root adjacent to glenoid cavity.

SURJIT KUMAR



General Features of right scapula (a) Costal Surface (b) Dorsal surface

SURAJ KUMAR



Attachments of right scapula (a) Coastal aspect
(b) Dorsal aspect

SURAJ KUMAR

HUMERUS

PARTS 2 ends + Shaft

UPPER END

1. Head : smooth rounded
   - forms glenohumeral joint
2. Neck
3. Greater tubercle
4. Lesser tubercle
5. Intertubercular sulcus.

SHAFT

- B/w upper & lower end
- Upper half - cylindrical
  Lower half - flattened
  - anteroposteriorly

LOWER END

1. Capitulum (lateral round convex projection)
2. Trochlea (medial pulley shaped)
3. Radial fossa (above capitulum)
4. Coronoid fossa (above trochlea)
5. Medial epicondyle
6. Lateral epicondyle
7. Olecranon fossa (deep hollow on posterior aspect above trochlea)

UPPER END

1. **HEAD :-** . Smooth, round, 1/3rd of sphere . Covered by articular hyaline cartilage
2. **NECK**
   - **Anatomical :-** . Provides attachment to capsular ligament of shoulder joint except, superiorly — for passage of tendon of long head of biceps brachii
     medially — capsule extends down for about 1 - 2 cm.
   - **Morphological :-** . B/w diaphysis and epiphysis . true junction of head with shaft
     . Represented by epiphyseal line
   - **Surgical :-** . Below greater and lesser tubercles / epiphyseal line
     . Related to axillary nerve and posterior & anterior circumflex humeral vessels.
     . One of sites of common humeral fractures.
3. **GREATER TUBERCLE :-** . Most lateral part of proximal end
   . Posterosuperior aspect — flattened facet like impressions
   - upper - supraspinatus muscle
   - middle - Infraspinatus muscle
   - lower - teres minor muscle
4. **LESSER TUBERCLE :-** . Small elevation on front of upper end (just above surgical neck)
   . Subscapularis muscle.
5. **INTERTUBERCULAR SULCUS :-** (Bicipital Groove)
   . Vertical groove
   . Contains
   1. long head of biceps in synovial sheath
   2. Ascending branch of anterior circumflex humeral artery.
   . Lateral lip of groove - Pectoralis major muscle
   Floor of groove - Latissimus dorsi muscle
   Medial lip of groove - Teres major muscle

SURAJ KUMAR

SHAFT

- Upper part – cylindrical
  lower part – triangular
- has 3 borders and 3 surface

SURFACE

- **Antero-lateral**
  - V-shaped deltoid tuberosity
  - Insertion to deltoid muscle.
- **Antero-medial**
  - Upper part – forms floor of intertubercular sulcus.
  - Nutrient foramen – about its middle and close to medial border – directing downward.
- **Posterior**
  - Upper 1/3rd – oblique ridge directing downward and laterally.
    - Origin to lateral head of triceps brachii.
  - Below & medial to oblique ridge – radial/spiral groove
    - Radial nerve & profunda brachii vessels.
  - Below spiral groove
    - origin to medial head of triceps brachii.

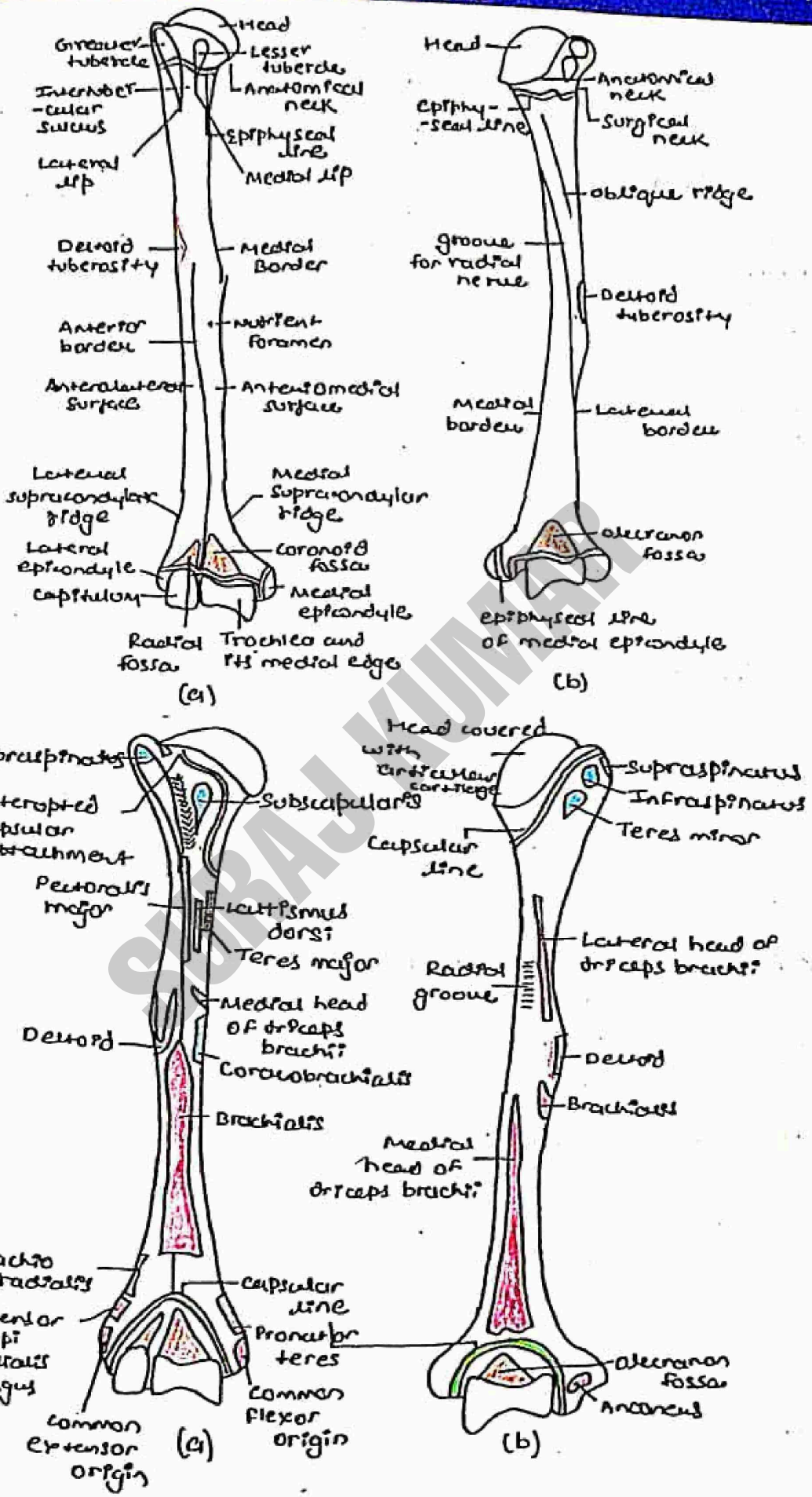
BORDERS

- **Anterior**
  - From lateral lip of intertubercular sulcus to anterior margin of deltoid tuberosity and ends in radial fossa
- **Medial**
  - Medial supracondylar ridge – medial intermuscular septum
  - Rough strip on middle – insertion to coracobrachialis muscle
  - Narrow area above medial epicondyle – origin to humeral head of pronator teres muscle.
- **Lateral**
  - Upper part – indistinct lower part – lateral supracondylar ridge
  - Above its middle – crossed by radial groove from behind.
  - lower part – lateral supracondylar ridge – lateral intermuscular septum.

LOWER END

- Capitulum articulate with head of radius
- Trochlea articulate with trochlear notch of ulna.
- Ulnar nerve – posterior surface of medial epicondyle.
- Anterior surface of medial epicondyle – common flexor origin of superficial flexors of forearm
- Anterolateral part of lateral epicondyle – common extensor origin
- Posterior surface of lateral epicondyle – Origin to anconeus muscle.
- Coronoid process of ulna – coronoid fossa (above trochlea) during full flexion
- Head of radius – radial fossa (above capitulum) during full flexion
- Olecranon process of ulna – olecranon fossa (posterion aspect above trochlea) during full extension
- Capsule of elbow joint is attached:
  Anteriorly: Above coronoid and radial fossae

SURAJ KUMAR



Attachments of right humerus

(a) Anterior view (b) Posterior view

SUBAJ KUMAR

RADIUS

- Lateral bone of forearm
- Homologous to tibia

PARTS - 2ends + Shaft

UPPER END
- Head + Neck + Radial Tuberosity
- Head : - disc shaped
  - articulate with capitulum of humerus
- Neck : - constricted
- Radial Tuberosity : - just below medial part of neck

SHAFT
- Presents lateral convexity
- Sharpest interosseous border on medial side

LOWER END
- widest part
- 5 surfaces
- Lateral surface – styloid process
- Dorsal surface – Lister's tubercle

UPPER END
- HEAD :- Humero-radial articulation, Annular ligament
- NECK :- Quadrate ligament
- RADIAL TUBEROSITY :- Insertion of tendon of biceps brachii

SHAFT - Has 3 borders & 3 surfaces.

SURFACE
- Anterior
  - Origin of Flexor policis longus — Upper 2/4th.
  - Insertion of Pronator quadratus — lower 1/4th.
  - Presence of nutrient foramen
    Nutrient artery — Anterior interosseous artery's branch.
- Posterior
  - Origin of Abductor policis Longus (APL) — middle 1/3rd.
  - Origin of Extensor policis brevis (EPB)
- Lateral
  - Insertion of Supinator — upper 1/3rd
  - Insertion of Pronator teres

BORDERS
- Anterior - Anterior oblique line — Origin of Flexor Digitorumsuperficialis (FDS)
- Posterior - Posterior oblique line
- Medial (Interosseous border) - Sharpest
  - Attachment of interrosseous membrane.

LOWER-END - Widest part with 5 surface

SURFACE
- Anterior - Palmar radio-carpal ligament of wrist joint
- Posterior - Dorsal tubercle of radius (Lister's tubercle) — for tendon of extensor policis longus.
  - Lateral to groove – ECRL & ECRB
  - Medial to groove – Extensor policis longus — passes tendons of extensor digitorum and extensor indicis.
- Medial - Articular disc of inferior radio-Ulnar joint
- Lateral - Tendons of Abductor pollicis longus & Extensor pollicis brevis.
  - Insertion of brachioradialis
- Inferior
  - Lateral Δr area — scaphoid
  - Medial □r area — lateral part of lunate

SURAJ KUMAR

ULNA

- Medial bone of forearm
- Homologous to fibula.

PARTS - 2 ends + Shaft

UPPER END
- Trochlear notch b/w
  - Above : Olecranon process
  - Below : Coronoid process

SHAFT
- Lateral border (interosseous border) is sharp crest-like.

LOWER END
- Has head & styloid process

UPPER END - Has 2 process & 2 notches.

PROCESS

OLECRANON - Has 5 surface

- **SURFACE**
  - **UPPER** - Insertion to triceps brachii - Capsular ligament of elbow jt
  - **ANTERIOR** - Upper part of trochlear notch.
  - **POSTERIOR** - Subcutaneous olecranon bursa
  - **MEDIAL** - Attachments to :-
    - Origin of Flexor carpi ulnaris (FCU)
    - Posterior & Oblique bands — Ulnar collateral ligament. (UCL)
  - **LATERAL**

CORONOID - Has 4 surface

- **SURFACE**
  - **SUPERIOR** - Lower part of trochlear notch
  - **ANTERIOR** - Ulnar tuberosity - Insertion of Brachialis muscle
    - Sublime tubercle :-
      - Ant band of UCL
      - Oblique band of UCL
      - Humero ulnar head of FDS
      - Pronator teres
      - Flexor pollicis longus
  - **MEDIAL** - Origin of Flexor digitorum Profundus.
  - **LATERAL** - Radial notch
    - Supinator fossa — Accommodates Radial tuberosity (during supination & pronation)
    - Origin to Supinator muscle

NOTCHES

- **TROCHLEAR** - Semilunar - With trochlea of humerus
  - Attachments to capsule of elbow joint
- **RADIAL** - Superior radio ulnar joint.

SHAFT - 3 surface + 3 borders

SURFACE

- **ANTERIOR**
  - Origin of Flexor digitorum profundus (Upper 3/4th)
  - Origin of Pronator quadratus (Lower 1/4th)
  - Presence of nutrient foramen
- **MEDIAL** - Origin of Flexor digitorum profundus (Upper 2/3rd)
- **POSTERIOR**
  - Insertion of Anconeus (above oblique line)
  - Below oblique line — larger medial, smaller lateral

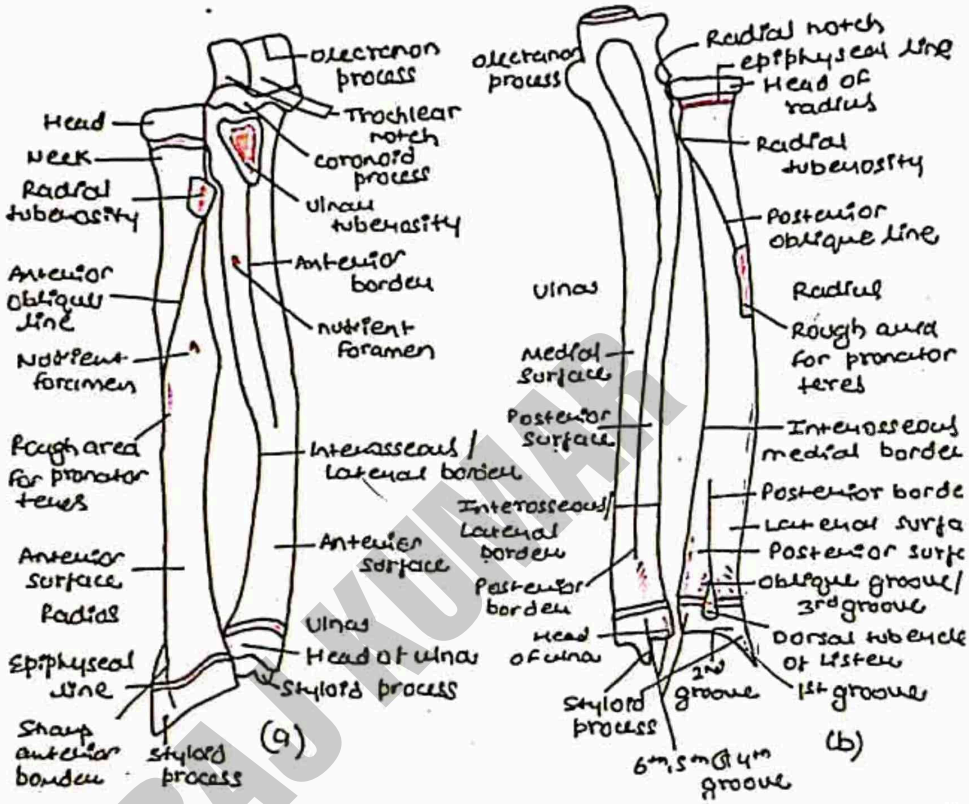
BORDERS

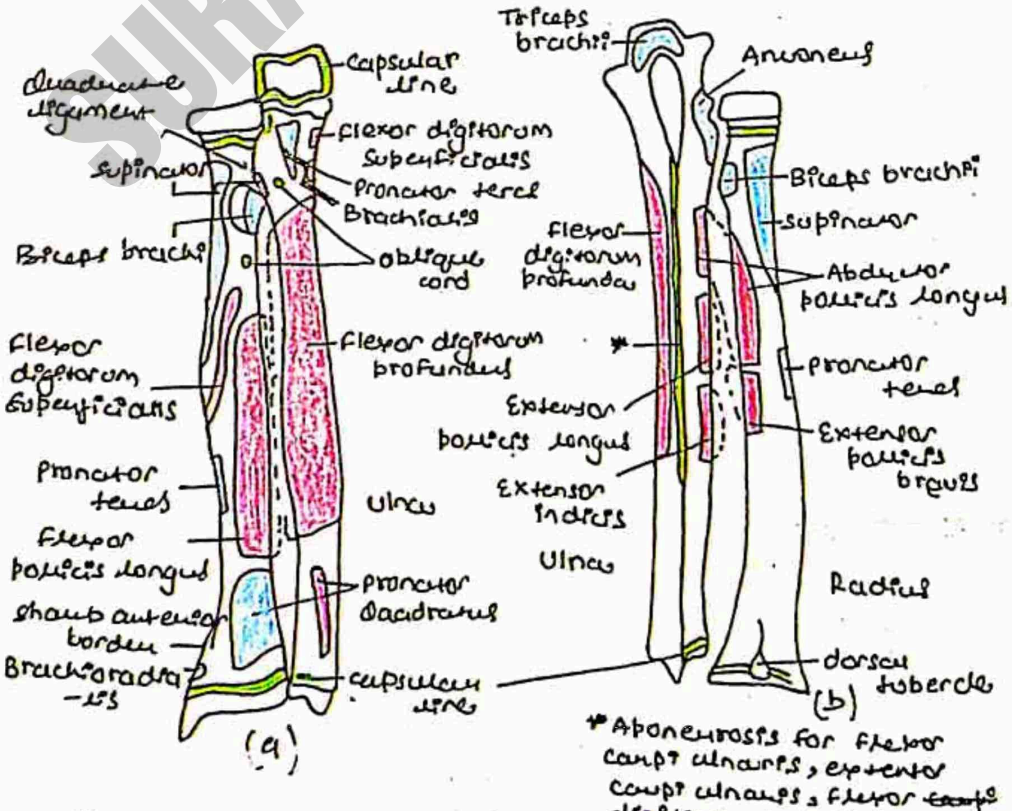
- **LATERAL (interosseous)** - Sharpest - Attachment to interosseous membrane
- **ANTERIOR** - Origin to Flexor digitorum profundus
- **POSTERIOR** - Subcutaneous
  - Attachment by common aponeurosis
    1. Flexor digitorum profundus
    2. Flexor carpi ulnaris
    3. Extensor carpi ulnaris

SURAJ KUMAR

LOWER END · Head + Styloid process

- **HEAD** · Ulnar notch of radius — Inferior radio ulnar joint
- **STYLOID PROCESS**
  - Tip — medial collateral ligament of wrist joint
  - Apex — depression b/w head & base of process
  - Groove — tendon of extensor carpi ulnaris





SURAJ KUMAR

HIP BONE

. Also known as 'Coxae', 'Innominatum' · Fusion of Ilium, Ischium & Pubis

ILIUM

. 2 ends + 3 borders + 3 surfaces.

→ ENDS → Upper (Iliac crest)
Anterior superior iliac spine → Above : Inguinal ligament
Below : Sartorius
Posterior superior iliac spine → Sacrotuberous ligament

A. VENTRAL SEGMENT — Outer lip + intermediate area + Inner lip

1. Outer lip – tubercle of iliac crest
– Attachments to
· origin of tensor fascia latae
· insertion of external oblique muscle
· origin of latissimus dorsi
2. Intermediate area · Origin of internal oblique muscles.
3. Inner lip · Anterior 2/3rd : Transversus abdominis muscle
· Posterior 1/3rd : Quadratus lumborum

B. DORSAL SEGMENT
1. Outer sloping surface : Gluteus maximus muscle
2. Inner sloping surface : Erector spinae muscle

→ Lower

BORDERS → Anterior · Anterior inferior iliac spine
upper : straight head of rectus femoris
lower : iliofemoral ligament

→ Posterior · Posterior inferior iliac spine
Greater sciatic notch → foramina
Ischial spine
Lesser sciatic notch → foramina
· Structures passing from dorsal aspect of ischial spine
(Lateral → medial)
– Pudendal nerve
– Internal pudendal vessels
– Nerve to obturator internus.

→ Medial · Arcuate line

SURFACE → Iliac fossa · Origin of iliacus

→ Gluteal · 4 areas by 3 gluteal lines.
· Origin to 4 muscles
Gluteus maximus behind posterior gluteal line
Gluteus medius b/w anterior and posterior gluteal line
Gluteus minimus b/w anterior and inferior gluteal line
Rectus femoris b/w inferior gluteal line and upper margin of acetabulum.

SURAJ KUMAR

SACROPELVIC
- A. SACRAL PART
  1. Iliac tuberosity – attachments to ligaments
     a. iliolumbar
     b. dorsal sacroiliac
     c. interosseous sacroiliac
  2. Auricular surface – sacroiliac joint
- B. PELVIC PART – Pre auricular sulcus
  – Origin to obturator internus.

ISCHIUM

- Body – 2 ends + 3 borders + 3 surface & Ramus

→ BODY

ENDS → Upper
↳ Lower

- Ischial tuberosity
  → Upper quadrilateral area → upper lateral
  ↳ lower medial
  ↳ lower triangular area → outer
  ↳ inner
- Origin to 4 muscles.
  1. Semimembranosus
  2. Semitendinosus
  3. Semitendinosus + long head of biceps femoris
  4. Adductor magnus
- Medial margin → lower end of sacrotuberous ligament.

BORDERS → ANTERIOR – Attachment to obturator membrane

→ POSTERIOR – Ischial spine → attachments to
  – sacrospinous ligament
  – levator ani
  – coccygeus muscle
- Lesser sciatic notch
  → upper end : Gemellus superior
  ↳ lower end : Gemellus inferior

↳ LATERAL – tendon of obturator externus.

SURFACE → Femoral
- Obturator externus – along obturator foramen
- Quadratus femoris – along lateral border of ischial tuberosity.

→ Dorsal – Tendon of obturator internus + 2 gemelli — traverses the groove

↳ Pelvic – Origin to obturator internus.

→ RAMUS – Conjoint ischiopubic ramus – 2 borders + 2 surface

BORDERS → Upper – Attachment to Obturator membrane
↳ Lower

SURFACE → Outer – Linear attachment to 4 muscles
1. Obturator externus
2. Adductor magnus
3. Adductor brevis
4. Gracilis

↳ Inner – 2 ridges + 3 areas
- Didnot → Honor ... hment to superior fascia of urogenital diaphragm
  hment to perineal membrane & falciform process of sacrotuberous ligament.

SURAJ KUMAR

- Area
  - Upper — origin to obturator internus
  - Middle — Attachment to sphincter urethrae & deep transversus perinei muscles.
  - Lower — Attachments to crus of penis, clitoris, ischiocavernosus & superficial transversus perinei muscles.

PUBIS

- 'Pubic bone / OS Pubis'.
- Body + superior ramus + inferior ramus.

→ **BODY**

- Pubic crest + Pubic tubercle + 3 surface.

. PUBIC CREST & PUBIC TUBERCLE

. SURFACE
- → Anterior
  - Attachments to 4 Muscles
    1. Adductor longus
    2. Gracilis
    3. Adductor brevis
    4. Obturator externus
- → Posterior (Pelvic) — Related to urinary bladder.
- → Medial (Symphyseal) — Symphysis pubis – 2° cartilaginous joint.

→ **SUPERIOR RAMUS** — 3 borders + 3 surfaces.

. BORDERS

→ Pectineal line (Posterior / superior)
- 'Pecten pubis'
- Attachments to
  - Medial
    - Lacunar ligament
    - inguinal ligament
    - conjoint tendon
    - Fascia transversalis
  - Lateral
    - Pectineal ligament (Ligament of Cooper)
    - Pectineus muscle
    - Pectineus fascia
    - Psoas minor (if present)

→ Obturator crest (Anterior)

↳ Inferior — Attachment to obturator membrane leaving a gap for obturator canal.

SURFACE
- → Pectineal — Origin of pectineus muscle.
- → Pelvic — Featureless & covered by peritoneum.
- ↳ Obturator — Obturator nerve & vessels traverses the groove.

→ **INFERIOR RAMUS** — form conjoint ischiopubic ramus.

OBTURATOR FORAMEN

- Attachment of obturator membrane except the groove to convert it into obturator canal
- Obturator canal transmits obturator nerves & vessels.

ACETABULUM

- Shallow vinegar cup
- Acetabular notch → acetabular foramen due to transverse acetabular ligament
- Articular – Lunate surface
- Non articular area – Acetabular fossa
- Acetabular labrum (fibrocartilaginous rim) ↳ margin of acetabulum
- Acetabular fossa – filled with pad of fat
- Lunate surface – covered by hyaline cartilage

SURAJ KUMAR

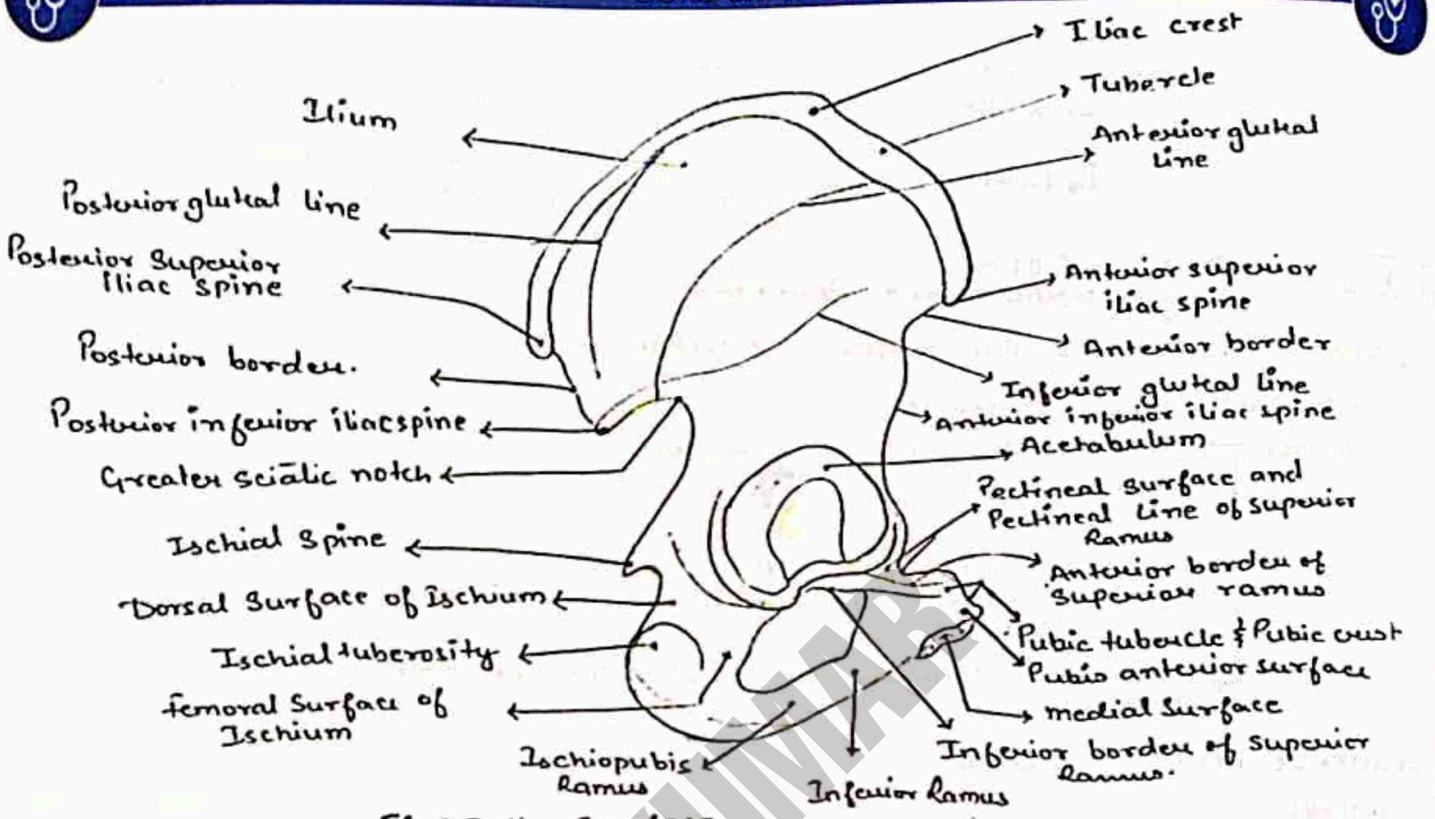


Fig: Outter Surface

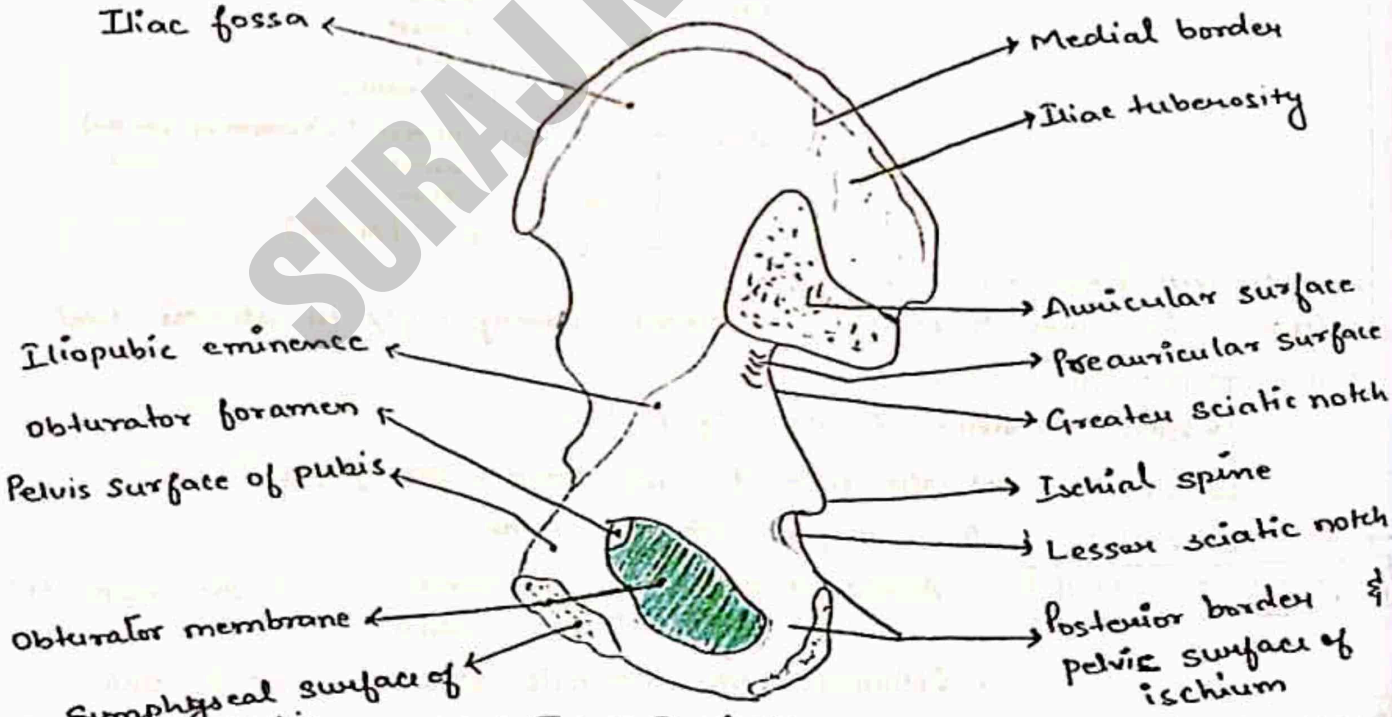


Fig: Inner Surface

SURAJ KUMAR

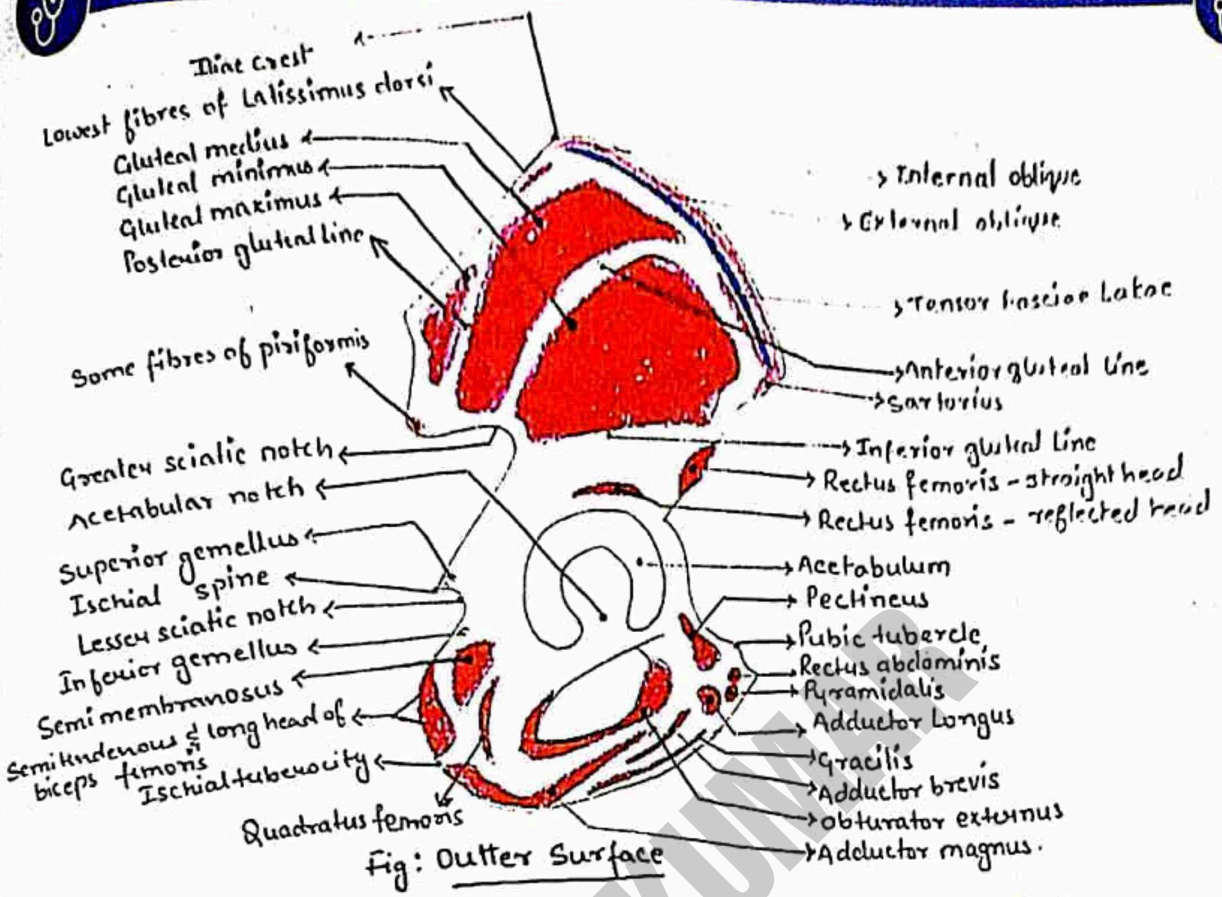


Fig: Outter Surface

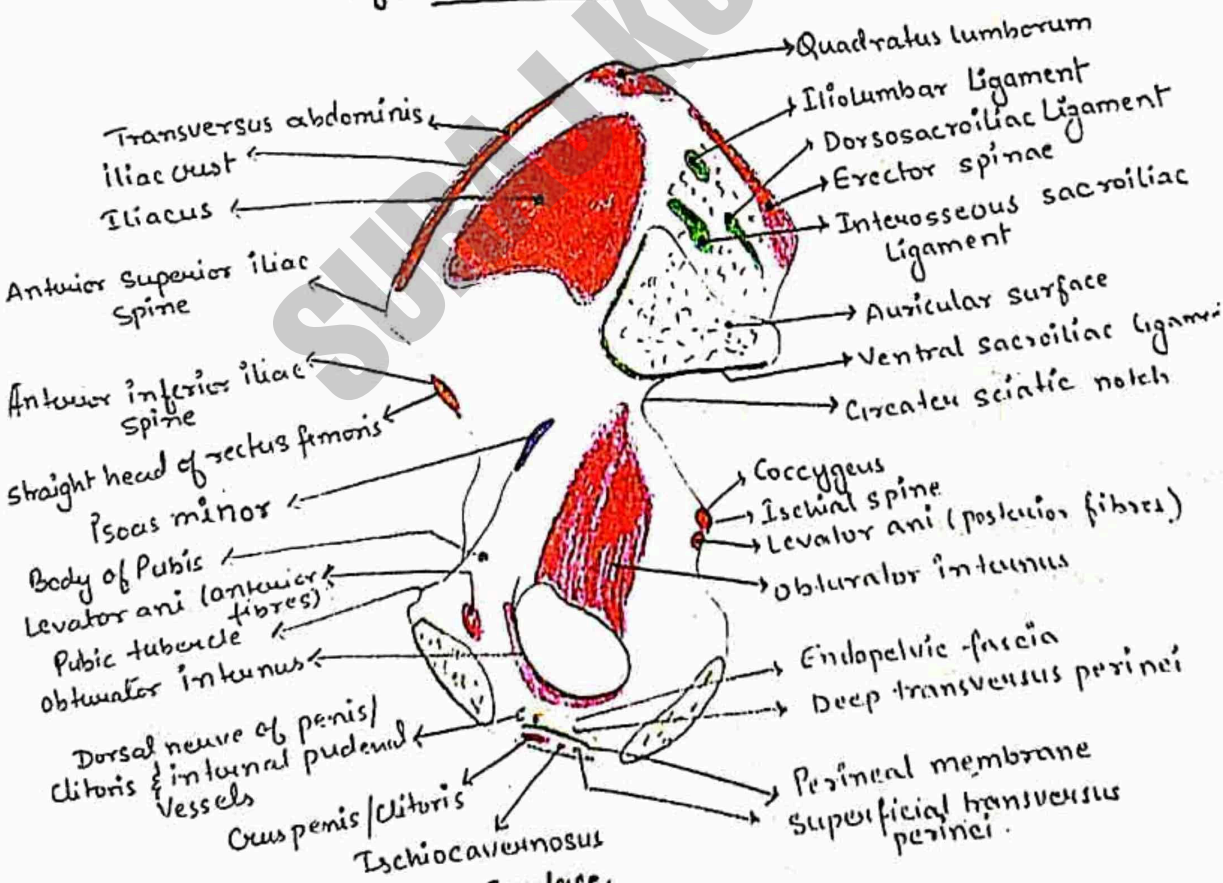


Fig: Inner Surface.

SUJIT KUMAR

⋗ FEMUR ⋖

- Longest & strongest

PARTS

UPPER END
- Head + Neck + Greater and Lesser trochanter

SHAFT
- Narrowest in middle

LOWER END
- Medial + Lateral condyles
- Intercondylar fossa

UPPER END

→ HEAD
- Acetabulam of hip bone — hip joint
- Fovea — ligamentum teres femoris.

→ NECK
- 2 borders and 2 surface

SURFACE → ANTERIOR
- Flat + no. of oblique bony ridges
- intertrochanteric line
- Completely intracapsular

★ INTERTROCHANTERIC LINE
- Spiral line
- Attachment to :
  1. Capsule of hip joint
  2. Iliofemoral ligament
  3. Upper end : Vastus lateralis
     Lower end : Vastus medialis

→ POSTERIOR
- intertrochanteric crest
- Medial half — intracapsular
- Tendon of obturator externus.

★ INTERTROCHANTERIC CREST
- Quadrate tubercle — Quadratus femoris

BORDER → UPPER
- Greater trochanter

→ LOWER
- Lesser trochanter.

→ GREATER TROCHANTER
- 1 border & 3 surface

BORDER → UPPER
- Piriformis

SURFACE → ANTERIOR
- Gluteus minimus - Lateral
- Trochanteric bursa of Gluteus minimus — Medial

→ MEDIAL
- Trochanteric fossa — obturator externus.
- Insertion of obturator internus + Gemellus superior and Gemellus inferior.

→ LATERAL
- Ridge — Gluteus medius muscle
- Ar areas — Anterior - Gluteus medius
  Posterior - Gluteus maximus

→ LESSER TROCHANTER
- Insertion to Psoas major
- Attachment of Iliacus.

SURAJ KUMAR

SHAFT

- It has 5 surface & 3 borders

SURFACE → ANTERIOR
→ MEDIAL
→ LATERAL
→ UPPER POSTERIOR
→ LOWER POSTERIOR (POPLITEAL SURFACE)

BORDERS → MEDIAL
→ LATERAL
→ POSTERIOR

LINEA ASPERA (Posterior border)

- Has inner lip + outer lip + intermediate area

UPPER END :–
⇒ Inner Medial Lip → spiral line
⇒ Outer Lateral lip → Gluteal tuberosity
⇒ Posterior surface

LOWER END :–
⇒ Inner Medial Lip → Medial Supracondylar line → Adductor tubercle
⇒ Outer Lateral lip → Lateral Supracondylar line
⇒ Popliteal surface

ATTACHMENTS → INTERMUSCULAR SEPTA
- Medial intermuscular septum – Medial lip
- Lateral intermuscular septum – Lateral lip

→ MUSCULAR
- Vastus intermedius
- Articularis genu
- Vastus lateralis
- Vastus medialis
- Gluteus maximus
- Adductor longus
- Pectineus
- Adductor brevis
- Adductor magnus
- Gastrocnemius
- Plantaris

LOWER END

MEDIAL CONDYLE
- Attachment to upper end of medial collateral ligament
- Adductor tubercle – ischial head of adductor magnus

INTERCONDYLAR FOSSA
- Medial wall – upper end of posterior cruciate ligament
- Lateral wall – upper end of anterior cruciate ligament [LAMP]

LATERAL CONDYLE
- Attachment to fibular collateral ligament of knee jt.
- Origin of lateral head of gastrocnemius.
- Attachment to popliteus

SURAJ KUMAR

Fig: Anterior aspect

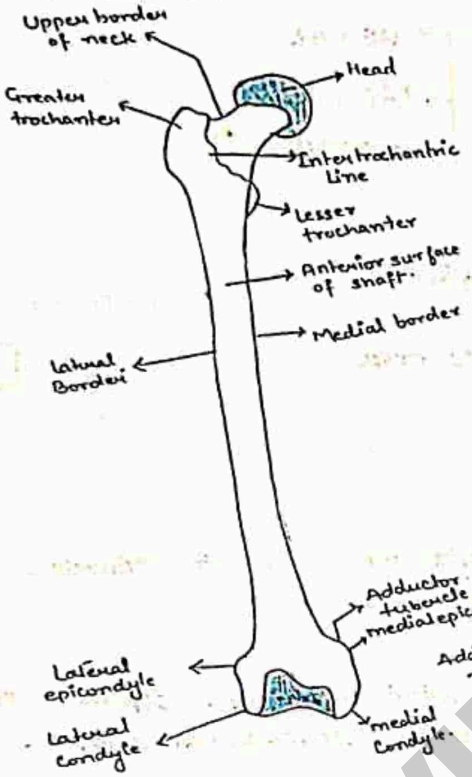
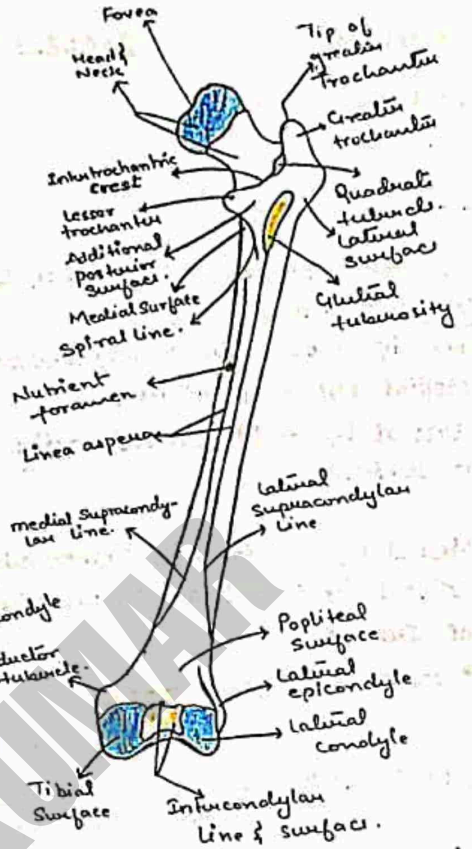
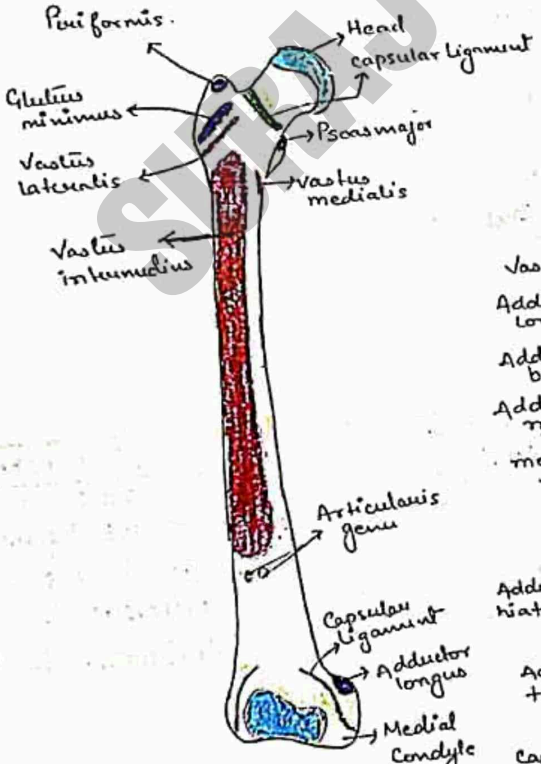


Fig: Posterior aspect





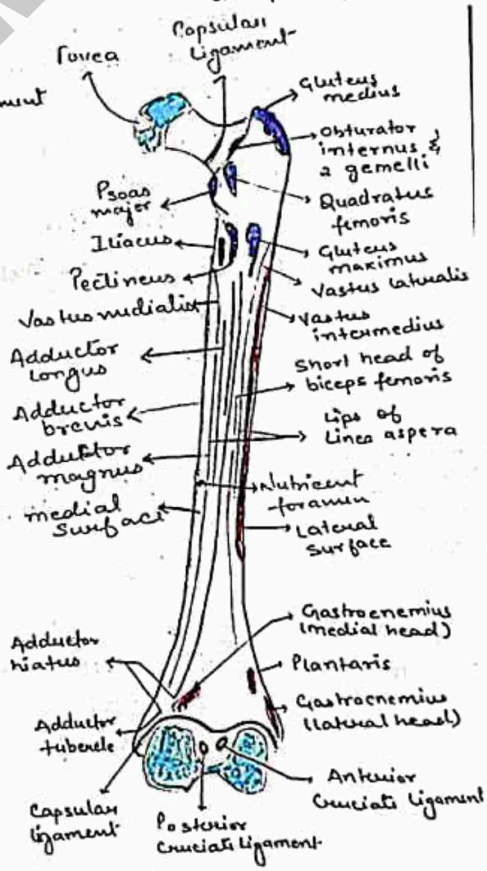


Fig: Anterior aspect

Fig: Posterior aspect

SURAJ KUMAR

FIBULA

- Homologous with ulna
- Pin-like bone

PARTS

UPPER END
- styloid process

SHAFT

LOWER END
- Malleolar fossa

UPPER END

→ HEAD
- Oval or circular articular facet
- styloid process → fibular collateral ligament
- sloping surface → insertion of biceps femoris

→ NECK
- Posterolateral → Common peroneal nerve
- Medial → Anterior tibial artery.

LOWER END

- 4 surface

SURFACE
- → ANTERIOR
  - Anterior talofibular ligament
  - Calcaneofibular ligament
- → POSTERIOR
  - tendons of peroneus brevis & longus.
- → MEDIAL
  - Malleolar fossa
  - Upper → posterior tibiofibular ligament
  - Lower → posterior talofibular ligament
- → LATERAL
  - Subcutaneous.

SHAFT

- 3 borders & 3 surface.

SURFACE
- → MEDIAL (Extensor)
  - Origin to extensor digitorum longus, extensor hallucis longus, peroneus tertius.
- → LATERAL (Peroneal)
  - Origin of Peroneus longus and brevis
- → POSTERIOR (Flexor)
  - Origin of tibialis posterior from medial concave part
  - Origin of soleus and flexor hallucis longus from lateral flattened part
  - Peroneal artery descends along medial crest
  - Presence of nutrient artery.

BORDERS
- → ANTERIOR
  - Attachment of superior extensor retinaculum and superior peroneal retinaculum
- → POSTERIOR
  - Attachment of posterior intermuscular septum.
- → MEDIAL (Interosseous)

SURAJ KUMAR

TIBIA

. Also known as 'Shinbone' . Second largest bone . Homologous to Radius

PARTS

UPPER END
. Medial & lateral condyles + tibial tuberosity

SHAFT
. 3 borders & 3 surface

LOWER END
. Medial malleolus

UPPER END . Has 2 condyles + intercondylar area + tibial tuberosity.

→ MEDIAL CONDYLE . 4 surface

- SURFACE → SUPERIOR
  - . Articulate with femoral condyle
  - . Medial meniscus (fibrocartilaginous plate)
- → POSTERIOR
  - . Insertion of tendon of semimembranosus.
- → ANTERIOR
- → MEDIAL
  - . Vascular foramina
  - . Attachment to medial patellar retinaculum.

→ LATERAL CONDYLE . 4 surface

- SURFACE → SUPERIOR
  - . Articulate with lateral femoral condyle
  - . Lateral meniscus
- → POSTERIOR
  - . Articular facet for head of fibula
  - . tendon of popliteus.
- → ANTERIOR
  - . Flat or facet – iliotibial tract
- → LATERAL
  - . Along with anterior surface – attachment to lateral patellar retinaculum.

→ INTERCONDYLAR AREA
- . Intercondylar eminence → intercondylar tubercles
- . Attachments to
  1. Anterior horn of Medial meniscus
  2. Anterior cruciate ligament
  3. Anterior horn of Lateral meniscus
  4. Posterior horn of Medial meniscus
  5. Posterior horn of Lateral meniscus
  6. Posterior Cruciate ligament

→ TIBIAL TUBEROSITY
- . Upper smooth – ligamentum patellae
  Lower rough – subcutaneous infrapatellar bursa
- . Deep infrapatellar bursa.

SHAFT . has 3 borders & 3 surfaces

BORDERS
- → ANTERIOR
  - . Sharpest, subcutaneous, sinous
  - . Tibial tuberosity → Anterior border of medial malleolus.
- → MEDIAL
  - . Medial condyle → Posterior border of medial malleolus
- → LATERAL (Interosseous border)
  - . Lateral condyle → Anterior border of fibular notch.
  - . Attachment to interosseous membrane uniting tibia & fibula
  - . Attachment of interosseous inferior tibiofibular ligament.

SURAJ KUMAR

SURFACE
- MEDIAL
  - Almost entirely subcutaneous
  - Upper part — insertion of
    - S - Sartorius
    - G - Gracilis
    - S - Semitendinous
- LATERAL
  - Upper 2/3 — origin to tibialis anterior
  - Lower 1/3 — medial → lateral
    1. Tibialis anterior
    2. Extensor hallucis longus
    3. Anterior tibial artery
    4. Anterior tibial nerve / Deep peroneal nerve
    5. Extensor digitorum longus
    6. Peroneus tertius.
- POSTERIOR
  - Soleal line — from above downward
    1. Fascia covering popliteus
    2. Fascia covering soleus
    3. Soleus
    4. Deep transverse fascia.
  - Ar area above soleal line — insertion to popliteus
  - Nutrient foramen — Nutrient artery branch of posterior tibial artery — largest nutrient artery of body.
  - Flexor digitorum longus — medial line below soleal line
  - Tibialis posterior — lateral area below soleal line.
  - Lower 1/4th - medial to lateral
    1. Tibialis posterior
    2. Flexor digitorum longus
    3. Posterior tibial artery
    4. Tibial nerve
    5. Flexor hallucis longus

LOWER END
- 5 surface & Medial Malleolus.

SURFACES
- ANTERIOR
- POSTERIOR
  - Groove for tendon of tibialis posterior
- MEDIAL
- LATERAL
  - Fibular notch — anterior and posterior tibiofibular ligaments.
    upper part — interosseous tibiofibular ligament
- INFERIOR
  - Articulate with body of talus.

MEDIAL MALLEOLUS
- Attachment to deltoid ligament.
- Lateral surface — medial surface of body of talus
- Posterior surface — vertical groove for tendon of tibialis posterior.
- Medial margin — flexor retinaculum.

SURAJ KUMAR

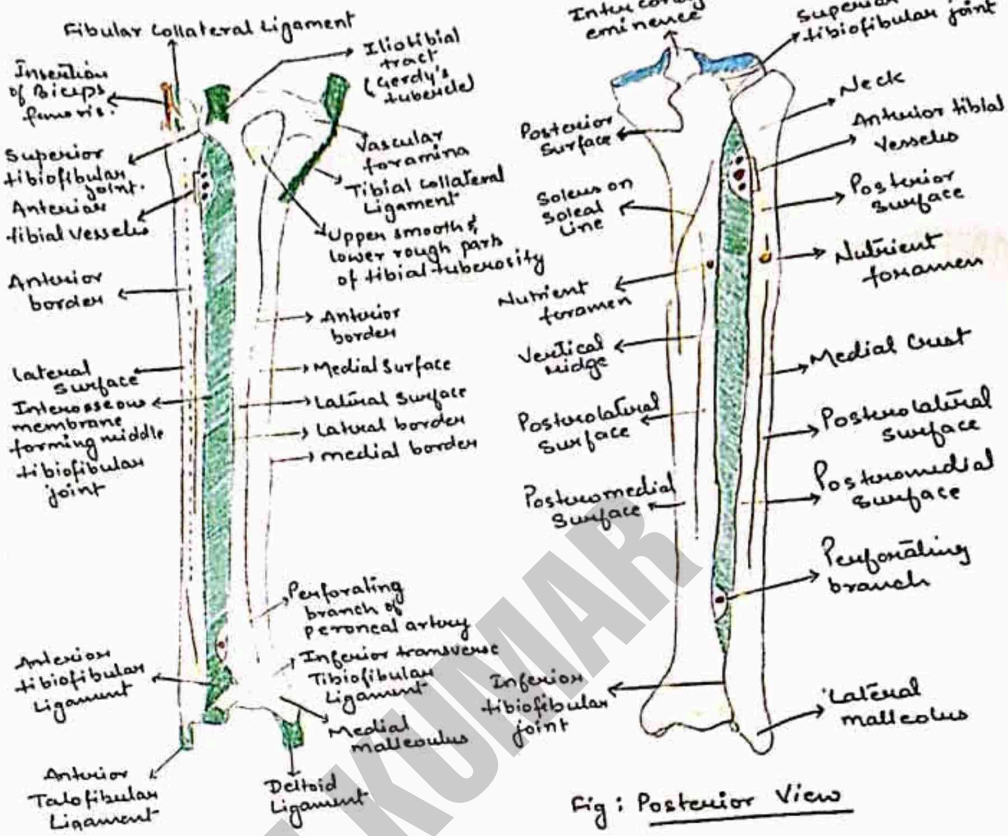


Fig : Posterior View

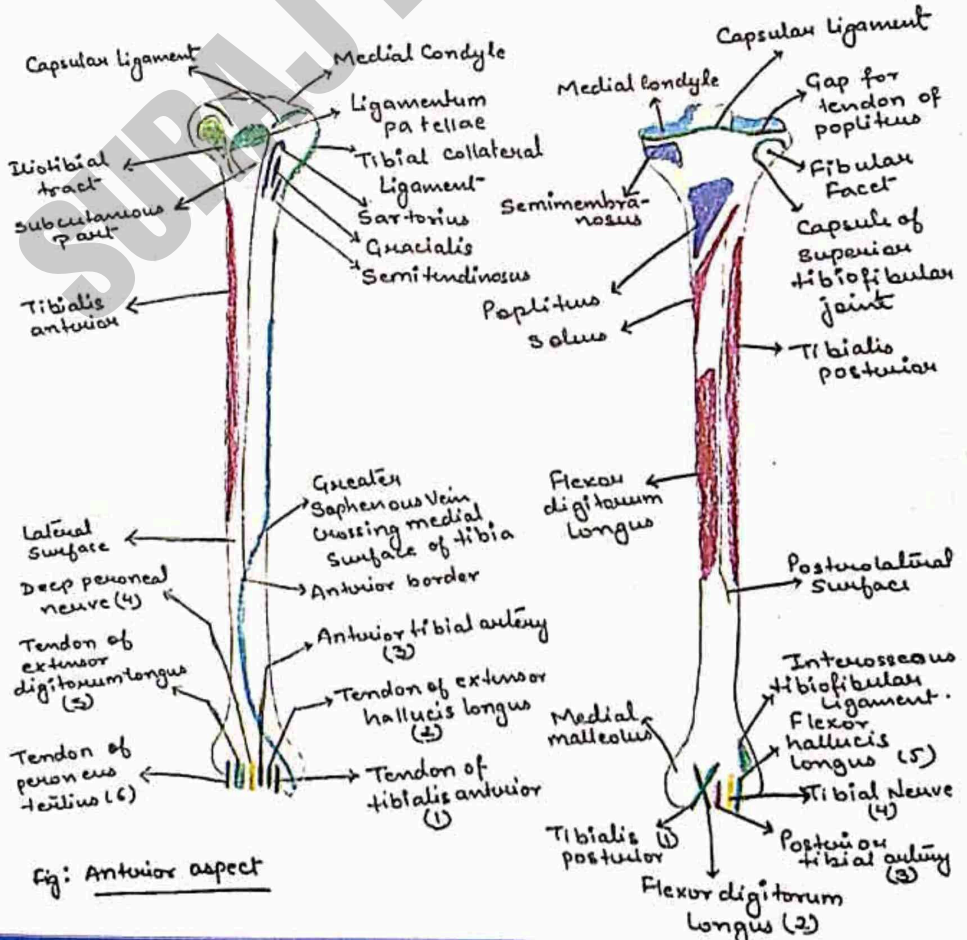
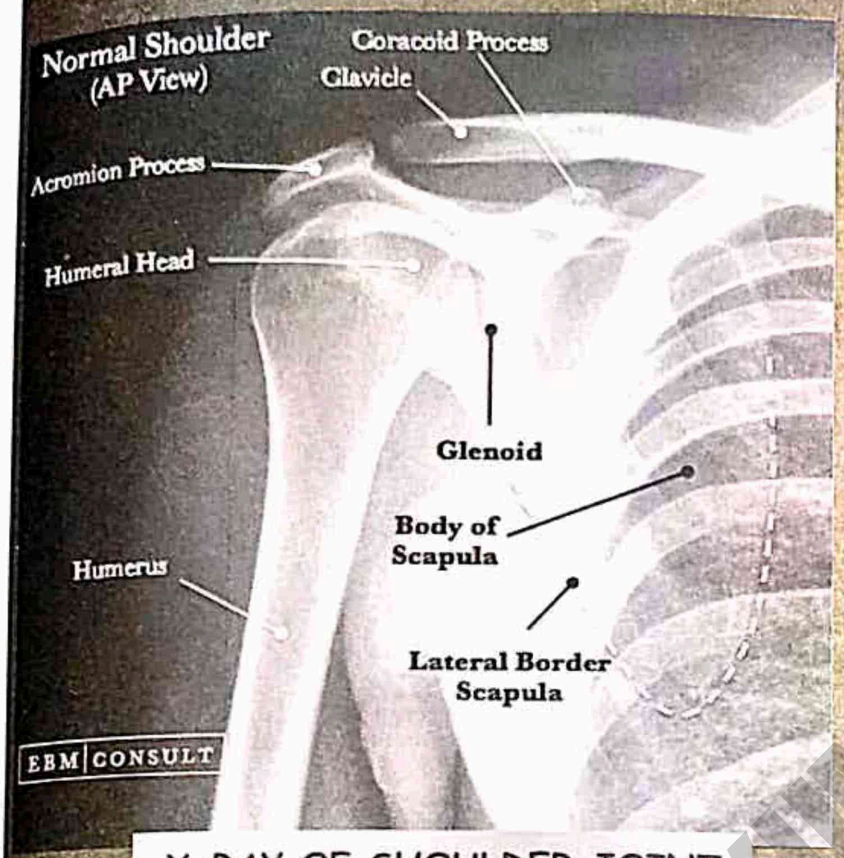


Fig: Anterior aspect

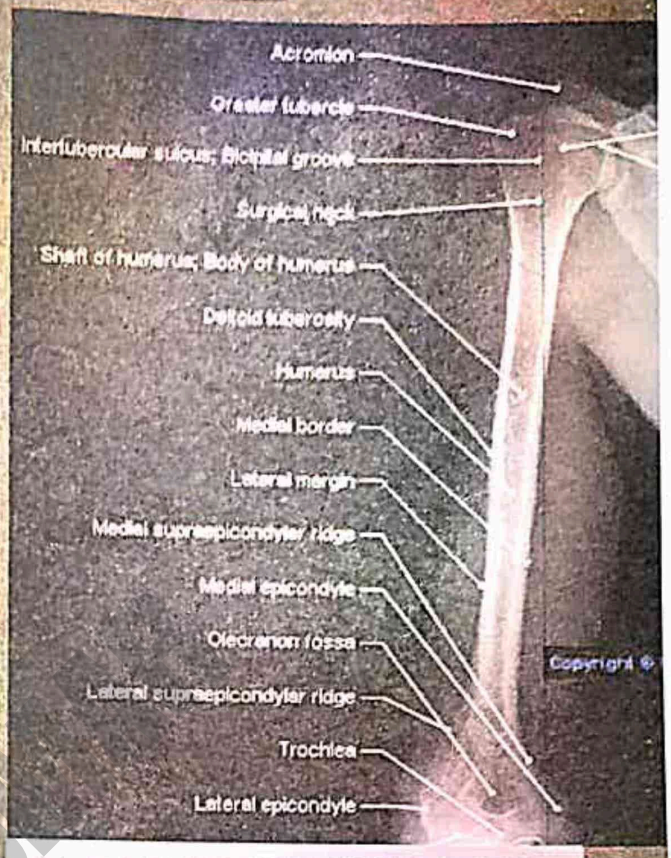
Fig : Po...

SUNIL KUMAR

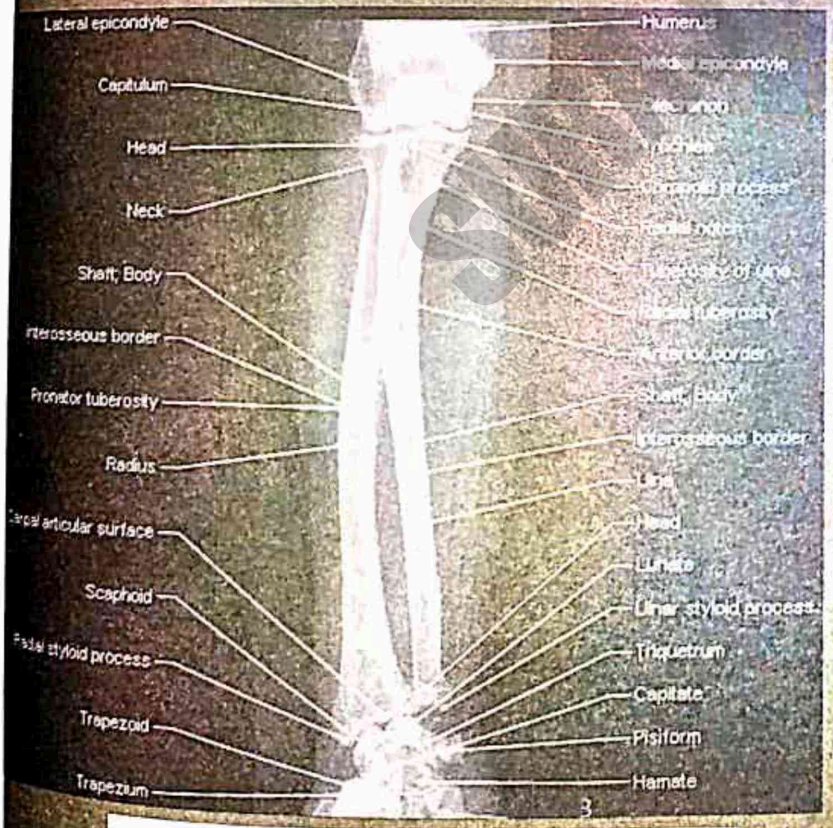
IMP X-RAY OF UPPER LIMB



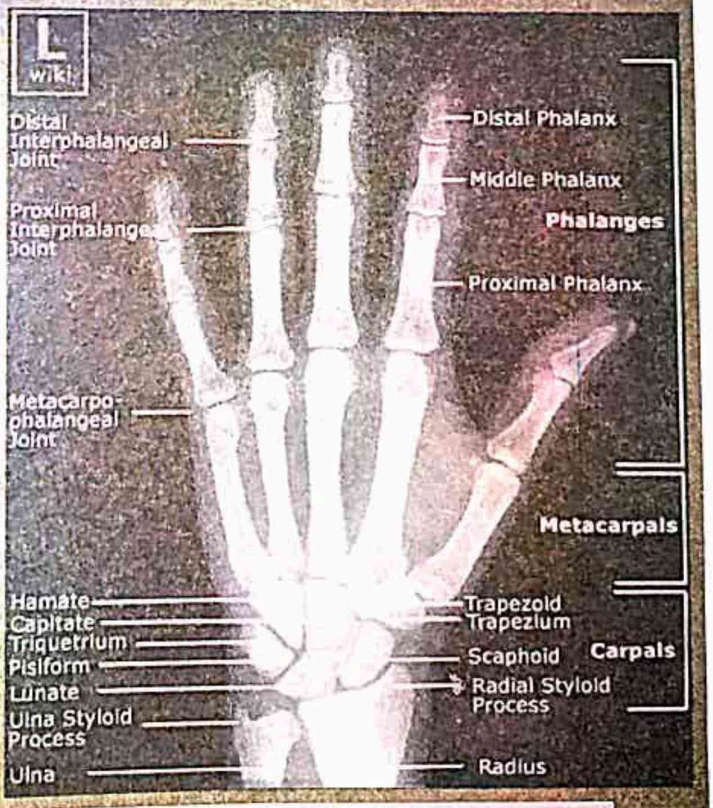
X-RAY OF SHOULDER JOINT



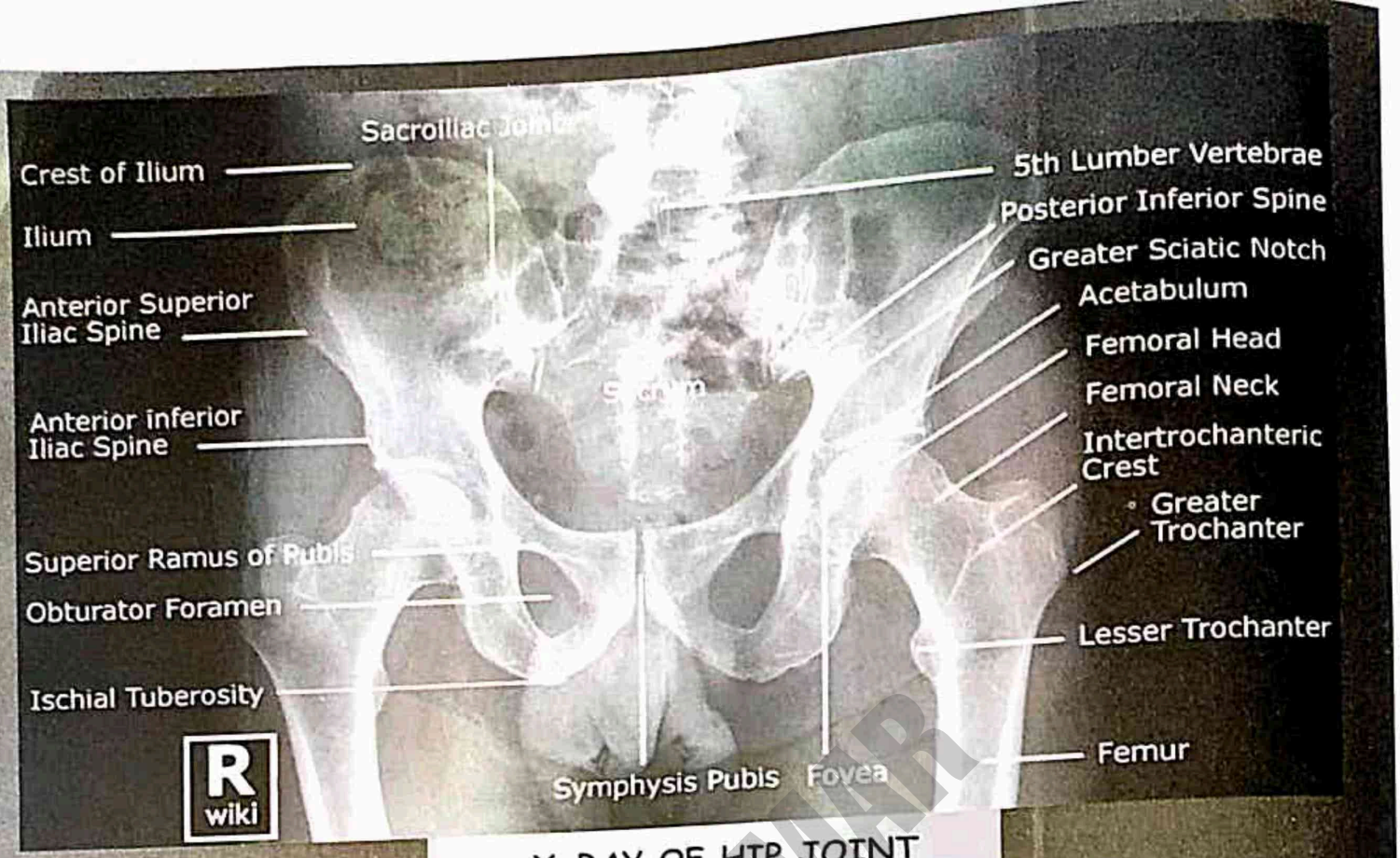
X-RAY OF HUMERUS BONE



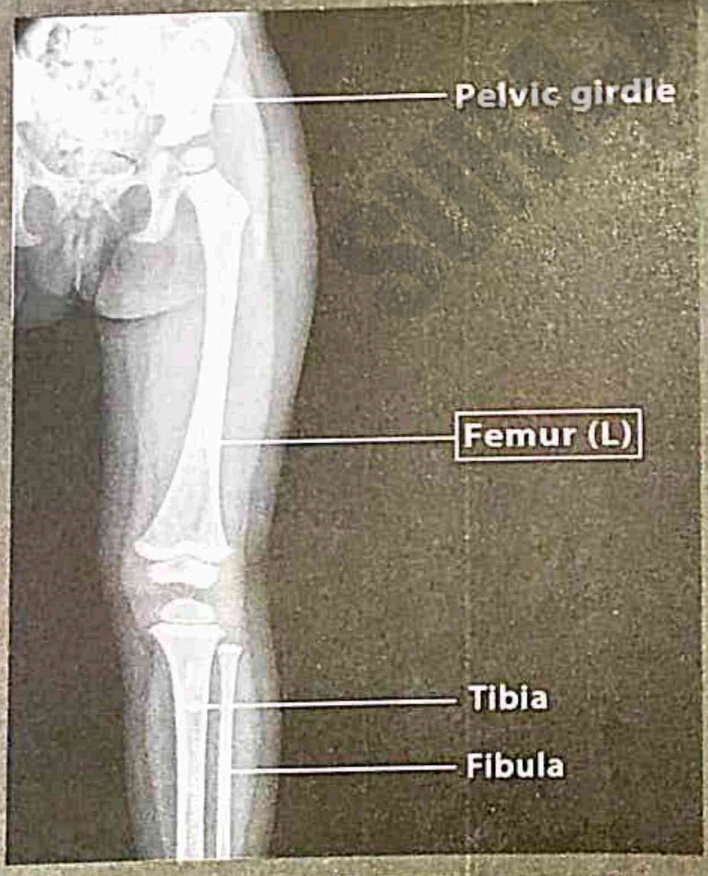
X-RAY OF FOREARM



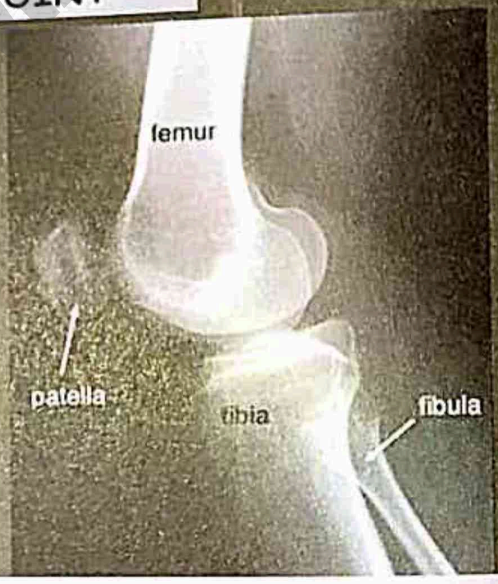
X-RAY OF HAND



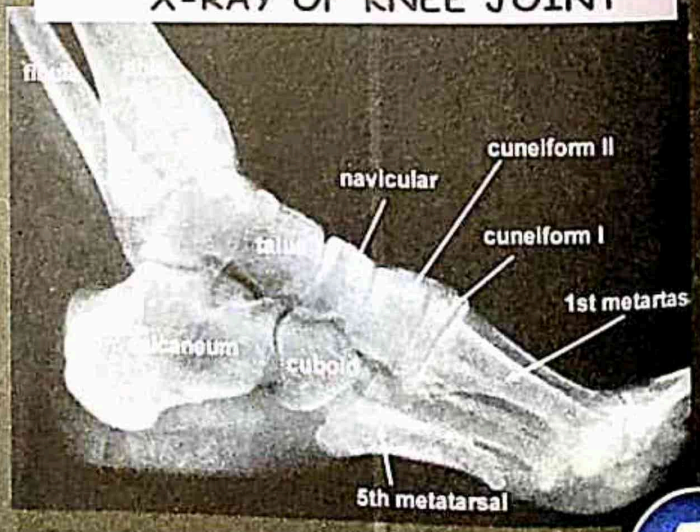
X-RAY OF HIP JOINT

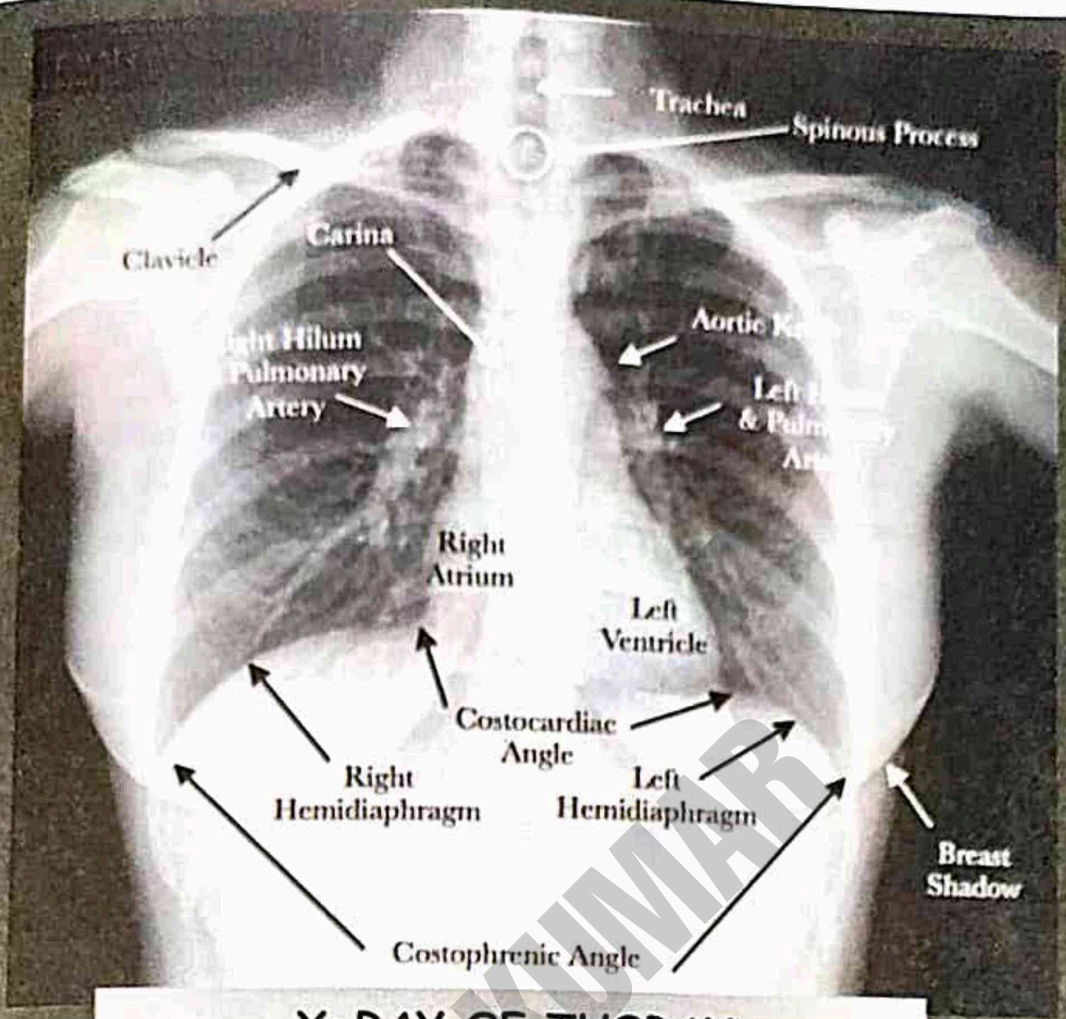


X-RAY OF LOWER LIMB

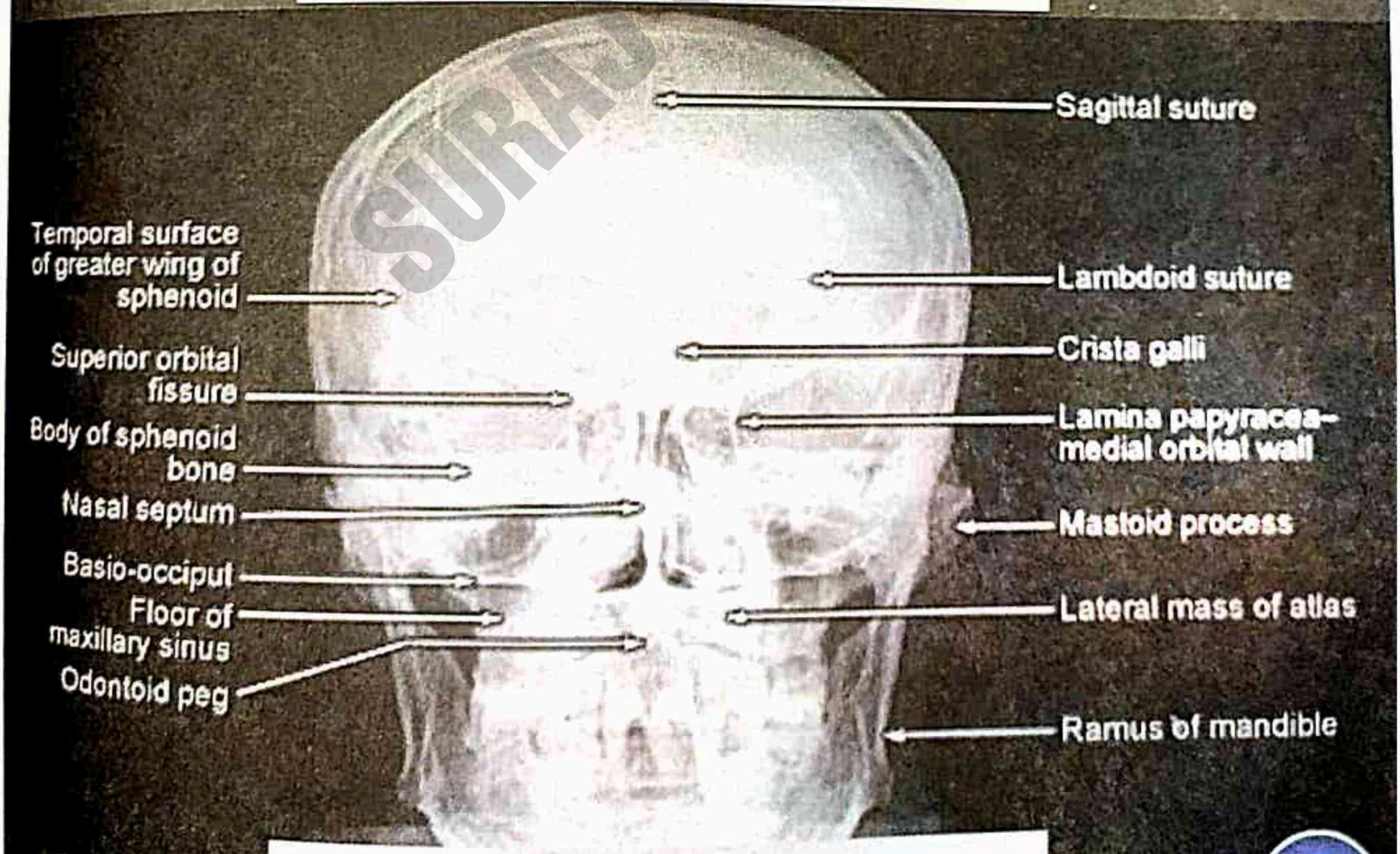


X-RAY OF KNEE JOINT





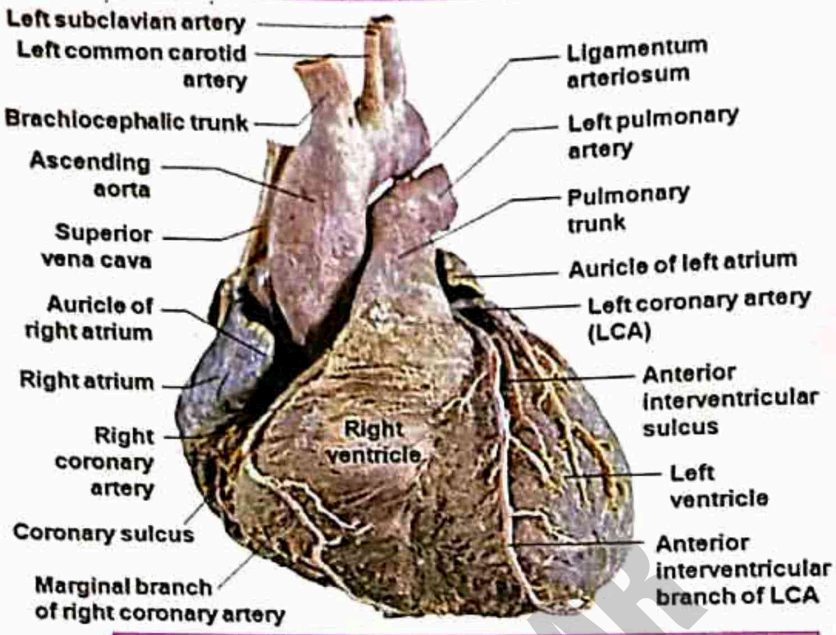
X-RAY OF THORAX



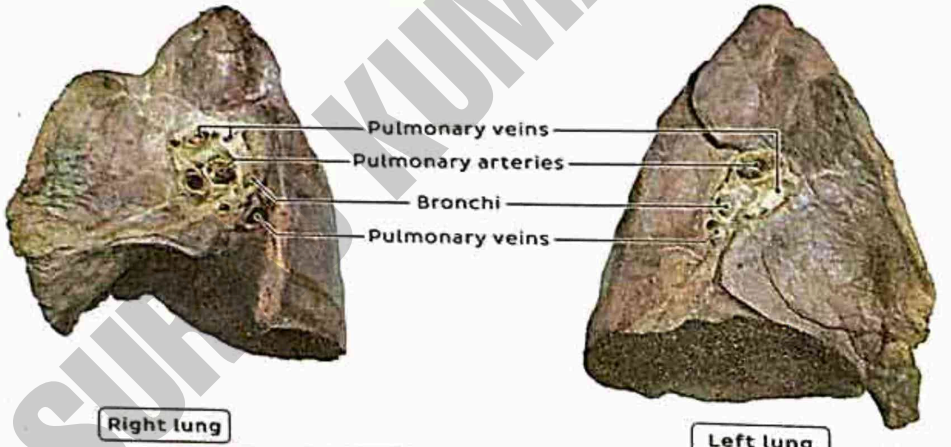
X-RAY OF SKULL

SURAJ KUMAR

IMP CADAVER ANATOMY IMAGES FOR VIVA



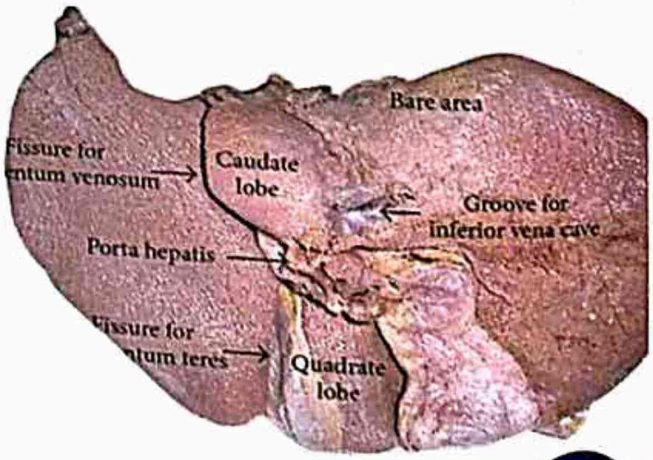
HEART CADAVER



LUNGS CADAVER

www.kenhub.com
KEN HUB

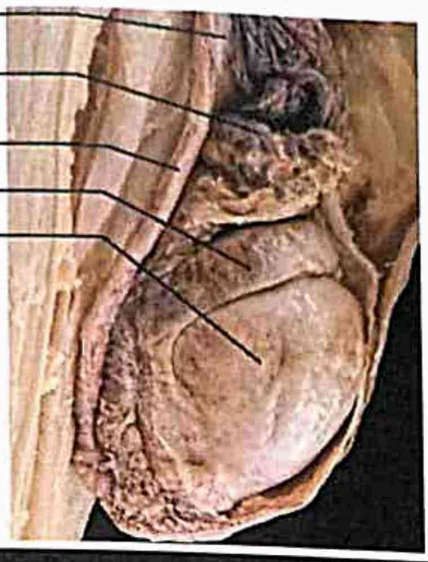
ABDOMEN CADAVER



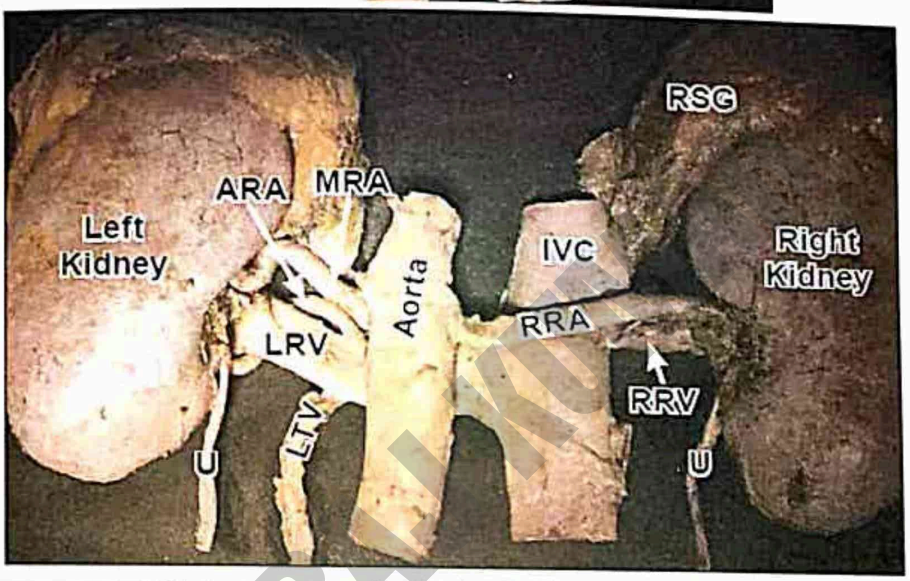
LIVER

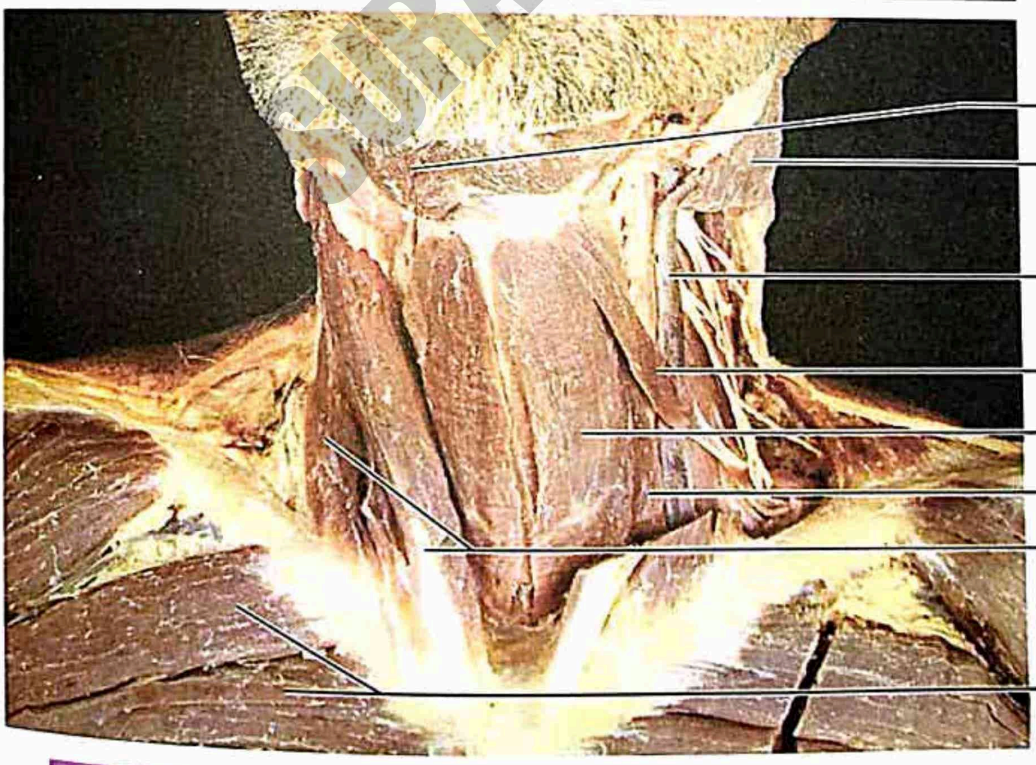
SURESH KUMAR

TESTIS



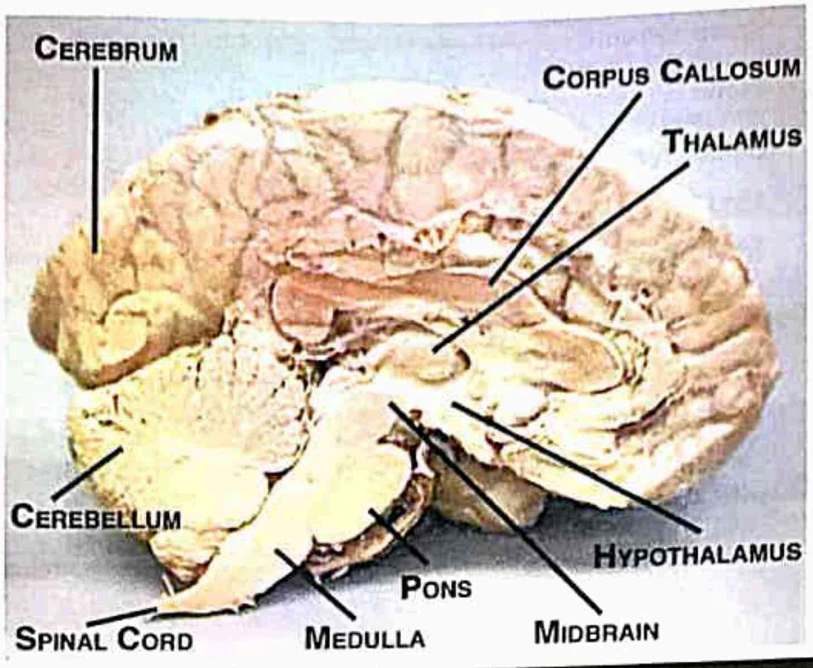
KIDNEY CADAVER



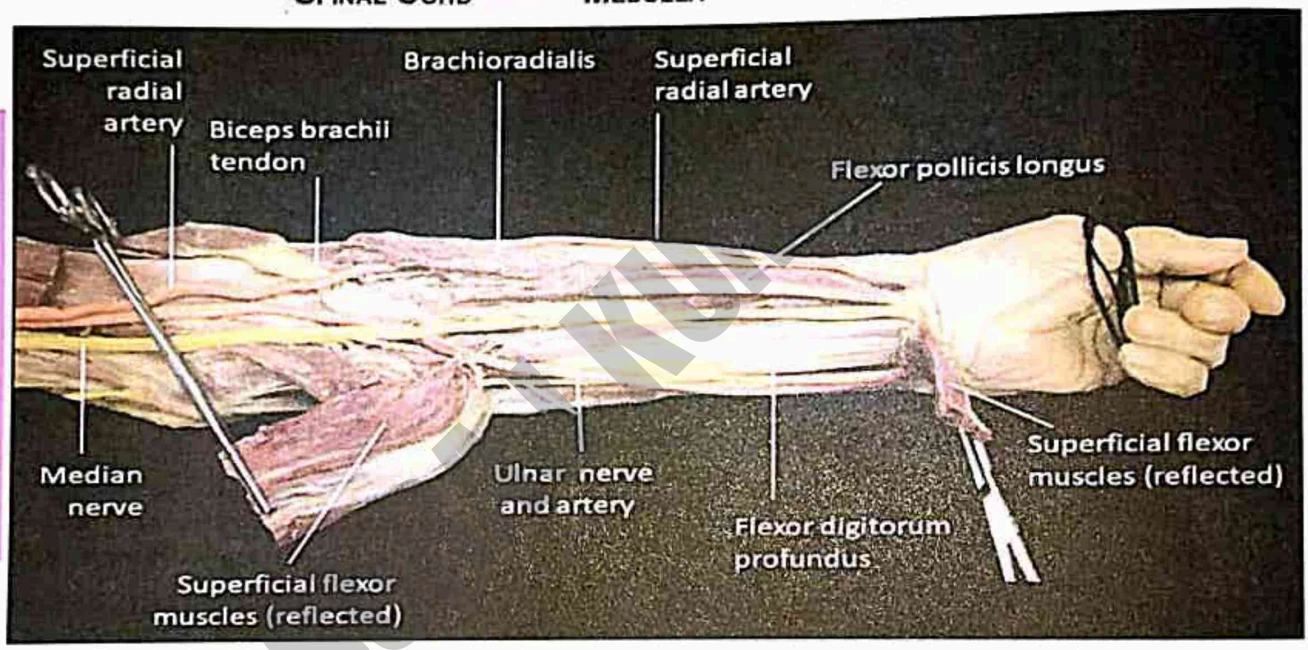


HEAD AND NECK CADAVER

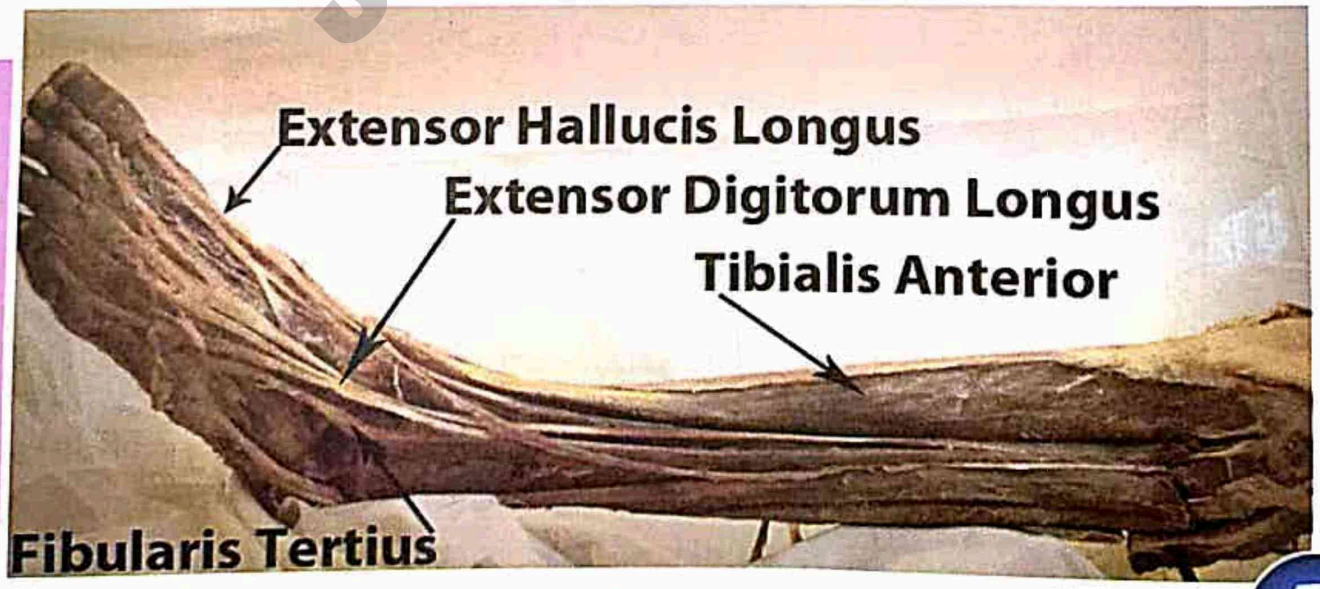
BRAIN CADAVER



UPPER LIMB

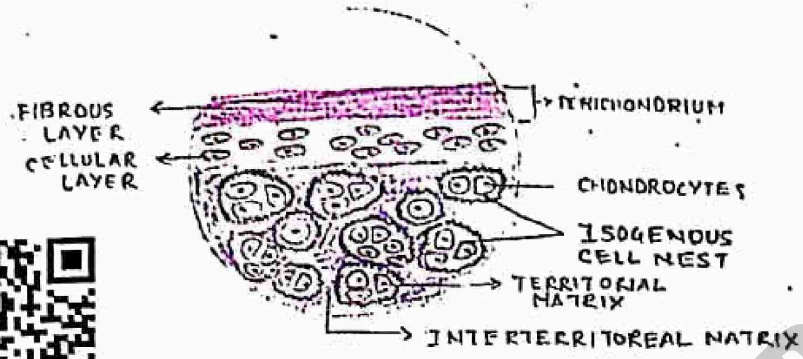


LOWER LIMB



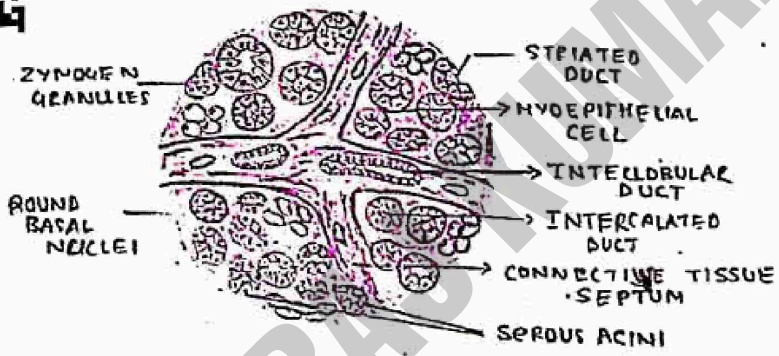
STUDY TIPS FOR MAXIMUM MARKS

1. ONLY ASK IN SHORT NOTES (DRAW A DIAGRAM OF HISTOLOGY)
2. DRAW HISTOLOGY SLIDE DIAGRAM IN ONE PAGE & WRITE 2-3 FEATURES
3. IMP. HISTOLOGY SLIDE DIAGRAMS ARE GIVEN BELOW



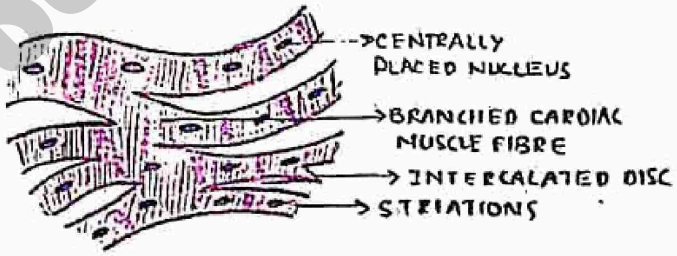
→ It has typical ground glass appearance with homogenous bluish matrix
→ chondrocytes lie within lacunae
→ It is surrounded by perichondrium which is two layered outer & inner
→ Matrix is basophilic

∵ HYALINE CARTILAGE



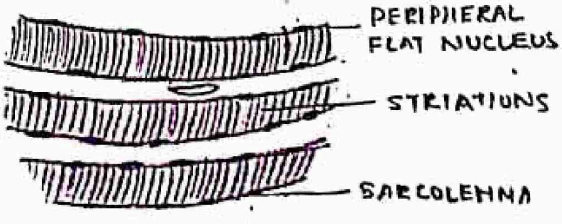
→ Serous acini is small with narrow lumen
→ capsular connective tissue sends septa, which divide glands into lobes & lobules
→ Intralobular ducts are lined by cuboidal cells
→ Interlobular ducts, lined by stratified columnar cells

∵ SEROUS SALIVARY GLAND



→ Found in Heart
→ Show cross-striations, less prominent than skeletal muscle
→ Made up of Cardiac myocytes
→ Has a centrally placed oval nucleus

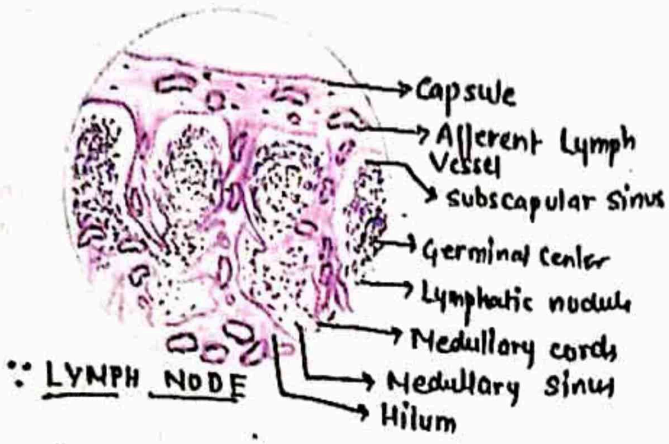
∵ CARDIAC MUSCLE



→ Sarcolemma contains numerous myofibrils
→ Striated appearence is due to regular arrays of actin & myosin
→ Each muscle fibre is variable ranging few mm to many cm.
→ Each muscle fibre is thou multinucleated flat oval

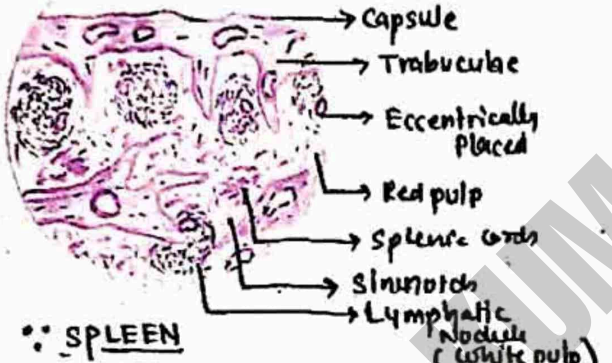
∵ SKELETOL MUSCLE

SURAJ KUMAR



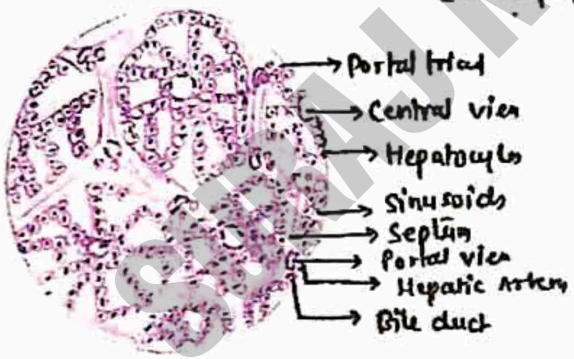
∵ LYNPH NODE

→ Lymph node has outer fibrous capsule
→ cortex contains lymphatic follicles
→ germinal center contain pale staining large, rapidly dividing B-lymphocytes
→ In medulla lymphocytes arranged like branching cords along the sinuses called "Medullary Cords"



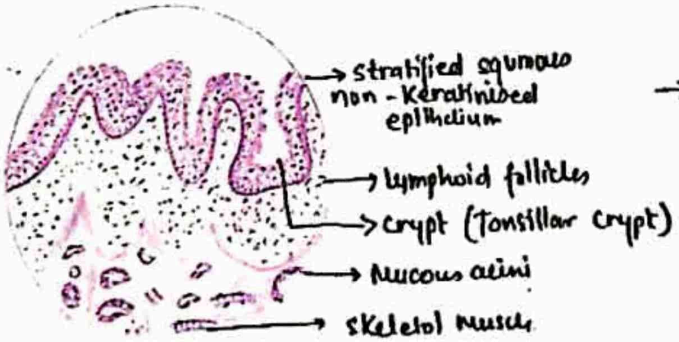
∵ SPLEEN

→ White pulp is aggregation of lymphocytes around arterial-peritarterial lymphatic sheath.
→ Red pulp form major part of Spleen, it is made up of lymphocytes arranged like branching & anastomosing cords.
→ Spleen is largest hemolymphoid organ
→ Spleen has thick capsule



∵ LIVER

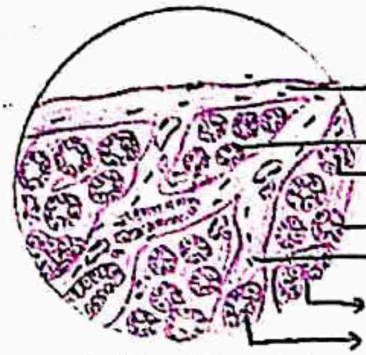
→ Hepatocytes are polyhedral in shape with round nuclei
→ Sinusoids is lined by epithelial cells
→ Portal triad consist of portal vein Hepatic artery & Bile duct
→ Center it contains central vein.



∵ Palatine Tonsil

→ Tonsils are large non-encapsulated mass of lymphoid tissue
→ Each lymph nodule has a germinal center where lymphocytes proliferates
→ Mucous acini is present
→ It is marked by stratified squamous non-keratinized epithelium.

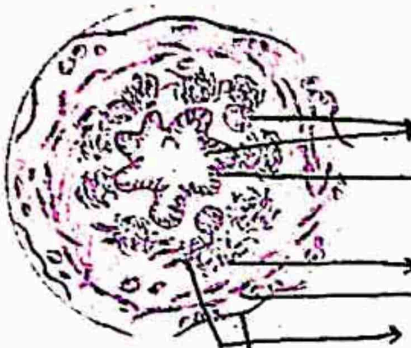
SHAH NUMAN



∴ PANCREAS

→ Pancreatic acini are compo arranged
→ It has thin capsule, septa from capsule divides glan into lobules
→ Islets of Langerhans forms endocrine part of gland
→ Centeroacinar cells are th beginning of intercalated ducts.

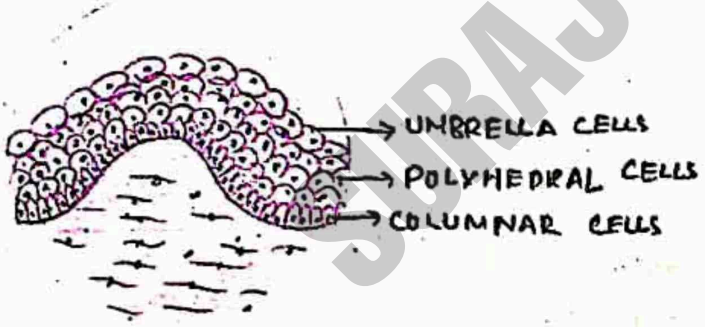
. . . . . × . . . . × . . .



∴ APPENDIX

→ It has 4 layers - mucosa, submucosa, muscularis exter serosa.
→ Mucosa lined by simple columnar cells with microvi
→ Submucosa is made up of loose connective tissue
→ Serosa --> peritoneal coveri made of flat squamous cell

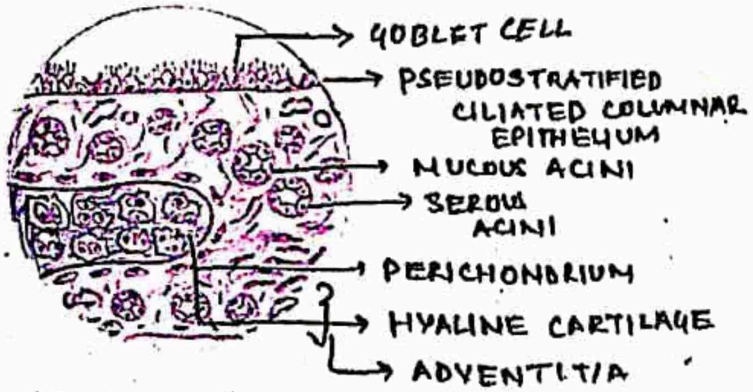
. - - - × - - - × - -



∴ UROTHELIUM

→ Highly specialized stratified epithelium
→ It form inner lining of urinar drainage system
→ It has property of expansion & contraction
→ Transitional / Urothelium is ma up of three types of cell layers: Basal, Intermediate & superficial

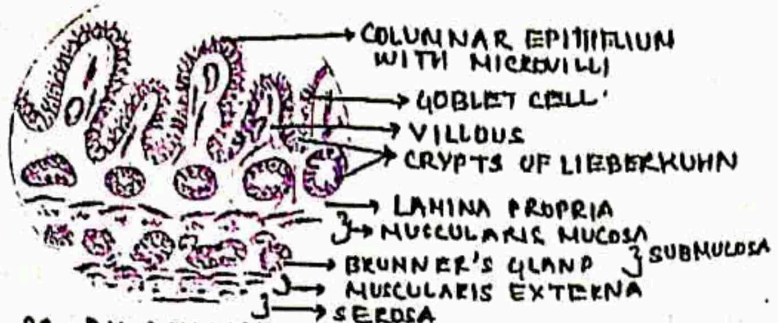
- - - - - × - - - → -



∴ RESPIRATORY EPITHELIUM

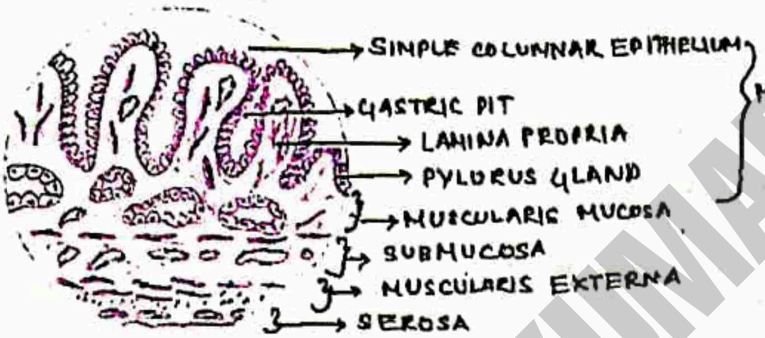
→ Mucosa is lined by pseudostrati ciliated columnar epithelium
→ Goblet cells & gland of submu secret mucus
→ Submucosa - made of CT with blood vessel & nerves
→ Adventitia - made of CT with blood vessel.

SURAJ KUMAR



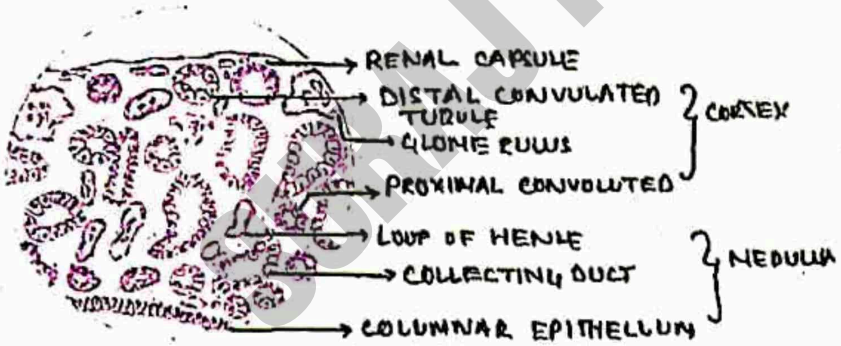
∴ DUODENUM

→ Mucosa - lined by simple columnar cells with microvilli
→ Mucosa thrown into folds called villi for increasing surface area
→ Lamina propria filled with intestinal glands - crypts of Lieberkühn
→ Brunner's gland is present in submucosa



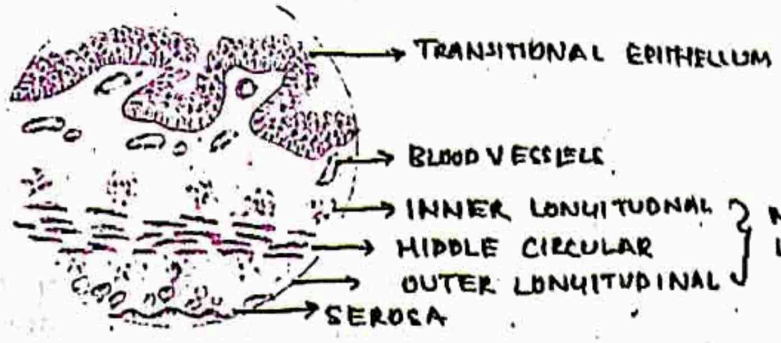
∴ STOMACH - PYLORUS

→ It has 4 layers - mucosa, submucosa, muscularis & serosa
→ Lining epithelium is of simple columnar type
→ It has shows depression Kla gastric pits
→ Pyloric glands is filled in the lamina propria



∴ KIDNEY

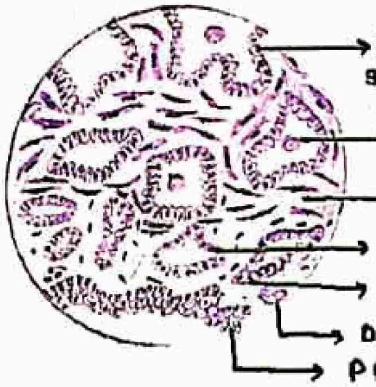
→ outer cortex, inner medulla seen
→ Cortex contains renal corpuscules - Bowman's capsule with glomerular plexus.
→ Section of DCT - pale staining cuboidal cells, large lumen
→ Section of PCT - eosinophil columnar cells with brush border, narrow lumen



∴ URINARY BLADDER

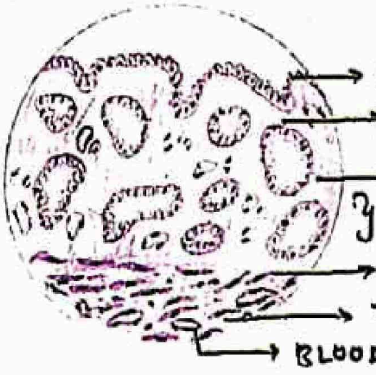
→ Mucosa line by transit epithelium
→ Empty bladder mucosa is thrown into folds. Distended bladder folds disappear
→ Thick muscle layer - inner longitudinal, middle circular & outer longitudinal
→ Serosa lined by flat squ

SURAJ KUMAR



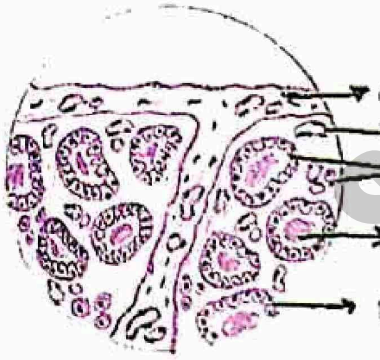
∴ PROSTATE GLAND

→ Tubuloalveolar glands separated by fibromuscular stroma
→ Gland lined by columnar epithelium
→ Lumen contains lamellated round masses of "corpora Amylacea"
→ Fibromuscular stroma is made up of collagen fibres & smooth muscles.



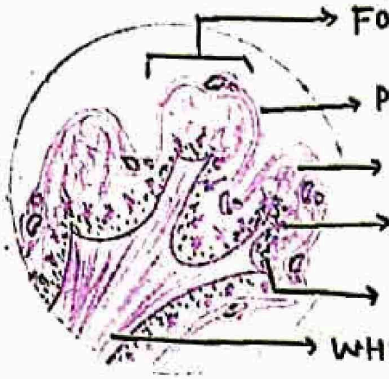
∴ UTERUS - PROLIFERATIVE PHASE

→ 3 layer - outer perimetrium middle myometrium & inner endometrium
→ Endometrium lined by columnar epithelium
→ Stroma is highly cellular contains simple tubular glands & spiral arteries
→ Myometrium is of smooth muscles.



∴ THYROID GLAND

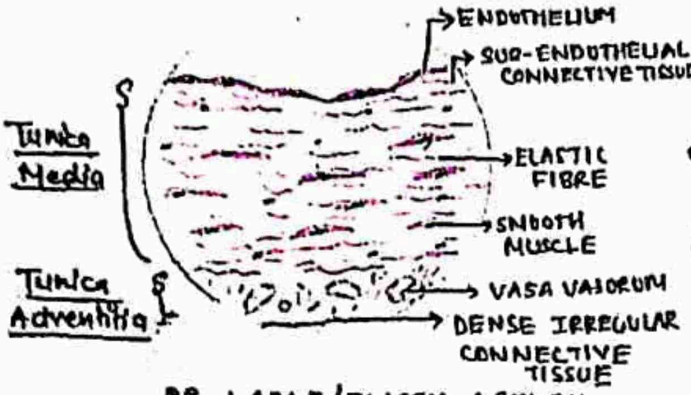
→ Fibrous capsule surrounds the gland
→ Each lobule contain cords/ cluster cells seperated by numerous sinusoids
→ Parafollicular cells are located blw follicular cells & basement membrane.
→ Blood vessels are present



∴ CEREBELLUM

→ Cortex is three layered structure
→ Outer molecular layer
  - Basket cells
  - stellate cells
  - Dendrite of purkinje cell
→ Middle purkinje cell is single layer
→ Inner granular cell layer made up of granule & Golgi cell & Synaptic glomeruli

SUBAJ KUMAR

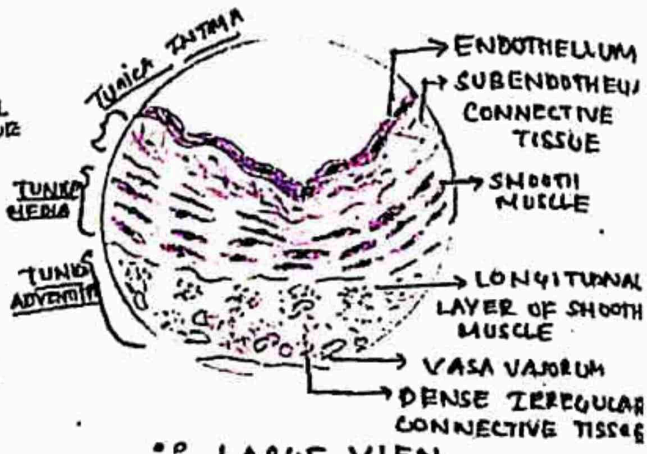


LARGE/ELASTIC ARTERY

→ Tunica intima contains endothelium flat squamous cells lines the lumen

→ Internal elastic lamina made of elastic fibres

→ Tunica media is thickest layer



LARGE VIEN

→ Tunica intima contains flat endothelium flat squamous cells lines the lumen

→ Internal elastic lamina is not well defined

→ Tunica media contain large amount of collagen fibers

SUBAJ KUMAR

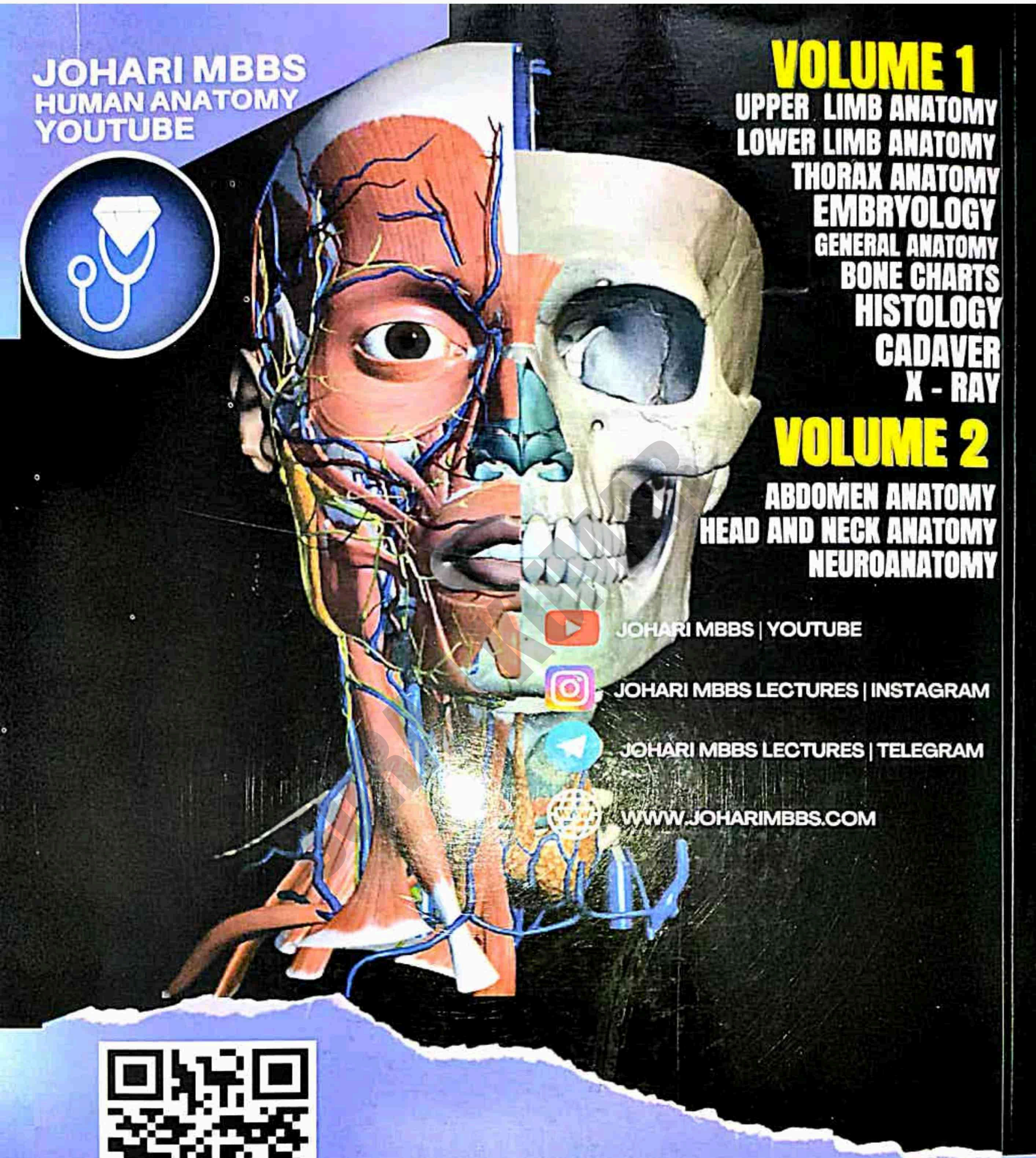
WWW.JOHARIMBBS.COM
FOR MORE BOOKS & OTHER PRODUCT INFORMATION

JOHARI MBBS | YOUTUBE
FOR VIDEO LECTURES

JOHARI MBBS LECTURES | INSTAGRAM
FOR 30SEC REVISION REELS AND NOTES

JOHARI MBBS LECTURES | TELEGRAM
FOR ALL BOOKS & QUESTION PAPER PDFS

SCAN QR CODES WITH QRBOT APP TO DIRECTLY OPEN LINK



JOHARI MBBS
HUMAN ANATOMY YOUTUBE

VOLUME 1
UPPER LIMB ANATOMY
LOWER LIMB ANATOMY
THORAX ANATOMY
EMBRYOLOGY
GENERAL ANATOMY
BONE CHARTS
HISTOLOGY
CADAVER
X - RAY

VOLUME 2
ABDOMEN ANATOMY
HEAD AND NECK ANATOMY
NEUROANATOMY

JOHARI MBBS | YOUTUBE

JOHARI MBBS LECTURES | INSTAGRAM

JOHARI MBBS LECTURES | TELEGRAM

WWW.JOHARIMBBS.COM

www.joharimbbs.com
SCAN QR CODE FOR MORE INFORMATION

For Further Queries and Support
Email :- joharimbbs@gmail.com
Address :- Pulse Computer
Nandwana loha bazar Vidisha (M.P)
YOUTUBE : JOHARIMBBS